Sudoku Variant Catalog

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Contents

1 Introduction

This report describes the Sudoku variants constraints that are currently implemented in our Sudoku solver. These constraints are grouped into different categories, and constraints in the same group are presented in alphabetical order. For each constraint we provide a short description of the constraint, which is also used by the application user interface, we describe the arguments of the constraints, and give an example of the JSON format in which the constraint are specified in their external representation. We then discuss the model used to implement the constraint in our solver, this considers the limited availability of global constraints, in the Choco solver used. We provide links to related constraints, and then give an example of the use of the constraint.

The examples given are collected from the Internet, we provide name and author of the puzzle where known. We also provide one or more links to a YouTube videos which explains the problem and the constraints used, and shows a human solution path. In this document we used the example to clarify the intended meaning of the constraints, we therefore show the initial input grid, and the solved grid showing the unique solution. Where possible, the examples only use a single variant constraint in addition to the standard Sudoku grid. Where this is not possible we indicate which other constraints are used in the example.

1.1 Acknowlegement

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2 Most Commonly Used Constraints

Table 1: Most Commonly Used Constraints

Name	Category	Nr Problems	Section
standard	Base	1085	3.13
cage	Area	81	11.4
arrow	Line	70	9.2
whiteDot	Pair	66	6.10
renban	Line	66	9.32
germanWhisper	Line	63	9.19
blackDot	Pair	56	6.1
thermo	Line	54	9.38
quad	Quadruple	42	8.8
box	Area	40	11.3
regionSumLine	Line	34	9.31
antiKnight	Global	30	4.2
even	Node	30	5.5
mainDiagonal	Global	28	4.9
minorDiagonal	Global	28	4.10
odd	Node	27	5.9
sumX	Pair	27	6.9
sumV	Pair	24	6.8
skyscraper	Outside	18	12.24
blackDotTriple	Triple	16	7.1
antiKing	Global	15	4.1
latin	Base	13	3.9
between	Line	13	9.9
nonConsecutive	Global	12	4.11
oddEvenPair	Pair	12	6.4
entropicLine	Line	12	9.18
palindrome	Line	11	9.30
strictWhiteDot	Global	10	4.31
clone	Area	10	11.8
8x8	Base	9	3.2
windoku	Global	9	4.34
slowThermo	Line	9	9.33
fortress	Area	9	11.10
littleKiller	Outside	9	12.10
strictSumXV	Global	7	4.29
dutchWhisper	Line	7	9.15
numberedRoom	Outside	7	12.13
oddEven	Line	6	9.29
squishdoku	Base	5	3.12
dutchFlatMates	Global	5	4.6
differencePair	Pair	5	6.2
checkerQuad	Quadruple	5	8.2
circleCount	Area	5	11.6

3 Base Constraints

3.1 Constraint 6x6

3.1.1 Category: Base

3.1.2 Description

The puzzle has 6 rows, 6 columns, and six 3x2 blocks, which are 3 cells long and 2 cells high. The cells are labelled with values from 1 to 6.

3.1.3 Arguments

none

3.1.4 JSON Format

```
"rules":["standard","6x6",...],
```

3.1.5 Model

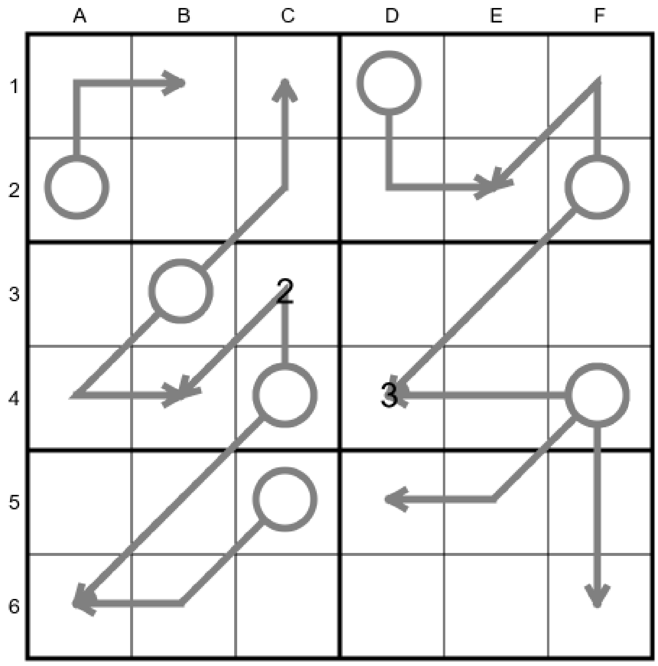
Model description not finished

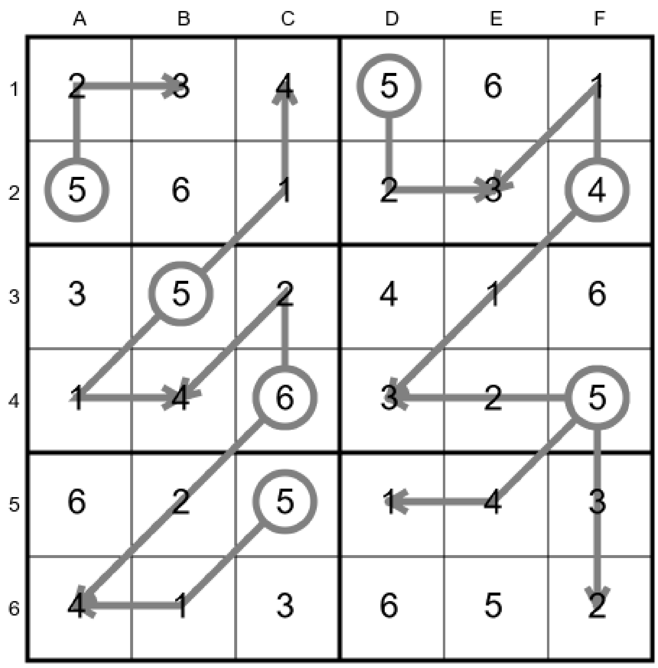
3.1.6 See Also

standard (Section 3.13)

3.1.7 6x6 example timotab84: "Buy None, Get Three (II)" by Philip Newman

Video: `https://youtu.be/mg_-GvgOqoQ?si=-ZkxZ3No5iSj8C15`





Rules: standard (Section 3.13) 6x6 (Section 3.1) arrow (Section 9.2)

3.2 Constraint 8x8

3.2.1 Category: Base

3.2.2 Description

The puzzle has 8 rows, 8 columns, and eight 4x2 blocks, which are 4 cells wide and 2 cells high. The cells are labelled with values from 1 to 8.

3.2.3 Arguments

none

3.2.4 JSON Format

```
"rules":["standard","8x8",...],
```

3.2.5 Model

Model description not finished

3.2.6 See Also

standard (Section 3.13)

3.2.7 8x8 example timotab97: "Nineless" by clover

Video: `https://www.youtube.com/watch?v=Tz9k-sEocPs` (11:18)

		5	2	2	5	4	3	3	2
		A	B	C	D	E	F	G	H
4	1								
4	2		1						
6	3						1		
3	4								
2	5								
5	6			8					
7	7							8	
3	8								

		5	2	2	5	4	3	3	2
		A	B	C	D	E	F	G	H
4	1	2	6	5	3	7	8	4	1
4	2	7	1	4	8	6	3	5	2
6	3	4	2	7	5	3	1	6	8
3	4	6	8	3	1	2	5	7	4
2	5	3	7	6	4	8	2	1	5
5	6	1	5	8	2	4	6	3	7
7	7	5	4	2	6	1	7	8	3
3	8	8	3	1	7	5	4	2	6

Rules: standard (Section 3.13) 8x8 (Section 3.2) sandwich18 (Section 12.22)

3.3 Constraint butterfly

3.3.1 Category: Base

3.3.2 Description

Four 9x9 Sudoku grids are arranged in the corners of a 12x12 grid. Normal Sudoku rules apply to each grid, overlapping cells are shared.

3.3.3 Arguments

none

3.3.4 Model

The grid of this problem type consists of a number of sub grids. In each sub grid, the usual row, column, and block constraints apply. Variables which belong to multiple sub grids are shared. Some cells of the complete grid may not belong to any sub grid, in that case they are ignored.

Table 2: Sub Grids

Name	X	Y	W	H
a	0	0	0	0
b	3	3	3	3
c	0	0	0	0
d	3	3	3	3

Model description not finished

3.3.5 See Also

3.3.6 butterfly example Sudoku Variants90: "Butterfly" by Steven Anderson

	A	B	C	D	E	F	G	H	I	J	K	L
1			1			7			6			3
2	8	9		4	6			1			8	
3		2			3		4	5		6		
4					4	5	8		7	3	9	
5	6							9	4	8		5
6				8			3				7	
7		4		7	8	6			1	4		2
8		8	3	5	1	4				9		
9	1					9		4		7	1	
10				4				1		5		9
11				9	5	2				1		
12			6			8	4			6		

	A	B	C	D	E	F	G	H	I	J	K	L
1	4	3	1	9	5	7	2	8	6	1	4	3
2	8	9	5	4	6	2	7	1	3	5	8	9
3	7	2	6	1	3	8	4	5	9	6	2	7
4	3	1	9	6	4	5	8	2	7	3	9	1
5	6	5	8	2	7	3	1	9	4	8	6	5
6	2	7	4	8	9	1	3	6	5	2	7	4
7	5	4	2	7	8	6	9	3	1	4	5	2
8	9	8	3	5	1	4	6	7	2	9	3	8
9	1	6	7	3	2	9	5	4	8	7	1	6
10	8	9	5	4	6	7	2	1	3	5	8	9
11	4	3	1	9	5	2	7	8	6	1	4	3
12	7	2	6	1	3	8	4	5	9	6	2	7

Rules: butterfly (Section 3.3)

3.4 Constraint cross

3.4.1 Category: Base

3.4.2 Description

Five 9x9 Sudoku grids are arranged in a +, so that the centre grid overlaps each outside grid in three blocks. Normal Sudoku rules apply to each grid, overlapping cells are shared.

3.4.3 Arguments

none

3.4.4 Model

The grid of this problem type consists of a number of sub grids. In each sub grid, the usual row, column, and block constraints apply. Variables which belong to multiple sub grids are shared. Some cells of the complete grid may not belong to any sub grid, in that case they are ignored.

Table 3: Sub Grids

Name	X	Y	W	H
a	6	6	6	6
b	0	0	0	0
c	6	6	6	6
d	6	6	6	6
e	12	12	12	12

Model description not finished

3.4.5 See Also

3.4.6 cross example Sudoku Variants91: "Cross" by Steven Anderson

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1											3		7								
2									9	4		8	3		5						
3							3			9					4						
4												6									
5								5		3	1	2	9								
6							9		8		4		6								
7		7	3		8		2					4	8			4		1	9	7	
8	9			1				4	7	6			2	1		8	3			6	5
9		6			9		8	1	3			9	4	6			5	2			
10	3			7		9		8	5			2	9		4		8			5	
11	7						6	3	2	9	4	7	1	8	5	7			3	4	
12		1	5		3	2		7		5		8					4	5			
13			7		2		9				2	5	3		8	5			7	2	9
14		9	1					5	8		9			2	6	3		4			8
15	6	3		9		5		2		8	3										
16							2	8		5	1	4	6	7							
17										3			4		2						
18							4		3		6	7		5	9						
19							5		2		7										
20							8			1			9		7						
21												8									

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1							4	8	2	5	3	1	7	9	6						
2							1	7	9	4	6	8	3	2	5						
3							3	6	5	9	2	7	1	8	4						
4							7	3	1	8	9	6	5	4	2						
5							6	5	4	3	1	2	9	7	8						
6							9	2	8	7	4	5	6	3	1						
7	1	7	3	5	8	4	2	9	6	1	7	4	8	5	3	4	6	1	9	7	2
8	9	8	2	1	6	3	5	4	7	6	8	3	2	1	9	8	3	7	4	6	5
9	5	6	4	2	9	7	8	1	3	2	5	9	4	6	7	9	5	2	1	8	3
10	3	2	6	7	4	9	1	8	5	3	6	2	9	7	4	6	8	3	2	5	1
11	7	4	9	8	5	1	6	3	2	9	4	7	1	8	5	7	2	9	3	4	6
12	8	1	5	6	3	2	4	7	9	5	1	8	6	3	2	1	4	5	8	9	7
13	4	5	7	3	2	8	9	6	1	7	2	5	3	4	8	5	1	6	7	2	9
14	2	9	1	4	7	6	3	5	8	4	9	1	7	2	6	3	9	4	5	1	8
15	6	3	8	9	1	5	7	2	4	8	3	6	5	9	1	2	7	8	6	3	4
16							2	8	9	5	1	4	6	7	3						
17							6	7	5	3	8	9	4	1	2						
18							4	1	3	2	6	7	8	5	9						
19							5	9	2	6	7	3	1	8	4						
20							8	4	6	1	5	2	9	3	7						
21							1	3	7	9	4	8	2	6	5						

Rules: cross (Section 3.4)

3.5 Constraint expanded

3.5.1 Category: Base

3.5.2 Description

The shape of the grid is not a rectangle, cells can be removed from consideration by marking them with 0.

3.5.3 Arguments

none

3.5.4 JSON Format

```
"rules":["standard","expanded"],
```

3.5.5 Model

Model description not finished

3.5.6 See Also

3.5.7 expanded example wpf gp2025r6c6: "Sudoku GP 2025 Round 6" by

	A	B	C	D	E	F	G	H	I	J	K	L
1						2	3					
2			1				4	5				
3		2		3				6	7			
4			4						8	9		
5										1	2	
6	6										3	4
7	4	8										5
8		9	5									
9			7	9						8		
10				2	8				9		7	
11					1	3				5		
12						4	6					

	A	B	C	D	E	F	G	H	I	J	K	L
1	7	4	6	8	5	2	3	9	1			
2	8	3	1	6	7	9	4	5	2			
3	5	2	9	3	4	1	8	6	7			
4	2	7	4				1	3	8	9	5	6
5	9	5	3				7	4	6	1	2	8
6	6	1	8				9	2	5	7	3	4
7	4	8	2	1	6	7				3	9	5
8	3	9	5	4	2	8				6	1	7
9	1	6	7	9	3	5				8	4	2
10				2	8	6	5	1	9	4	7	3
11				7	1	3	2	8	4	5	6	9
12				5	9	4	6	7	3	2	8	1

Rules: standard (Section 3.13) expanded (Section 3.5)

3.6 Constraint flower

3.6.1 Category: Base

3.6.2 Description

Five 9x9 Sudoku grids are placed in an + shape, overlapping in 3 blocks each. Normal Sudoku rules apply to each grid, overlapping cells are shared.

3.6.3 Arguments

none

3.6.4 Model

The grid of this problem type consists of a number of sub grids. In each sub grid, the usual row, column, and block constraints apply. Variables which belong to multiple sub grids are shared. Some cells of the complete grid may not belong to any sub grid, in that case they are ignored.

Table 4: Sub Grids

Name	X	Y	W	H
a	0	0	0	0
b	3	3	3	3
c	6	6	6	6
d	3	3	3	3
e	3	3	3	3

Model description not finished

3.6.5 See Also

3.6.6 flower example Sudoku Variants87: "Flower" by Steven Anderson

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1					6		5	4		3	7				
2							7	6	3	9	2	8			
3				8		3	2		1	5	4	6			
4		3	8										2	9	
5	6	2	9											5	7
6	5	7	4											6	3
7		5	1											8	
8	7		3										5		9
9		9											4	7	1
10	4		5										7	1	
11	9	8	7										6	3	2
12		1	2										9	4	5
13					1		9	5	7		8				
14				7	2				4	1	9				
15				9	8	6	1	3	2	4	5	7			

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1				2	6	9	5	4	8	3	7	1			
2				4	5	1	7	6	3	9	2	8			
3				8	7	3	2	9	1	5	4	6			
4	1	3	8	9	2	4	6	5	7	1	8	3	2	9	4
5	6	2	9	7	8	5	1	3	4	6	9	2	8	5	7
6	5	7	4	3	1	6	9	8	2	4	5	7	1	6	3
7	2	5	1	6	3	8	4	7	9	2	1	5	3	8	6
8	7	4	3	5	9	2	8	1	6	7	3	4	5	2	9
9	8	9	6	1	4	7	3	2	5	8	6	9	4	7	1
10	4	6	5	8	7	1	2	9	3	5	4	6	7	1	8
11	9	8	7	2	6	3	5	4	1	9	7	8	6	3	2
12	3	1	2	4	5	9	7	6	8	3	2	1	9	4	5
13				3	1	4	9	5	7	6	8	2			
14				7	2	5	6	8	4	1	9	3			
15				9	8	6	1	3	2	4	5	7			

Rules: flower (Section 3.6)

3.7 Constraint gattai-3

3.7.1 Category: Base

3.7.2 Description

Three 9x9 Sudoku grids partially overlap. Normal Sudoku rules apply to each grid, overlapping cells are shared.

3.7.3 Arguments

none

3.7.4 Model

The grid of this problem type consists of a number of sub grids. In each sub grid, the usual row, column, and block constraints apply. Variables which belong to multiple sub grids are shared. Some cells of the complete grid may not belong to any sub grid, in that case they are ignored.

Table 5: Sub Grids

Name	X	Y	W	H
a	3	3	3	3
b	6	6	6	6
c	0	0	0	0

Model description not finished

3.7.5 See Also

3.7.6 gattai-3 example Sudoku Variants92: "Gattai-3" by Steven Anderson

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1						1		7		9		8			
2				8			2		4		5	6			
3						5	8				1				
4				5		4	7		1	2	6	9	5		
5						2			5	3	8	7			2
6							3							7	
7			1		5				7	1				3	
8	6	2	3		4			5						9	
9	5	7		1	2	6				8				1	
10		3	4	5			2				1			5	7
11		1					5		4	9	2	3	6		1
12		6	5			2		1		7	5		4		9
13	1				3	9	7								
14	9	5		6	1	4	3								
15		4	2	7	8	5	9	6	1						

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1				4	6	1	5	7	3	9	2	8			
2				8	9	3	2	1	4	7	5	6			
3				2	7	5	8	9	6	4	1	3			
4				5	3	4	7	8	1	2	6	9	5	4	3
5				6	1	2	9	4	5	3	8	7	1	6	2
6				7	8	9	3	6	2	5	4	1	9	7	8
7	4	9	1	3	5	8	6	2	7	1	9	4	8	3	5
8	6	2	3	9	4	7	1	5	8	6	3	2	7	9	4
9	5	7	8	1	2	6	4	3	9	8	7	5	2	1	6
10	8	3	4	5	7	1	2	9	6	4	1	8	3	5	7
11	2	1	9	8	6	3	5	7	4	9	2	3	6	8	1
12	7	6	5	4	9	2	8	1	3	7	5	6	4	2	9
13	1	8	6	2	3	9	7	4	5						
14	9	5	7	6	1	4	3	8	2						
15	3	4	2	7	8	5	9	6	1						

Rules: gattai-3 (Section 3.7)

3.8 Constraint kazaguruma

3.8.1 Category: Base

3.8.2 Description

Five 9x9 Sudoku grids are arranged so that the centre grid overlaps each outside grid by two blocks. Normal Sudoku rules apply to each grid, overlapping cells are shared.

3.8.3 Arguments

none

3.8.4 Model

The grid of this problem type consists of a number of sub grids. In each sub grid, the usual row, column, and block constraints apply. Variables which belong to multiple sub grids are shared. Some cells of the complete grid may not belong to any sub grid, in that case they are ignored.

Table 6: Sub Grids

Name	X	Y	W	H
a	3	3	3	3
b	12	12	12	12
c	0	0	0	0
d	6	6	6	6
e	9	9	9	9

Model description not finished

3.8.5 See Also

3.8.6 kazaguruma example Sudoku Variants89: "Kazaguruma" by Steven Anderson

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1					1		6		8												
2				2		7					3	8									
3				5	6	8	3	7	2												
4				1		9			4						7		3	9			
5						4			5		8		8				2	6			
6					5			6	3	1	9		9	3	2	7		4		6	
7					2	5	8		6	9			2		4		1	8	7	9	
8						1	4		7	8	6				3		7			4	2
9						6	9				4	2	7		6						
10			7						4	2	8					9	4			2	8
11	8				5	7						9	4	7		2	6	3	9	5	1
12				2					5										4		7
13	9			1	7	4			8	5	1	3	9			6	2	8			
14	7	2					1		3							3					
15	4					3	5						8			5		7			
16	1			7	8	5				6	5	4	7			8					
17		4	2	3						8	7		4	1				6			
18	5		8	4			9			9			6			7		4			
19										2	4						8	9			
20										3	8		1		4	2	6				
21												5	2	8		4	7	3			

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1				9	1	3	6	4	8	5	2	7									
2				2	4	7	5	1	9	6	3	8									
3				5	6	8	3	7	2	4	1	9									
4				1	3	9	2	8	4	7	5	6	5	6	7	1	3	9	2	8	4
5				6	7	4	1	9	5	2	8	3	8	4	1	5	2	6	3	7	9
6				8	5	2	7	6	3	1	9	4	9	3	2	7	8	4	1	6	5
7				4	2	5	8	3	6	9	7	1	2	5	4	3	1	8	7	9	6
8				3	9	1	4	2	7	8	6	5	1	9	3	6	7	5	8	4	2
9				7	8	6	9	5	1	3	4	2	7	8	6	4	9	2	5	1	3
10	2	5	7	8	3	1	6	9	4	2	8	7	3	1	5	9	4	7	6	2	8
11	8	6	4	9	5	7	3	1	2	6	5	9	4	7	8	2	6	3	9	5	1
12	3	1	9	2	4	6	7	8	5	1	3	4	6	2	9	8	5	1	4	3	7
13	9	3	5	1	7	4	2	6	8	5	1	3	9	4	7	6	2	8			
14	7	2	6	5	9	8	1	4	3	7	9	8	5	6	2	3	4	1			
15	4	8	1	6	2	3	5	7	9	4	2	6	8	3	1	5	9	7			
16	1	9	3	7	8	5	4	2	6	6	5	4	7	9	3	8	1	2			
17	6	4	2	3	1	9	8	5	7	8	7	2	4	1	5	9	3	6			
18	5	7	8	4	6	2	9	3	1	9	3	1	6	2	8	7	5	4			
19										2	4	7	3	5	6	1	8	9			
20										3	8	9	1	7	4	2	6	5			
21										1	6	5	2	8	9	4	7	3			

Rules: kazaguruma (Section 3.8)

3.9 Constraint latin

3.9.1 Category: Base

3.9.2 Description

The puzzle must be a Latin square, the value in each row and each column must be pairwise different.

3.9.3 Arguments

none

3.9.4 JSON Format

```
"rules":["latin",...],
```

3.9.5 Model

The grid forms a *Latin Square*. This means that within every row and every column the variables are pairwise different. This can be expressed by a set of alldifferent constraints for the rows, and another for the columns.

$$\forall_{R \in Rows}: \quad \text{alldifferent}([x|x \in R]) \tag{1}$$

$$\forall_{C \in Columns}: \quad \text{alldifferent}([x|x \in C]) \tag{2}$$

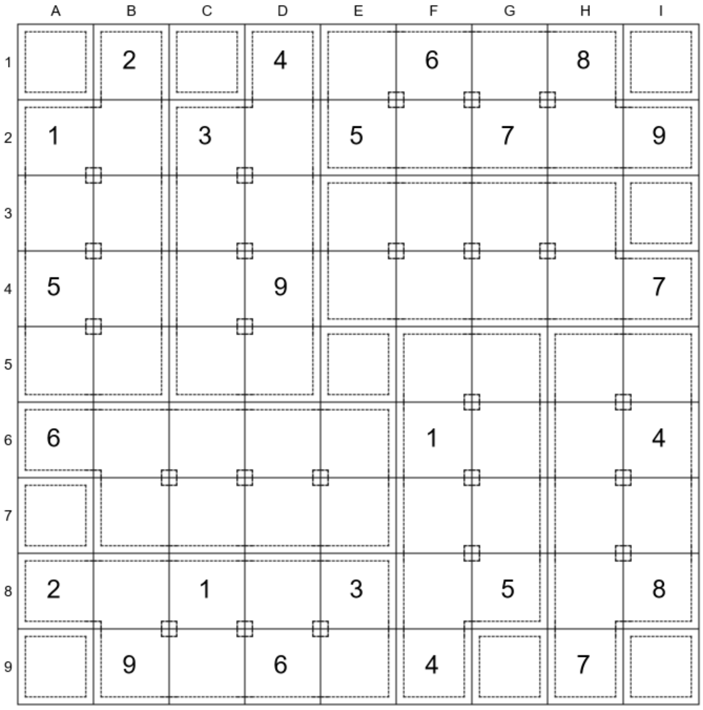
The difference to the standard Sudoku problem is that we do not have the constraints on the regular blocks. These are often replaced by constraints on irregular regions, and are typically expressed by a set of box constraints.

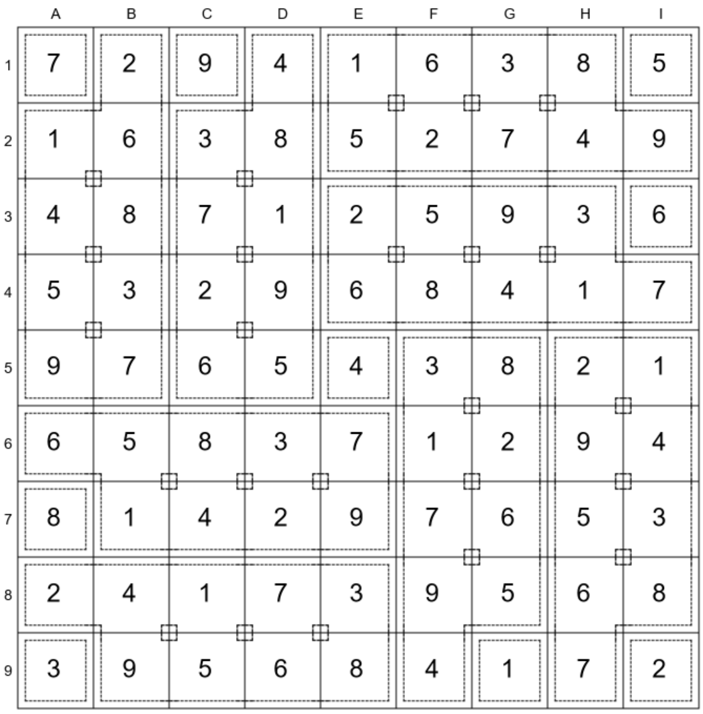
3.9.6 See Also

standard (Section 3.13)

3.9.7 latin example gas65: "Feeling Scattered" by clover

Video: `https://www.youtube.com/watch?v=P8s33SqPctk`





Rules: latin (Section 3.9) box (Section 11.3)

3.10 Constraint samurai

3.10.1 Category: Base

3.10.2 Description

Five 9x9 Sudoku grids are placed in an X shape, overlapping in one block. Normal Sudoku rules apply to each grid, overlapping cells are shared.

3.10.3 Arguments

none

3.10.4 Model

The grid of this problem type consists of a number of sub grids. In each sub grid, the usual row, column, and block constraints apply. Variables which belong to multiple sub grids are shared. Some cells of the complete grid may not belong to any sub grid, in that case they are ignored.

Table 7: Sub Grids

Name	X	Y	W	H
a	0	0	0	0
b	12	12	12	12
c	6	6	6	6
d	0	0	0	0
e	12	12	12	12

Model description not finished

3.10.5 See Also

3.10.6 samurai example Sudoku Variants86: "Samurai" by Steven Anderson

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1			3		4	8			9				6	3	4	7				8	
2	4			5				3													
3		9		6		2		7	8						1	6		9	3	7	2
4		3											3	2	6						8
5	9			4	2									4		9	8				
6	5		2	9			8									1	3			6	
7			7	2	5					7	2	1			8						
8					6				7						3	4	9		8		
9	3			8	1				5			6				8	6	3	2		
10									9				6		1						
11							8			1		7			2						
12											6	2			9						
13	9	8					6						8	9	4						
14			5				4		8	9		5			6		3		5	4	9
15	1			4						6						4	6	9	7		
16	8				4			1					6	7		9	5		8		
17	2		7										9		2		4	3			
18		5			2								4			1	7				6
19	6			2			3	7					2						3	9	8
20		7	1	6			9	2	4				3		8			7			
21				5		9		8	6					6		3	8		2		1

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	6	2	3	7	4	8	5	1	9				6	3	4	7	2	1	9	8	5
2	4	7	8	5	9	1	6	3	2				7	9	2	3	5	8	6	4	1
3	1	9	5	6	3	2	4	7	8				5	8	1	6	4	9	3	7	2
4	7	3	6	1	8	5	2	9	4				3	2	6	5	7	4	1	9	8
5	9	8	1	4	2	6	7	5	3				1	4	5	9	8	6	7	2	3
6	5	4	2	9	7	3	8	6	1				8	7	9	1	3	2	5	6	4
7	8	1	7	2	5	9	3	4	6	7	2	1	9	5	8	2	1	7	4	3	6
8	2	5	9	3	6	4	1	8	7	4	5	9	2	6	3	4	9	5	8	1	7
9	3	6	4	8	1	7	9	2	5	3	8	6	4	1	7	8	6	3	2	5	9
10							2	7	9	5	3	4	6	8	1						
11							8	6	3	1	9	7	5	4	2						
12							5	1	4	8	6	2	3	7	9						
13	9	8	4	7	3	2	6	5	1	2	7	3	8	9	4	7	2	5	6	1	3
14	7	2	5	9	6	1	4	3	8	9	1	5	7	2	6	8	3	1	5	4	9
15	1	3	6	4	5	8	7	9	2	6	4	8	1	3	5	4	6	9	7	8	2
16	8	6	9	3	4	5	2	1	7				6	7	1	9	5	2	8	3	4
17	2	1	7	8	9	6	5	4	3				9	8	2	6	4	3	1	5	7
18	4	5	3	1	2	7	8	6	9				4	5	3	1	7	8	9	2	6
19	6	9	8	2	1	4	3	7	5				2	4	7	5	1	6	3	9	8
20	5	7	1	6	8	3	9	2	4				3	1	8	2	9	7	4	6	5
21	3	4	2	5	7	9	1	8	6				5	6	9	3	8	4	2	7	1

Rules: samurai (Section 3.10)

3.11 Constraint sohei

3.11.1 Category: Base

3.11.2 Description

Four 9x9 Sudoku grids are placed in a + shape, overlapping in a single block. Normal Sudoku rules apply to each grid, overlapping cells are shared.

3.11.3 Arguments

none

3.11.4 Model

The grid of this problem type consists of a number of sub grids. In each sub grid, the usual row, column, and block constraints apply. Variables which belong to multiple sub grids are shared. Some cells of the complete grid may not belong to any sub grid, in that case they are ignored.

Table 8: Sub Grids

Name	X	Y	W	H
a	0	0	0	0
b	6	6	6	6
c	12	12	12	12
d	6	6	6	6

Model description not finished

3.11.5 See Also

3.11.6 sohei example Sudoku Variants88: "Sohei" by Steven Anderson

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1							2					1	6		7						
2							5		6	9	2			8							
3									8				2								
4								4	9	7	5	2			8						
5							8		1		3		7								
6									2	1	8			3							
7						7		8		4		3	9	1		5	8		6		3
8	2		8	5	9		1					8	5	4	6	7	1	3			
9	5			1	8	6			4			6	8		3						4
10	1	5	4			3	8	9						8			2			6	
11	6	9	2		5	8		7						6		4	5				8
12	7	8			6		4	5					1	3	4					2	5
13		2				9				2		5	6	9	8	2	4	5	3		7
14	9	7	6		4		2	1	8	6	3	9		5	7			6	2		9
15			1				5			4		7		2	1	8	9	7			6
16							4				2		9	7	3						
17								9	2		5	4		8							
18							1			3	9				2						
19							6	5	4	8	7		2	1							
20							8		1			2		6							
21							9	2		5			8	3	4						

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1							2	9	3	8	4	1	6	5	7						
2							5	1	6	9	2	7	3	8	4						
3							4	7	8	3	6	5	2	9	1						
4							3	4	9	7	5	2	1	6	8						
5							8	5	1	6	3	4	7	2	9						
6							7	6	2	1	8	9	4	3	5						
7	4	1	9	2	3	7	6	8	5	4	7	3	9	1	2	5	8	4	6	7	3
8	2	6	8	5	9	4	1	3	7	2	9	8	5	4	6	7	1	3	8	9	2
9	5	3	7	1	8	6	9	2	4	5	1	6	8	7	3	9	6	2	1	5	4
10	1	5	4	7	2	3	8	9	6				7	8	5	3	2	9	4	6	1
11	6	9	2	4	5	8	3	7	1				2	6	9	4	5	1	7	3	8
12	7	8	3	9	6	1	4	5	2				1	3	4	6	7	8	9	2	5
13	8	2	5	6	1	9	7	4	3	2	1	5	6	9	8	2	4	5	3	1	7
14	9	7	6	3	4	5	2	1	8	6	3	9	4	5	7	1	3	6	2	8	9
15	3	4	1	8	7	2	5	6	9	4	8	7	3	2	1	8	9	7	5	4	6
16							4	8	5	1	2	6	9	7	3						
17							3	9	2	7	5	4	1	8	6						
18							1	7	6	3	9	8	5	4	2						
19							6	5	4	8	7	3	2	1	9						
20							8	3	1	9	4	2	7	6	5						
21							9	2	7	5	6	1	8	3	4						

Rules: sohei (Section 3.11)

3.12 Constraint squishdoku

3.12.1 Category: Base

3.12.2 Description

This is a non-standard Sudoku where the blocks overlap by one row/column. The resulting grid is 7x7 for nine distinct values, rows/columns and blocks (indicated by dotted lines) must contain each value at most once.

3.12.3 Arguments

none

3.12.4 JSON Format

```
"rules":["squishdoku"],
```

3.12.5 Model

Model description not finished

3.12.6 See Also

3.12.7 squishdoku example timotab77: "Squishdoku" by Bill Murphy

Video: `https://www.youtube.com/watch?v=7oQzA2w6Uos`

	A	B	C	D	E	F	G
1				5		1	
2			2		4		
3		1		3			
4							
5				6		9	
6			5		8		
7		8		7			

	A	B	C	D	E	F	G
1	3	4	6	5	9	1	8
2	7	9	2	1	4	3	5
3	5	1	8	3	7	6	2
4	6	3	9	2	5	8	4
5	2	7	4	6	1	9	3
6	1	6	5	9	8	4	7
7	9	8	3	7	2	5	6

Rules: squishdoku (Section 3.12)

3.13 Constraint standard

3.13.1 Category: Base

3.13.2 Description

This is a standard Sudoku, each row, each column, and each major block (outlined with a wider line) must consist of pairwise different values. The grid has k rows, k columns, and k blocks. The cells are labelled with values from 1 to k.

3.13.3 Arguments

none

3.13.4 JSON Format

```
"rules":["standard",...],
```

3.13.5 Model

In a standard Sudoku of order n, we have n^2 rows, columns and blocks, and cells labelled with values from $1..n^2$. Each block is a square of length n.

We denote the set of rows as *Rows*, the set of columns as *Columns*, and the set of blocks as *Blocks*. We call the union of these sets *Houses*.

The values in each house must be pairwise different. We can express this with an alldifferent constraint for each row, column and block.

$$\forall_{R \in \text{Rows}} : \quad \text{alldifferent}([x|x \in R]) \tag{3}$$

$$\forall_{C \in \text{Columns}} : \quad \text{alldifferent}([x|x \in C]) \tag{4}$$

$$\forall_{B \in \text{Blocks}} : \quad \text{alldifferent}([x|x \in B]) \tag{5}$$

By default, we use *domain consistent* propagators for the alldifferent constraints, this can be overridden in the user interface of the application.

3.13.6 See Also

latin (Section 3.9)

3.13.7 standard example timotab34: "A Sudoku Themed Sudoku" by Philip Newman

Video: `https://www.youtube.com/watch?v=fjuEPxTXFBE`

	A	B	C	D	E	F	G	H	I
1	1		2			4	5		
2	3		4						
3	5	6	7	8					
4			9						
5			1			2	3	4	5
6	4					6			
7		5		4		7	8	9	1
8					5				2
9						3	4	5	6

	A	B	C	D	E	F	G	H	I
1	1	9	2	6	7	4	5	8	3
2	3	8	4	5	2	1	6	7	9
3	5	6	7	8	3	9	2	1	4
4	8	2	9	3	4	5	1	6	7
5	6	7	1	9	8	2	3	4	5
6	4	3	5	7	1	6	9	2	8
7	2	5	3	4	6	7	8	9	1
8	9	4	6	1	5	8	7	3	2
9	7	1	8	2	9	3	4	5	6

Rules: standard (Section 3.13)

3.14 Constraint tripledoku1x2

3.14.1 Category: Base

3.14.2 Description

Three 9x9 Sudoku grids overlap in 1x2 blocks. Normal Sudoku rules apply to each grid, overlapping cells are shared.

3.14.3 Arguments

none

3.14.4 Model

The grid of this problem type consists of a number of sub grids. In each sub grid, the usual row, column, and block constraints apply. Variables which belong to multiple sub grids are shared. Some cells of the complete grid may not belong to any sub grid, in that case they are ignored.

Table 9: Sub Grids

Name	X	Y	W	H
a	0	0	0	0
b	3	3	3	3
c	6	6	6	6

Model description not finished

3.14.5 See Also

3.14.6 tripledoku1x2 example Sudoku Variants96: "Tripledoku" by Steven Anderson

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	1	9			4	3		7	8						
2					2										
3		2	4	9	6			3							
4				4	3										
5	4	8	3		9	6	1		5						
6	7	5	9				3	4	6						
7		3	8		5	9	4	1	7	6		3			
8	9	1		3				5	2			9			
9	5					1		9	3						
10					3			6				7			
11					2	7	9	3							
12					6	5			8	3	9				
13				8	9				1	4				3	
14				7				4	6	9		8			1
15				5		6	2	8	9		3			5	
16								6	8	2	1	3	4	7	5
17							4	3	7					6	
18								5						8	
19							7		4			5	3	9	6
20										1		6			
21										3	7	4	2		8

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	1	9	6	5	4	3	2	7	8						
2	3	7	5	1	2	8	9	6	4						
3	8	2	4	9	6	7	5	3	1						
4	2	6	1	4	3	5	7	8	9						
5	4	8	3	7	9	6	1	2	5						
6	7	5	9	8	1	2	3	4	6						
7	6	3	8	2	5	9	4	1	7	6	8	3			
8	9	1	7	3	8	4	6	5	2	1	7	9			
9	5	4	2	6	7	1	8	9	3	2	4	5			
10				9	3	8	1	6	4	5	2	7			
11				4	2	7	9	3	5	8	1	6			
12				1	6	5	7	2	8	3	9	4			
13				8	9	3	5	7	1	4	6	2	8	3	9
14				7	1	2	3	4	6	9	5	8	7	2	1
15				5	4	6	2	8	9	7	3	1	6	5	4
16							9	6	8	2	1	3	4	7	5
17							4	3	7	5	8	9	1	6	2
18							1	5	2	6	4	7	9	8	3
19							7	1	4	8	2	5	3	9	6
20							8	2	3	1	9	6	5	4	7
21							6	9	5	3	7	4	2	1	8

Rules: tripledoku1x2 (Section 3.14)

3.15 Constraint tripledoku2x2

3.15.1 Category: Base

3.15.2 Description

Three 9x9 Sudoku grids overlap in 2x2 blocks. Normal Sudoku rules apply to each grid, overlapping cells are shared.

3.15.3 Arguments

none

3.15.4 Model

The grid of this problem type consists of a number of sub grids. In each sub grid, the usual row, column, and block constraints apply. Variables which belong to multiple sub grids are shared. Some cells of the complete grid may not belong to any sub grid, in that case they are ignored.

Table 10: Sub Grids

Name	X	Y	W	H
a	0	0	0	0
b	3	3	3	3
c	6	6	6	6

Model description not finished

3.15.5 See Also

3.15.6 tripledoku2x2 example Sudoku Variants97: "Tripledoku" by Steven Anderson

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1		5			4	7		6							
2		6		5	3			9							
3	4	3					7		8						
4	5	8	2		1			7			5	8			
5		7									7				
6	1		4		5	6	8		3	9		1			
7					6	3	5		1		9		8		
8		1	3		9	5	2	8		6			4	9	
9	8		5	2	7	1	6	3	9	4		5	2	7	
10				6			7			1			3	6	9
11				5	2	7			4	8	3				
12					4				2	5	6	7			
13									5			6	9		2
14							4	2		9	5	1	7		
15							9	1	3		7	8			

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	9	5	8	1	4	7	3	6	2						
2	2	6	7	5	3	8	1	9	4						
3	4	3	1	6	2	9	7	5	8						
4	5	8	2	3	1	4	9	7	6	2	5	8			
5	3	7	6	9	8	2	4	1	5	3	7	6			
6	1	9	4	7	5	6	8	2	3	9	4	1			
7	7	2	9	8	6	3	5	4	1	7	9	2	8	3	6
8	6	1	3	4	9	5	2	8	7	6	1	3	4	9	5
9	8	4	5	2	7	1	6	3	9	4	8	5	2	7	1
10				6	3	9	7	5	8	1	2	4	3	6	9
11				5	2	7	1	6	4	8	3	9	5	2	7
12				1	4	8	3	9	2	5	6	7	1	4	8
13							8	7	5	3	4	6	9	1	2
14							4	2	6	9	5	1	7	8	3
15							9	1	3	2	7	8	6	5	4

Rules: tripledoku2x2 (Section 3.15)

3.16 Constraint twodoku1x1

3.16.1 Category: Base

3.16.2 Description

Two 9x9 Sudoku grids overlap in 1x1 blocks. Normal Sudoku rules apply to each grid, overlapping cells are shared.

3.16.3 Arguments

none

3.16.4 Model

The grid of this problem type consists of a number of sub grids. In each sub grid, the usual row, column, and block constraints apply. Variables which belong to multiple sub grids are shared. Some cells of the complete grid may not belong to any sub grid, in that case they are ignored.

Table 11: Sub Grids

Name	X	Y	W	H
a	0	0	0	0
b	6	6	6	6

Model description not finished

3.16.5 See Also

3.16.6 twodoku1x1 example Sudoku Variants93: "Twodoku" by Steven Anderson

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	3				8		2		9						
2			2	6		4									
3	7	1		3	2			6	4						
4		4		8	7			1							
5	9	2	7	1		6	4	8							
6	1		3	9											
7							3	2					7		
8		3			6		1								
9				2			6		7	9				8	
10								3		1		9			7
11							4	1	9			6	5		
12													6	9	
13							9		4	8	1			5	
14							2	5					8		
15							8				2	7	9	1	4

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	3	6	4	7	8	1	2	5	9						
2	8	5	2	6	9	4	7	3	1						
3	7	1	9	3	2	5	8	6	4						
4	6	4	5	8	7	3	9	1	2						
5	9	2	7	1	5	6	4	8	3						
6	1	8	3	9	4	2	5	7	6						
7	4	7	6	5	1	9	3	2	8	4	5	1	7	6	9
8	2	3	8	4	6	7	1	9	5	7	6	8	3	4	2
9	5	9	1	2	3	8	6	4	7	9	3	2	1	8	5
10							5	3	6	1	8	9	4	2	7
11							4	1	9	2	7	6	5	3	8
12							7	8	2	3	4	5	6	9	1
13							9	7	4	8	1	3	2	5	6
14							2	5	1	6	9	4	8	7	3
15							8	6	3	5	2	7	9	1	4

Rules: twodoku1x1 (Section 3.16)

3.17 Constraint twodoku1x2

3.17.1 Category: Base

3.17.2 Description

Two 9x9 Sudoku grids overlap in 1x2 blocks. Normal Sudoku rules apply to each grid, overlapping cells are shared.

3.17.3 Arguments

none

3.17.4 Model

The grid of this problem type consists of a number of sub grids. In each sub grid, the usual row, column, and block constraints apply. Variables which belong to multiple sub grids are shared. Some cells of the complete grid may not belong to any sub grid, in that case they are ignored.

Table 12: Sub Grids

Name	X	Y	W	H
a	0	0	0	0
b	3	3	3	3

Model description not finished

3.17.5 See Also

3.17.6 twodoku1x2 example Sudoku Variants94: "Twodoku" by Steven Anderson

	A	B	C	D	E	F	G	H	I	J	K	L
1									3			
2		2		7	8	9			4			
3			7	1								
4	2							1	7			
5	1	7	5			6		4				
6		4				1	2		5			
7		1	2		4		9		6	1		
8	7								1			
9		5	9			3	7	8	2			5
10					8	1				6		
11						5				4		
12					3		6	7		8	1	9
13					7					3	8	
14					2	6			8			
15					5	8	1	6				

	A	B	C	D	E	F	G	H	I	J	K	L
1	9	8	4	2	6	5	1	7	3			
2	3	2	1	7	8	9	5	6	4			
3	5	6	7	1	3	4	8	2	9			
4	2	9	3	4	5	8	6	1	7			
5	1	7	5	9	2	6	3	4	8			
6	6	4	8	3	7	1	2	9	5			
7	8	1	2	5	4	7	9	3	6	1	2	8
8	7	3	6	8	9	2	4	5	1	7	6	3
9	4	5	9	6	1	3	7	8	2	9	4	5
10				9	8	1	2	4	3	6	5	7
11				7	6	5	8	1	9	4	3	2
12				2	3	4	6	7	5	8	1	9
13				1	7	9	5	2	4	3	8	6
14				4	2	6	3	9	8	5	7	1
15				3	5	8	1	6	7	2	9	4

Rules: twodoku1x2 (Section 3.17)

3.18 Constraint twodoku2x2

3.18.1 Category: Base

3.18.2 Description

Two 9x9 Sudoku grids overlap in 2x2 blocks. Normal Sudoku rules apply to each grid, overlapping cells are shared.

3.18.3 Arguments

none

3.18.4 JSON Format

```
"rules":["twodoku2x2"],
```

3.18.5 Model

The grid of this problem type consists of a number of sub grids. In each sub grid, the usual row, column, and block constraints apply. Variables which belong to multiple sub grids are shared. Some cells of the complete grid may not belong to any sub grid, in that case they are ignored.

Table 13: Sub Grids

Name	X	Y	W	H
a	0	0	0	0
b	3	3	3	3

Model description not finished

3.18.6 See Also

3.18.7 twodoku2x2 example gas74: "Overlapping Sudoku" by Bill Murphy

Video: `https://youtu.be/SRdYdt7Cw0E?si=9v6jrHnNk5ch_Zry`

	A	B	C	D	E	F	G	H	I	J	K	L
1												
2		9	8	7		1	2	5				
3		7		9	3	2		6				
4		5	3					4				
5			7					3	8	2	6	
6		1	4								8	
7		2								7	9	
8		3	6	4	1					5		
9					2					1	4	
10					4		8	7	3		1	
11					5	8	9		1	3	2	
12												

	A	B	C	D	E	F	G	H	I	J	K	L
1	3	4	2	6	5	8	9	7	1			
2	6	9	8	7	4	1	2	5	3			
3	1	7	5	9	3	2	8	6	4			
4	9	5	3	2	8	6	1	4	7	9	5	3
5	2	6	7	1	9	4	5	3	8	2	6	7
6	8	1	4	3	7	5	6	2	9	4	8	1
7	7	2	9	8	6	3	4	1	5	7	9	2
8	5	3	6	4	1	9	7	8	2	5	3	6
9	4	8	1	5	2	7	3	9	6	1	4	8
10				9	4	2	8	7	3	6	1	5
11				7	5	8	9	6	1	3	2	4
12				6	3	1	2	5	4	8	7	9

Rules: twodoku2x2 (Section 3.18)

4 Global Constraints

4.1 Constraint antiKing

4.1.1 Category: Global

4.1.2 Description

All cell connected by a king's move in chess (including diagonals) cannot be consecutive.

4.1.3 Arguments

none

4.1.4 JSON Format

```
"antiKing":"true",
```

4.1.5 Model

Model description not finished

4.1.6 See Also

antiKnight (Section 4.2) , superKing (Section 4.32)

4.1.7 antiKing example gas61: "engy engy Revolution" by Philip Newman

Video: `https://www.youtube.com/watch?v=YNEMiUR9lUQ`
Video2: `https://www.youtube.com/watch?v=c2dr8HtVeJg`

	A	B	C	D	E	F	G	H	I
1		1	2						
2		3					6		
3		4				5		7	
4	5					4		8	
5	6				3				9
6		7		2					1
7		8		1				2	
8			9					3	
9							4	5	

	A	B	C	D	E	F	G	H	I
1	7	1	2	6	4	3	8	9	5
2	9	3	5	8	7	2	6	1	4
3	8	4	6	9	1	5	3	7	2
4	5	9	1	7	6	4	2	8	3
5	6	2	8	5	3	1	7	4	9
6	3	7	4	2	9	8	5	6	1
7	4	8	3	1	5	6	9	2	7
8	2	5	9	4	8	7	1	3	6
9	1	6	7	3	2	9	4	5	8

Rules: standard (Section 3.13) antiKing (Section 4.1)

4.2 Constraint antiKnight

4.2.1 Category: Global

4.2.2 Description

All cell pairs connected by a Knight's move in chess cannot take the same value.

4.2.3 Arguments

none

4.2.4 JSON Format

```
"antiKnight":"true",
```

4.2.5 Model

This is a global constraint, affecting the whole grid. The values for any two cells which are a Knight's move apart must be different.

$$\forall_{i,j \in \text{Grid s.t. KM}(i,j)} : \quad x_i \neq x_j \tag{6}$$

4.2.6 See Also

antiKing (Section 4.1)

4.2.7 antiKnight example gas54: "engy's Negativity 2-ur" by Philip Newman

Video: `https://www.youtube.com/watch?v=TNz3NKxwlp4`

	A	B	C	D	E	F	G	H	I
1					1	8			
2			2	7				9	
3	3	6							
4			4				1		
5	5				2				8
6			3				9		
7								7	1
8		4				6	2		
9				5	3				

	A	B	C	D	E	F	G	H	I
1	9	5	7	2	1	8	3	4	6
2	4	1	2	7	6	3	8	9	5
3	3	6	8	4	9	5	7	1	2
4	6	8	4	9	5	7	1	2	3
5	5	7	9	3	2	1	4	6	8
6	1	2	3	6	8	4	9	5	7
7	2	3	6	8	4	9	5	7	1
8	8	4	5	1	7	6	2	3	9
9	7	9	1	5	3	2	6	8	4

Rules: standard (Section 3.13) antiKnight (Section 4.2)

4.3 Constraint argyle

4.3.1 Category: Global

4.3.2 Description

The cells on selected side diagonals must be pairwise different.

4.3.3 Arguments

none

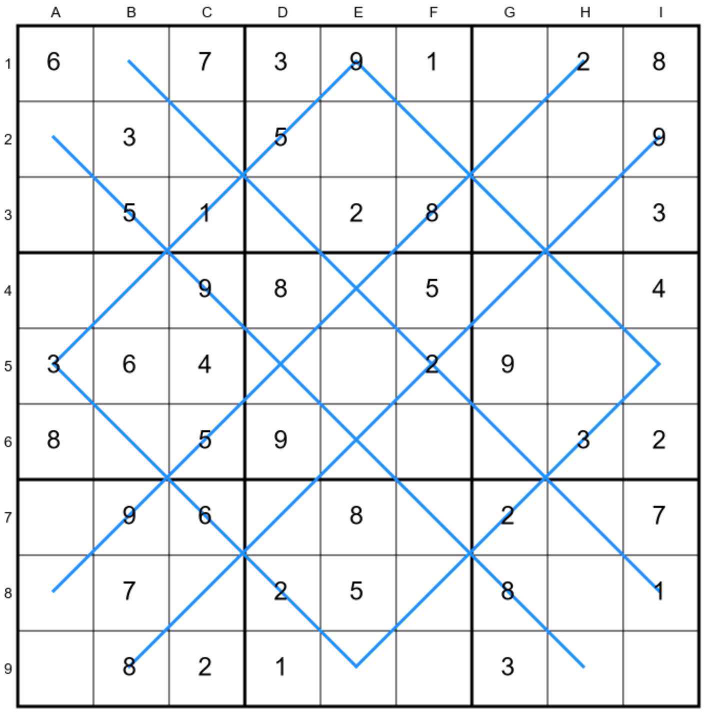
4.3.4 Model

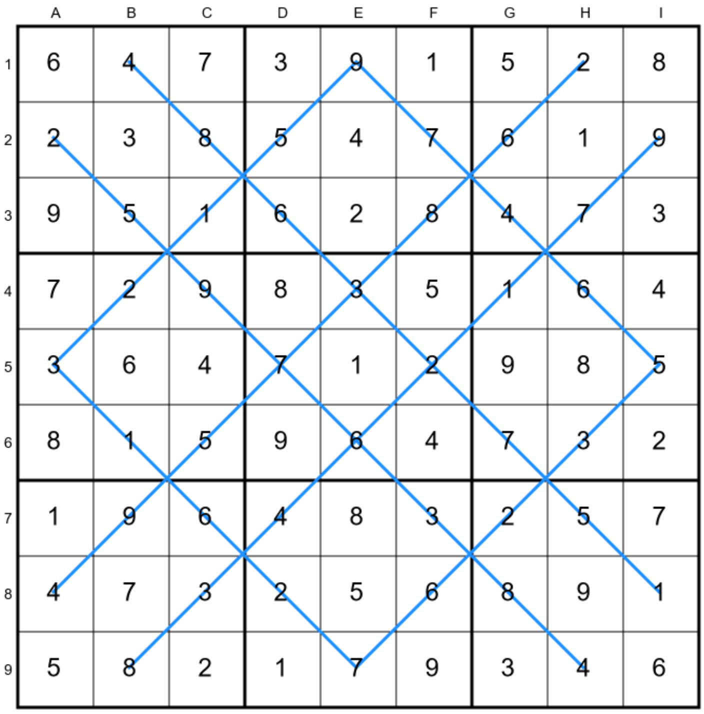
Model description not finished

4.3.5 See Also

mainDiagonal (Section 4.9) , minorDiagonal (Section 4.10)

4.3.6 argyle example Sudoku Variants28: "Argyle" by Steven Anderson





Rules: standard (Section 3.13) argyle (Section 4.3)

4.4 Constraint blockRowColumnRenban

4.4.1 Category: Global

4.4.2 Description

advance background, not yet implemented.

4.4.3 Arguments

none

4.4.4 JSON Format

```
"blockRowColumnRenban":"true",
```

4.4.5 Model

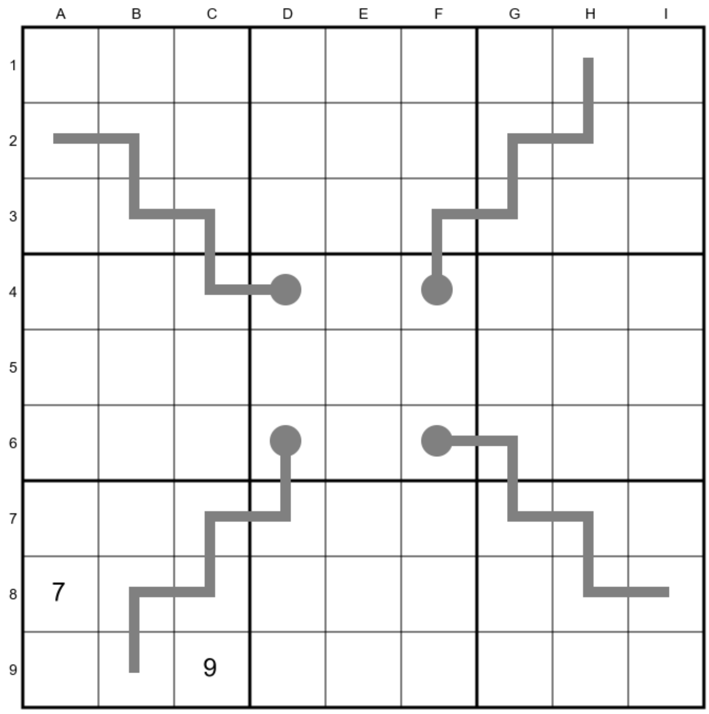
Model description not finished

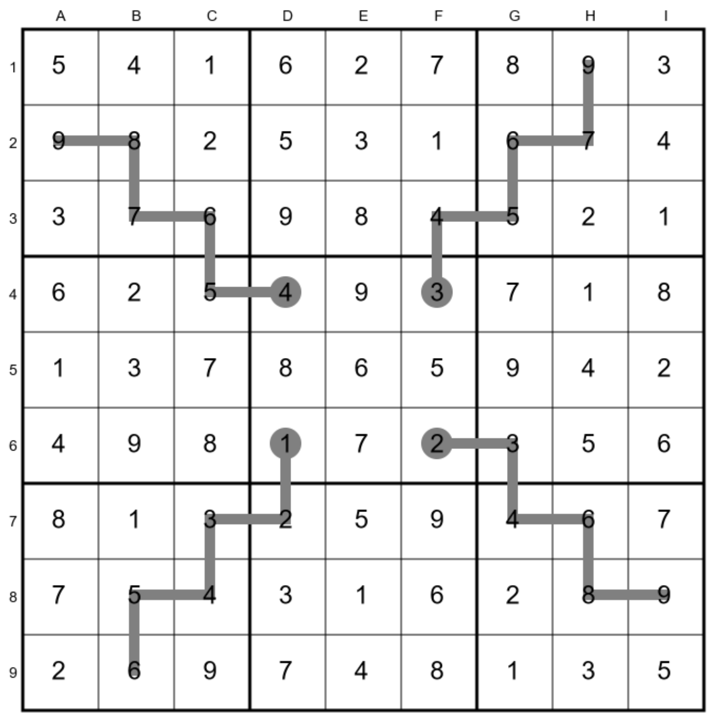
4.4.6 See Also

renban (Section 9.32)

4.4.7 blockRowColumnRenban example ctc13: "Stepped Thermos" by Aad van de Wetering

Video: `https://www.youtube.com/watch?v=AdSOJQ3huN0`





Rules: standard (Section 3.13) thermo (Section 9.38) blockRowColumnRenban (Section 4.4)

4.5 Constraint boxKiller

4.5.1 Category: Global

4.5.2 Description

All boxes inside a block add up to the same value, but the value is different for every block.

4.5.3 Arguments

none

4.5.4 JSON Format

```
"boxKiller":"true",
```

4.5.5 Model

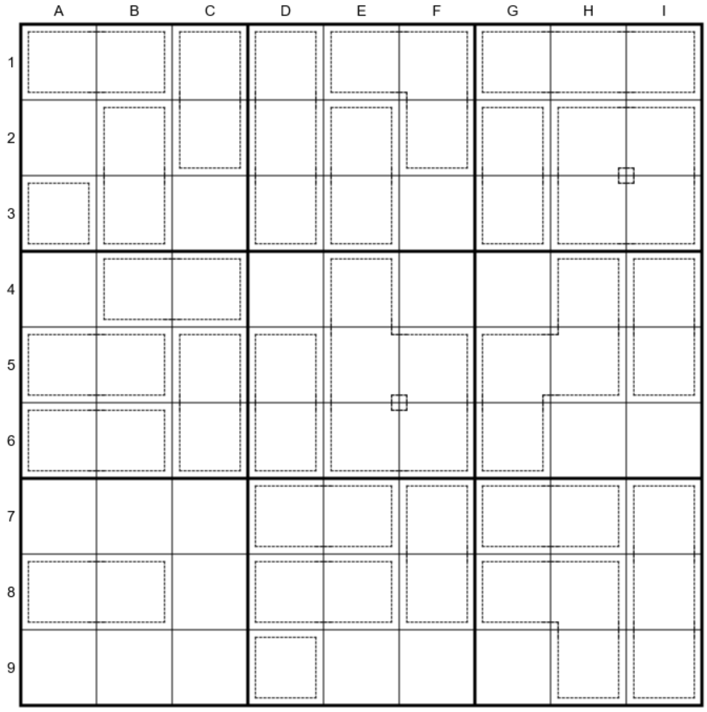
Model description not finished

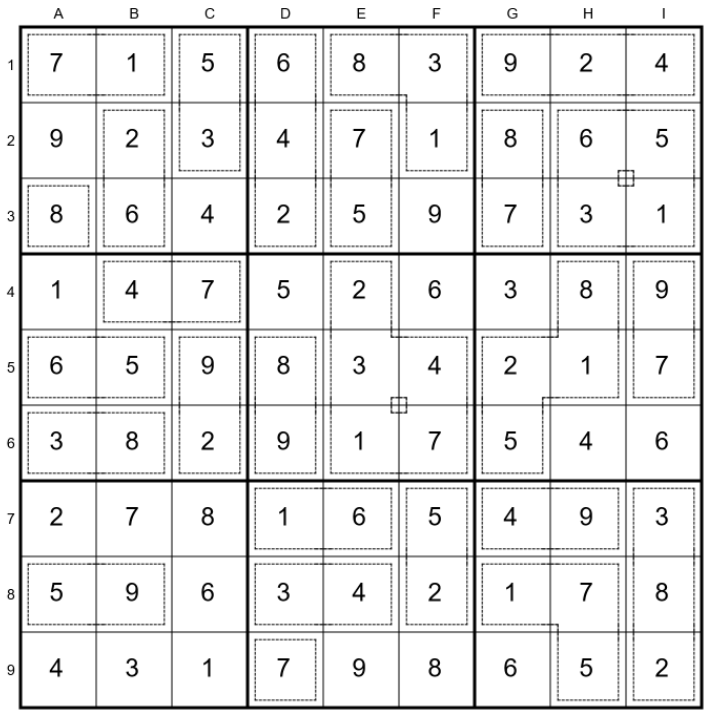
4.5.6 See Also

box (Section 11.3)

4.5.7 boxKiller example ctc32: "Box Killer" by udukos

Video: `https://www.youtube.com/watch?v=M7HOmpeYW0O`





Rules: standard (Section 3.13) boxKiller (Section 4.5)

4.6 Constraint dutchFlatMates

4.6.1 Category: Global

4.6.2 Description

Every 5 in the grid must have a 1 above it, or a 9 below it. It can have both a 1 above and a 9 below.

4.6.3 Arguments

none

4.6.4 JSON Format

```
"dutchFlatMates":"true",
```

4.6.5 Model

Model description not finished

4.6.6 See Also

dutchWhisper (Section 9.15)

4.6.7 dutchFlatMates example gas86: "Dutch Flatmates" by clover

Video: `https://www.youtube.com/watch?v=YwpBBGq1FAE`

	A	B	C	D	E	F	G	H	I
1					8		9		2
2				5	6	7			
3					1		7		3
4	7	8							
5									
6								3	4
7	3		7		2				
8				3	4	5			
9	8		1		9				

	A	B	C	D	E	F	G	H	I
1	1	7	5	4	8	3	9	6	2
2	2	3	9	5	6	7	4	1	8
3	4	6	8	9	1	2	7	5	3
4	7	8	3	1	5	4	2	9	6
5	5	2	4	6	3	9	1	8	7
6	9	1	6	2	7	8	5	3	4
7	3	5	7	8	2	1	6	4	9
8	6	9	2	3	4	5	8	7	1
9	8	4	1	7	9	6	3	2	5

Rules: standard (Section 3.13) dutchFlatMates (Section 4.6)

4.7 Constraint dutchRows

4.7.1 Category: Global

4.7.2 Description

All pairs of connected cells in each row must be at least four values apart.

4.7.3 Arguments

none

4.7.4 JSON Format

```
"dutchRows":"true",
```

4.7.5 Model

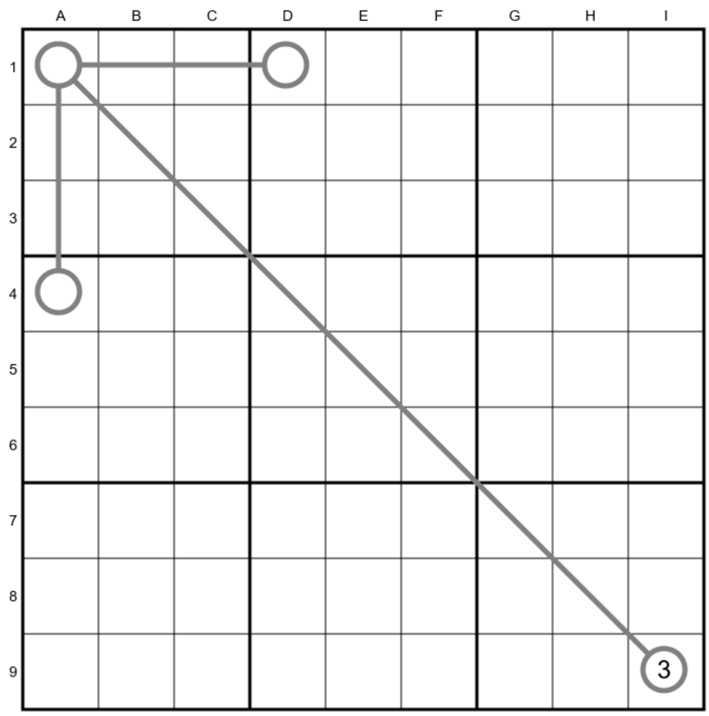
Model description not finished

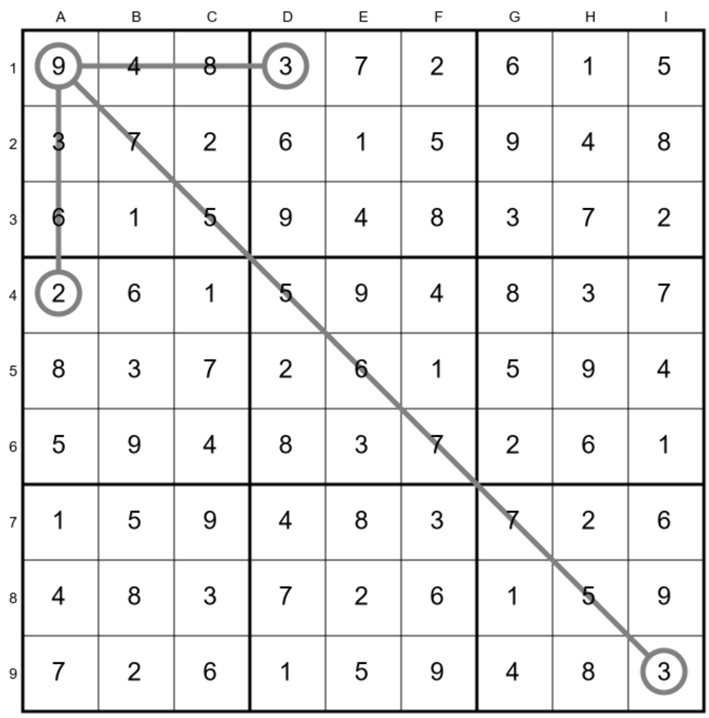
4.7.6 See Also

dutchWhisper (Section 9.15)

4.7.7 dutchRows example ctc12: "Is that a 3 in the corner?" by James Kopp

Video: `https://www.youtube.com/watch?v=MMw_SKG7pbA`





Rules: standard (Section 3.13) between (Section 9.9) dutchRows (Section 4.7)

4.8 Constraint inequalityTriple

4.8.1 Category: Global

4.8.2 Description

For every border between two blocks in the grid, we have that all three variables on one side of the border are greater or smaller together than the matching three variables in the other block.

4.8.3 Arguments

none

4.8.4 JSON Format

```
"inequalityTriple":"true",
```

4.8.5 Model

Model description not finished

4.8.6 See Also

4.8.7 inequalityTriple example wpf gp2025r6c13: "Sudoku GP 2025 Round 6" by

	A	B	C	D	E	F	G	H	I
1	5	1			9			3	2
2	3	2			8				9
3									
4									
5	6							4	1
6									
7									
8	8							7	6
9	9	6			4			5	3

	A	B	C	D	E	F	G	H	I
1	5	1	6	4	9	7	8	3	2
2	3	2	4	1	8	5	7	6	9
3	7	8	9	2	6	3	5	1	4
4	1	3	8	9	7	4	6	2	5
5	6	7	2	3	5	8	9	4	1
6	4	9	5	6	2	1	3	8	7
7	2	5	1	7	3	6	4	9	8
8	8	4	3	5	1	9	2	7	6
9	9	6	7	8	4	2	1	5	3

Rules: standard (Section 3.13) inequalityTriple (Section 4.8)

4.9 Constraint mainDiagonal

4.9.1 Category: Global

4.9.2 Description

Each value on the main diagonal (bottom-left to top-right) can occur only once.

4.9.3 Arguments

none

4.9.4 JSON Format

```
"mainDiagonal":"true",
```

4.9.5 Model

Let $[x_1, x_2, ..., x_n]$ be the variables corresponding to the cells on the main diagonal of the grid (from bottom-left to top-right). The constraint states that these variables are pairwise different, i.e.

$$\text{alldifferent}([x_1, x_2, ..., x_n]) \tag{7}$$

This alldifferent constraint may interact with other alldifferent constraints in the grid, so that further deductions can be made.

4.9.6 See Also

minorDiagonal (Section 4.10)

4.9.7 mainDiagonal example timotab101: "Oh No I Lost A Box" by Bill Murphy

Video: `https://www.youtube.com/watch?v=-tx7OHHxJiw` (17:07)

	A	B	C	D	E	F	G	H	I
1									
2		2	3	4		8	6	1	
3		1		5		7		8	
4		8	7	6		5	3	9	
5									
6		3	9	8					
7		5		3					
8		6	8	7					
9									

	A	B	C	D	E	F	G	H	I
1	8	9	5	1	6	3	4	2	7
2	7	2	3	4	9	8	6	1	5
3	6	1	4	5	2	7	9	8	3
4	1	8	7	6	4	5	3	9	2
5	5	4	6	9	3	2	1	7	8
6	2	3	9	8	7	1	5	4	6
7	9	5	2	3	8	4	7	6	1
8	3	6	8	7	1	9	2	5	4
9	4	7	1	2	5	6	8	3	9

Rules: standard (Section 3.13) mainDiagonal (Section 4.9) minorDiagonal (Section 4.10)

4.10 Constraint minorDiagonal

4.10.1 Category: Global

4.10.2 Description

Each value on the minor diagonal (top-left to bottom-right) can occur only once.

4.10.3 Arguments

none

4.10.4 JSON Format

```
"minorDiagonal":"true",
```

4.10.5 Model

Let $[x_1, x_2, ..., x_n]$ be the variables corresponding to the cells on the minor diagonal of the grid (from top-left to bottom-right). The constraint states that these variables are pairwise different, i.e.

$$\text{alldifferent}([x_1, x_2, ..., x_n]) \tag{8}$$

This alldifferent constraint may interact with other alldifferent constraints in the grid, so that further deductions can be made.

4.10.6 See Also

mainDiagonal (Section 4.9)

4.10.7 minorDiagonal example timotab101: "Oh No I Lost A Box" by Bill Murphy

Video: `https://www.youtube.com/watch?v=-tx7OHHxJiw` (17:07)

	A	B	C	D	E	F	G	H	I
1									
2		2	3	4		8	6	1	
3		1		5		7		8	
4		8	7	6		5	3	9	
5									
6		3	9	8					
7		5		3					
8		6	8	7					
9									

	A	B	C	D	E	F	G	H	I
1	8	9	5	1	6	3	4	2	7
2	7	2	3	4	9	8	6	1	5
3	6	1	4	5	2	7	9	8	3
4	1	8	7	6	4	5	3	9	2
5	5	4	6	9	3	2	1	7	8
6	2	3	9	8	7	1	5	4	6
7	9	5	2	3	8	4	7	6	1
8	3	6	8	7	1	9	2	5	4
9	4	7	1	2	5	6	8	3	9

Rules: standard (Section 3.13) mainDiagonal (Section 4.9) minorDiagonal (Section 4.10)

4.11 Constraint nonConsecutive

4.11.1 Category: Global

4.11.2 Description

All orthogonally connected cell pairs cannot take consecutive values.

4.11.3 Arguments

none

4.11.4 JSON Format

```
"nonConsecutive":"true",
```

4.11.5 Model

Model description not finished

4.11.6 See Also

strictConsecutive (Section 4.23) , whiteDot (Section 6.10) , strictWhiteDot (Section 4.31)

4.11.7 nonConsecutive example timotab99: "Thinking Inside And Outside Of The Box" by Bill Murphy

Video: `https://www.youtube.com/watch?v=w5s25-xwADU` (9:31)

	A	B	C	D	E	F	G	H	I
1									
2		6	4	2	8	5	3	1	
3		2						9	
4		9						3	
5		4						5	
6		1						7	
7		8						2	
8		5	2	9	6	3	7	4	
9									

	A	B	C	D	E	F	G	H	I
1	5	3	1	7	4	9	6	8	2
2	9	6	4	2	8	5	3	1	7
3	7	2	8	6	3	1	5	9	4
4	2	9	5	1	7	4	8	3	6
5	8	4	7	3	9	6	2	5	1
6	6	1	3	8	5	2	4	7	9
7	3	8	6	4	1	7	9	2	5
8	1	5	2	9	6	3	7	4	8
9	4	7	9	5	2	8	1	6	3

Rules: standard (Section 3.13) nonConsecutive (Section 4.11)

4.12 Constraint nonKropki

4.12.1 Category: Global

4.12.2 Description

All orthogonally connected cells cannot be consecutive or satisfy a X:2X ratio.

4.12.3 Arguments

none

4.12.4 JSON Format

```
"nonKropki":"true",
```

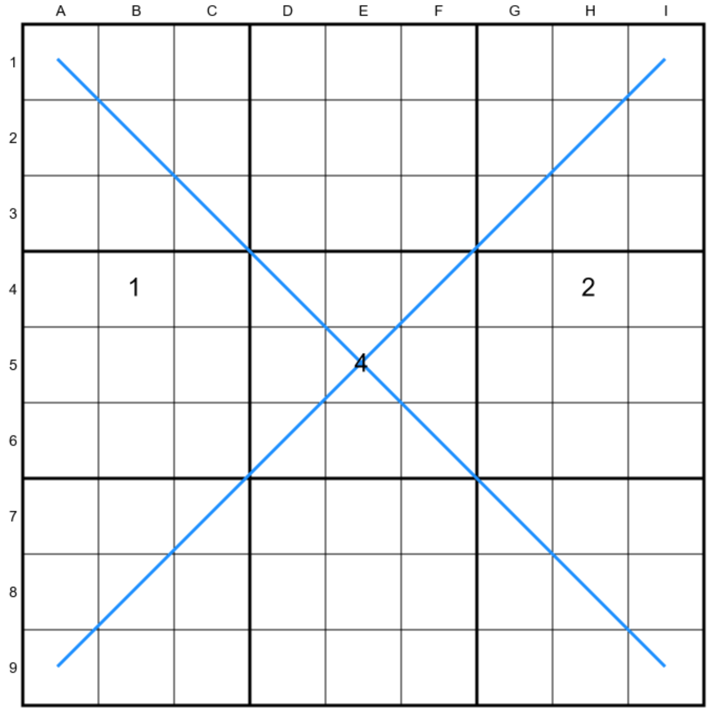
4.12.5 Model

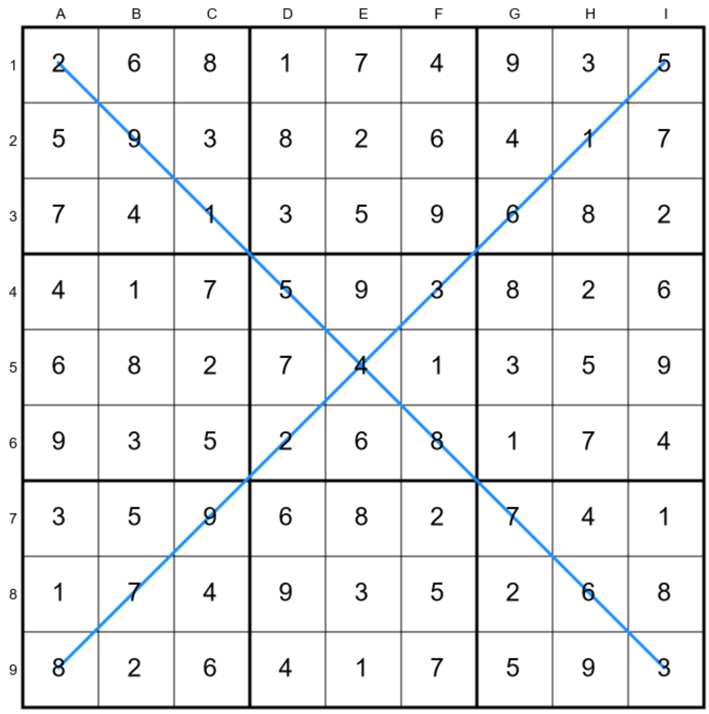
Model description not finished

4.12.6 See Also

4.12.7 nonKropki example ctc28: "Dotless Kropki Sudoku X" by Phistomefel

Video: `https://www.youtube.com/watch?v=1QP7yviZYTU`





Rules: standard (Section 3.13) mainDiagonal (Section 4.9) minorDiagonal (Section 4.10) nonKropki (Section 4.12)

4.13 Constraint orthoLE5

4.13.1 Category: Global

4.13.2 Description

The maximum difference between any two orthogonally connected cells can be 5.

4.13.3 Arguments

none

4.13.4 JSON Format

```
"orthoLE5":"true",
```

4.13.5 Model

Model description not finished

4.13.6 See Also

4.13.7 orthoLE5 example ctc92: "<= 5 Sudoku" by Aad van de Wetering

Video: `https://www.youtube.com/watch?v=ZU5fSDHJq8k`

	A	B	C	D	E	F	G	H	I
1									1
2		8							
3									
4	8				1				
5			3		6	2			
6									
7								9	
8	4		7	9					
9									

	A	B	C	D	E	F	G	H	I
1	2	4	9	6	7	5	8	3	1
2	3	8	5	1	4	9	6	7	2
3	7	6	1	2	3	8	9	5	4
4	8	9	6	3	1	4	5	2	7
5	5	7	3	8	6	2	1	4	9
6	1	2	4	5	9	7	3	8	6
7	6	1	2	4	8	3	7	9	5
8	4	3	7	9	5	1	2	6	8
9	9	5	8	7	2	6	4	1	3

Rules: standard (Section 3.13) orthoLE5 (Section 4.13)

4.14 Constraint posDiagAtmost11

4.14.1 Category: Global

4.14.2 Description

Consider all positive diagonals (bottom left to top right). The sum of two consecutive entries on each diagonal can be at most 11.

4.14.3 Arguments

none

4.14.4 JSON Format

```
"posDiagAtmost11":"true",
```

4.14.5 Model

Model description not finished

4.14.6 See Also

4.14.7 posDiagAtmost11 example ctc34: "Miracle Of Eleven" by Aad van de Wetering

Video: `https://www.youtube.com/watch?v=OzzuJUU6g84`

	A	B	C	D	E	F	G	H	I
1									
2									
3									
4					3				
5									
6					8				
7									
8									
9									

	A	B	C	D	E	F	G	H	I
1	7	5	2	6	9	3	1	8	4
2	1	8	4	2	7	5	3	6	9
3	3	6	9	4	1	8	5	2	7
4	5	2	7	9	3	6	8	4	1
5	8	4	1	7	5	2	6	9	3
6	6	9	3	1	8	4	2	7	5
7	2	7	5	3	6	9	4	1	8
8	4	1	8	5	2	7	9	3	6
9	9	3	6	8	4	1	7	5	2

Rules: standard (Section 3.13) nonConsecutive (Section 4.11) posDiagAtmost11 (Section 4.14)

4.15 Constraint posDiagDiff1or8

4.15.1 Category: Global

4.15.2 Description

Consider all positive diagonals (bottom left to top right). The distance between two consecutive entries on each diagonal must be either 1 or 8.

4.15.3 Arguments

none

4.15.4 JSON Format

```
"posDiagDiff1or8":"true",
```

4.15.5 Model

Model description not finished

4.15.6 See Also

4.15.7 posDiagDiff1or8 example ctc42: "Simple Miracle" by Aad van de Wetering

Video: `https://www.youtube.com/watch?v=4fwsRuKC6EY`

	A	B	C	D	E	F	G	H	I
1									
2									
3									
4									
5									
6									
7									
8		1							
9						2			

	A	B	C	D	E	F	G	H	I
1	8	3	7	2	6	1	5	9	4
2	2	6	1	5	9	4	8	3	7
3	5	9	4	8	3	7	2	6	1
4	1	5	9	4	8	3	7	2	6
5	4	8	3	7	2	6	1	5	9
6	7	2	6	1	5	9	4	8	3
7	3	7	2	6	1	5	9	4	8
8	6	1	5	9	4	8	3	7	2
9	9	4	8	3	7	2	6	1	5

Rules: standard (Section 3.13) antiKnight (Section 4.2) posDiagDiff1or8 (Section 4.15)

4.16 Constraint positionAlldifferent

4.16.1 Category: Global

4.16.2 Description

We group corresponding cells in all blocks, for example all cells in the top left corner of each block. Each such group, the value sets of digits place in the cells must be pairwise different.

4.16.3 Arguments

none

4.16.4 JSON Format

```
"positionAlldifferent":"true",
```

4.16.5 Model

Model description not finished

4.16.6 See Also

4.16.7 positionAlldifferent example gas1: "Battery Death" by Bill

Video: `https://www.youtube.com/watch?v=ED5Omzwu1Lk`

	A	B	C	D	E	F	G	H	I
1		1	2						
2		3	9					6	4
3								9	3
4				4		5			
5									
6				1		8			
7	7	9							
8	8	2					9	7	
9							6	5	

	A	B	C	D	E	F	G	H	I
1	6	1	2	9	3	4	5	8	7
2	5	3	9	7	8	2	1	6	4
3	4	8	7	5	1	6	2	9	3
4	1	7	3	4	6	5	8	2	9
5	2	5	8	3	9	7	4	1	6
6	9	6	4	1	2	8	7	3	5
7	7	9	6	2	5	1	3	4	8
8	8	2	5	6	4	3	9	7	1
9	3	4	1	8	7	9	6	5	2

Rules: standard (Section 3.13) positionAlldifferent (Section 4.16)

4.17 Constraint positiveDiagonalsAlldifferent

4.17.1 Category: Global

4.17.2 Description

The cells on each positive diagonal, running from bottom left to top right, must be different from each other.

4.17.3 Arguments

none

4.17.4 JSON Format

```
"positiveDiagonalsAlldifferent":"true",
```

4.17.5 Model

Model description not finished

4.17.6 See Also

4.17.7 positiveDiagonalsAlldifferent example ctc89: "The Dutch Miracle" by Aad van de Wetering

Video: `https://www.youtube.com/watch?v=wUnnXwLTbnA&t=279s`

	A	B	C	D	E	F	G	H	I
1									
2									
3									
4									
5									
6									
7									
8									
9	1		2						

	A	B	C	D	E	F	G	H	I
1	8	7	4	1	9	2	3	5	6
2	3	9	6	5	7	8	1	2	4
3	5	2	1	3	4	6	7	9	8
4	7	6	8	9	2	3	5	4	1
5	2	4	5	7	8	1	9	6	3
6	9	1	3	4	6	5	2	8	7
7	6	8	9	2	1	7	4	3	5
8	4	5	7	6	3	9	8	1	2
9	1	3	2	8	5	4	6	7	9

Rules: standard (Section 3.13) positiveDiagonalsAlldifferent (Section 4.17) positiveDiagonalsDutch-Whisper (Section 4.18)

4.18 Constraint positiveDiagonalsDutchWhisper

4.18.1 Category: Global

4.18.2 Description

The cells on each positive diagonal, running from bottom left to top right, must form a Dutch Whisper line, i.e. consecutive values differ by at least 4.

4.18.3 Arguments

none

4.18.4 JSON Format

```
"positiveDiagonalsDutchWhisper":"true",
```

4.18.5 Model

Model description not finished

4.18.6 See Also

dutchWhisper (Section 9.15)

4.18.7 positiveDiagonalsDutchWhisper example ctc89: "The Dutch Miracle" by Aad van de Wetering

Video: `https://www.youtube.com/watch?v=wUnnXwLTbnA&t=279s`

	A	B	C	D	E	F	G	H	I
1									
2									
3									
4									
5									
6									
7									
8									
9	1		2						

	A	B	C	D	E	F	G	H	I
1	8	7	4	1	9	2	3	5	6
2	3	9	6	5	7	8	1	2	4
3	5	2	1	3	4	6	7	9	8
4	7	6	8	9	2	3	5	4	1
5	2	4	5	7	8	1	9	6	3
6	9	1	3	4	6	5	2	8	7
7	6	8	9	2	1	7	4	3	5
8	4	5	7	6	3	9	8	1	2
9	1	3	2	8	5	4	6	7	9

Rules: standard (Section 3.13) positiveDiagonalsAlldifferent (Section 4.17) positiveDiagonalsDutchWhisper (Section 4.18)

4.19 Constraint sameBox

4.19.1 Category: Global

4.19.2 Description

The sum the values inside each box must be the same. The values inside each box must be different from each other.

4.19.3 Arguments

none

4.19.4 JSON Format

```
"sameBox":"true",
```

4.19.5 Model

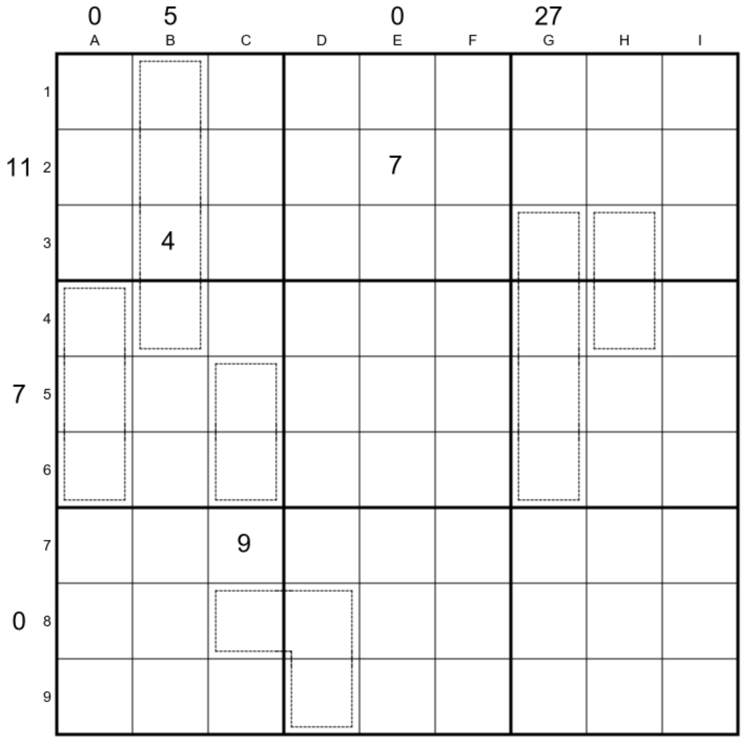
Model description not finished

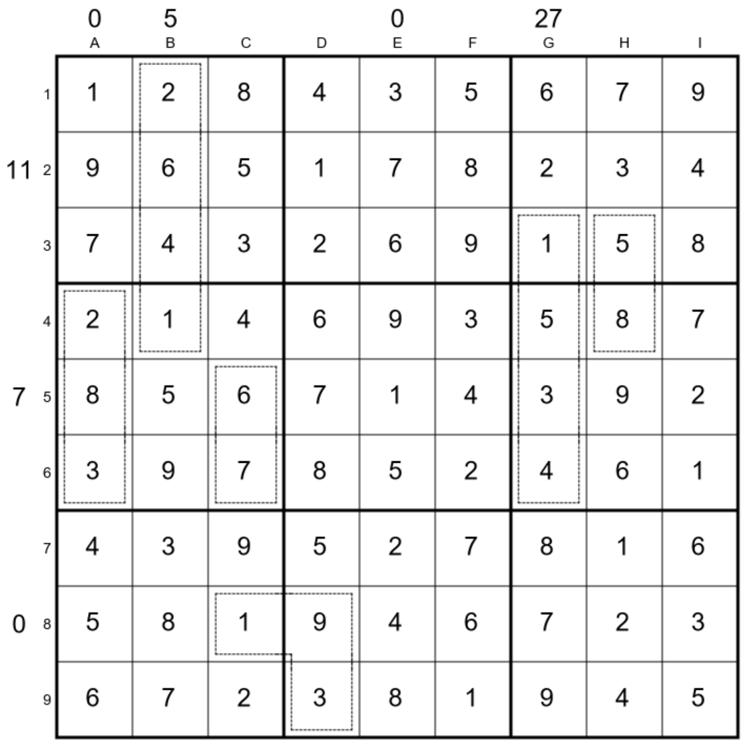
4.19.6 See Also

box (Section 11.3)

4.19.7 sameBox example ctc79: "The Knight's Last Meal" by Dook

Video: `https://www.youtube.com/watch?v=eXuzsrXhYmI`





Rules: standard (Section 3.13) antiKnight (Section 4.2) sameBox (Section 4.19) box (Section 11.3) sandwich (Section 12.21)

4.20 Constraint strictBlackDot

4.20.1 Category: Global

4.20.2 Description

Any pair of orthogonally connected cells which are not marked as doubler by a black dot cannot be in 1:2 ratio.

4.20.3 Arguments

none

4.20.4 JSON Format

```
"strictBlackDot":"true",
```

4.20.5 Model

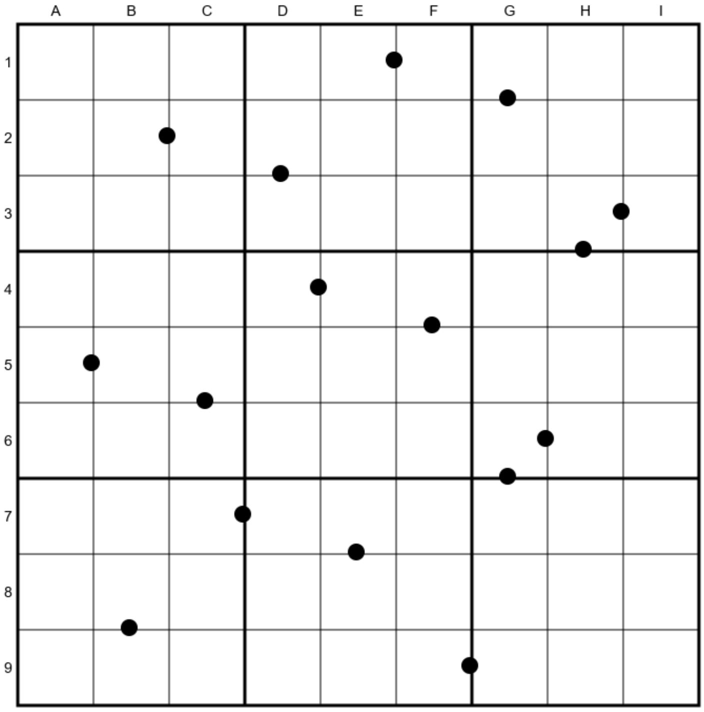
Model description not finished

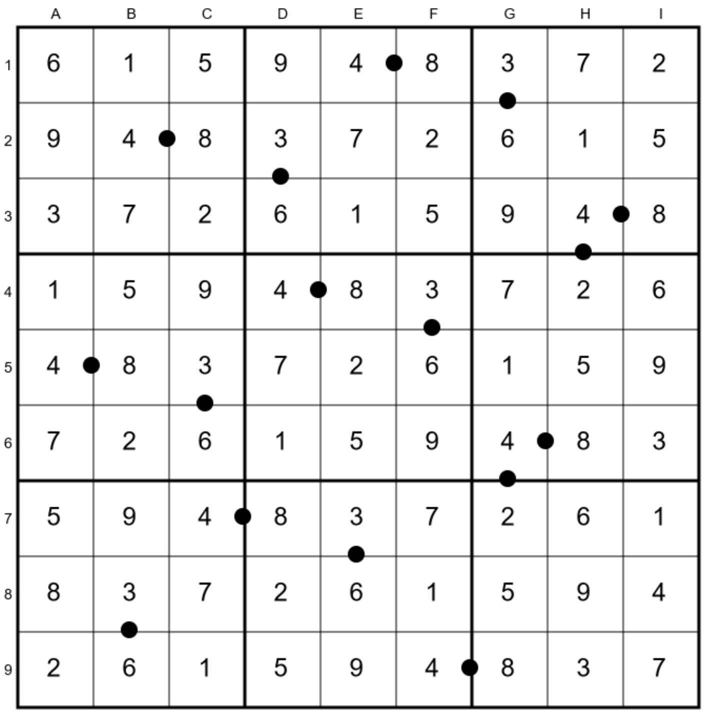
4.20.6 See Also

blackDot (Section 6.1)

4.20.7 strictBlackDot example logiclemurgaming26: "Ratios are our Friends" by The Bard

Video: `https://www.youtube.com/watch?v=09-zn0g0Pzo`





Rules: standard (Section 3.13) blackDot (Section 6.1) nonConsecutive (Section 4.11) strictBlackDot (Section 4.20)

4.21 Constraint strictBlockKropki

4.21.1 Category: Global

4.21.2 Description

For the blocks indicated, orthogonally connected cell pairs fully contained in the block not marked with a dot must not be consecutive or in a 1:2 ratio.

4.21.3 Arguments

strictBlockKropki List(Integer)

4.21.4 JSON Format

```
"strictBlockKropki":[2,4,9],
```

4.21.5 Model

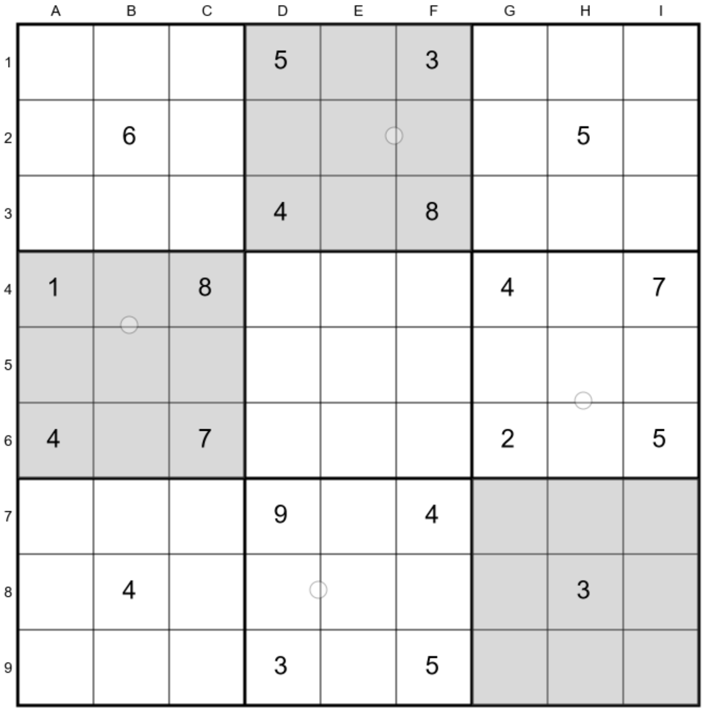
Model description not finished

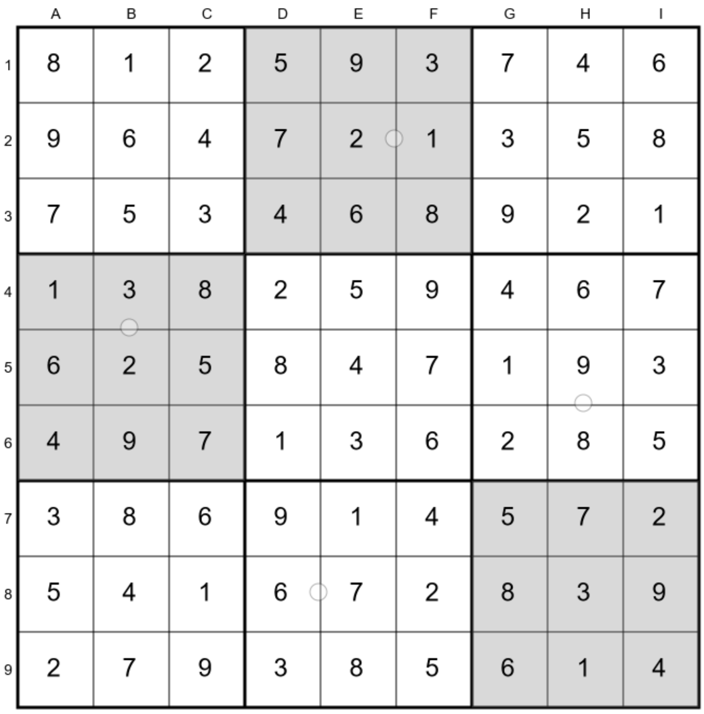
4.21.6 See Also

blackDot (Section 6.1) , whiteDot (Section 6.10)

4.21.7 strictBlockKropki example timotab55: "CAPTCHA Sudoku" by Bill Murphy

Video: `https://www.youtube.com/watch?v=JKaaroYNZLQ`





Rules: standard (Section 3.13) whiteDot (Section 6.10) strictBlockKropki (Section 4.21)

4.22 Constraint strictCheckerQuad

4.22.1 Category: Global

4.22.2 Description

Opposing corners of the quad must have the same parity (odd or even), while orthogonally adjacent cells in the quad must have opposite parity. All possible clues are given, i.e. quads not marked cannot have a checkerboard configuration.

4.22.3 Arguments

none

4.22.4 JSON Format

```
"strictCheckerQuad":"true",
```

4.22.5 Model

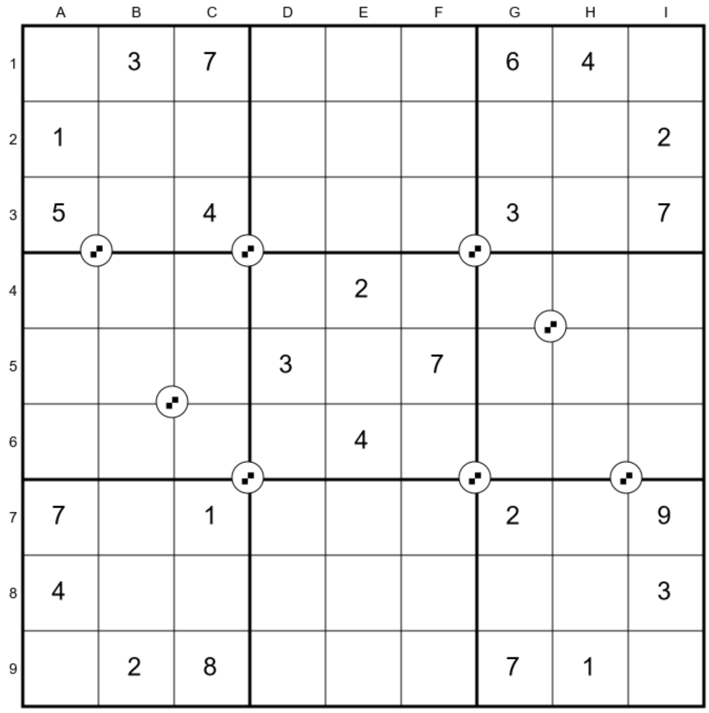
Model description not finished

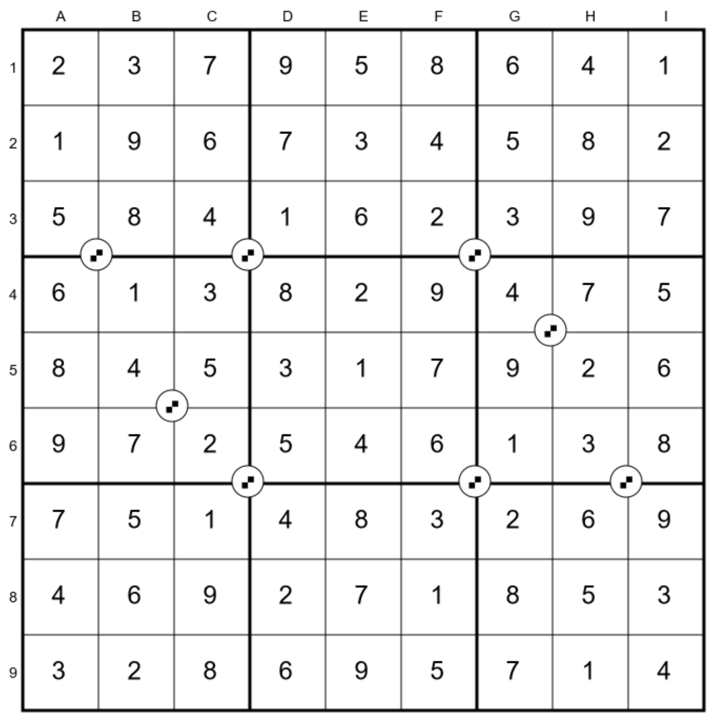
4.22.6 See Also

checkerQuad (Section 8.2)

4.22.7 strictCheckerQuad example gas17: "Accentuate the Negative" by clover

Video: `https://www.youtube.com/watch?v=7t3cuZOCavQ&list=PLkqs9GgCgVT6oKvb8KGhVlwS2rYpZwlIb&index=64`





Rules: standard (Section 3.13) checkerQuad (Section 8.2) strictCheckerQuad (Section 4.22)

4.23 Constraint strictConsecutive

4.23.1 Category: Global

4.23.2 Description

Any pair of orthogonally connected cells which are not marked by a white dot or a black dot must be non consecutive.

4.23.3 Arguments

none

4.23.4 JSON Format

```
"strictConsecutive":"true",
```

4.23.5 Model

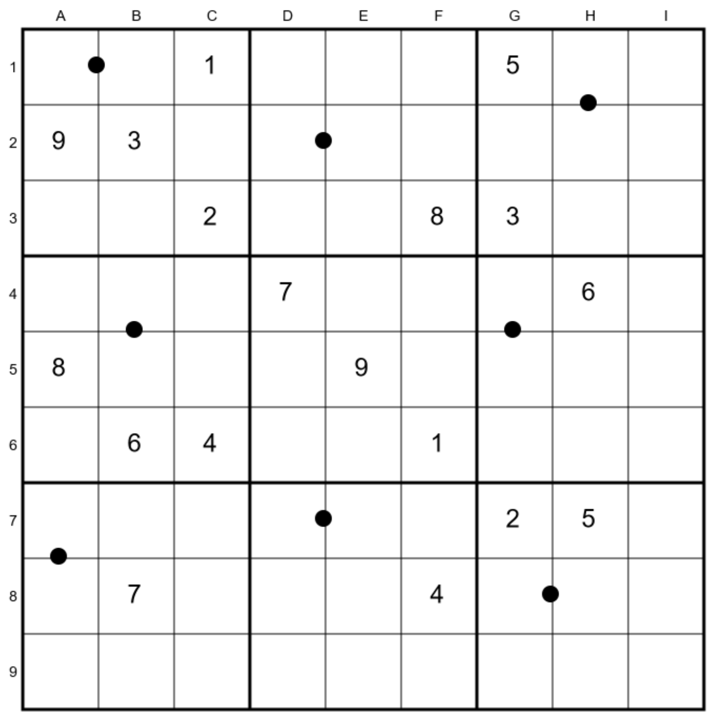
Model description not finished

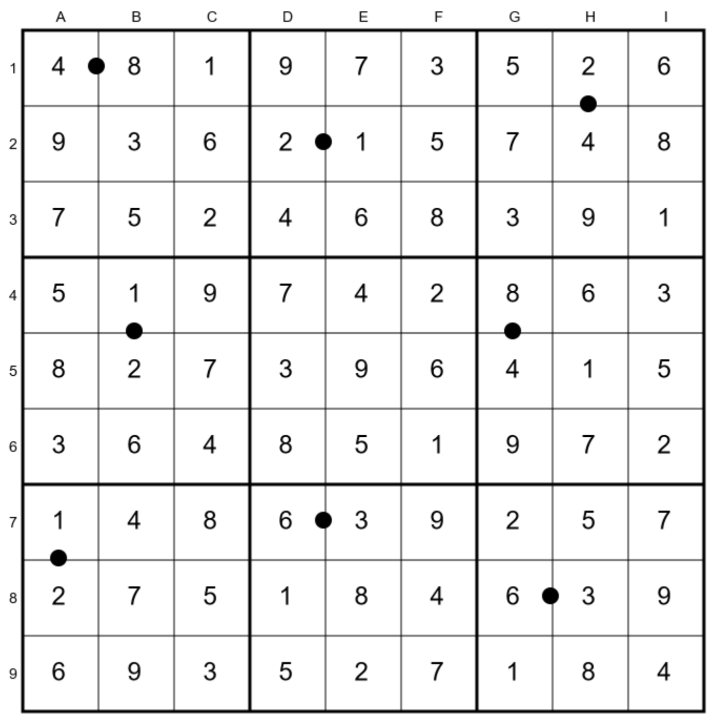
4.23.6 See Also

whiteDot (Section 6.10) , blackDot (Section 6.1) , strictWhiteDot (Section 4.31)

4.23.7 strictConsecutive example timotab50: "Partial Pairs" by Philip Newman

Video: `https://www.youtube.com/watch?v=Pb-14z8wd9U`





Rules: standard (Section 3.13) blackDot (Section 6.1) strictConsecutive (Section 4.23)

4.24 Constraint strictDifferencePair

4.24.1 Category: Global

4.24.2 Description

For every orthogonally connected pair for which there is not explicit difference pair constraint given, the absolute difference of the cell values cannot be any of the indicated values.

4.24.3 Arguments

strictDifferencePair List(Integer)

4.24.4 JSON Format

```
"strictDifferencePair":[4,8],
```

4.24.5 Model

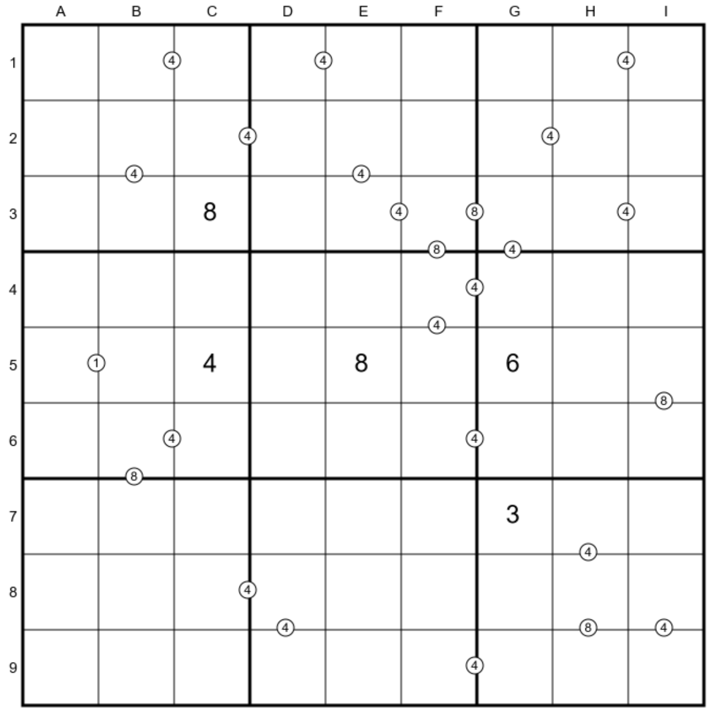
Model description not finished

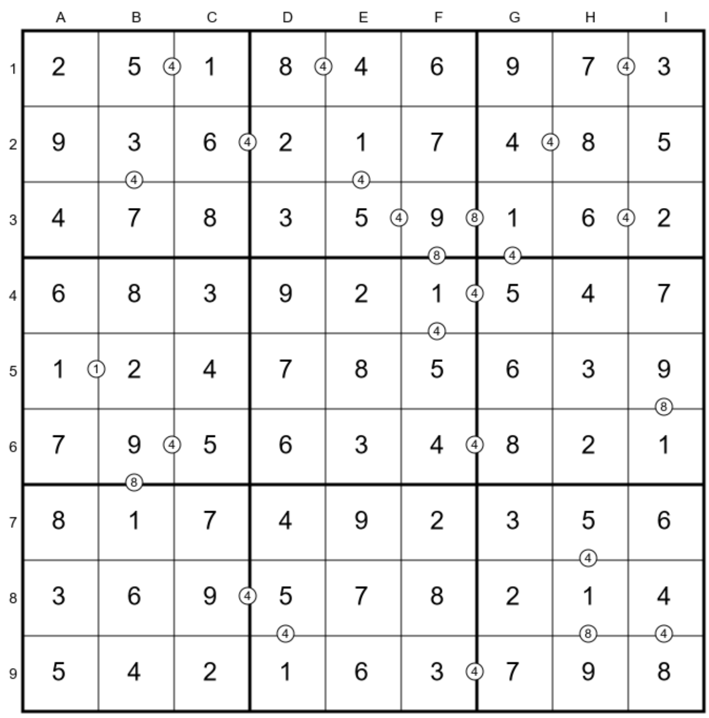
4.24.6 See Also

differencePair (Section 6.2)

4.24.7 strictDifferencePair example ctc119: "Orwellian" by The Chiropractor

Video: `https://www.youtube.com/watch?v=20ZlHGHO9DY` (21:34)





Rules: standard (Section 3.13) differencePair (Section 6.2) strictDifferencePair (Section 4.24)

4.25 Constraint strictIncDec

4.25.1 Category: Global

4.25.2 Description

For a row/column where no incDec constraint is given, the first three digits in the row/column next to the missing clue cannot be in either strictly increasing or strictly decreasing order.

4.25.3 Arguments

none

4.25.4 Model

Model description not finished

4.25.5 See Also

4.25.6 strictIncDec example Sudoku Variants79: "Rossini" by Steven Anderson

	A	B	C	D	E	F	G	H	I
1								6	4
2									
3	4				5	3			8
4					9			2	
5	7					2		8	
6		8		5	3		1		
7	1	7		3				4	2
8		2			6	9		5	1
9			6				8		

	A	B	C	D	E	F	G	H	I
1	5	9	3	1	2	8	7	6	4
2	2	1	8	4	7	6	9	3	5
3	4	6	7	9	5	3	2	1	8
4	6	4	1	8	9	7	5	2	3
5	7	3	5	6	1	2	4	8	9
6	9	8	2	5	3	4	1	7	6
7	1	7	9	3	8	5	6	4	2
8	8	2	4	7	6	9	3	5	1
9	3	5	6	2	4	1	8	9	7

Rules: standard (Section 3.13) incDec (Section 12.8) strictIncDec (Section 4.25)

4.26 Constraint strictKropki

4.26.1 Category: Global

4.26.2 Description

Any pair of orthogonally connected cells which are not marked as doubler or consecutive by a white/black dot must be non consecutive and cannot be in 1:2 ratio.

4.26.3 Arguments

none

4.26.4 JSON Format

```
"strictKropki":"true",
```

4.26.5 Model

Model description not finished

4.26.6 See Also

whiteDot (Section 6.10) , blackDot (Section 6.1)

4.26.7 strictKropki example gas35: ":O" by Bill Murphy

Video: `https://youtu.be/AU4-Yde1gZ8?si=7fOsflCJ-rIdwLpQ`

	A	B	C	D	E	F	G	H	I
1			5				4		
2									
3	1			3	5	9			2
4			1				8		
5			8				3		
6			3				7		
7	5			6	2	8			7
8									
9			4				2		

	A	B	C	D	E	F	G	H	I
1	3	9	5	2	8	6	4	7	1
2	8	6	2	7	1	4	9	3	5
3	1	4	7	3	5	9	6	8	2
4	4	7	1	5	9	3	8	2	6
5	6	2	8	1	4	7	3	5	9
6	9	5	3	8	6	2	7	1	4
7	5	3	9	6	2	8	1	4	7
8	2	8	6	4	7	1	5	9	3
9	7	1	4	9	3	5	2	6	8

Rules: standard (Section 3.13) strictKropki (Section 4.26)

4.27 Constraint strictSumV

4.27.1 Category: Global

4.27.2 Description

No pair of orthogonally connected cells which is not explicitly marked by a sumV constraint can add up to 5.

4.27.3 Arguments

none

4.27.4 JSON Format

```
"strictSumV":"true",
```

4.27.5 Model

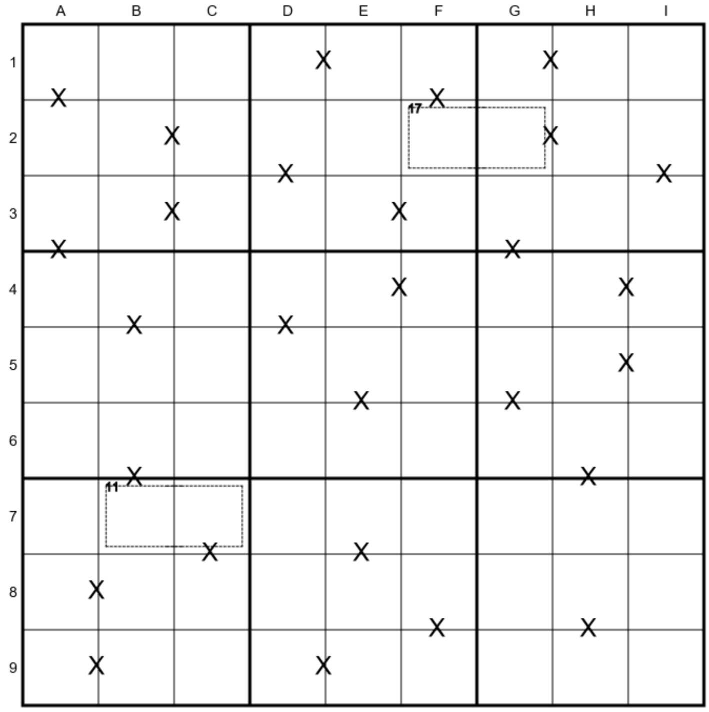
Model description not finished

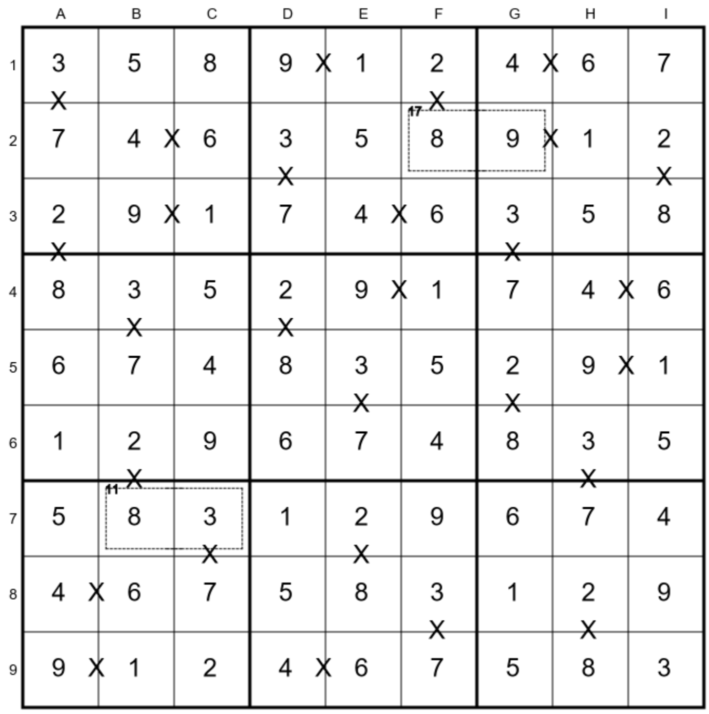
4.27.6 See Also

sumX (Section 6.9) , sumV (Section 6.8) , strictSumX (Section 4.28) , strictSumXV (Section 4.29)

4.27.7 strictSumV example logiclemurgaming5: "eXciting colours" by DarthSillious72

Video: `https://www.youtube.com/watch?v=43A9zO1A0HY&t=357s`





Rules: standard (Section 3.13) sumX (Section 6.9) antiKing (Section 4.1) antiKnight (Section 4.2) strictSumV (Section 4.27) cage (Section 11.4)

4.28 Constraint strictSumX

4.28.1 Category: Global

4.28.2 Description

No pair of orthogonally connected cells which is not explicitly marked by a sumX constraint can add up to 10.

4.28.3 Arguments

none

4.28.4 JSON Format

```
"strictSumX":"true",
```

4.28.5 Model

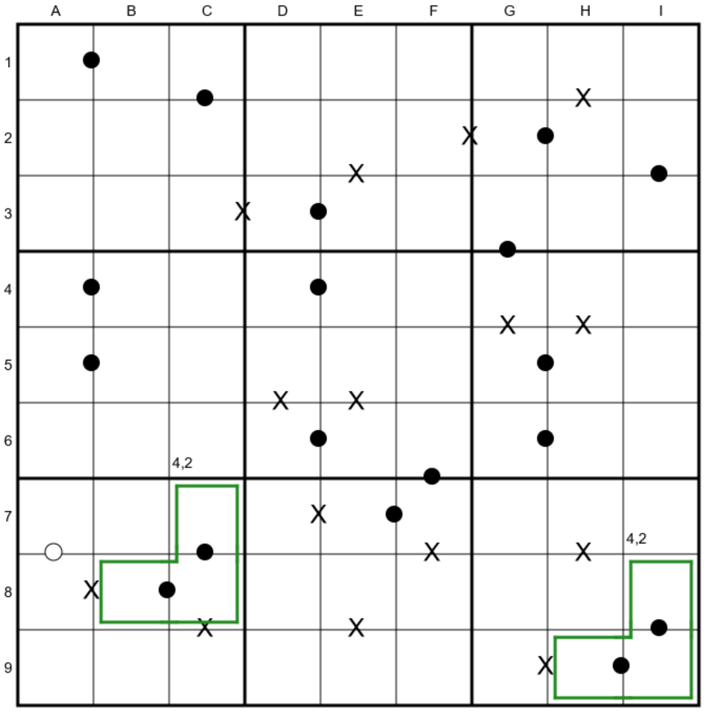
Model description not finished

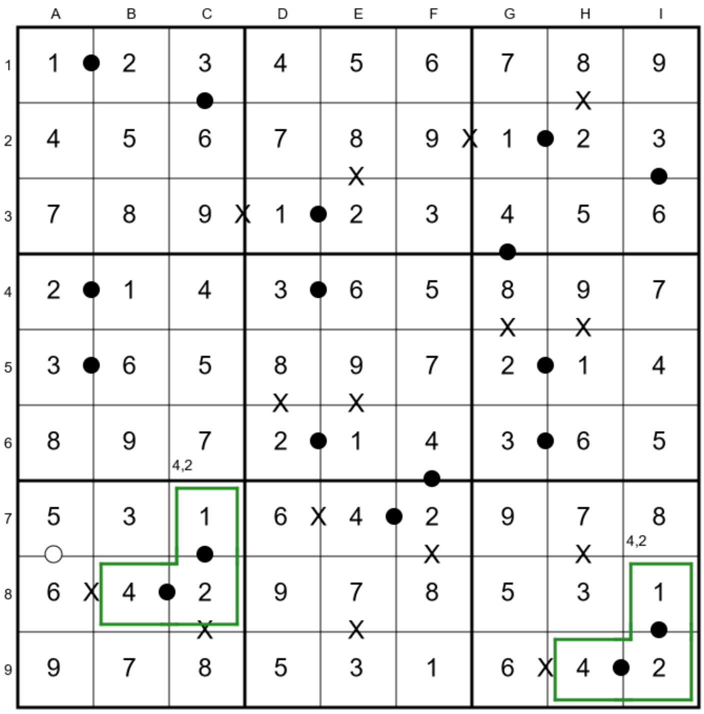
4.28.6 See Also

sumX (Section 6.9) , sumV (Section 6.8) , strictSumXV (Section 4.29) , strictSumV (Section 4.27)

4.28.7 strictSumX example timotab35: "Primogenesis" by Fool on Hill

Video: `https://www.youtube.com/watch?v=cnyhOcMmzuE`





Rules: standard (Section 3.13) blackDot (Section 6.1) whiteDot (Section 6.10) sumX (Section 6.9) strictBlackDot (Section 4.20) strictSumX (Section 4.28) blackDotTriple (Section 7.1)

4.29 Constraint strictSumXV

4.29.1 Category: Global

4.29.2 Description

No pair of orthogonally connected cells which is not explicitly marked by a sumX or sumV constraint can add up to 5 or 10. Both values are excluded.

4.29.3 Arguments

none

4.29.4 JSON Format

```
"strictSumXV":"true",
```

4.29.5 Model

Model description not finished

4.29.6 See Also

sumX (Section 6.9) , sumV (Section 6.8) , strictSumX (Section 4.28) , strictSumV (Section 4.27)

4.29.7 strictSumXV example gas67: "Superstar" by Bill Murphy

Video: `https://www.youtube.com/watch?v=mOSQcTkQYE4`

	A	B	C	D	E	F	G	H	I
1	5	2	7				4	3	6
2	4								2
3	8								9
4									
5									
6									
7	6								5
8	7								1
9	9	5	2				3	6	8

	A	B	C	D	E	F	G	H	I
1	5	2	7	1	8	9	4	3	6
2	4	9	6	7	5	3	8	1	2
3	8	3	1	6	2	4	5	7	9
4	3	4	5	2	1	8	6	9	7
5	1	7	8	9	3	6	2	5	4
6	2	6	9	5	4	7	1	8	3
7	6	1	3	8	9	2	7	4	5
8	7	8	4	3	6	5	9	2	1
9	9	5	2	4	7	1	3	6	8

Rules: standard (Section 3.13) strictSumXV (Section 4.29)

4.30 Constraint strictTwinDetector

4.30.1 Category: Global

4.30.2 Description

An arrow in a cell indicates that the sum of the nearest neighbours of the cell in the given direction is equal to the value of the cell. All possible arrows are given.

4.30.3 Arguments

none

4.30.4 JSON Format

```
"strictTwinDetector":"true"
```

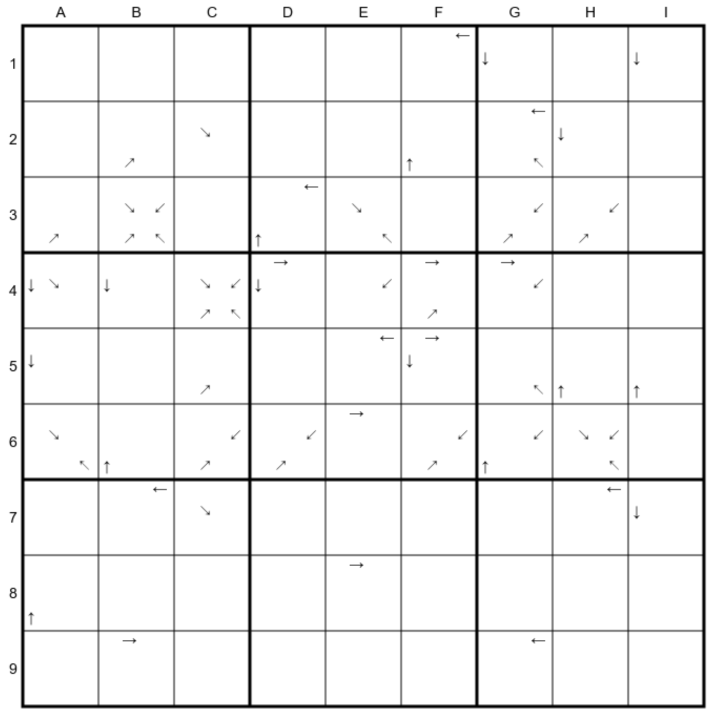
4.30.5 Model

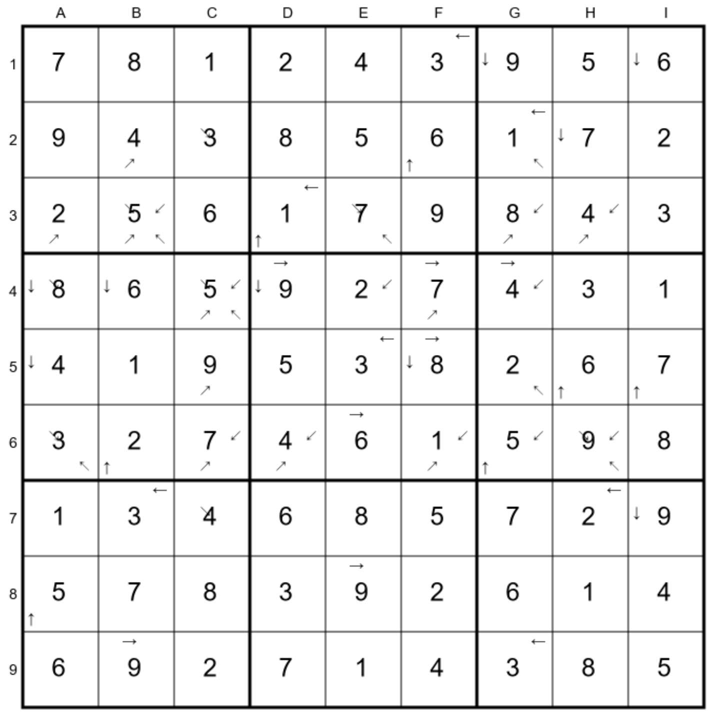
Model description not finished

4.30.6 See Also

twinDetector (Section 5.14)

4.30.7 strictTwinDetector example wpf gp2025r5c9: "Sudoku GP 2025 Round 5" by





Rules: standard (Section 3.13) strictTwinDetector (Section 4.30)

4.31 Constraint strictWhiteDot

4.31.1 Category: Global

4.31.2 Description

Any pair of orthogonally connected cells which are not marked as consecutive by a white dot must be non consecutive.

4.31.3 Arguments

none

4.31.4 JSON Format

```
"strictWhiteDot":"true",
```

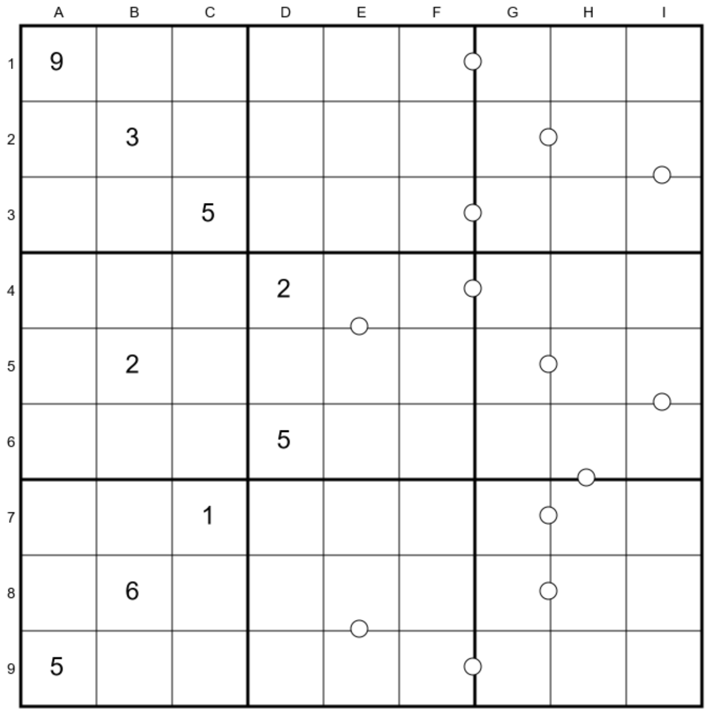
4.31.5 Model

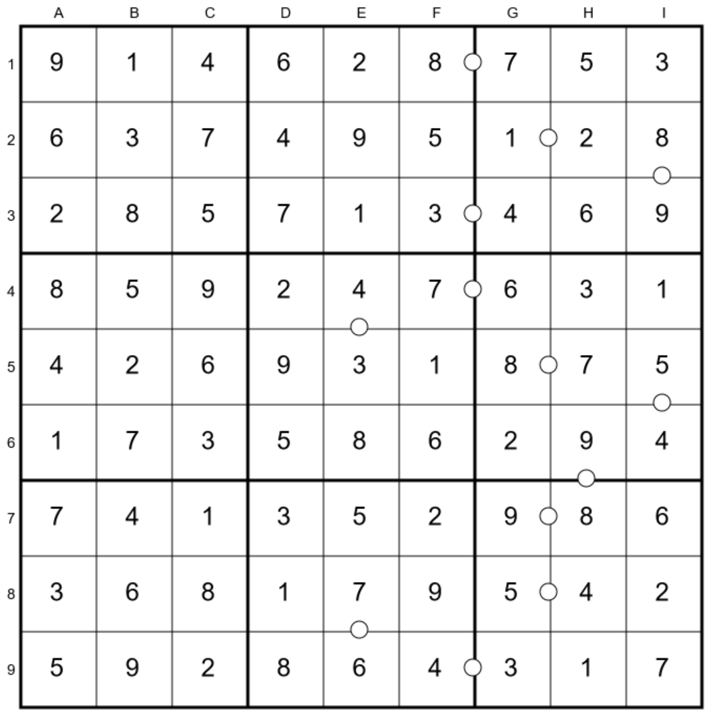
Model description not finished

4.31.6 See Also

whiteDot (Section 6.10) , strictConsecutive (Section 4.23)

4.31.7 strictWhiteDot example wpf gp2025r5c11: "Sudoku GP 2025 Round 5" by





Rules: standard (Section 3.13) strictWhiteDot (Section 4.31)

4.32 Constraint superKing

4.32.1 Category: Global

4.32.2 Description

Cells with are connected diagonally must be at least 2 apart.

4.32.3 Arguments

none

4.32.4 JSON Format

```
"superKing":"true",
```

4.32.5 Model

Model description not finished

4.32.6 See Also

antiKing (Section 4.1)

4.32.7 superKing example ctc93: "Superking" by Aad van de Wetering

Video: `https://www.youtube.com/watch?v=Sd8tPBYj16E`

	A	B	C	D	E	F	G	H	I
1									
2									
3	7								8
4									
5									
6									
7		5						8	
8				6		7			
9									

	A	B	C	D	E	F	G	H	I
1	8	4	9	5	1	6	2	7	3
2	1	3	2	7	9	8	4	6	5
3	7	6	5	4	3	2	1	9	8
4	4	9	8	1	6	5	7	3	2
5	5	1	6	2	7	3	8	4	9
6	3	2	7	9	8	4	6	5	1
7	6	5	4	3	2	1	9	8	7
8	9	8	1	6	5	7	3	2	4
9	2	7	3	8	4	9	5	1	6

Rules: standard (Section 3.13) superKing (Section 4.32) mainDiagonal (Section 4.9) minorDiagonal (Section 4.10)

4.33 Constraint uniqueRenban

4.33.1 Category: Global

4.33.2 Description

All renban lines or areas in the grid must be different from each other, i.e. they cannot have the same min and max value.

4.33.3 Arguments

none

4.33.4 JSON Format

```
"uniqueRenban":"true",
```

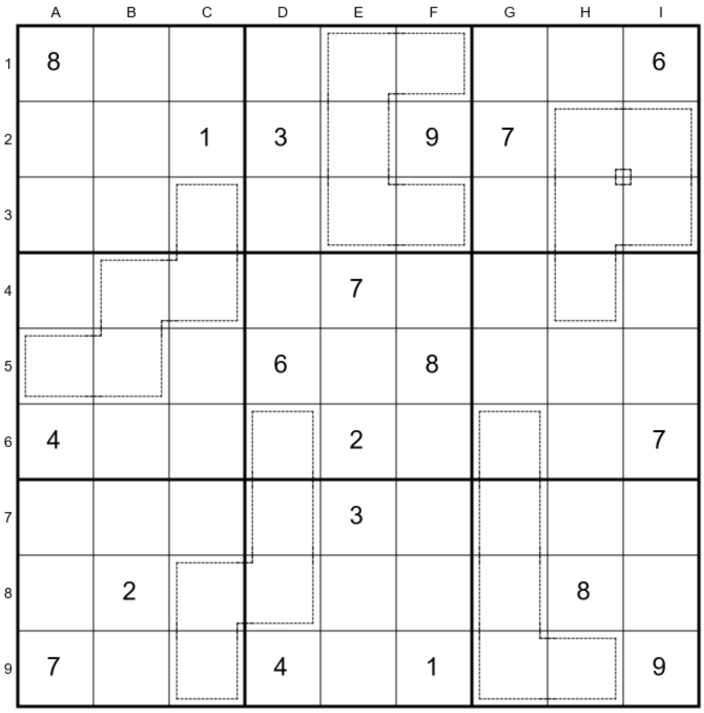
4.33.5 Model

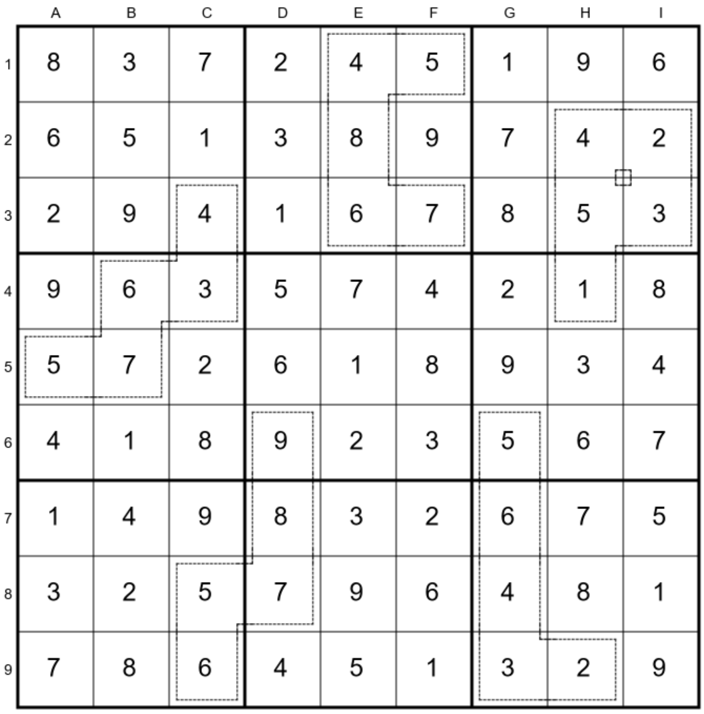
Model description not finished

4.33.6 See Also

renban (Section 9.32)

4.33.7 uniqueRenban example wpf gp2025r7c8: "Sudoku GP 2025 Round 7" by





Rules: standard (Section 3.13) renbanArea (Section 11.14) uniqueRenban (Section 4.33)

4.34 Constraint windoku

4.34.1 Category: Global

4.34.2 Description

The cells on the four 3x3 blocks offset by one from the corners must be alldifferent. This implies a number of additional alldifferent constraints, which are also imposed.

4.34.3 Arguments

none

4.34.4 JSON Format

```
"windoku":"true",
```

4.34.5 Model

Model description not finished

4.34.6 See Also

box (Section 11.3)

4.34.7 windoku example timotab14: "Hedron" by Bill Murphy

Video: `https://www.youtube.com/watch?v=tjqtQtdKXvI`

	A	B	C	D	E	F	G	H	I
1									
2	7		3		6		5		9
3		9		1		2		7	
4									
5									
6									
7	2		6		5		9		8
8		8		3		4		6	
9									

	A	B	C	D	E	F	G	H	I
1	4	6	1	5	7	9	3	8	2
2	7	2	3	4	6	8	5	1	9
3	5	9	8	1	3	2	6	7	4
4	8	5	7	6	2	3	4	9	1
5	9	3	4	8	1	5	7	2	6
6	6	1	2	9	4	7	8	5	3
7	2	4	6	7	5	1	9	3	8
8	1	8	5	3	9	4	2	6	7
9	3	7	9	2	8	6	1	4	5

Rules: standard (Section 3.13) windoku (Section 4.34)

5 Node Constraints

5.1 Constraint closestNine

5.1.1 Category: Node

5.1.2 Description

The arrow in the cell indicates the direction of a nine in the row, column or diagonal. The value of the cell gives the distance to the closest nine in that direction.

5.1.3 Arguments

cell cell

dir Direction

5.1.4 JSON Format

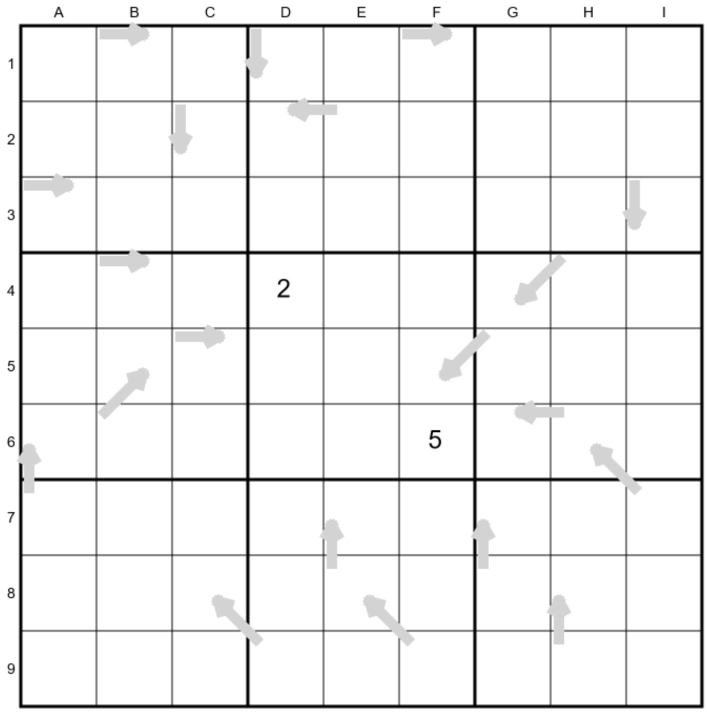
```
"closestNine":[{"cell":2,"dir":"e"},...],
```

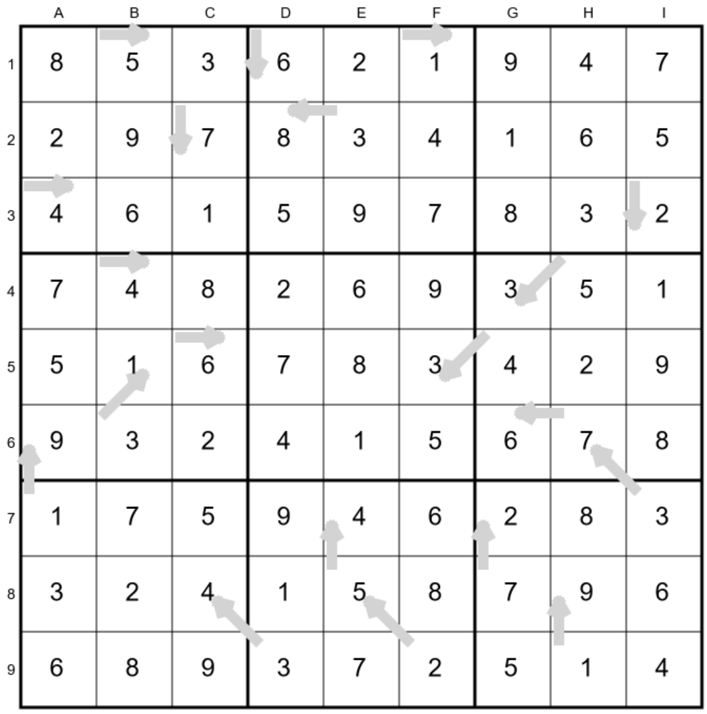
5.1.5 Model

Model description not finished

5.1.6 See Also

5.1.7 closestNine example wpf gp2025r8c9: "Sudoku GP 2025 Round 8" by





Rules: standard (Section 3.13) closestNine (Section 5.1)

5.2 Constraint entropicHigh

5.2.1 Category: Node

5.2.2 Description

The value of the cell indicated by the red rectangle must be high, values 7, 8, or 9.

5.2.3 Arguments

cell Cell

5.2.4 JSON Format

```
"entropicHigh":[11,15,21,25,26,38,48,52,53,58,59,65,69],
```

5.2.5 Model

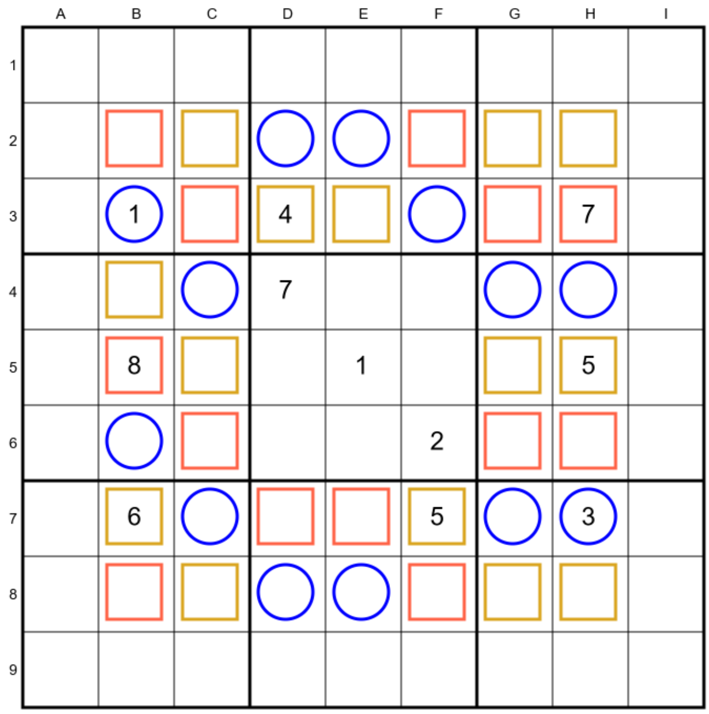
Model description not finished

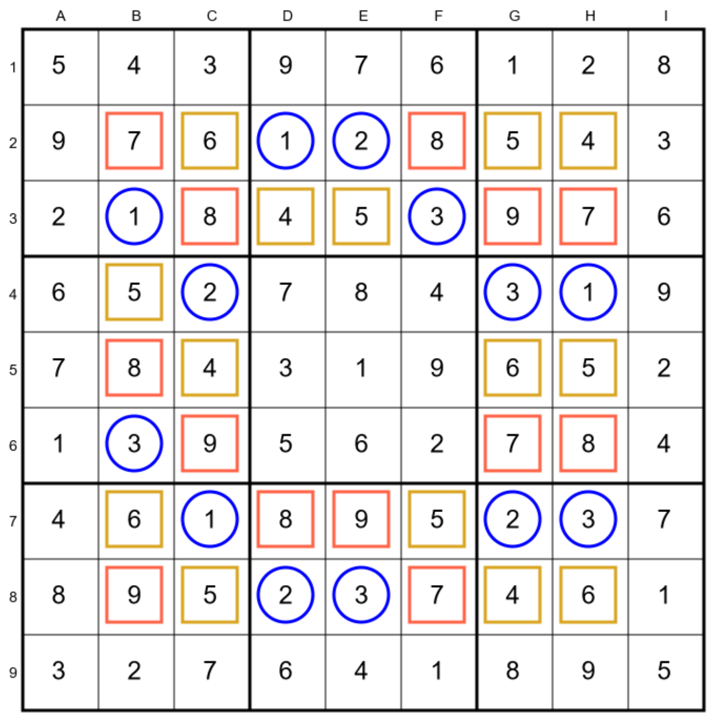
5.2.6 See Also

entropicLow (Section 5.3) , entropicMid (Section 5.4)

5.2.7 entropicHigh example gas47: "Partial Trio" by Philip Newman

Video: https://www.youtube.com/watch?v=GhYkeaaJkyA
Video2: https://www.youtube.com/watch?v=inhIL3EyBWA





Rules: standard (Section 3.13) entropicLow (Section 5.3) entropicMid (Section 5.4) entropicHigh (Section 5.2)

5.3 Constraint entropicLow

5.3.1 Category: Node

5.3.2 Description

The value of the cell indicated by the circle must be low, values 1, 2, or 3.

5.3.3 Arguments

cell Cell

5.3.4 JSON Format

```
"entropicLow":[1,2,9,10,14,18,22,23,25,30,31,36,37,39,42,51,52,53,56,60,62,65,66,71,76,77,79],
```

5.3.5 Model

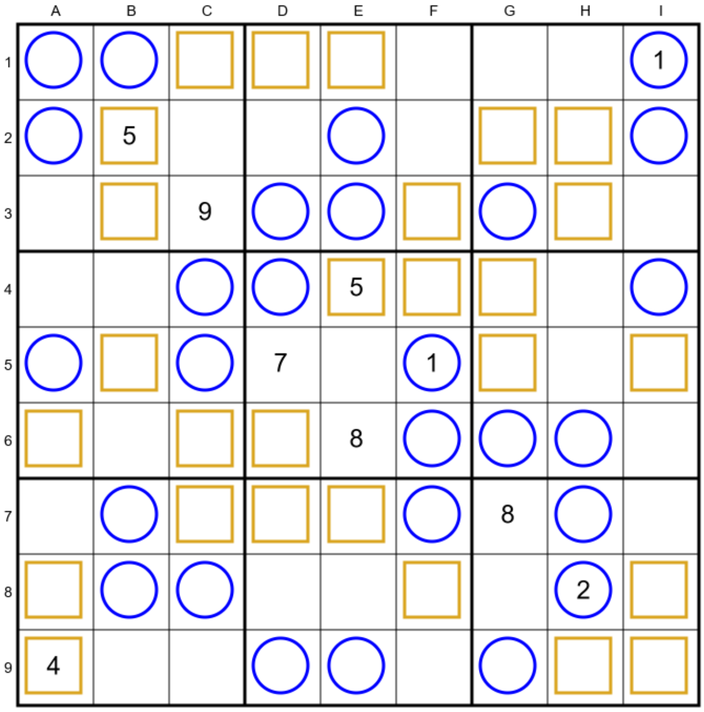
Model description not finished

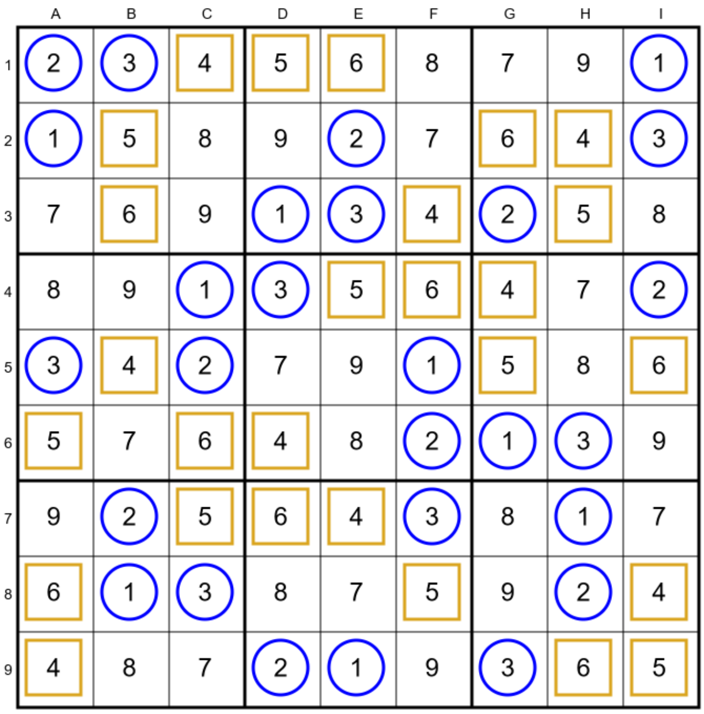
5.3.6 See Also

entropicMid (Section 5.4) , entropicHigh (Section 5.2)

5.3.7 entropicLow example gas31: "Three Rolled Into One" by clover

Video: `https://youtu.be/M4Pn-J1BoUA?si=GVu7FBeJJnbnuN9A`





Rules: standard (Section 3.13) entropicLow (Section 5.3) entropicMid (Section 5.4)

5.4 Constraint entropicMid

5.4.1 Category: Node

5.4.2 Description

The value of the cell indicated by the rectangle must be mid range, values 4, 5, or 6.

5.4.3 Arguments

cell Cell

5.4.4 JSON Format

`"entropicMid":[3,4,5,11,16,17,20,24,26,32,33,34,38,43,45,46,48,49,57,58,59,64,69,72,73,80,81],`

5.4.5 Model

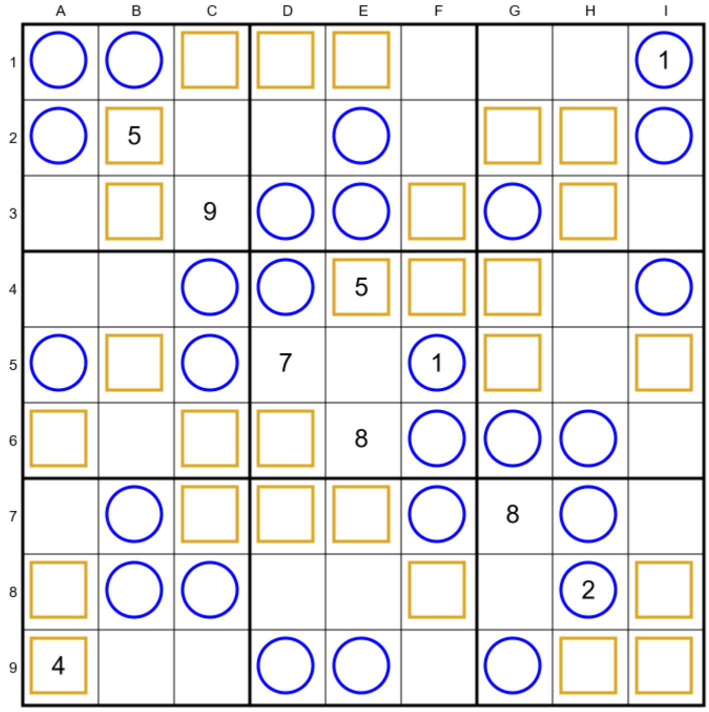
Model description not finished

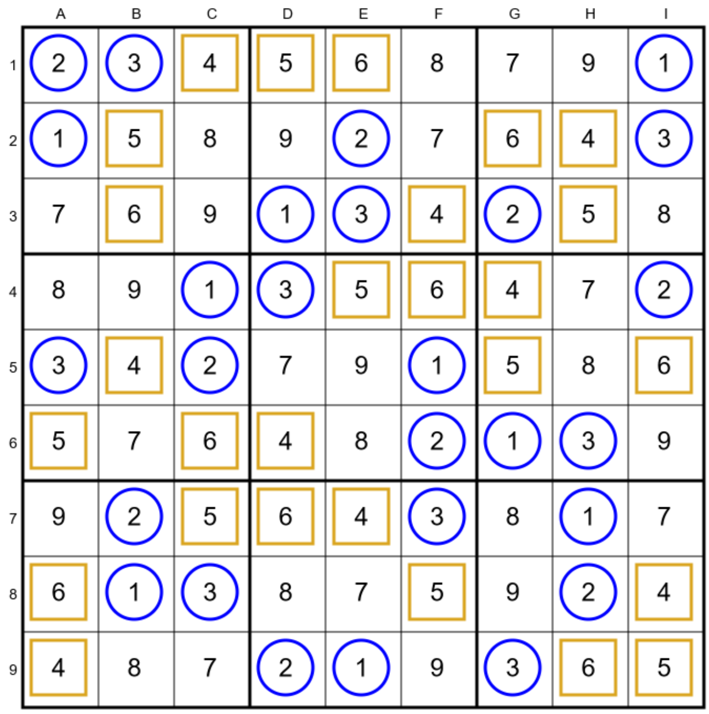
5.4.6 See Also

entropicLow (Section 5.3) , entropicHigh (Section 5.2)

5.4.7 entropicMid example gas31: "Three Rolled Into One" by clover

Video: `https://youtu.be/M4Pn-J1BoUA?si=GVu7FBeJJnbnuN9A`





Rules: standard (Section 3.13) entropicLow (Section 5.3) entropicMid (Section 5.4)

5.5 Constraint even

5.5.1 Category: Node

5.5.2 Description

The cell marked by a rectangle must be even.

5.5.3 Arguments

cell Cell

5.5.4 JSON Format

```
"even":[11,12,16,17,19,22,24,27,28,32,36,37,45,47,53,57,61,67,69,77],
```

5.5.5 Model

Let x be the variable associated with the cell indicated in the constraint. The value of x must be even, i.e.

$$0 \equiv x \bmod 2 \tag{9}$$

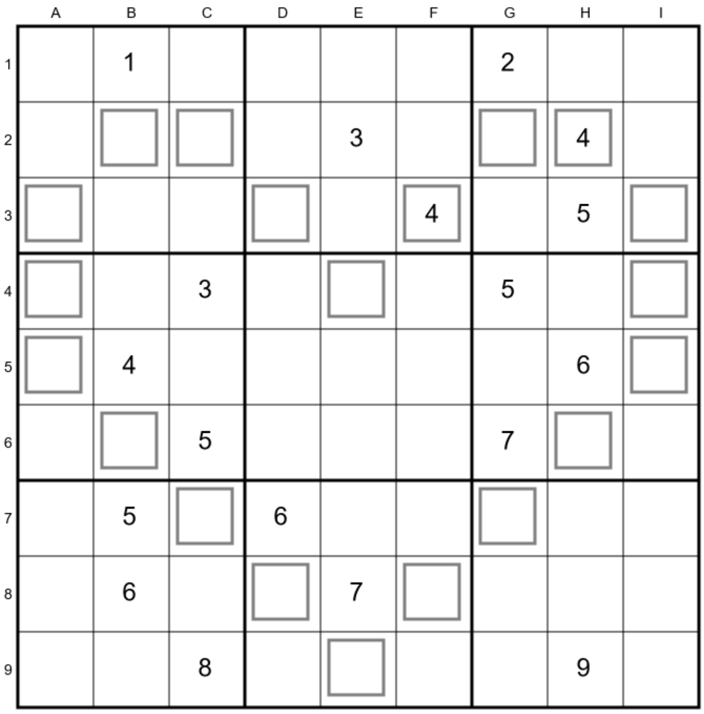
We can avoid the modulo operator by restricting the domain of x to only even numbers.

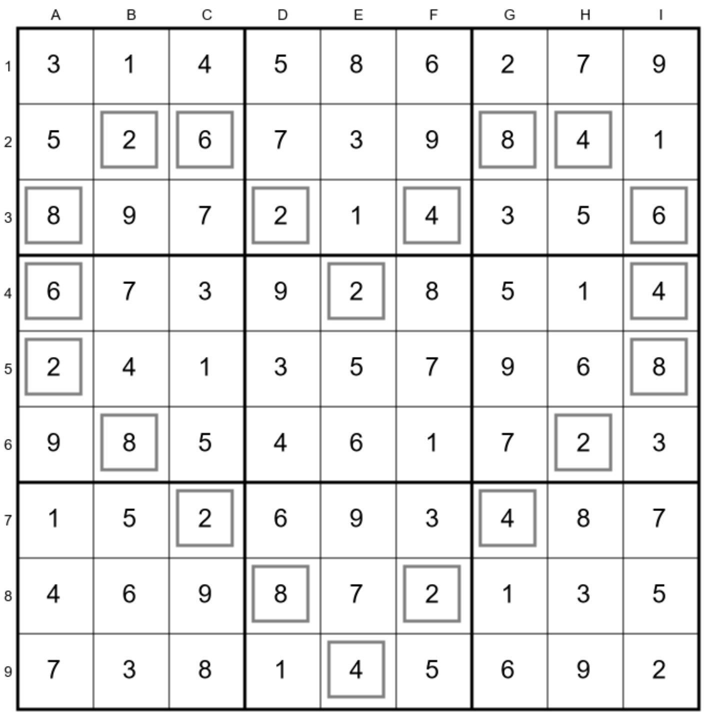
5.5.6 See Also

odd (Section 5.9)

5.5.7 even example timotab15: "Valentine's, Even" by Philip Newman

Video: `https://www.youtube.com/watch?v=SfbdL1VsEPA`
Video2: `https://youtu.be/0j0r3IjoYwI?si=Dbny8Pl0zPILxae1`





Rules: standard (Section 3.13) even (Section 5.5)

5.6 Constraint indexed

5.6.1 Category: Node

5.6.2 Description

The value in the cell must be equal to either the row, column or block number of the cell.

5.6.3 Arguments

indexed List(Cell)

5.6.4 JSON Format

"indexed":[4,7,9,16,18,20,23,26,27,28,31,33,35,37,38,39,46,47,58,59,60,62,65,68,70,76,77],

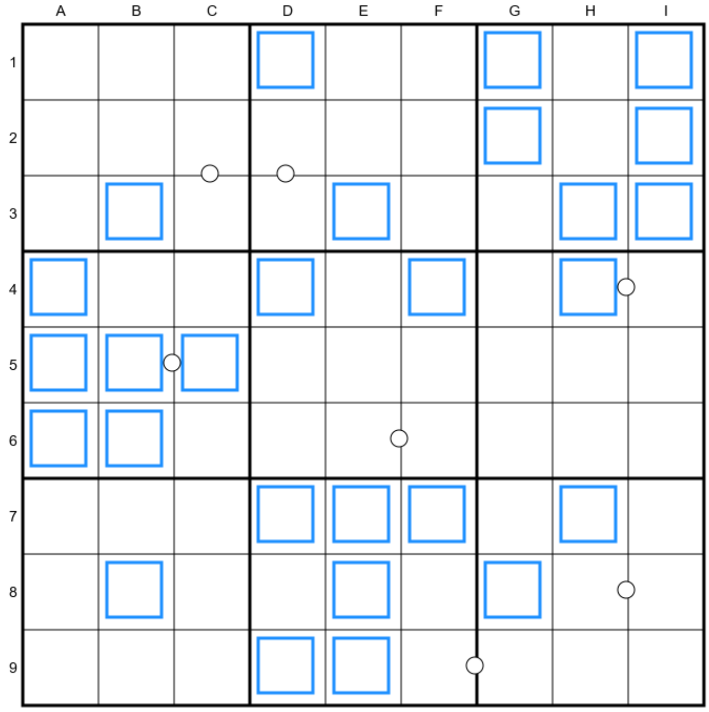
5.6.5 Model

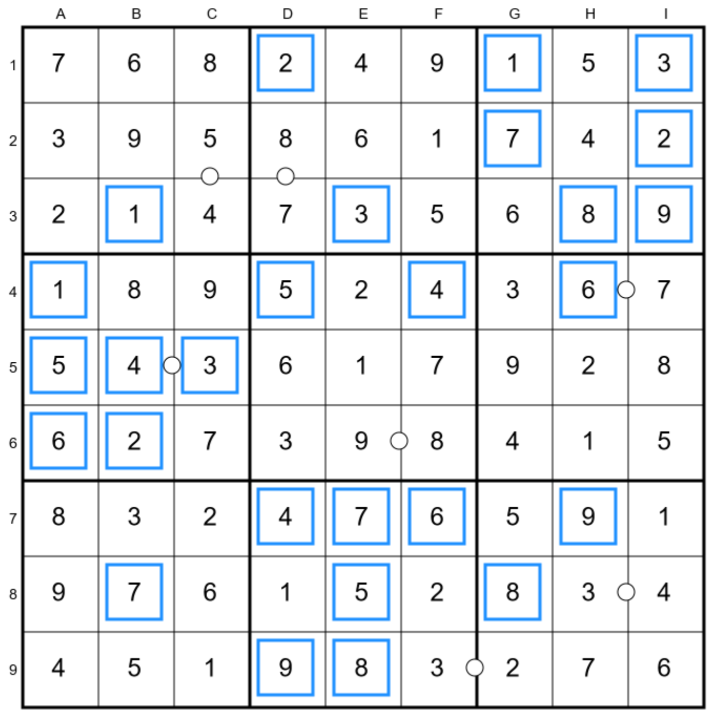
Model description not finished

5.6.6 See Also

5.6.7 indexed example logiclemurgaming7: "Fields of Clover" by balpha

Video: `https://www.youtube.com/watch?v=Cf2vSZ3D3SE&t=150s`





Rules: standard (Section 3.13) indexed (Section 5.6) whiteDot (Section 6.10)

5.7 Constraint nValue

5.7.1 Category: Node

5.7.2 Description

The value in the cell tells how many different digits occur in the neighbouring eight cells.

5.7.3 Arguments

cell Cell

value Integer

5.7.4 JSON Format

```
"nValue":[{"cell":21,"value":3},{"cell":25,"value":4},...],
```

5.7.5 Model

Model description not finished

5.7.6 See Also

5.7.7 nValue example timotab3: "Count Different" by Bill Murphy

Video: `https://www.youtube.com/watch?v=dgVN6scJm4c`

	A	B	C	D	E	F	G	H	I
1		6							
2		4	1				8	3	9
3		2	3				4	6	
4									
5									
6									
7		8	4				3	9	
8	2	7	6				5	8	
9								1	

	A	B	C	D	E	F	G	H	I
1	8	6	5	9	3	4	1	2	7
2	7	4	1	2	5	6	8	3	9
3	9	2	3	1	8	7	4	6	5
4	5	1	2	4	9	3	6	7	8
5	6	9	8	5	7	1	2	4	3
6	4	3	7	6	2	8	9	5	1
7	1	8	4	7	6	5	3	9	2
8	2	7	6	3	1	9	5	8	4
9	3	5	9	8	4	2	7	1	6

Rules: standard (Section 3.13) nValue (Section 5.7)

5.8 Constraint neighborSum

5.8.1 Category: Node

5.8.2 Description

The value in a circled cell indicates the sum of the orthogonally connected neighbours of that cell.

5.8.3 Arguments

cell Cell

5.8.4 JSON Format

```
"neighborSum":[3,5,7,9,13,19,21,29,36,37,55,61,73,76,81],
```

5.8.5 Model

Model description not finished

5.8.6 See Also

5.8.7 neighborSum example gas70: "Circle of Summands" by clover

Video: `https://youtu.be/E--VB9JBDEo?si=pi1MfRPAcULfCaPL`

	A	B	C	D	E	F	G	H	I
1			7						6
2				8					
3	8		6						
4		7							9
5									
6									
7							9		5
8									
9	6			9			4		8

	A	B	C	D	E	F	G	H	I
1	3	4	7	1	9	5	8	2	6
2	5	9	2	8	3	6	1	7	4
3	8	1	6	2	7	4	5	9	3
4	2	7	1	3	4	8	6	5	9
5	9	3	8	5	6	2	7	4	1
6	4	6	5	7	1	9	3	8	2
7	7	2	4	6	8	1	9	3	5
8	1	8	9	4	5	3	2	6	7
9	6	5	3	9	2	7	4	1	8

Rules: standard (Section 3.13) neighborSum (Section 5.8)

5.9 Constraint odd

5.9.1 Category: Node

5.9.2 Description

The cell marked by a circle must be odd.

5.9.3 Arguments

cell Cell

5.9.4 JSON Format

```
"odd":[11,12,16,17,19,22,24,27,28,32,36,37,45,47,53,57,61,67,69,77],
```

5.9.5 Model

Let x be the variable associated with the cell indicated in the constraint. The value of x must be odd, i.e.

$$1 \equiv x \bmod 2 \tag{10}$$

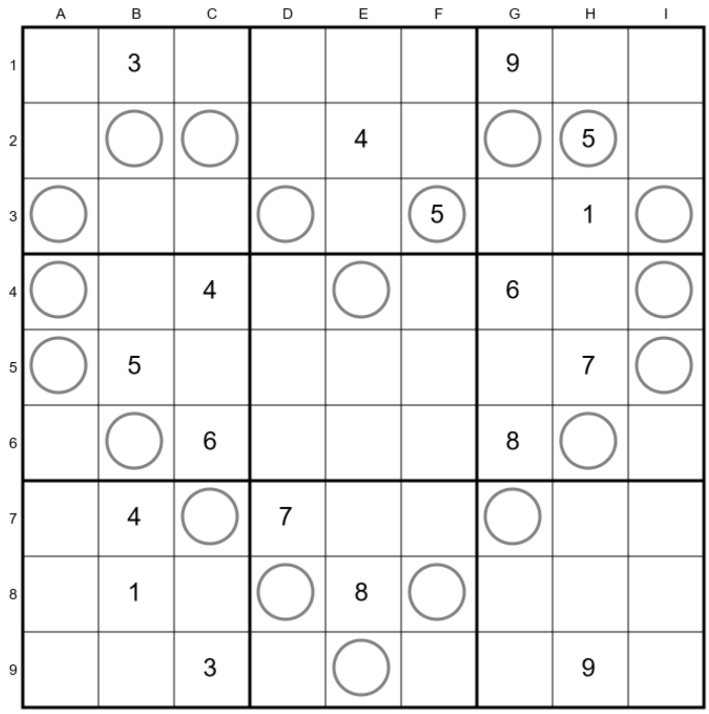
We can avoid the modulo operator by restricting the domain of x to only odd numbers.

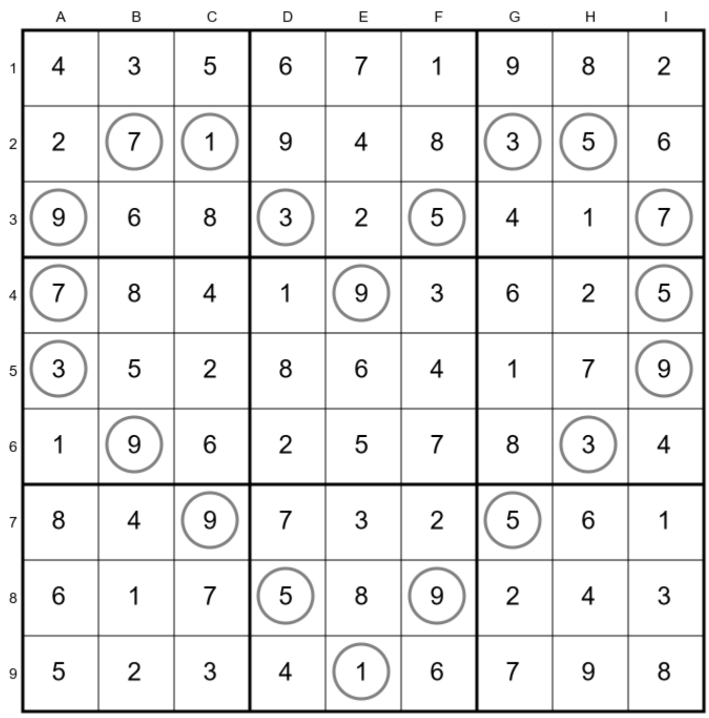
5.9.6 See Also

even (Section 5.5)

5.9.7 odd example timotab18: "An Odd Coincidence" by Philip Newman

Video: `https://www.youtube.com/watch?v=3qfYRMQSlzI`





Rules: standard (Section 3.13) odd (Section 5.9)

5.10 Constraint oddNeighbors

5.10.1 Category: Node

5.10.2 Description

The value in the cell indicates how many of the orthogonal neighbour cells are odd.

5.10.3 Arguments

cell Cell

5.10.4 JSON Format

```
"oddNeighbors":[9,11,33,37,54,61],
```

5.10.5 Model

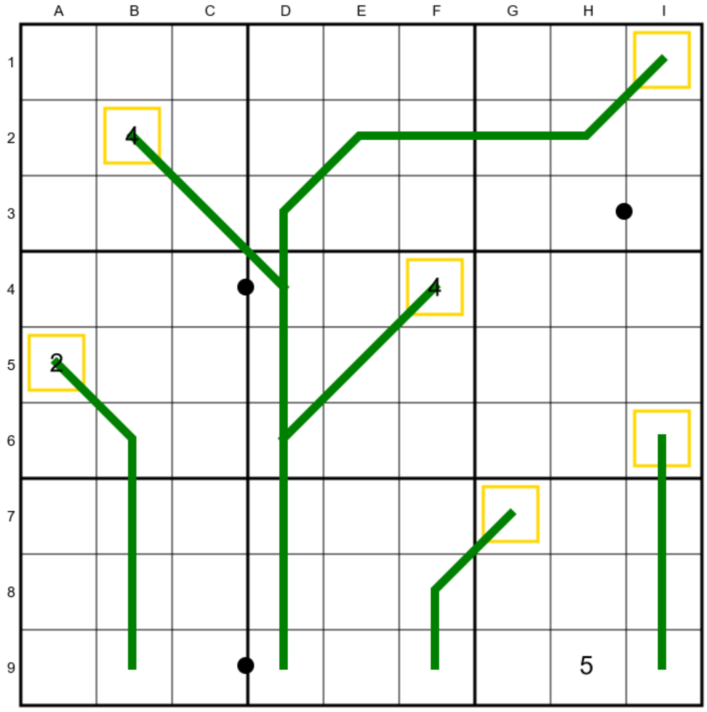
Model description not finished

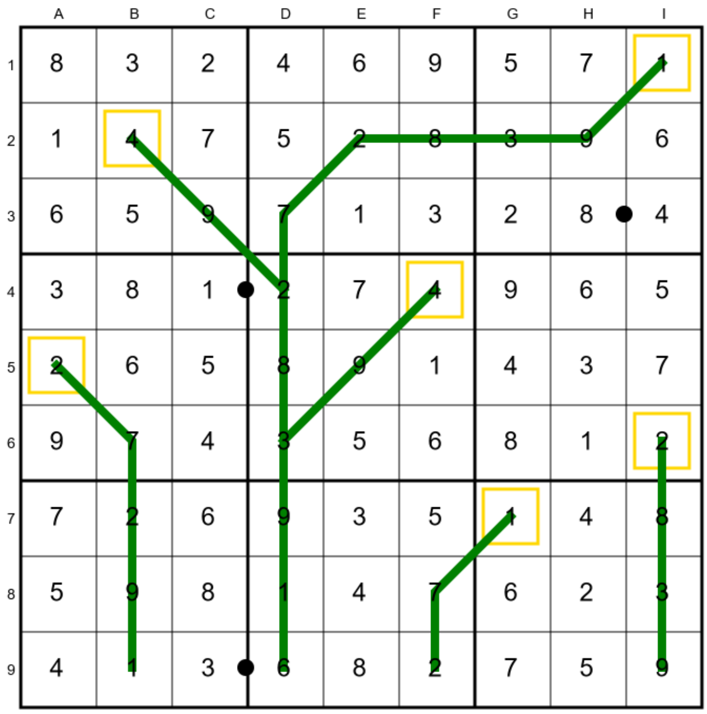
5.10.6 See Also

evenNeighbors (Section **??**)

5.10.7 oddNeighbors example ctc63: "Flower Petals" by Gregui

Video: `https://www.youtube.com/watch?v=X2e6JsSPvWQ`





Rules: standard (Section 3.13) germanWhisper (Section 9.19) blackDot (Section 6.1) oddNeighbors (Section 5.10)

5.11 Constraint offByOne

5.11.1 Category: Node

5.11.2 Description

If a cell contains a hint, the value in the cell is either exactly one larger or exactly one smaller than the hint.

5.11.3 Arguments

cell Cell

value Integer

5.11.4 JSON Format

```
"offByOne":[{"cell":1,"value":2},{"cell":2,"value":2},...],
```

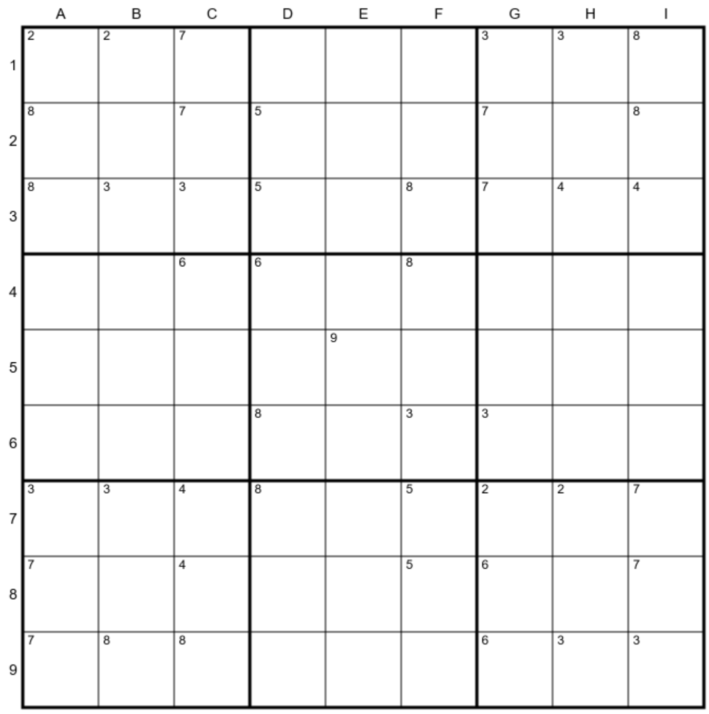
5.11.5 Model

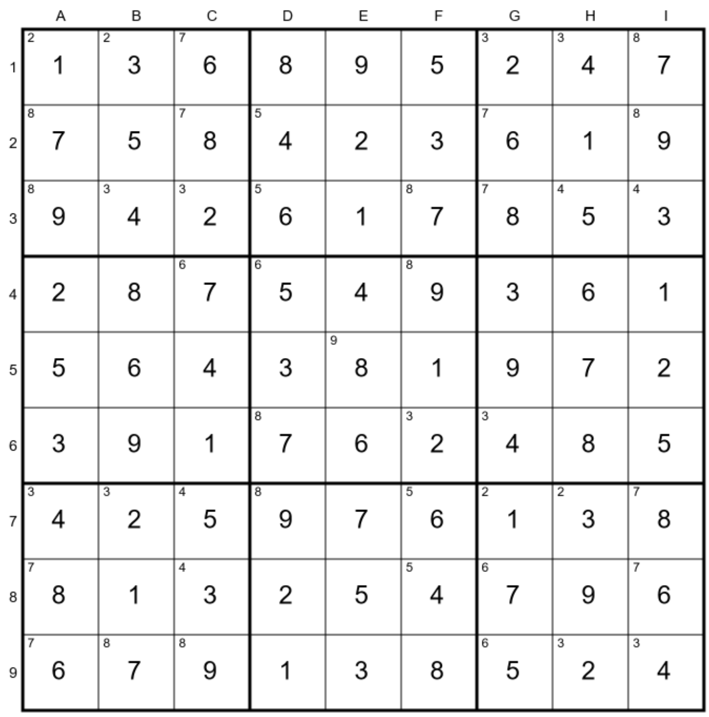
Model description not finished

5.11.6 See Also

5.11.7 offByOne example timotab21: "Stretching the Truth" by clover

Video: `https://www.youtube.com/watch?v=AfiJCQ0eEaU`
Video2: `https://www.youtube.com/watch?v=E7RDDTbd26I&t=14s`





Rules: standard (Section 3.13) offByOne (Section 5.11)

5.12 Constraint partialOrder

5.12.1 Category: Node

5.12.2 Description

For cells containing an arrow, its value must be higher than the value of all cells pointed to by the arrow.

5.12.3 Arguments

cell cell

dir Direction

5.12.4 JSON Format

```
"partialOrder":[{"cell":2,"dir":"se"},...],
```

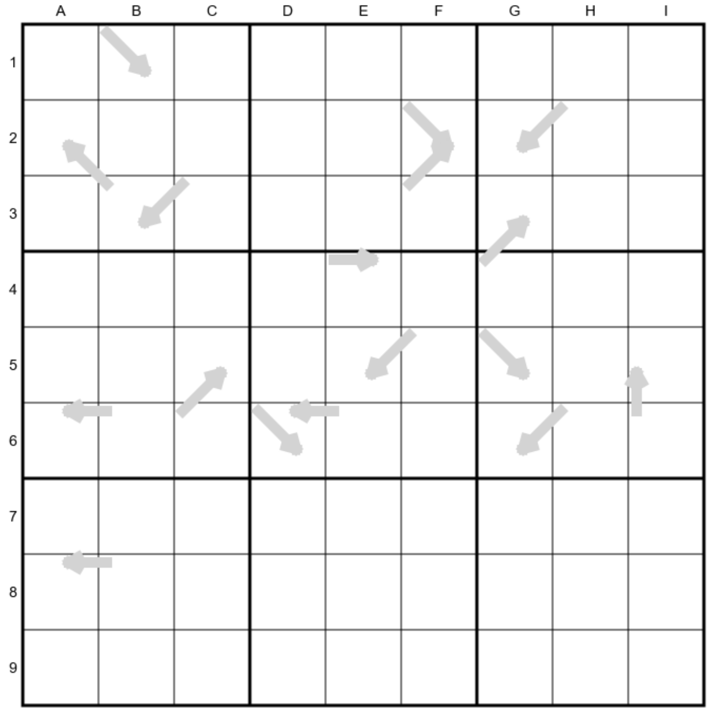
5.12.5 Model

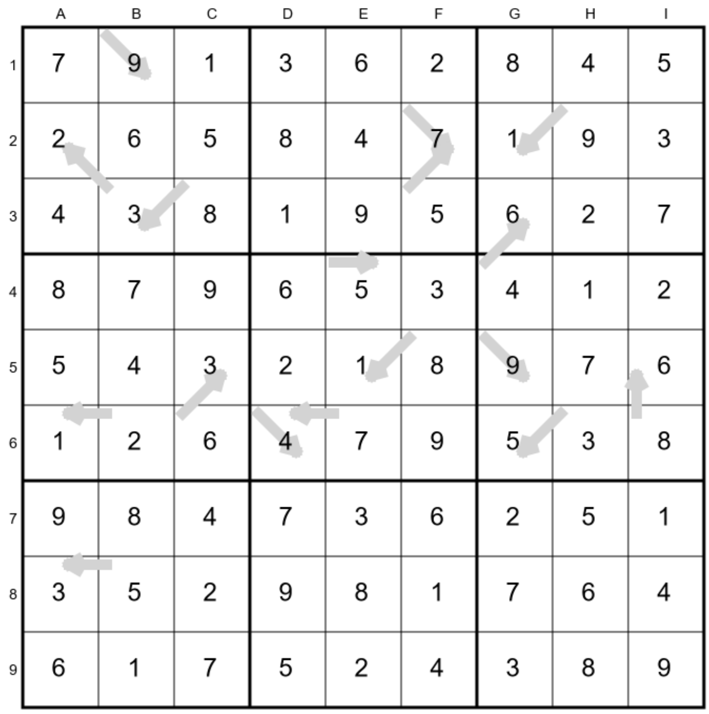
Model description not finished

5.12.6 See Also

5.12.7 partialOrder example ctc83: "Take The Low Road" by Chilly

Video: `https://www.youtube.com/watch?v=wSPu8nH59iY`





	A	B	C	D	E	F	G	H	I
1	7	9	1	3	6	2	8	4	5
2	2	6	5	8	4	7	1	9	3
3	4	3	8	1	9	5	6	2	7
4	8	7	9	6	5	3	4	1	2
5	5	4	3	2	1	8	9	7	6
6	1	2	6	4	7	9	5	3	8
7	9	8	4	7	3	6	2	5	1
8	3	5	2	9	8	1	7	6	4
9	6	1	7	5	2	4	3	8	9

Rules: standard (Section 3.13) partialOrder (Section 5.12)

5.13 Constraint repeated

5.13.1 Category: Node

5.13.2 Description

The arrow in the cell points in the direction of a repeated occurrence of the value n the cell.

5.13.3 Arguments

cell cell

dir Direction

5.13.4 JSON Format

```
"repeated":[{"cell":16,"dir":"se"},...]
```

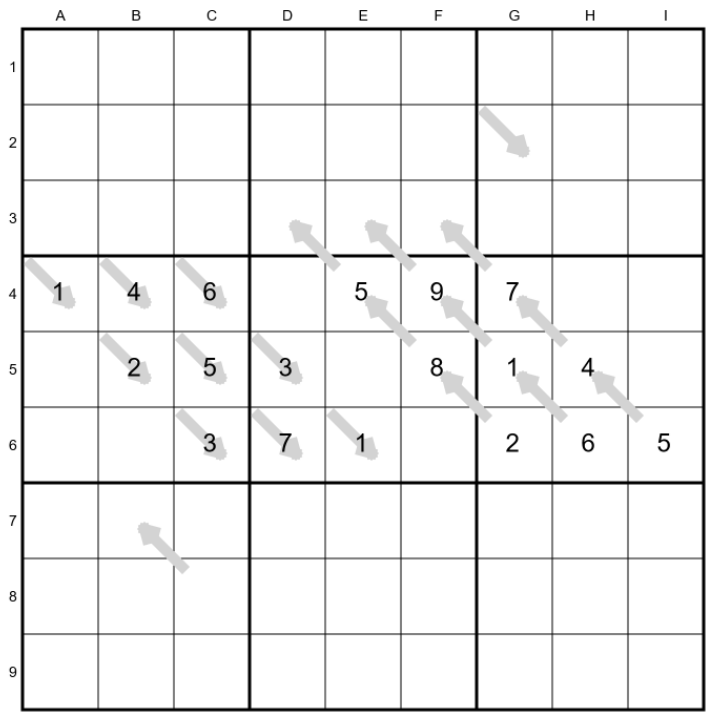
5.13.5 Model

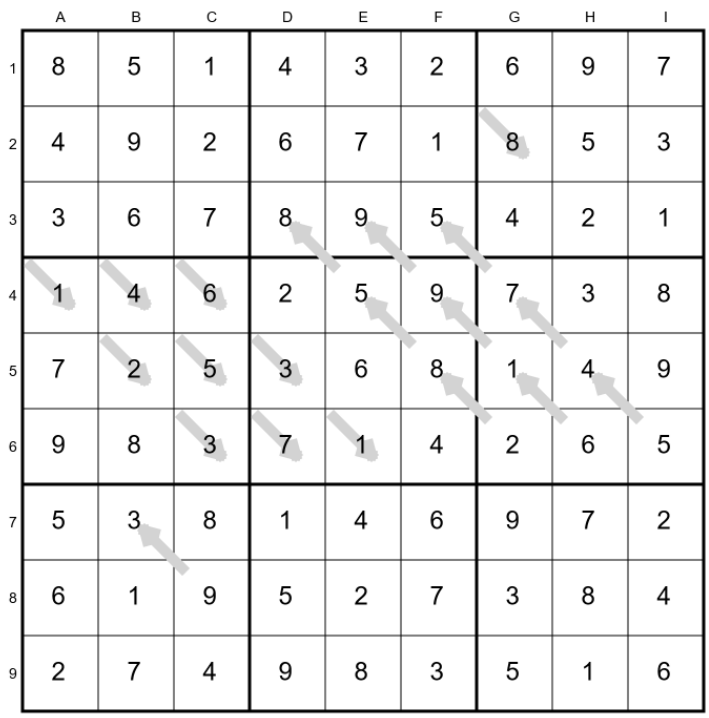
Model description not finished

5.13.6 See Also

5.13.7 repeated example gas44: "Playing Classics" by Bill Murphy

Video: `https://youtu.be/-V_8d0Qkez0?si=4M9Of0MnBH6v3dNz`





Rules: standard (Section 3.13) repeated (Section 5.13)

5.14 Constraint twinDetector

5.14.1 Category: Node

5.14.2 Description

An arrow in a cell indicates that the sum of the nearest neighbours of the cell in the given direction is equal to the value of the cell.

5.14.3 Arguments

cell cell

directions List(Direction)

5.14.4 JSON Format

```
"twinDetector":[{"cell":12,"directions":["ne","se"]},...]
```

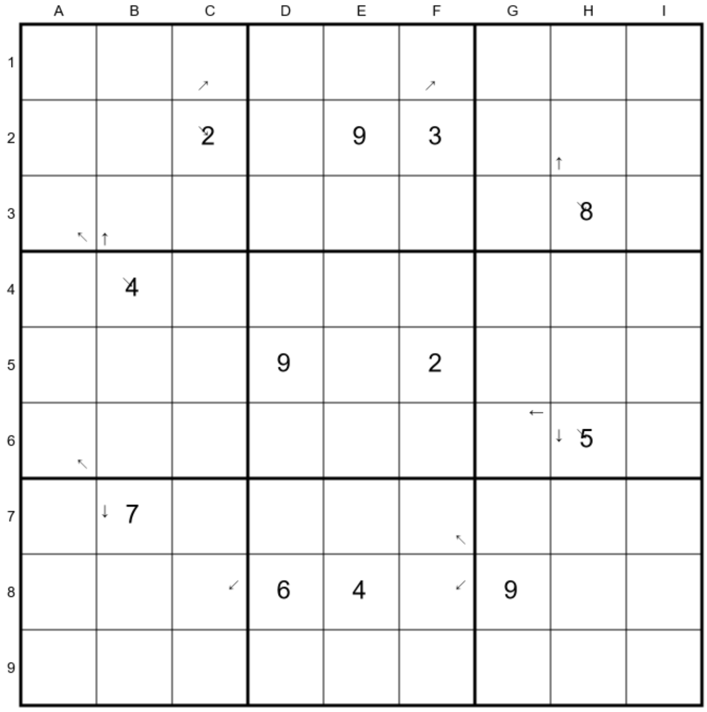
5.14.5 Model

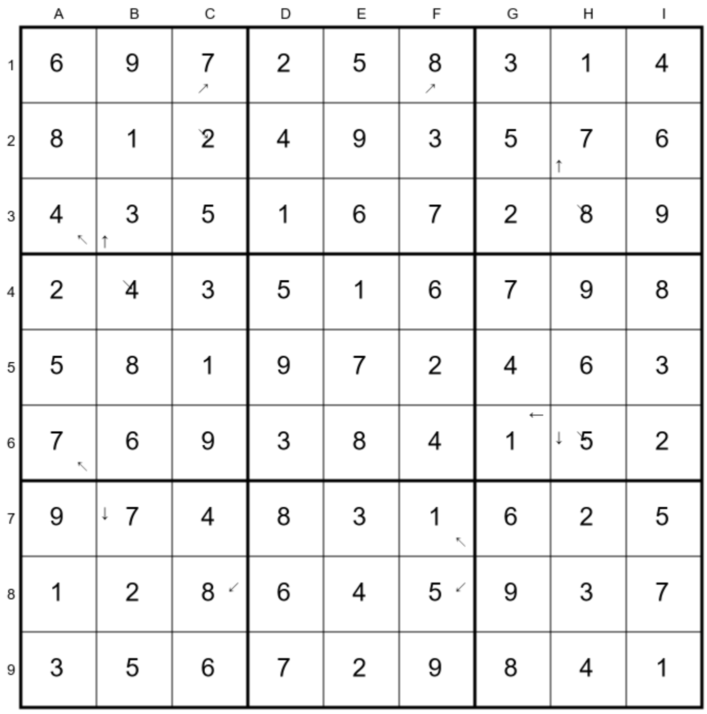
Model description not finished

5.14.6 See Also

5.14.7 twinDetector example gas28: "Twin Detector" by Bill Murphy

Video: `https://youtu.be/_WnXiBz6FpY?si=7VSxEQ6RsCI0kg05`





Rules: standard (Section 3.13) twinDetector (Section 5.14)

5.15 Constraint xValue

5.15.1 Category: Node

5.15.2 Description

The value in the cell tells how many different digits occur in the four diagonally neighbouring cells.

5.15.3 Arguments

cell Cell

5.15.4 JSON Format

```
"xValue":[12,13,25,26,35,47,56,57,69,70],
```

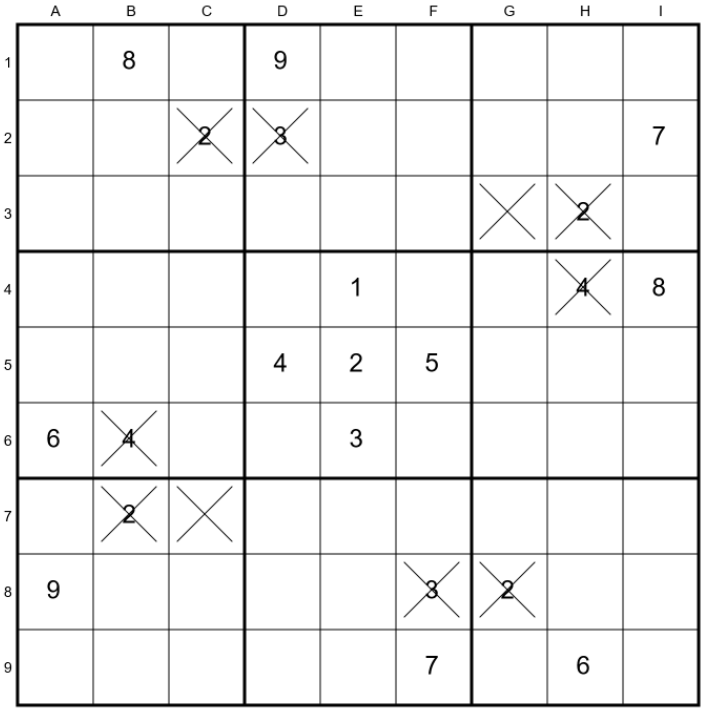
5.15.5 Model

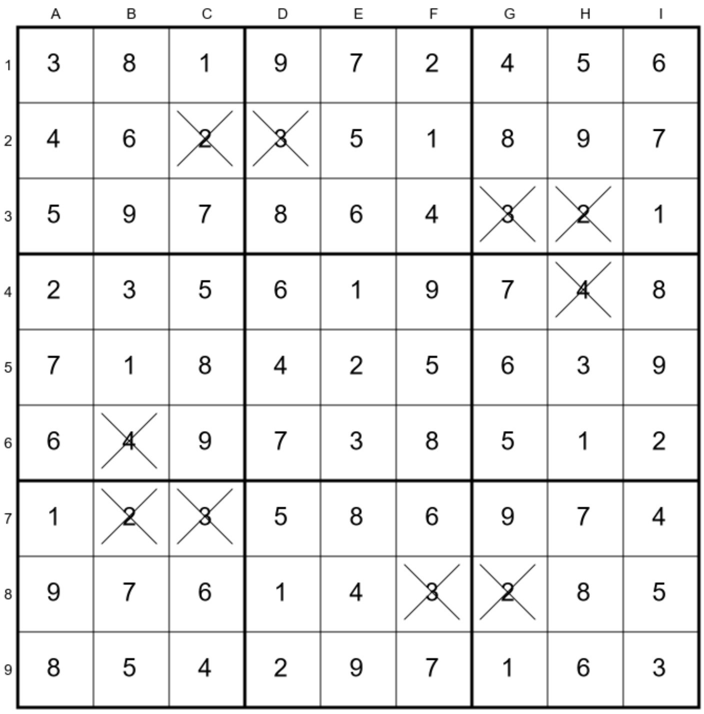
Model description not finished

5.15.6 See Also

5.15.7 xValue example timotab74: "X Marks The Spot" by clover

Video: `https://www.youtube.com/watch?v=wh4Q25DKVJU`





Rules: standard (Section 3.13) xValue (Section 5.15)

6 Pair Constraints

6.1 Constraint blackDot

6.1.1 Category: Pair

6.1.2 Description

The values in two cells marked by a black dot must be X and 2X, in any order.

6.1.3 Arguments

cell1 Cell

cell2 Cell

6.1.4 JSON Format

```
"blackDot":[[1,2],[8,17],[13,14],[29,38],[34,43],[55,64],[58,59],[70,71]],
```

6.1.5 Model

Let x_1 and x_2 be the variables corresponding to the two given cells of the pair. The variables must be in 1 1:2 ratio, i.e. one is the double of the other. This can be expressed by a constraint.

$$x_1 = 2x_2 \lor x_2 = 2x_1 \tag{11}$$

We can express the constraint with a table constraint

$$\text{table}([x_1, x_2], \text{blackDot}_2) \tag{12}$$

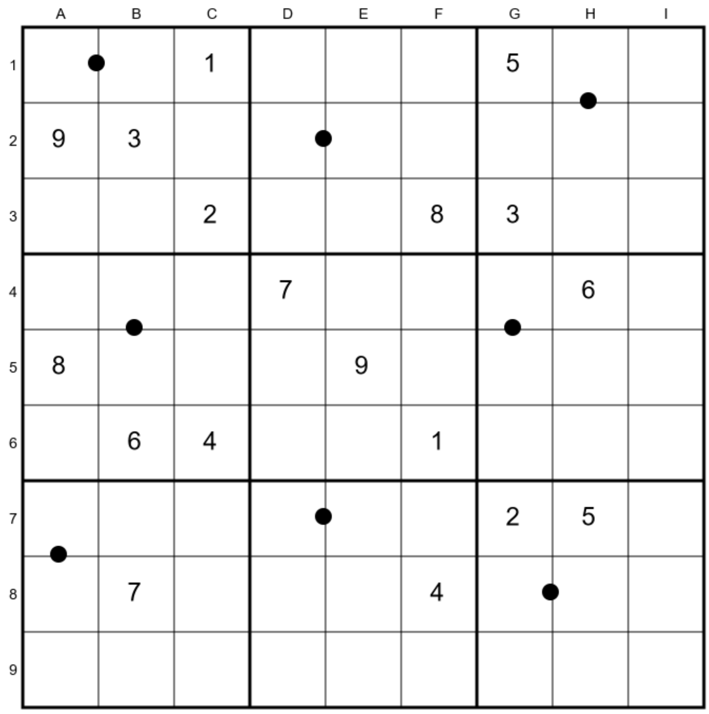
with tuples blackDot_2 which enumerates all combinations of values that satisfy constraint (11).

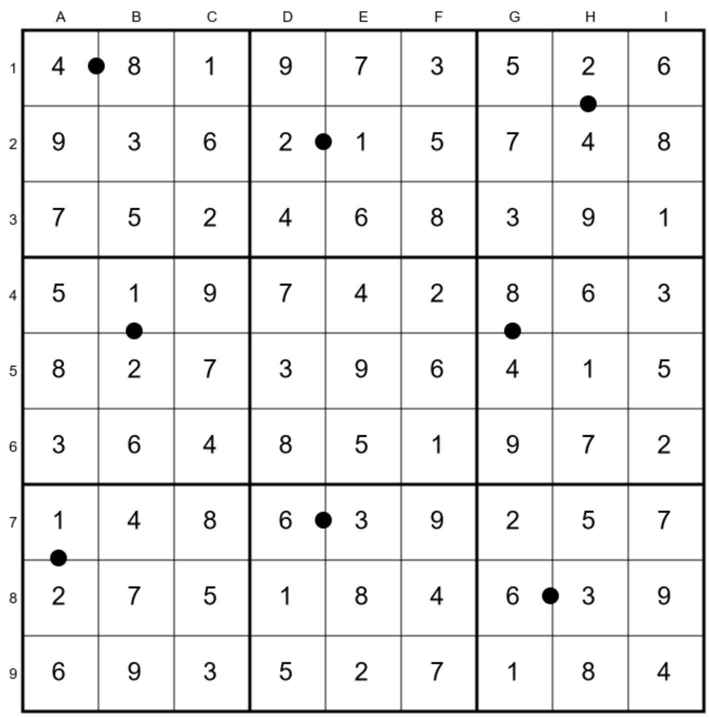
6.1.6 See Also

whiteDot (Section 6.10) , strictBlackDot (Section 4.20) , strictKropki (Section 4.26) , ratioPair (Section 6.6) , blackDotTriple (Section 7.1)

6.1.7 blackDot example timotab50: "Partial Pairs" by Philip Newman

Video: `https://www.youtube.com/watch?v=Pb-14z8wd9U`





Rules: standard (Section 3.13) blackDot (Section 6.1) strictConsecutive (Section 4.23)

6.2 Constraint differencePair

6.2.1 Category: Pair

6.2.2 Description

The absolute difference between the two cells in the pair equals the difference given as the hint in the white dot between the cells.

6.2.3 Arguments

cell1 Cell

cell2 Cell

Value Integer

6.2.4 JSON Format

```
"differencePair":[{"cell1":12,"cell2":13,"value":3},...]
```

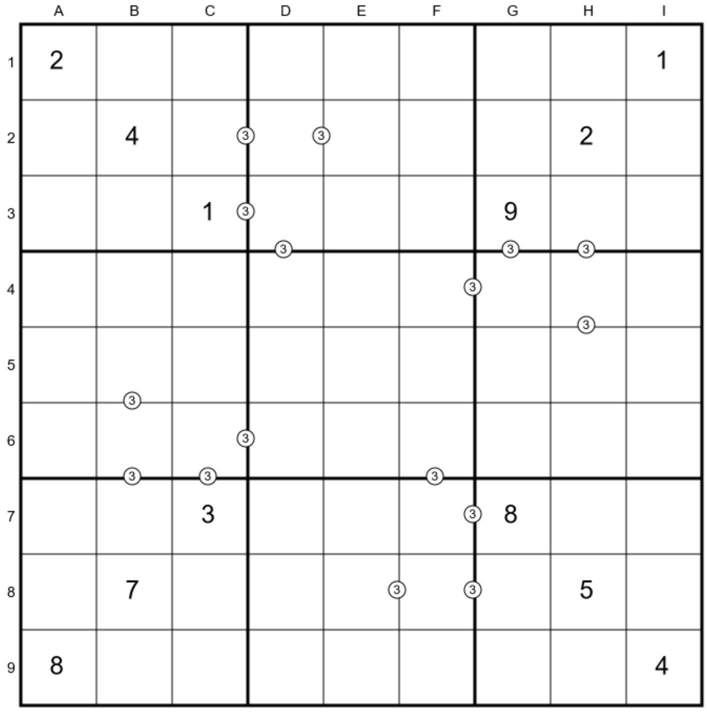
6.2.5 Model

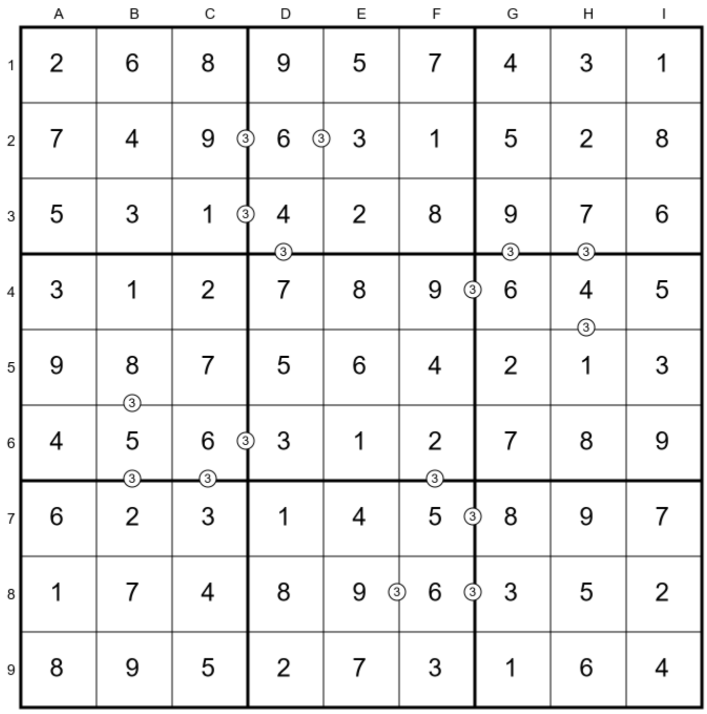
Model description not finished

6.2.6 See Also

6.2.7 differencePair example gas87: "Fotomat" by Bill Murphy

Video: `https://www.youtube.com/watch?v=LEh3U_QdUZQ`





Rules: standard (Section 3.13) differencePair (Section 6.2)

6.3 Constraint lt

6.3.1 Category: Pair

6.3.2 Description

An inequality sign between a pair of cells indicates that the value in one cell is smaller than the value in the other.

6.3.3 Arguments

cell1 Cell

cell2 Cell

6.3.4 JSON Format

```
"lt":[[10,11],[11,12],[13,14],[14,15],[16,17],[17,18],...],
```

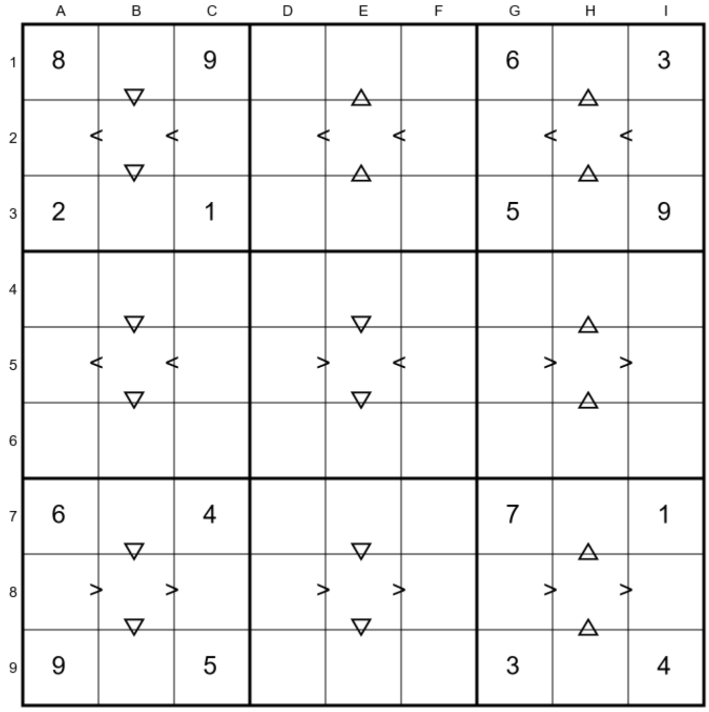
6.3.5 Model

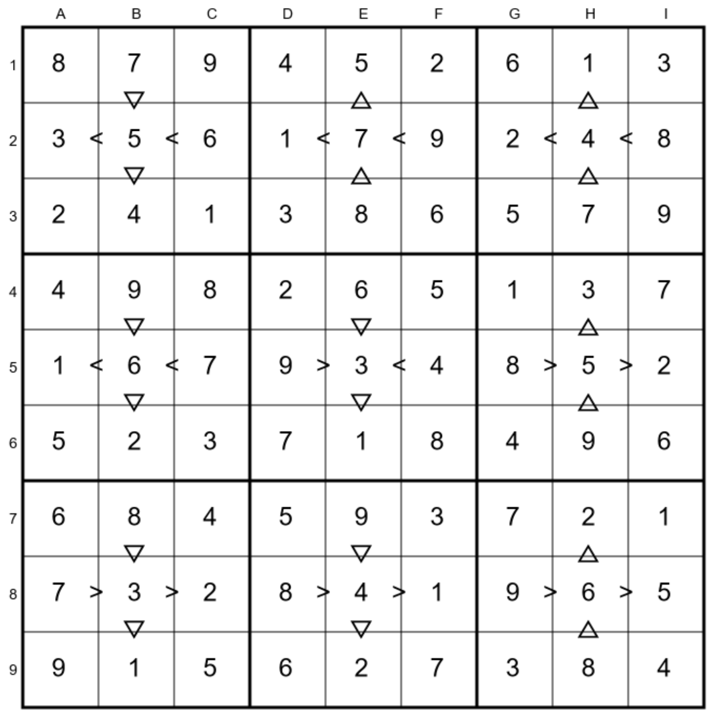
Model description not finished

6.3.6 See Also

6.3.7 lt example gas34: "Flower Garden" by clover

Video: `https://youtu.be/ipLsV-lvhZw?si=-HbirHjSTAzuGqfG`





Rules: standard (Section 3.13) lt (Section 6.3)

6.4 Constraint oddEvenPair

6.4.1 Category: Pair

6.4.2 Description

One of the two cells in the pair is odd, the other is even.

6.4.3 Arguments

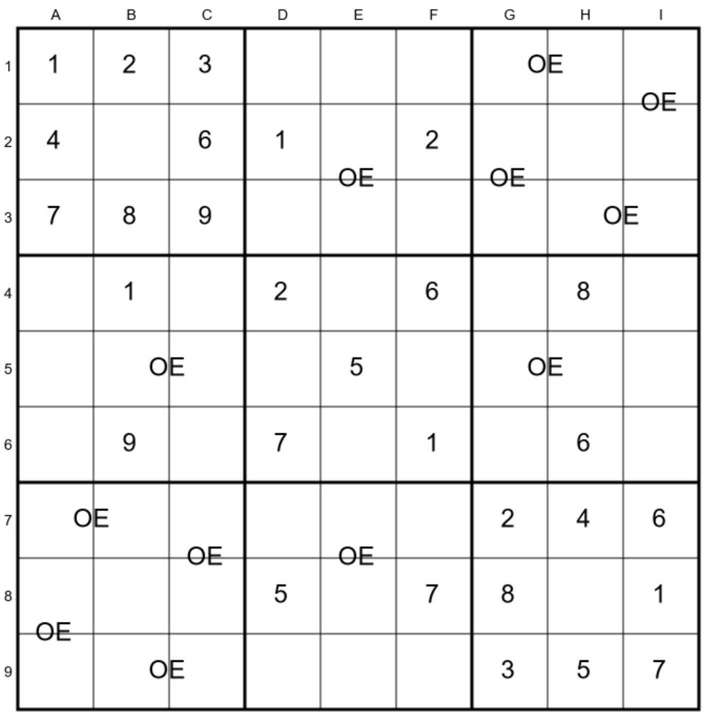
none

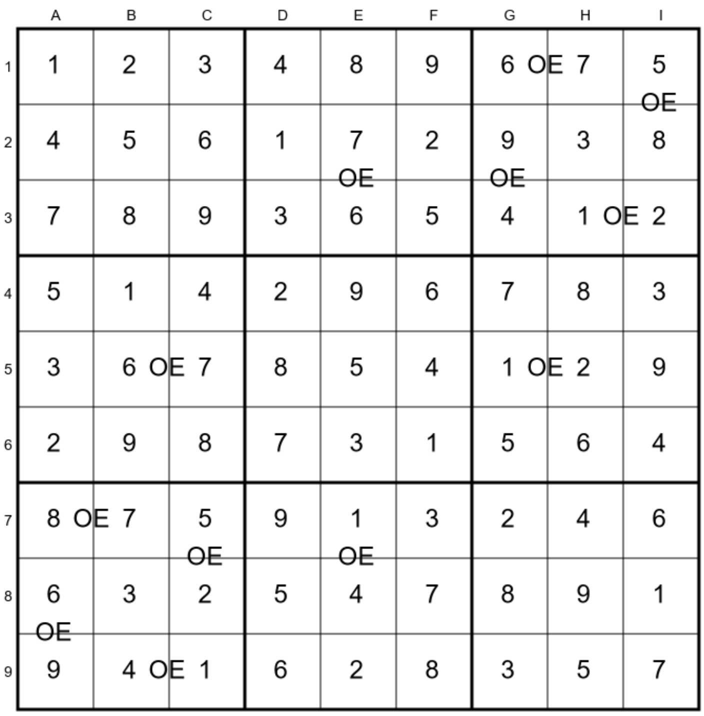
6.4.4 Model

Model description not finished

6.4.5 See Also

6.4.6 oddEvenPair example World of Sudoku vol. 81: "Capsule Sudoku 01" by Akash Doulani





Rules: standard (Section 3.13) oddEvenPair (Section 6.4)

6.5 Constraint oddNeighborPair

6.5.1 Category: Pair

6.5.2 Description

The value in the circled hint between two cells gives the number of odd values in the orthogonal and diagonal neighbours of the two cells in the pair. The two pair cells are not included in the count.

6.5.3 Arguments

cell1 Cell

cell2 Cell

Value Integer

6.5.4 JSON Format

```
"oddNeighborPair":[{"cell1":12,"cell2":13,"value":10},...],
```

6.5.5 Model

Model description not finished

6.5.6 See Also

6.5.7 oddNeighborPair example gas77: "Ten and Two" by clover

Video: `https://www.youtube.com/watch?v=cn7-XNawaCg`

	A	B	C	D	E	F	G	H	I
1	6			3					8
2								3	
3			5						
4	4				1			6	
5									
6		9			2				7
7							5		
8		2							
9	5					3			9

	A	B	C	D	E	F	G	H	I
1	6	1	9	3	5	4	2	7	8
2	8	7	4	2	9	6	1	3	5
3	2	3	5	1	7	8	4	9	6
4	4	5	7	8	1	9	3	6	2
5	1	8	2	6	3	7	9	5	4
6	3	9	6	4	2	5	8	1	7
7	7	6	3	9	4	2	5	8	1
8	9	2	8	5	6	1	7	4	3
9	5	4	1	7	8	3	6	2	9

Rules: standard (Section 3.13) oddNeighborPair (Section 6.5)

6.6 Constraint ratioPair

6.6.1 Category: Pair

6.6.2 Description

The values in the two cells connected by the black dot are in the ratio given by the clue, i.e. a black dot with a three inside means that one cells is three times the value in the other cell.

6.6.3 Arguments

cell1 Cell

cell2 Cell

Value Integer

6.6.4 JSON Format

```
"ratioPair":[{"cell1":1,"cell2":10,"value":5},
```

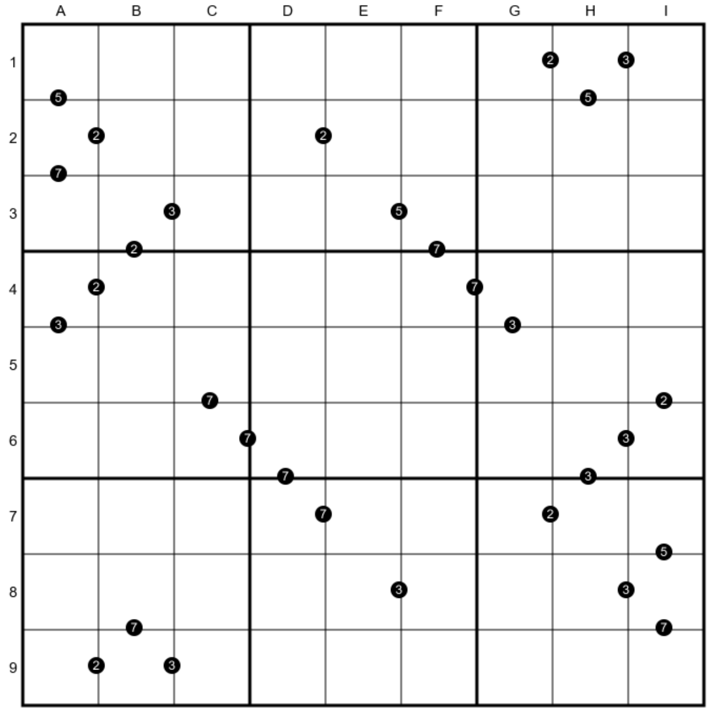
6.6.5 Model

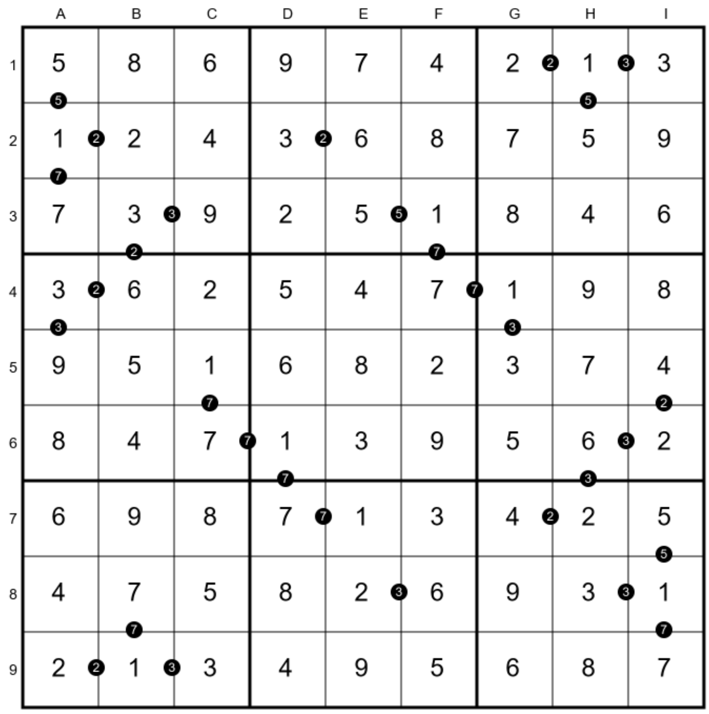
Model description not finished

6.6.6 See Also

6.6.7 ratioPair example gas48: "Prime Time" by clover

Video: `https://www.youtube.com/watch?v=iwlLKcr87Kw`





Rules: standard (Section 3.13) ratioPair (Section 6.6)

6.7 Constraint roundingError

6.7.1 Category: Pair

6.7.2 Description

The clue gives the tens digit of the product of the two cells of the pair. A pair 8-9 fits clue 7 (product 72), a pair 6-8 fits the clue 4 (product 48).

6.7.3 Arguments

cell1 Cell

cell2 Cell

Value Integer

6.7.4 JSON Format

```
"roundingError":[{"cell1":1,"cell2":2,"value":1},...]
```

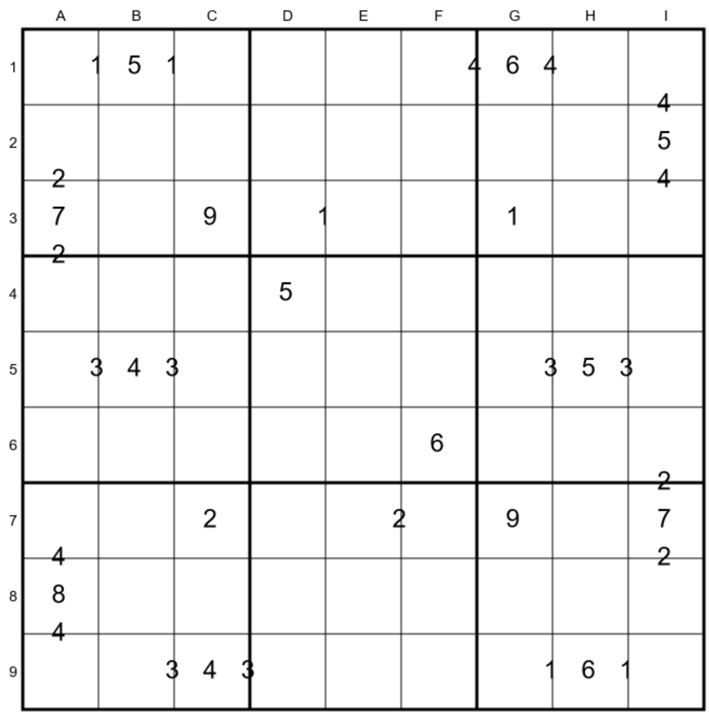
6.7.5 Model

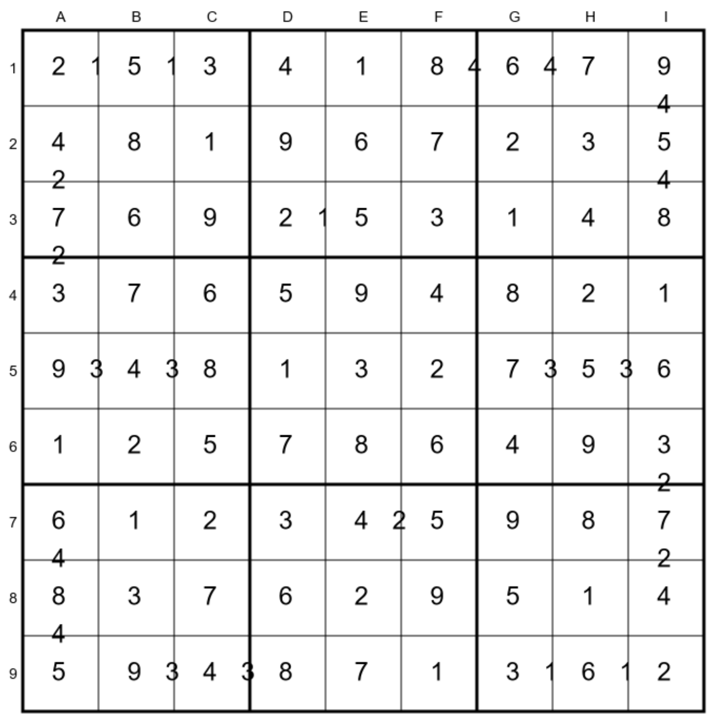
Model description not finished

6.7.6 See Also

6.7.7 roundingError example gas18: "Rounding Error" by clover

Video: `https://youtu.be/0j0r3IjoYwI?si=Dbny8Pl0zPILxae1`





Rules: standard (Section 3.13) roundingError (Section 6.7)

6.8 Constraint sumV

6.8.1 Category: Pair

6.8.2 Description

The two values in cells marked by a V must add to five.

6.8.3 Arguments

cell1 Cell

cell2 Cell

6.8.4 JSON Format

```
"sumV":[[5,14],[17,26],[25,34],[34,35],[31,32],[37,46],[38,47],[45,54],[51,60],[65,66],[75,76]],
```

6.8.5 Model

Let x_1 and x_2 be the variables corresponding to the two given cells of the pair. The variables must sum to five (Roman numeral V) This can be expressed by a constraint.

$$x_1 + x_2 = 5 \tag{13}$$

We can express the constraint with a table constraint

$$\text{table}([x_1, x_2], \text{sumV}_2) \tag{14}$$

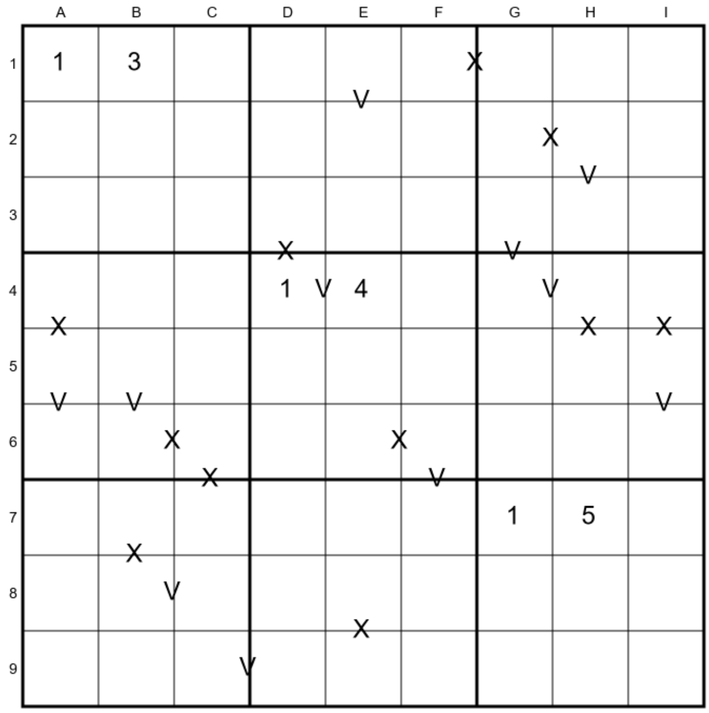
with tuples sumV_2 which enumerates all combinations of values that satisfy constraint (13).

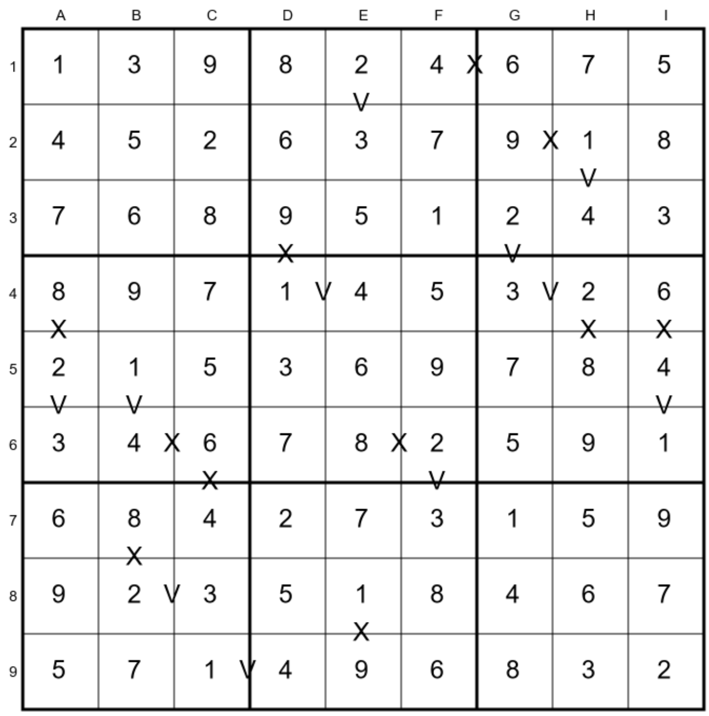
6.8.6 See Also

sumX (Section 6.9) , strictSumV (Section 4.27) , strictSumXV (Section 4.29)

6.8.7 sumV example gas15: "Knoxvillian Fluxvalve" by Philip Newman

Video: `https://www.youtube.com/watch?v=ssfKoRGOLEQ&list=PLkqs9GgCgVT6oKvb8KGhVlwS2rYpZwlIb&index=62`





Rules: standard (Section 3.13) sumV (Section 6.8) sumX (Section 6.9)

6.9 Constraint sumX

6.9.1 Category: Pair

6.9.2 Description

The two values in the cells marked by an X must add to ten.

6.9.3 Arguments

cell1 Cell

cell2 Cell

6.9.4 JSON Format

"sumX":[[6,7],[16,17],[22,31],[28,37],[35,44],[36,45],[47,48],[48,57],[50,51],[56,65],[68,77]],

6.9.5 Model

Let x_1 and x_2 be the variables corresponding to the two given cells of the pair. The variables must sum to 10 (Roman numeral X) This can be expressed by a constraint.

$$x_1 + x_2 = 10 \tag{15}$$

We can express the constraint with a table constraint

$$\mathrm{table}([x_1, x_2], \mathrm{sumX}_2) \tag{16}$$

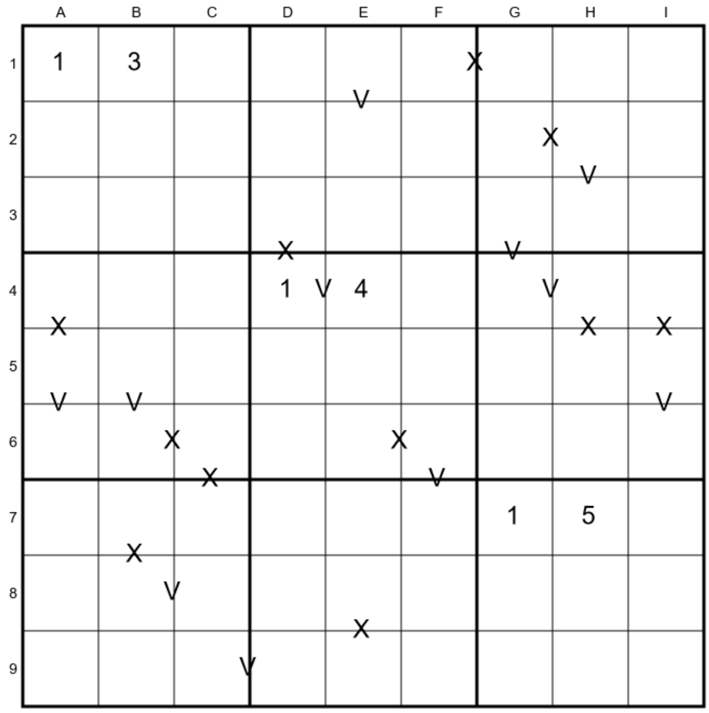
with tuples sumX_2 which enumerates all combinations of values that satisfy constraint (15).

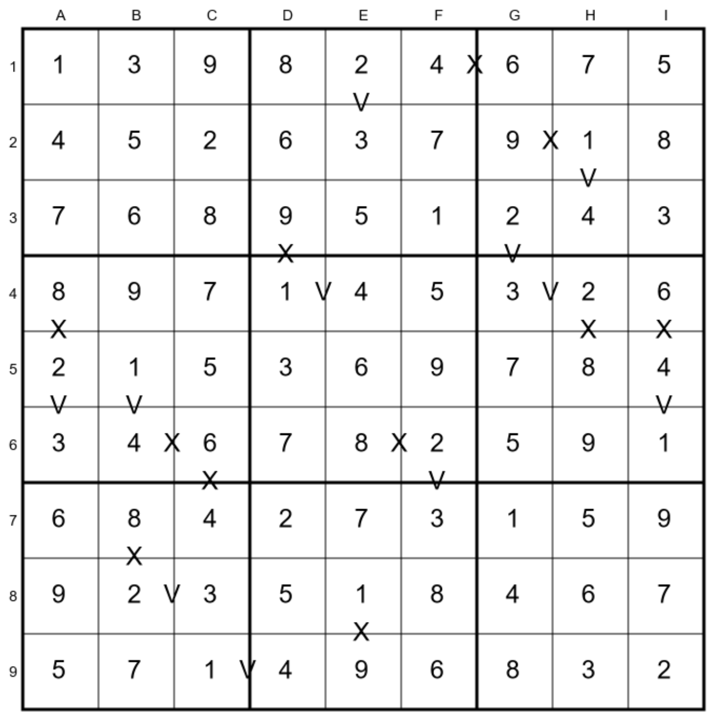
6.9.6 See Also

sumV (Section 6.8) , strictSumX (Section 4.28) , strictSumXV (Section 4.29)

6.9.7 sumX example gas15: "Knoxvillian Fluxvalve" by Philip Newman

Video: `https://www.youtube.com/watch?v=ssfKoRGOLEQ&list=PLkqs9GgCgVT6oKvb8KGhVlwS2rYpZwlIb&index=62`





Rules: standard (Section 3.13) sumV (Section 6.8) sumX (Section 6.9)

6.10 Constraint whiteDot

6.10.1 Category: Pair

6.10.2 Description

The values in the cells marked by a white dot must be consecutive, in any order.

6.10.3 Arguments

cell1 Cell

cell2 Cell

6.10.4 JSON Format

```
"whiteDot":[[1,2],[4,5],[7,8],[12,21],[18,27],[22,23],[31,40],...],
```

6.10.5 Model

Let x_1 and x_2 be the variables corresponding to the two given cells of the pair. The variables must be consecutive, i.e. their absolute difference must be one.

$$|x_1 - x_2| = 1 \tag{17}$$

We can express the constraint with a table constraint

$$\text{table}([x_1, x_2], \text{consecutive}_2) \tag{18}$$

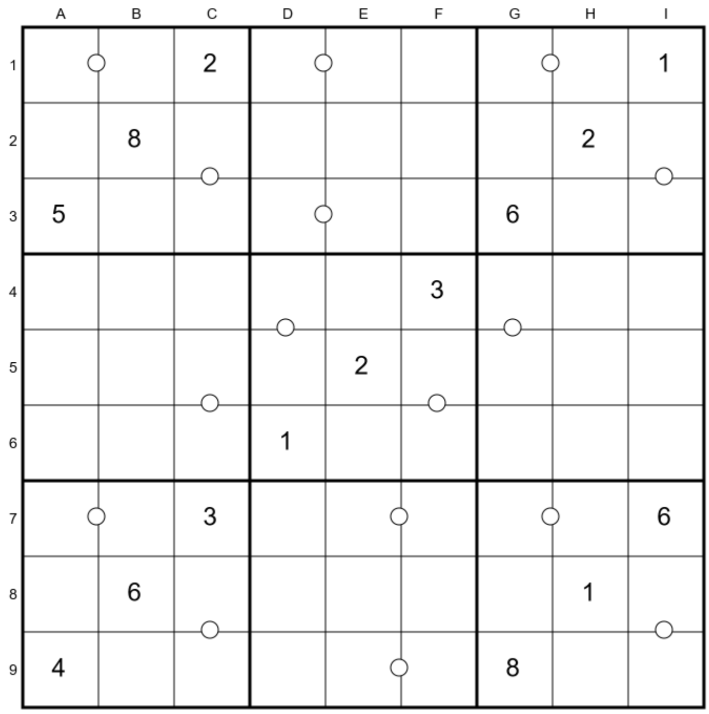
with tuples consecutive_2 which enumerates all combinations of values that satisfy constraint (17).

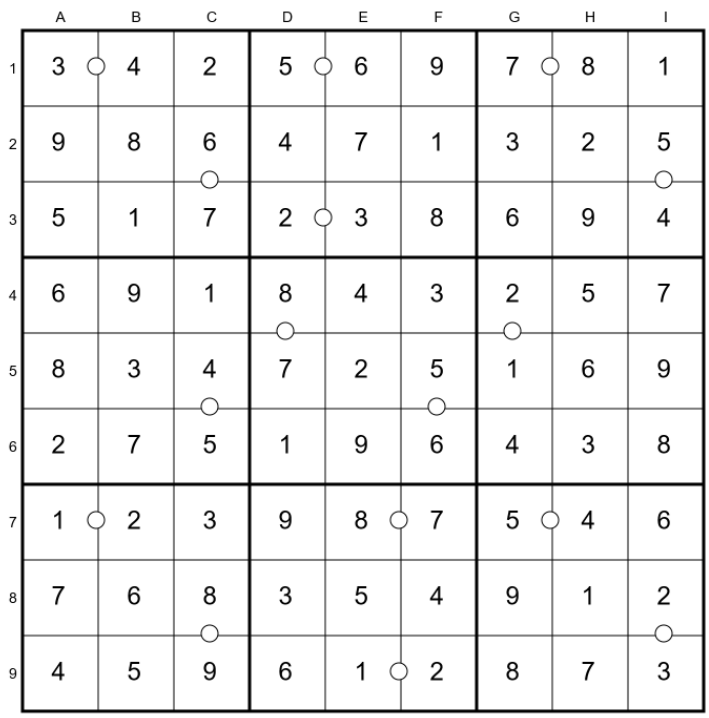
6.10.6 See Also

blackDot (Section 6.1) , differencePair (Section 6.2) , strictWhiteDot (Section 4.31) , strictKropki (Section 4.26)

6.10.7 whiteDot example timotab24: "Depreston" by Bill Murphy

Video: `https://www.youtube.com/watch?v=IFTU200HkqU&t=64s`





Rules: standard (Section 3.13) whiteDot (Section 6.10)

7 Triple Constraints

7.1 Constraint blackDotTriple

7.1.1 Category: Triple

7.1.2 Description

A doubler triple is formed by two connected black dot pairs. If within the same house, the only valid sequences are <1, 2, 4> or <2, 4, 8>.

7.1.3 Arguments

cell1 Cell

cell2 Cell

cell3 Cell

7.1.4 JSON Format

```
"blackDotTriple":[[32,41,50]],
```

7.1.5 Model

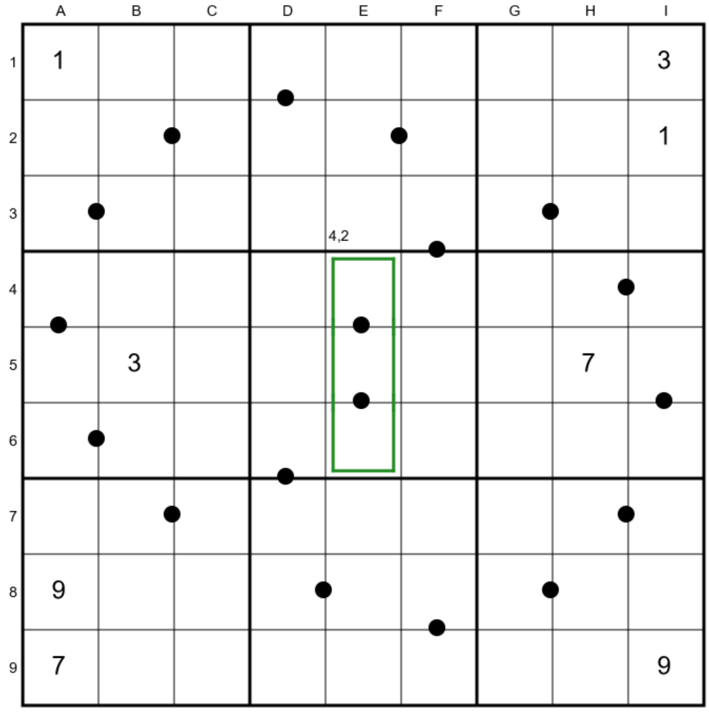
Model description not finished

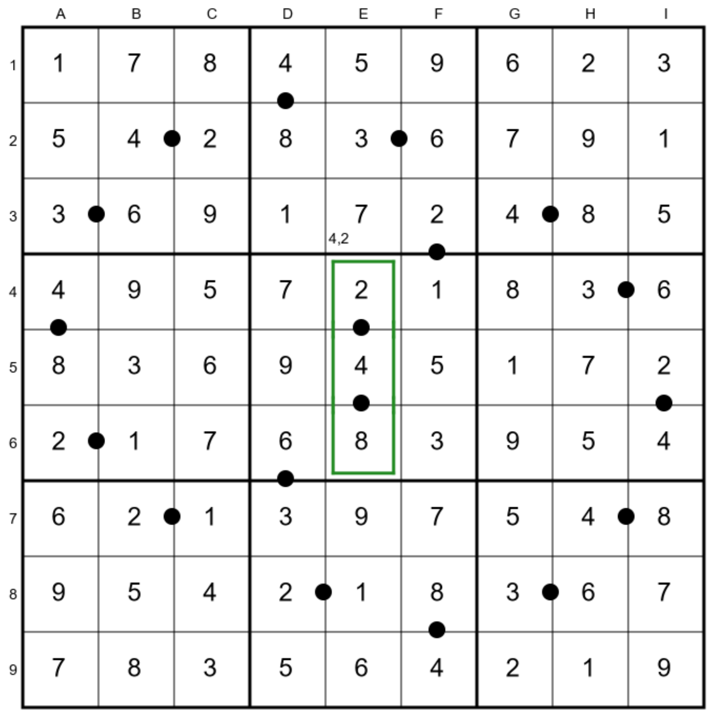
7.1.6 See Also

blackDot (Section 6.1)

7.1.7 blackDotTriple example gas8: "Half!" by Philip Newman

Video: `https://youtu.be/ftDSAmeEc3k?si=w3c3swQspE85FQ2q`





Rules: standard (Section 3.13) blackDot (Section 6.1) blackDotTriple (Section 7.1)

7.2 Constraint whiteDotTriple

7.2.1 Category: Triple

7.2.2 Description

A consecutive triple is formed by two connected consecutive (white dot) pairs. If within the same same house, there is additional reasoning that can be applied to the triple.

7.2.3 Arguments

cell1 Cell

cell2 Cell

cell3 Cell

7.2.4 JSON Format

```
"whiteDotTriple":[[28,37,46]],
```

7.2.5 Model

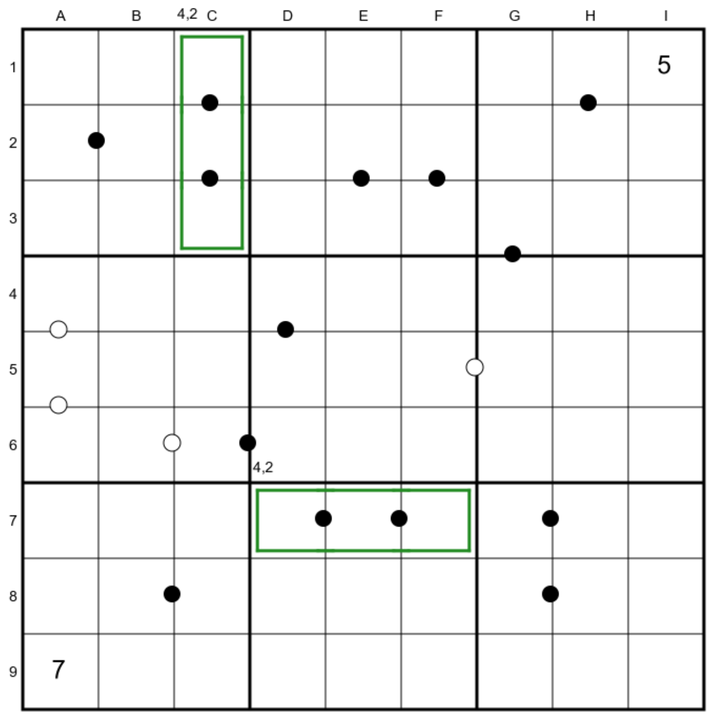
Model description not finished

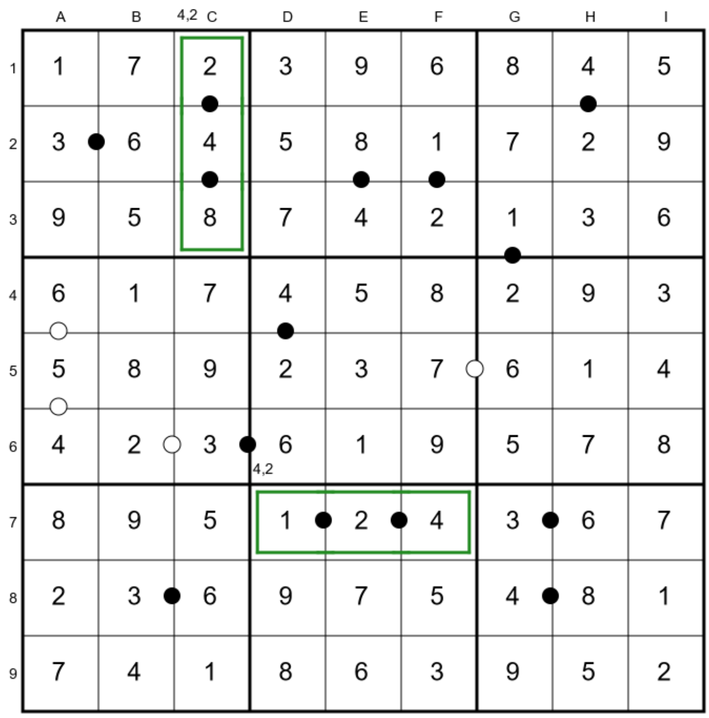
7.2.6 See Also

whiteDot (Section 6.10)

7.2.7 whiteDotTriple example ctc2: "December Colors" by Walter Grönholm

Video: `https://www.youtube.com/watch?v=Bq_I16-UJAg`





Rules: standard (Section 3.13) blackDot (Section 6.1) whiteDot (Section 6.10) blackDotTriple (Section 7.1) whiteDotTriple (Section 7.2)

8 Quadruple Constraints

8.1 Constraint antiQuad

8.1.1 Category: Quadruple

8.1.2 Description

The values in the circle may not appear in any of the four connected cells.

8.1.3 Arguments

cell Cell

values List(Integer)

8.1.4 JSON Format

```
"antiQuad":[{"cell":11,"values":[4,6,8]},...],
```

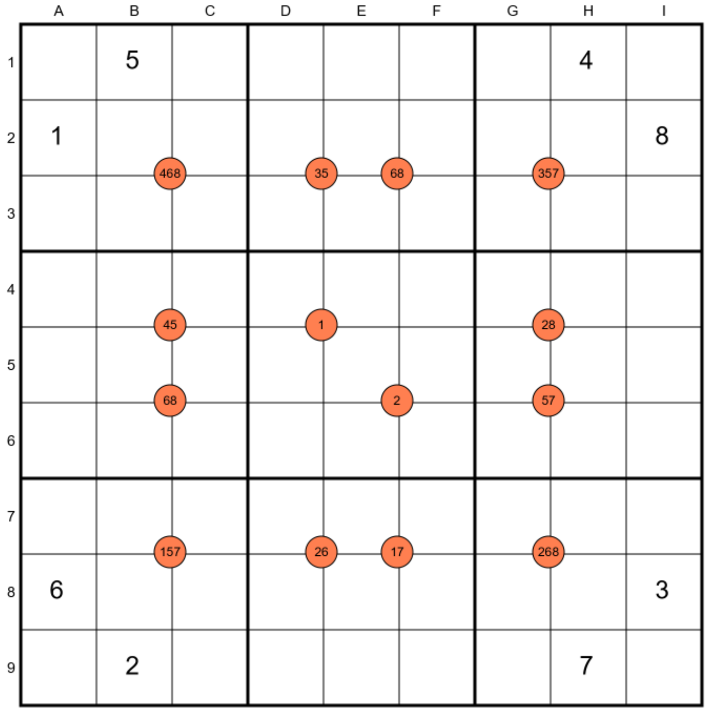
8.1.5 Model

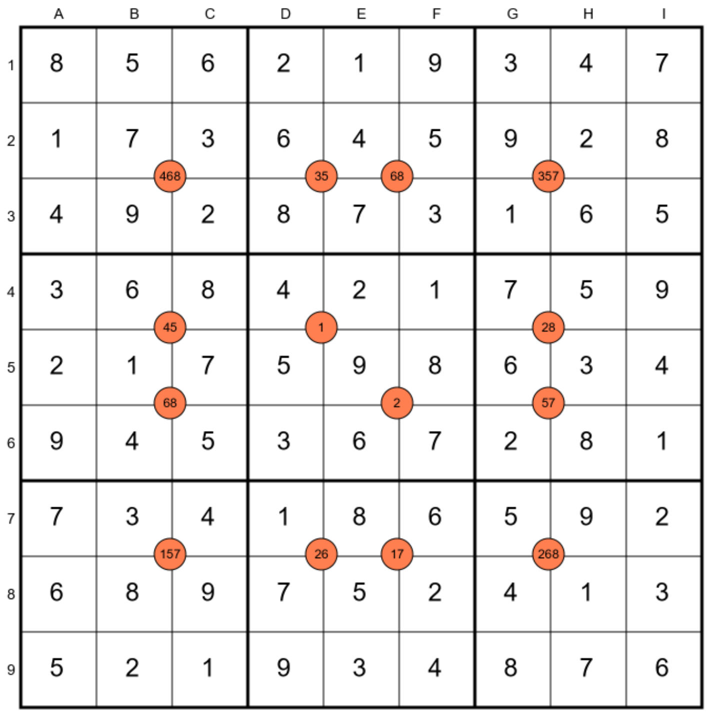
Model description not finished

8.1.6 See Also

8.1.7 antiQuad example gas10: "Exclusion Sudoku" by Bill Murphy

Video: `https://youtu.be/zYMlV2FS4ok?si=hNOcSAoPX4lnRs_m`





Rules: standard (Section 3.13) antiQuad (Section 8.1)

8.2 Constraint checkerQuad

8.2.1 Category: Quadruple

8.2.2 Description

Opposing corners of the quad must have the same parity (odd or even), while orthogonally adjacent cells in the quad must have opposite parity.

8.2.3 Arguments

cell Cell

8.2.4 JSON Format

```
"checkerQuad":[1,8,24,31,41,48,64,71],
```

8.2.5 Model

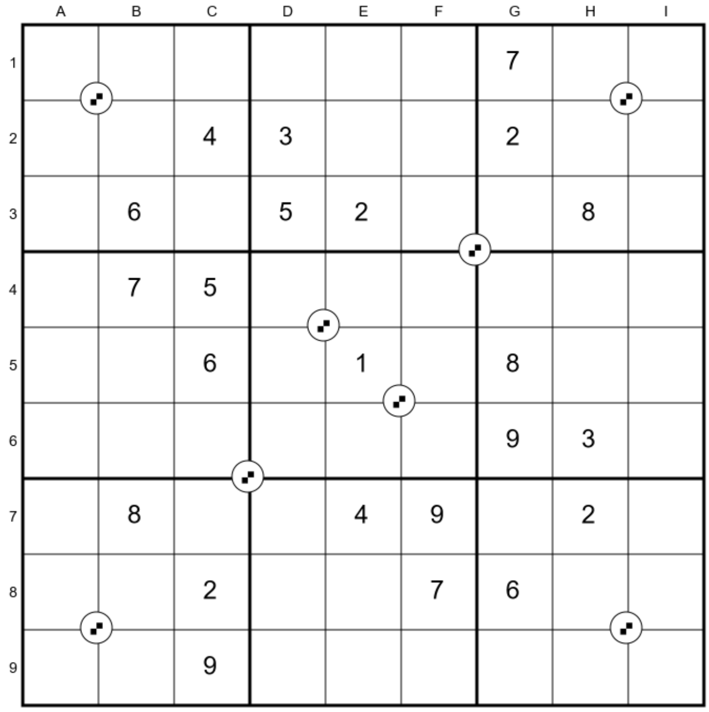
Model description not finished

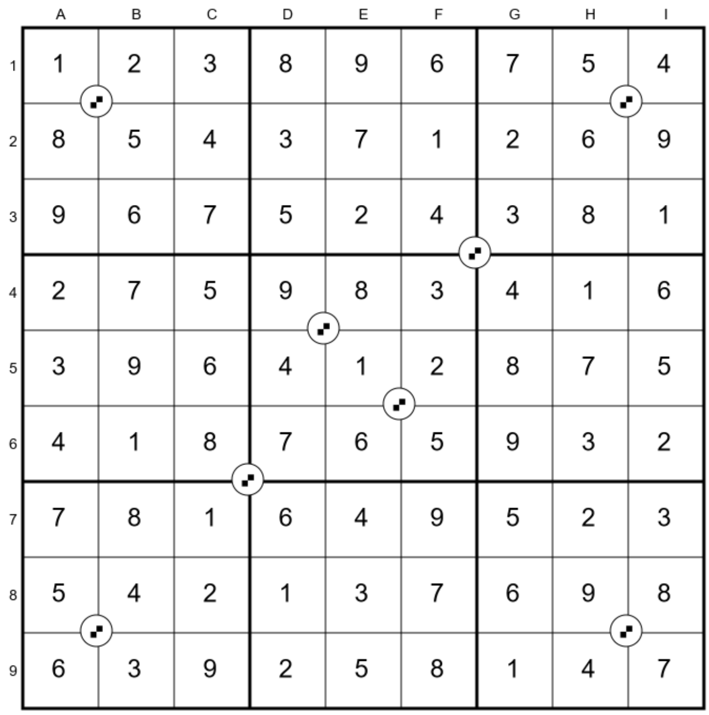
8.2.6 See Also

strictCheckerQuad (Section 4.22)

8.2.7 checkerQuad example timotab91: "Accentuate the Positive" by clover

Video: `https://youtu.be/m2RGvR7HpxM?si=GXk2DmlMihLeTj83` (11:30)





Rules: standard (Section 3.13) checkerQuad (Section 8.2)

8.3 Constraint clockwiseQuad

8.3.1 Category: Quadruple

8.3.2 Description

The four cells around the circle mark must increase in a clockwise manner. The position of the lowest value must be determined by the solver.

8.3.3 Arguments

cell Cell

8.3.4 JSON Format

```
"clockwiseQuad":[1,7,11,17,24,31,41,48,55,61,65,71],
```

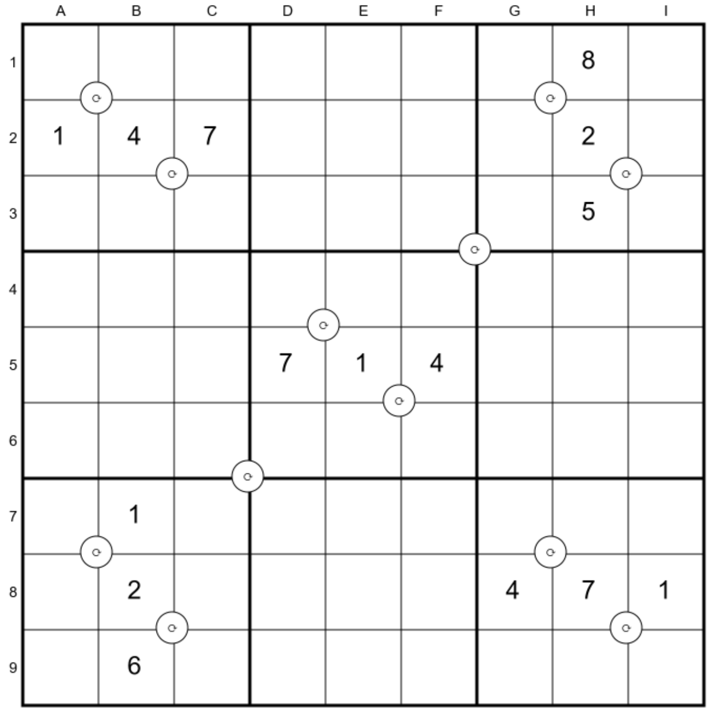
8.3.5 Model

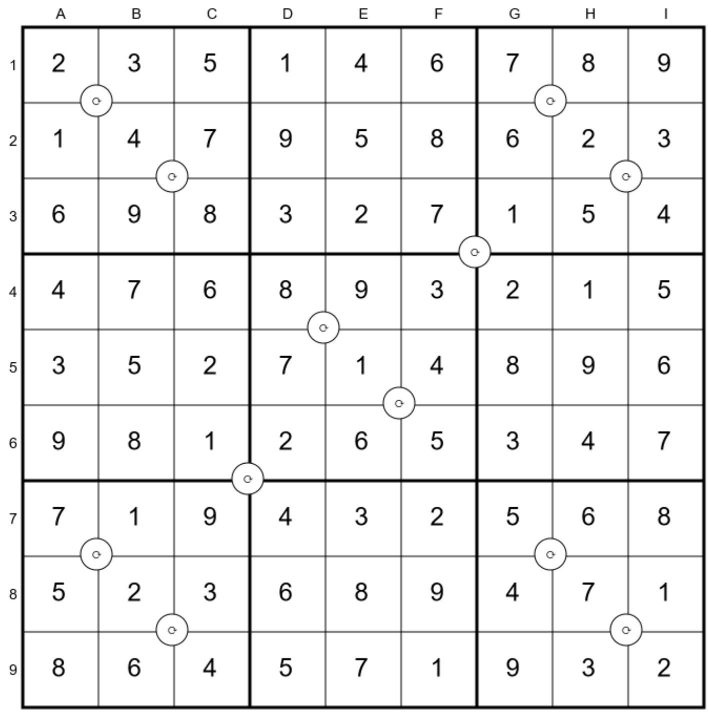
Model description not finished

8.3.6 See Also

8.3.7 clockwiseQuad example timotab66: "Clock Wise" by clover

Video: `https://www.youtube.com/watch?v=lOMRjVh4iN4`





Rules: standard (Section 3.13) clockwiseQuad (Section 8.3)

8.4 Constraint crossSum

8.4.1 Category: Quadruple

8.4.2 Description

Both diagonal pairs in the quad sum to the same value. This is strict, all possible quads satisfying the property are given.

8.4.3 Arguments

cell Cell

8.4.4 JSON Format

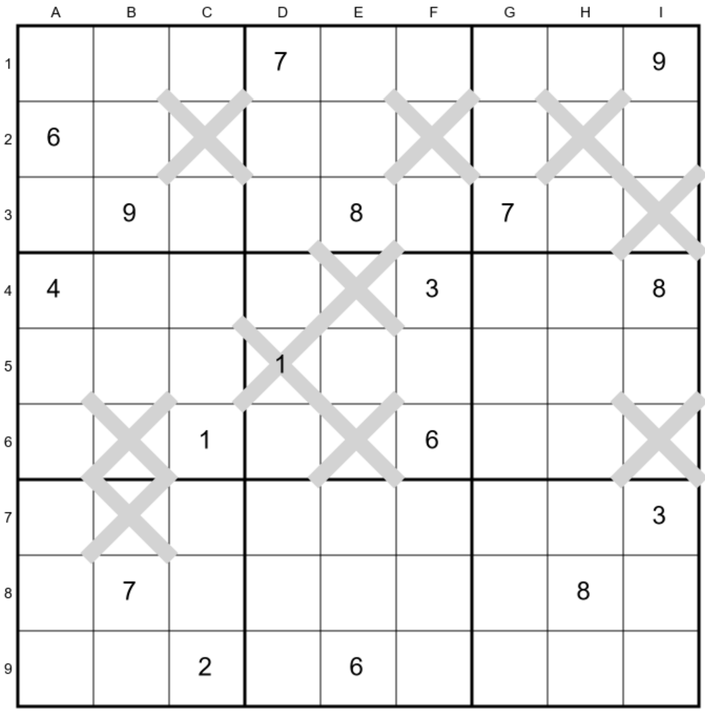
```
"crossSum":[2,5,7,17,22,30,37,40,44,46],
```

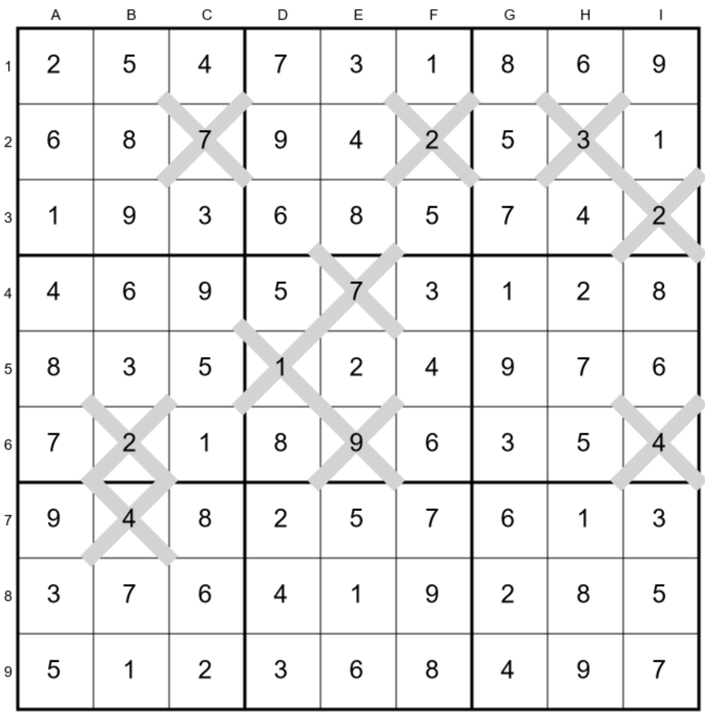
8.4.5 Model

Model description not finished

8.4.6 See Also

8.4.7 crossSum example wpf gp2025r5c12: "Sudoku GP 2025 Round 5" by





Rules: standard (Section 3.13) crossSum (Section 8.4)

8.5 Constraint halfOddQuad

8.5.1 Category: Quadruple

8.5.2 Description

Two of the cells around the square mark are odd, and two are even.

8.5.3 Arguments

cell Cell

8.5.4 JSON Format

```
"halfOddQuad":[4,10,15,16,17,55,56,57,62,68],
```

8.5.5 Model

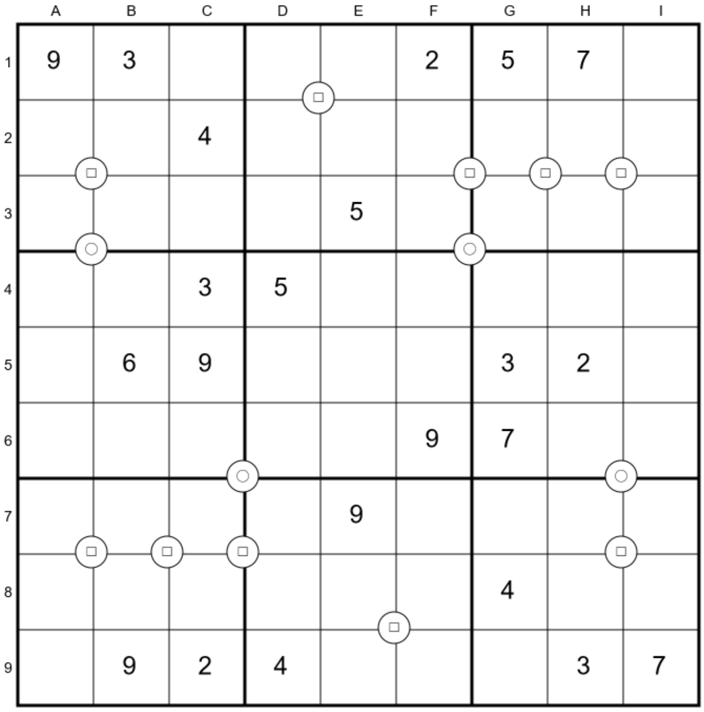
Model description not finished

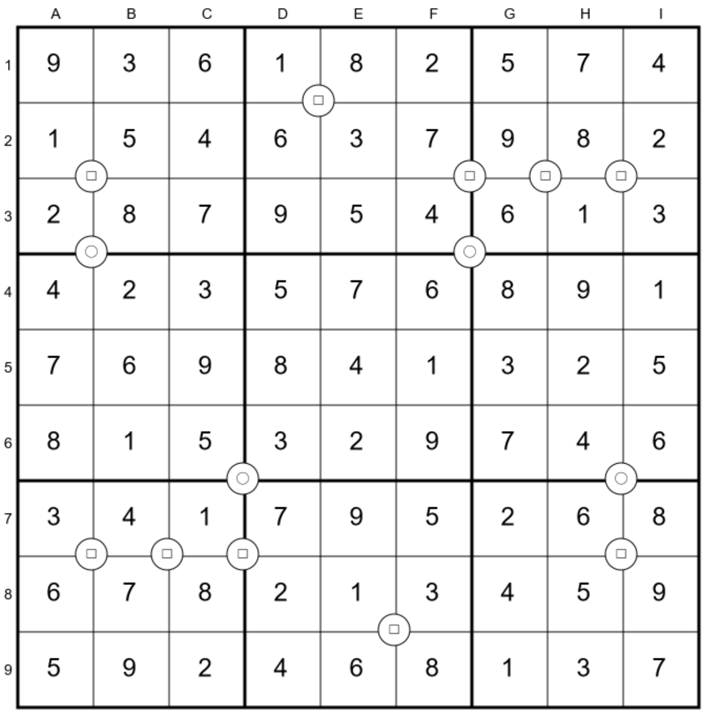
8.5.6 See Also

odd (Section 5.9) , even (Section 5.5) , oddOrEvenQuad (Section 8.7)

8.5.7 halfOddQuad example BremSter68: "Full or Half Sudoku" by

Video: `https://www.youtube.com/watch?v=Ib5EwDMJgi8`





Rules: standard (Section 3.13) oddOrEvenQuad (Section 8.7) halfOddQuad (Section 8.5)

8.6 Constraint mathrax

8.6.1 Category: Quadruple

8.6.2 Description

The diagonally opposed pairs of variables must satisfy an arithmetic constraint.

8.6.3 Arguments

cell Cell

expr Expression

8.6.4 JSON Format

```
"mathrax":[{"cell":3,"expr":"36x"},...],
```

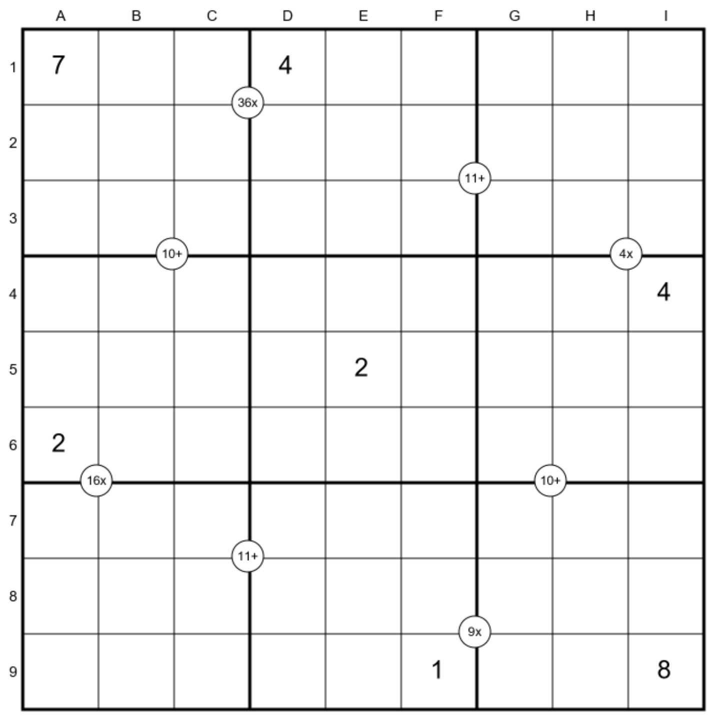
8.6.5 Model

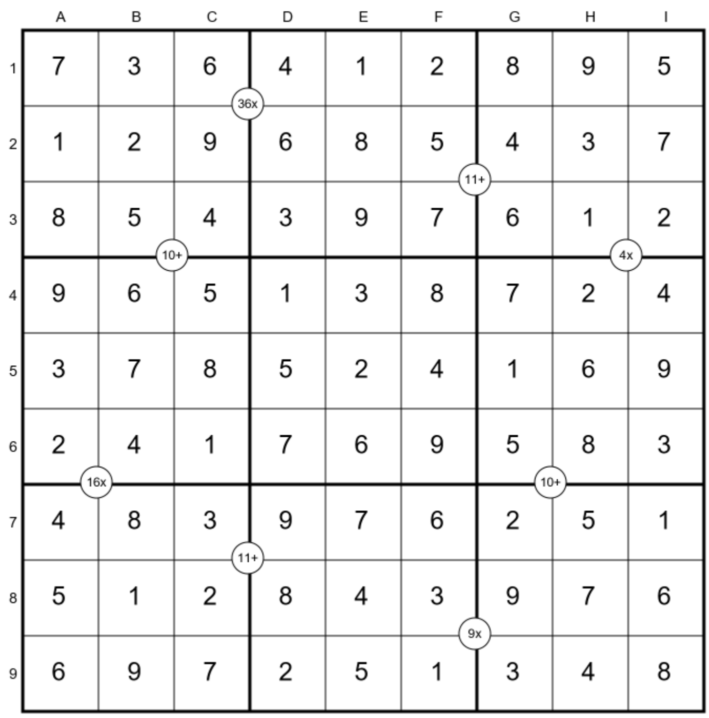
Model description not finished

8.6.6 See Also

8.6.7 mathrax example BremSter14: "Mathrax" by Bill Murphy

Video: `https://www.youtube.com/watch?v=spw0gOYhlT4`





Rules: standard (Section 3.13) mathrax (Section 8.6)

8.7 Constraint oddOrEvenQuad

8.7.1 Category: Quadruple

8.7.2 Description

The four cells around the circle mark are either all odd or all even.

8.7.3 Arguments

cell Cell

8.7.4 JSON Format

```
"oddOrEvenQuad":[19,24,48,53],
```

8.7.5 Model

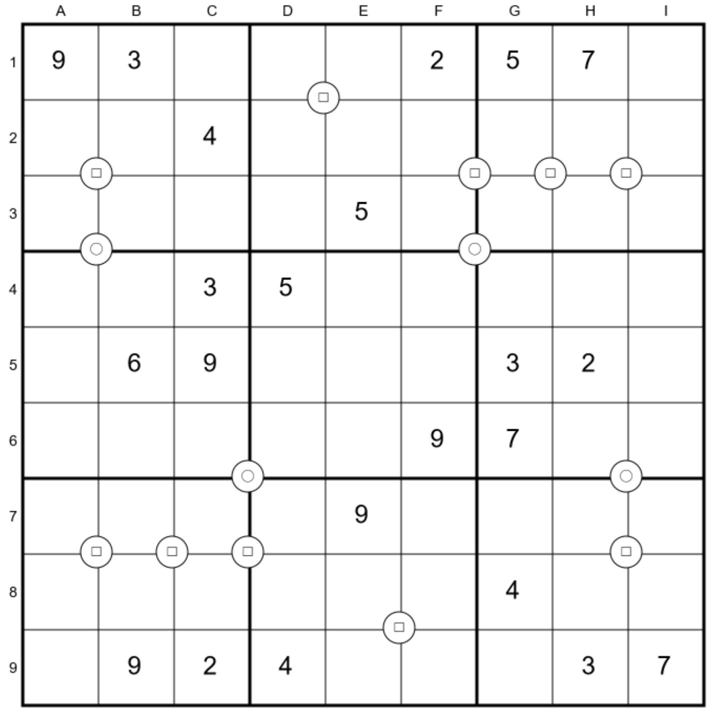
Model description not finished

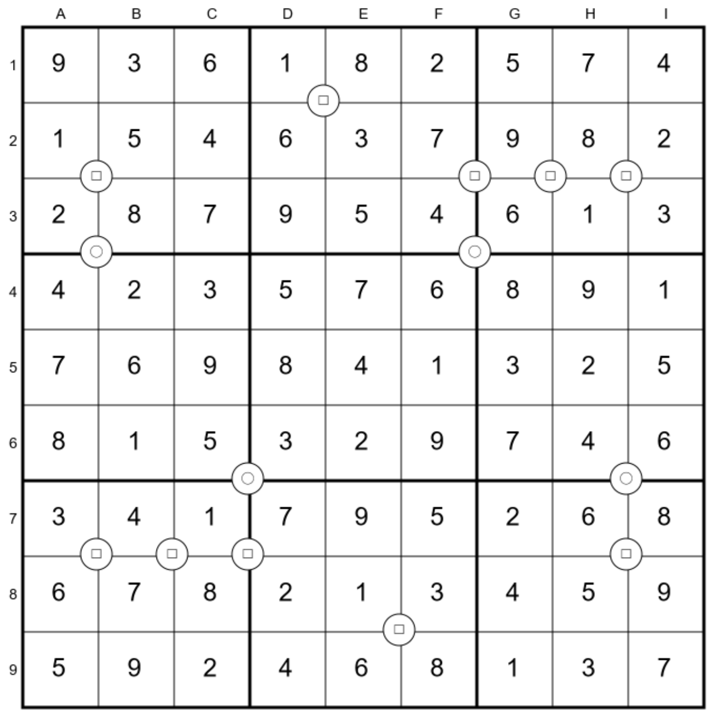
8.7.6 See Also

odd (Section 5.9) , even (Section 5.5) , halfOddQuad (Section 8.5)

8.7.7 oddOrEvenQuad example BremSter68: "Full or Half Sudoku" by

Video: `https://www.youtube.com/watch?v=Ib5EwDMJgi8`





Rules: standard (Section 3.13) oddOrEvenQuad (Section 8.7) halfOddQuad (Section 8.5)

8.8 Constraint quad

8.8.1 Category: Quadruple

8.8.2 Description

The values in the circle must appear at least once in the four connected cells.

8.8.3 Arguments

cell Cell

values List(Integer)

8.8.4 JSON Format

```
"quad":[{"cell":3,"values":[1,1,2,3]},{"cell":15,"values":[5,5]},...],
```

8.8.5 Model

A quad is indicated by a cell which identifies the top-left corner of the quad. Let $X = [x_1, x_2, x_3, x_4]$ be the four variables corresponding to the cells of the quad. Let $V = [v_1, ..., v_j]$ be the multi-set of values given for the quad, values can be repeated in the argument, and j can also be smaller than four. The constraint is expressed with a instance specific table constraint

$$\mathrm{table}([x_1, x_2, x_3, x_4], \mathrm{quad}_4(V)) \tag{19}$$

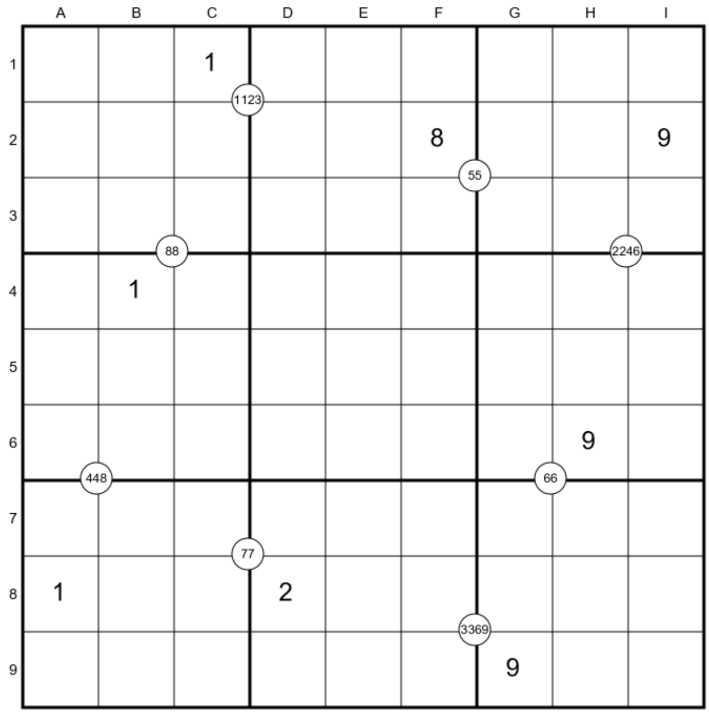
where the tuples $\mathrm{quad}_4(V)$ consist of all combination of four values that contain the multi-set V.

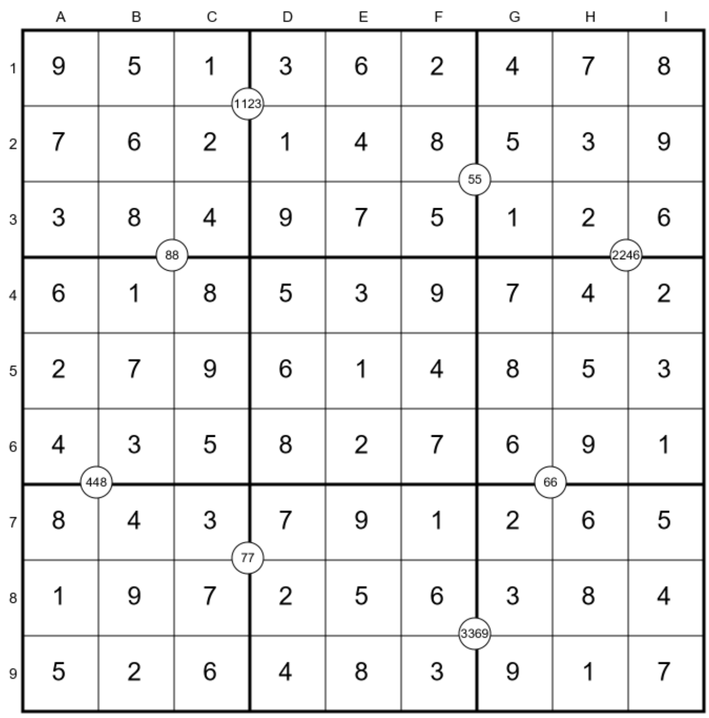
8.8.6 See Also

antiQuad (Section 8.1) , wheel (Section 8.9) , clockwiseQuad (Section 8.3)

8.8.7 quad example timotab65: "Tetrapod Quads" by Philip Newman

Video: `https://www.youtube.com/watch?v=fmFZpajZdAQ`





Rules: standard (Section 3.13) quad (Section 8.8)

8.9 Constraint wheel

8.9.1 Category: Quadruple

8.9.2 Description

The values in the circle must appear in the four connected cells in the same order, but the assignment can be rotated.

8.9.3 Arguments

cell Cell

values List(Integer)

8.9.4 JSON Format

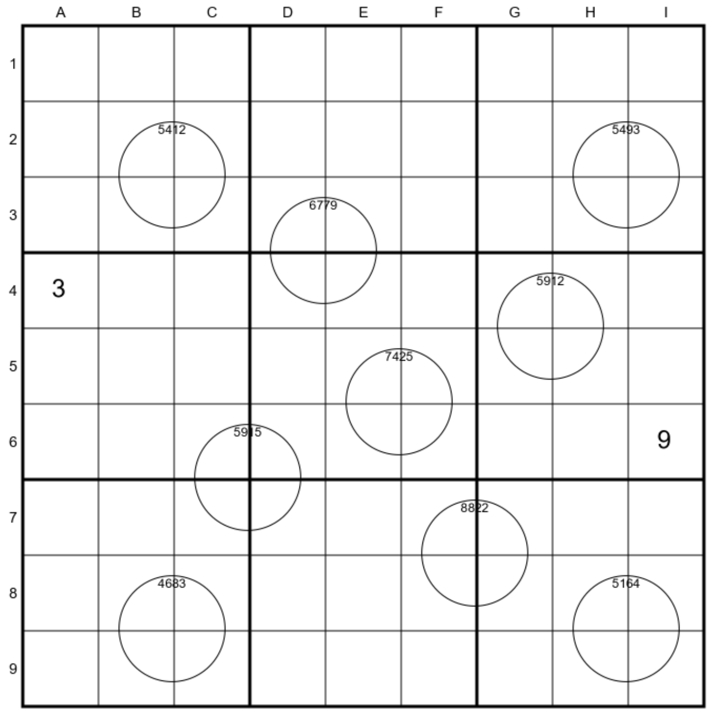
```
"wheel":[{"cell":11,"values":[5,4,1,2]},...],
```

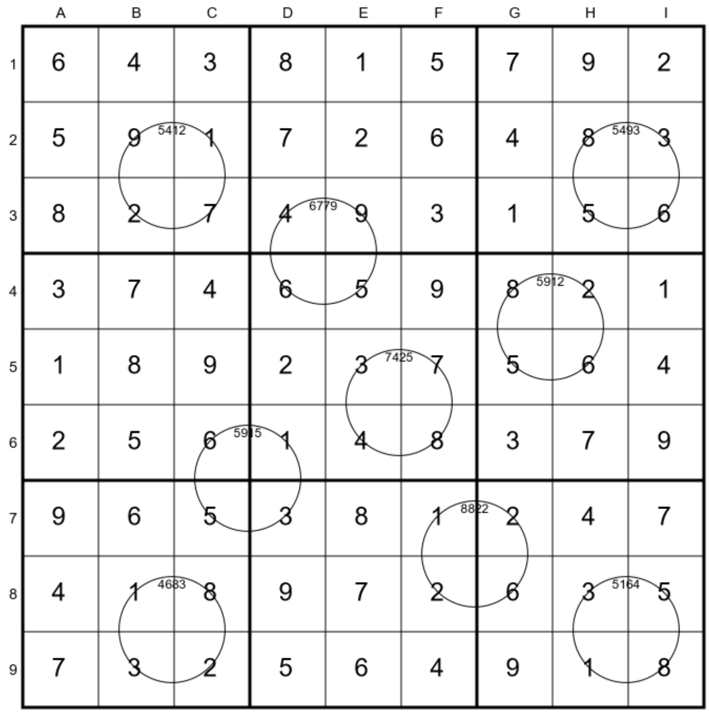
8.9.5 Model

Model description not finished

8.9.6 See Also

8.9.7 wheel example wpf gp2025r3c10: "Sudoku GP 2025 Round 3" by





Rules: standard (Section 3.13) wheel (Section 8.9)

9 Line Constraints

9.1 Constraint alternating

9.1.1 Category: Line

9.1.2 Description

The values on the line must alternative between being smaller and larger than the previous cell. Digits may not repeat on the line.

9.1.3 Arguments

alternating List(Cell)

9.1.4 JSON Format

```
"lines":[{"alternating":[1,10,19,28,37,46,55]},...],
```

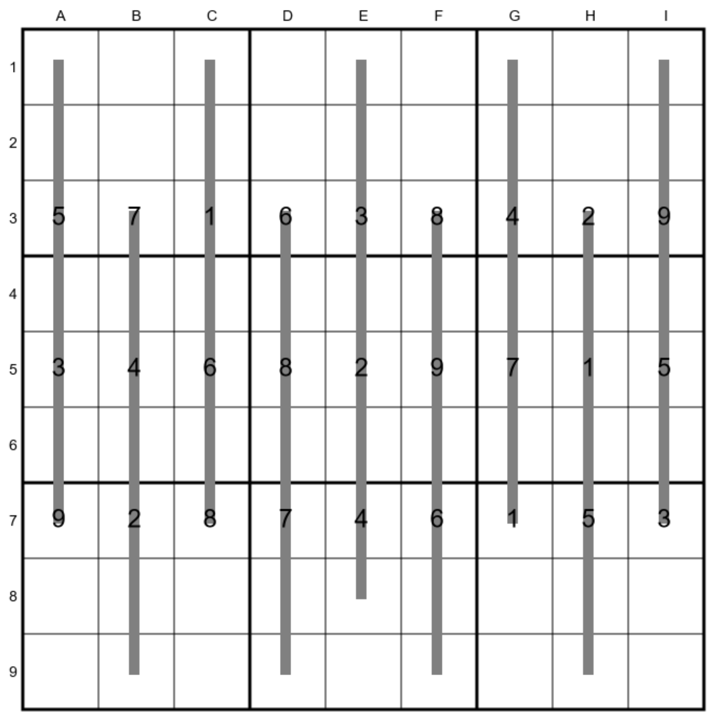
9.1.5 Model

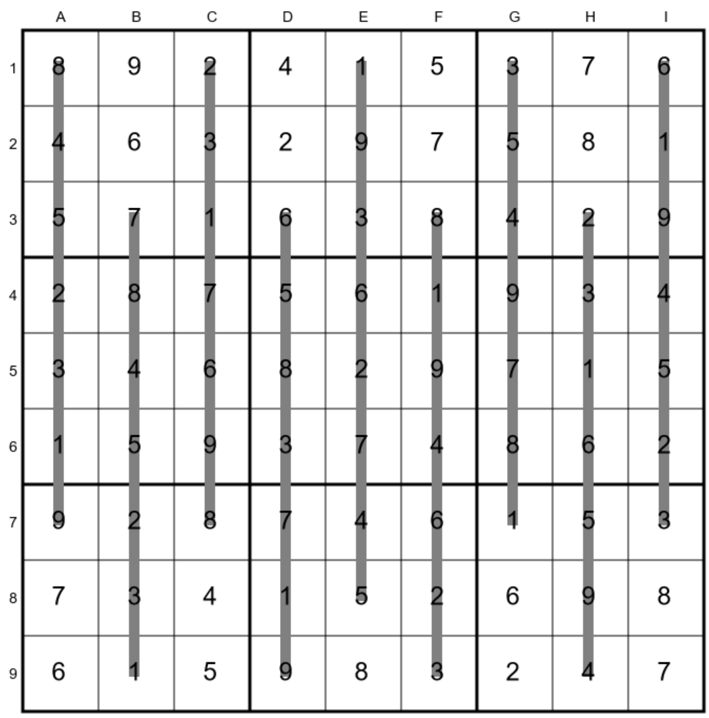
Model description not finished

9.1.6 See Also

9.1.7 alternating example timotab72: "Alternating Stripes Sudoku" by Bill Murphy

Video: `https://www.youtube.com/watch?v=EaekALXJUVo`





Rules: standard (Section 3.13) alternating (Section 9.1)

9.2 Constraint arrow

9.2.1 Category: Line

9.2.2 Description

The sum of the digits on the arrow must sum to the value in the circle. The sequence <7, 4, 2, 1> would form a valid arrow of length four. The value in the circle is always larger than any one of the values on the arrow.

9.2.3 Arguments

arrow List(Cell)

9.2.4 JSON Format

```
"lines":[{"arrow":[1,2,3]},{"arrow":[1,11,21]},...],
```

9.2.5 Model

Let $[x_1, x_2, ..., x_k]$ be the set variables corresponding to the given list of cells. The value of the first cell, indicated by the circle, is equal to the sum of the other variables.

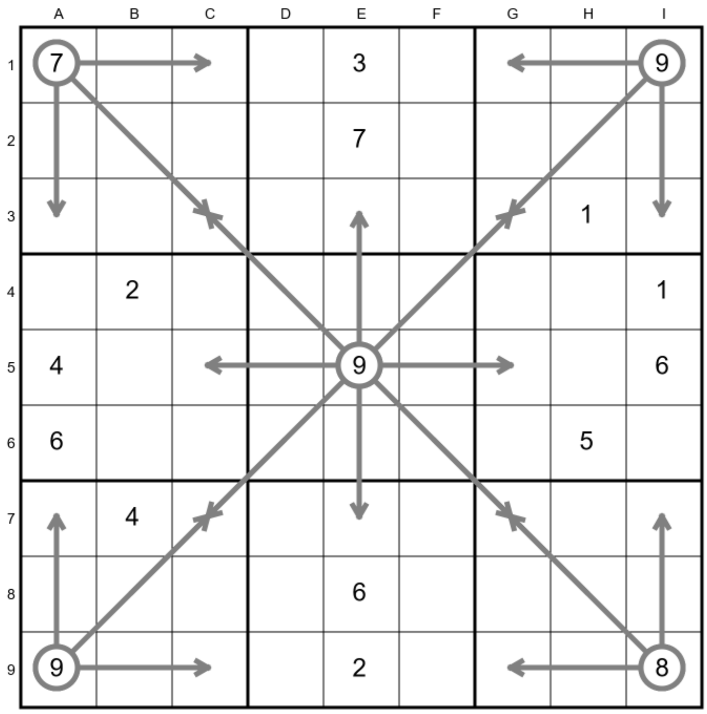
$$x_1 = \sum_{i=2}^{k} x_i \tag{20}$$

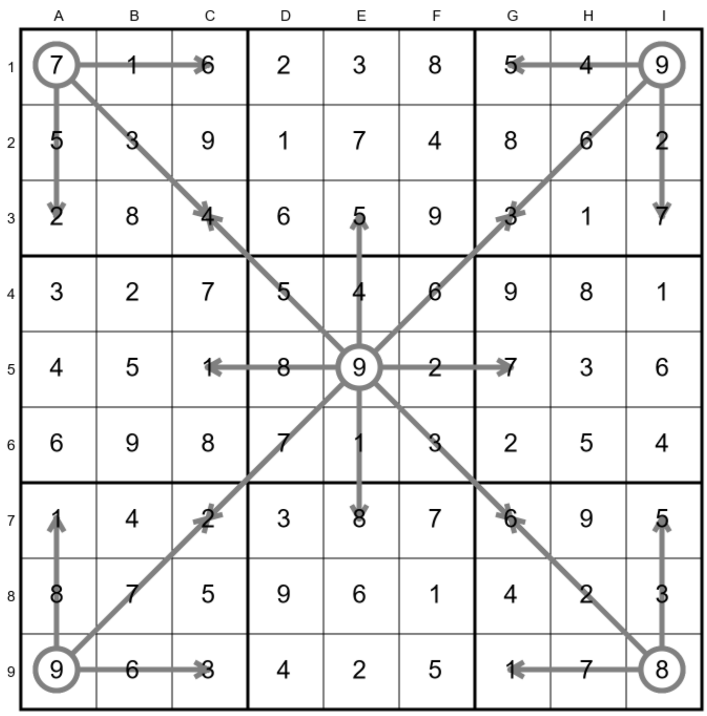
9.2.6 See Also

arrow2 (Section 9.3) , arrowPlusMinus1 (Section 9.6)

9.2.7 arrow example gas19: "Here Comes The Sum" by clover

Video: `https://youtu.be/WEgNqwCck4Q?si=br_5SsMXwOWm9upb`





Rules: standard (Section 3.13) arrow (Section 9.2)

9.3 Constraint arrow2

9.3.1 Category: Line

9.3.2 Description

The sum of the digits on the arrow must sum to the value in the pill formed by two cells.

9.3.3 Arguments

arrow2 List(Cell)

9.3.4 JSON Format

```
"lines":[{"arrow2":[78,79,80,81]}],
```

9.3.5 Model

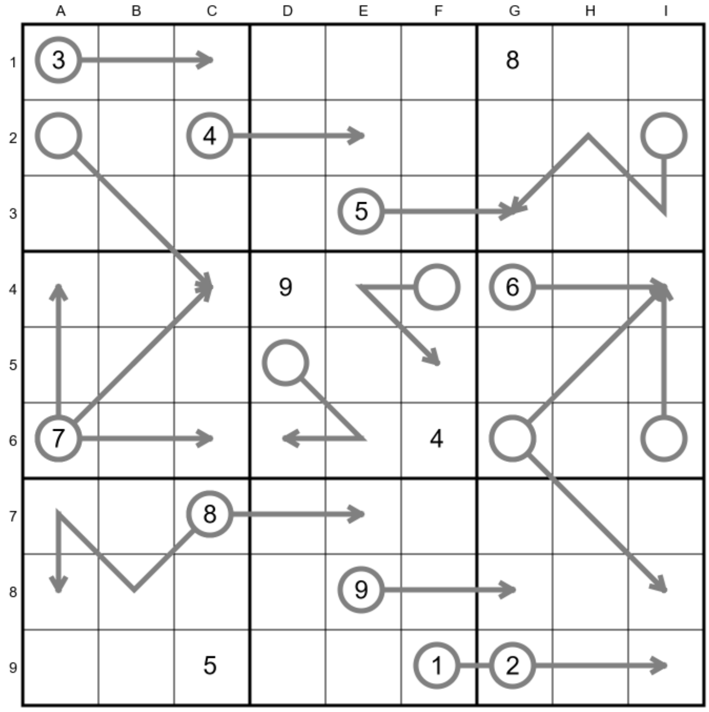
Model description not finished

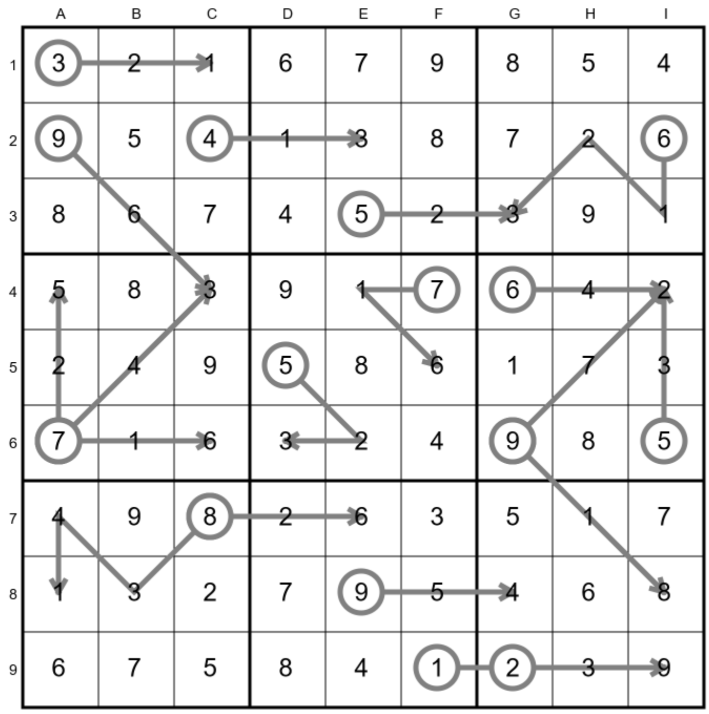
9.3.6 See Also

arrow (Section 9.2) , arrowPlusMinus1 (Section 9.6)

9.3.7 arrow2 example timotab27: "Arrow There" by Philip Newman

Video: `https://youtu.be/eNCrFTNxFIE?si=JRjLaTqBumWmqaBS`





Rules: standard (Section 3.13) arrow (Section 9.2) arrow2 (Section 9.3)

9.4 Constraint arrowLE

9.4.1 Category: Line

9.4.2 Description

The sum of the digits along the arrow shaft must be less than or equal to the digit in the circle at the base of the arrow.

9.4.3 Arguments

arrowLE List(Cell)

9.4.4 JSON Format

```
"lines":[{"arrowLE":[1,10,19]},...],
```

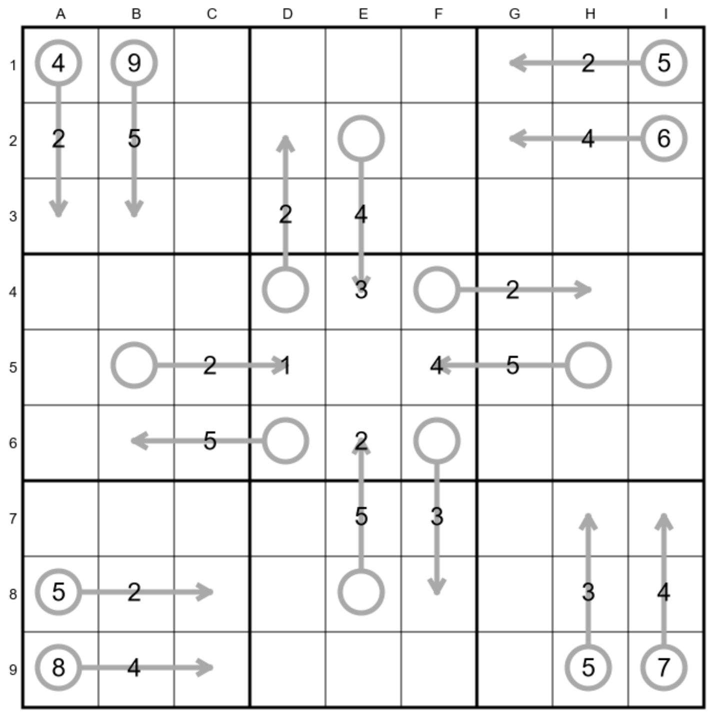
9.4.5 Model

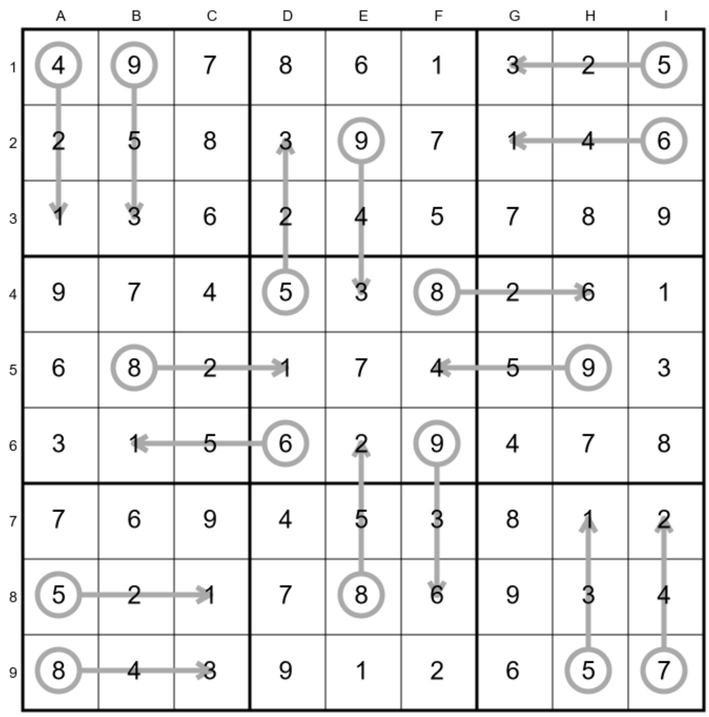
Model description not finished

9.4.6 See Also

9.4.7 arrowLE example timotab63: "Still Not An Arrow Sudoku" by clover

Video: `https://www.youtube.com/watch?v=vB5aZx1iu2A`
Video2: `https://www.youtube.com/watch?v=CXAjOiQkKPE`





Rules: standard (Section 3.13) arrowLE (Section 9.4)

9.5 Constraint arrowLess

9.5.1 Category: Line

9.5.2 Description

The digits along the line are each less than the digit in the attached circle.

9.5.3 Arguments

arrowLess List(Cell)

9.5.4 JSON Format

```
"lines":[{"arrowLess":[6,15,16,25,26,35,36]},...],
```

9.5.5 Model

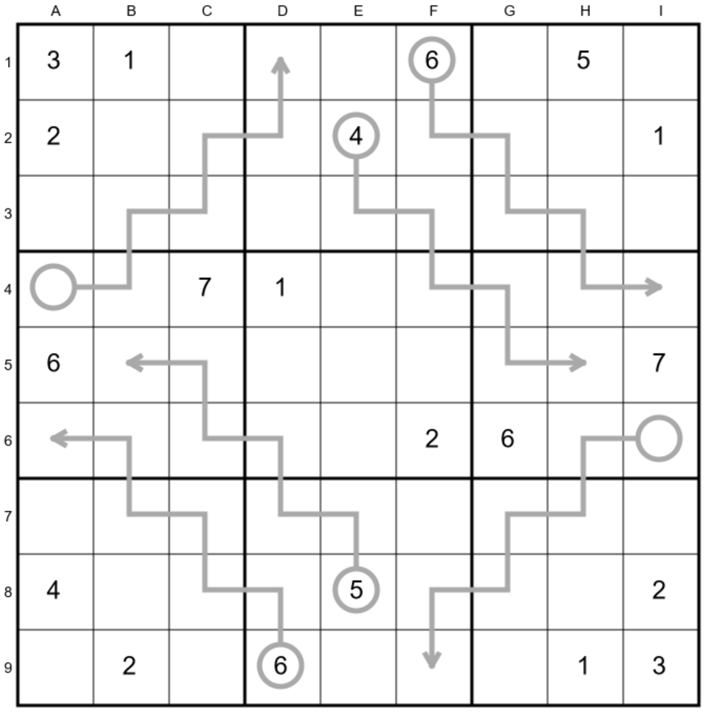
Model description not finished

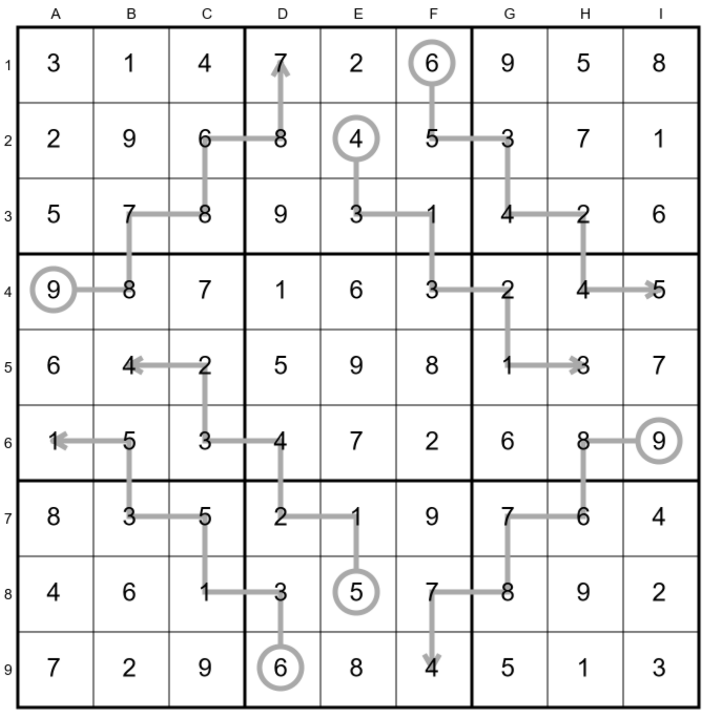
9.5.6 See Also

arrow (Section 9.2)

9.5.7 arrowLess example timotab71: "Once Again, Not An Arrow Sudoku" by clover

Video: `https://www.youtube.com/watch?v=qJ0CH_2XfbA`





Rules: standard (Section 3.13) arrowLess (Section 9.5)

9.6 Constraint arrowPlusMinus1

9.6.1 Category: Line

9.6.2 Description

The sum of the digits on the arrow must sum to a value which is either one lower or one higher than the hint in the circle.

9.6.3 Arguments

arrowPlusMinus1 List(Cell)

9.6.4 JSON Format

```
"lines":[{"arrowPlusMinus1":[8,9,18]},...],
```

9.6.5 Model

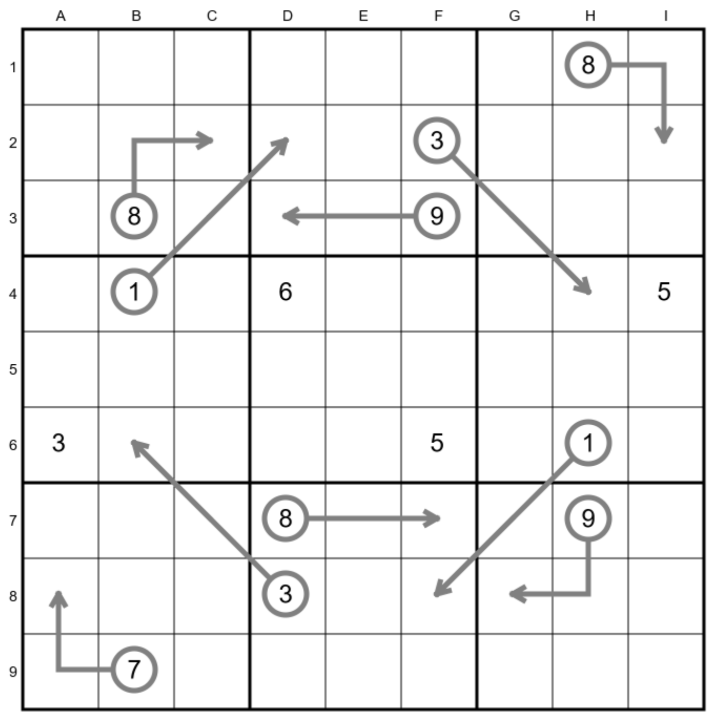
Model description not finished

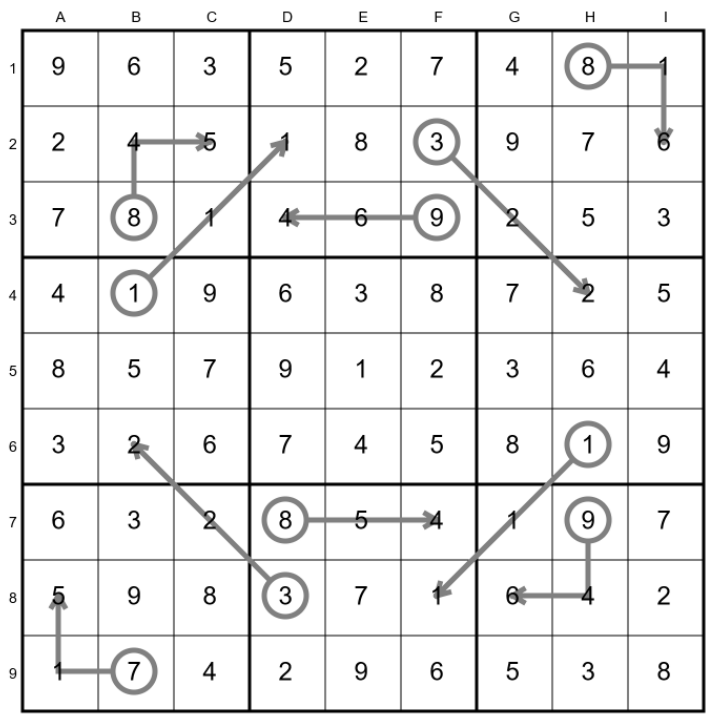
9.6.6 See Also

arrow (Section 9.2) , arrow2 (Section 9.3)

9.6.7 arrowPlusMinus1 example timotab30: "Knapp Daneben Arrows" by Bill Murphy

Video: `https://youtu.be/-5HqaUvDtsk?si=mnHfIQVqxHJdc-5K`





Rules: standard (Section 3.13) arrowPlusMinus1 (Section 9.6)

9.7 Constraint autoGlyph

9.7.1 Category: Line

9.7.2 Description

A line in the shape of a digit must contain that digit.

9.7.3 Arguments

autoGlyph List(Cell)

value Integer

9.7.4 JSON Format

```
"lines":[{"autoGlyph":[2,11,20],"value":1},...],
```

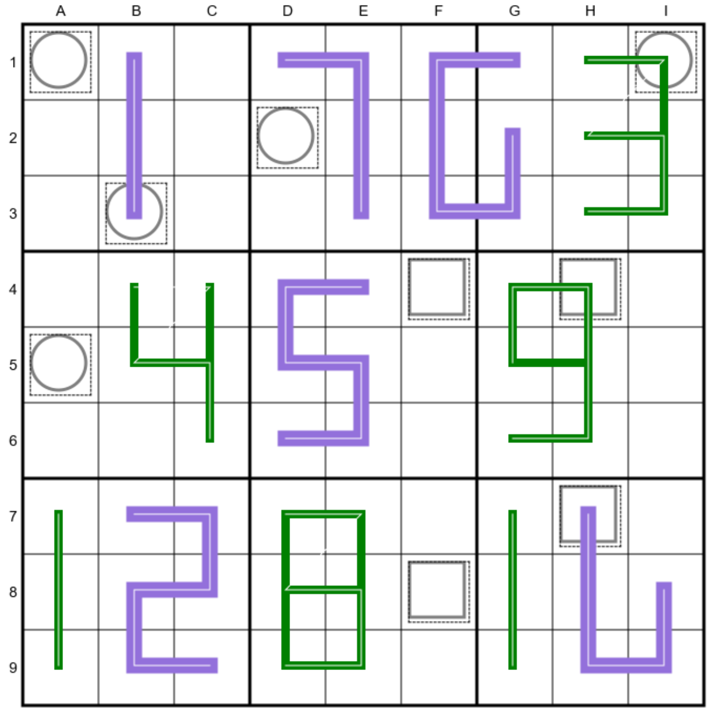
9.7.5 Model

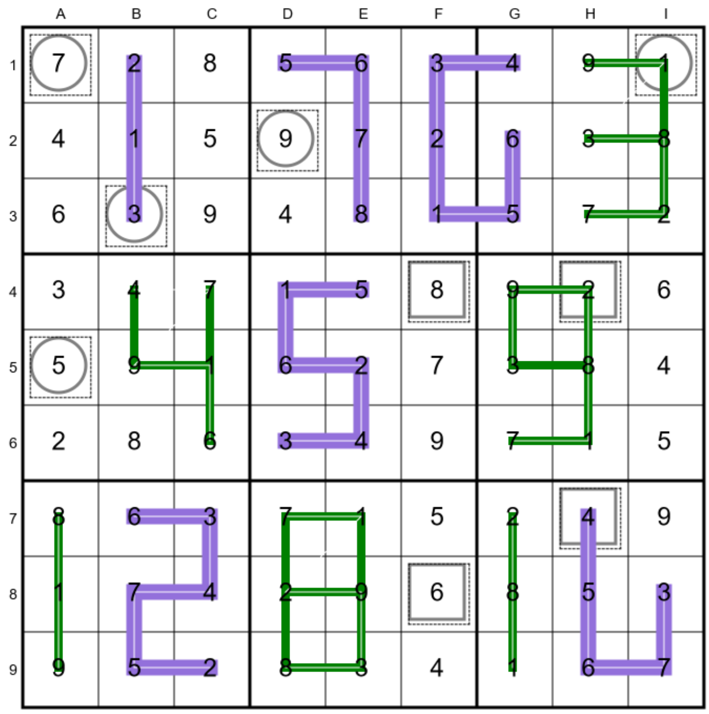
Model description not finished

9.7.6 See Also

9.7.7 autoGlyph example logiclemurgaming17: "Autoglyphs 6" by juddimal

Video: `https://www.youtube.com/watch?v=fPnCPbX37-o`





Rules: standard (Section 3.13) renban (Section 9.32) germanWhisper (Section 9.19) odd (Section 5.9) even (Section 5.5) box (Section 11.3) autoGlyph (Section 9.7)

9.8 Constraint averageArrow

9.8.1 Category: Line

9.8.2 Description

The number in the bulb is the average of the values on the arrow. This must be exact, no rounding of the average value is required.

9.8.3 Arguments

averageArrow List(Cell)

9.8.4 JSON Format

```
"lines":[{"averageArrow":[4,3,2]},...],
```

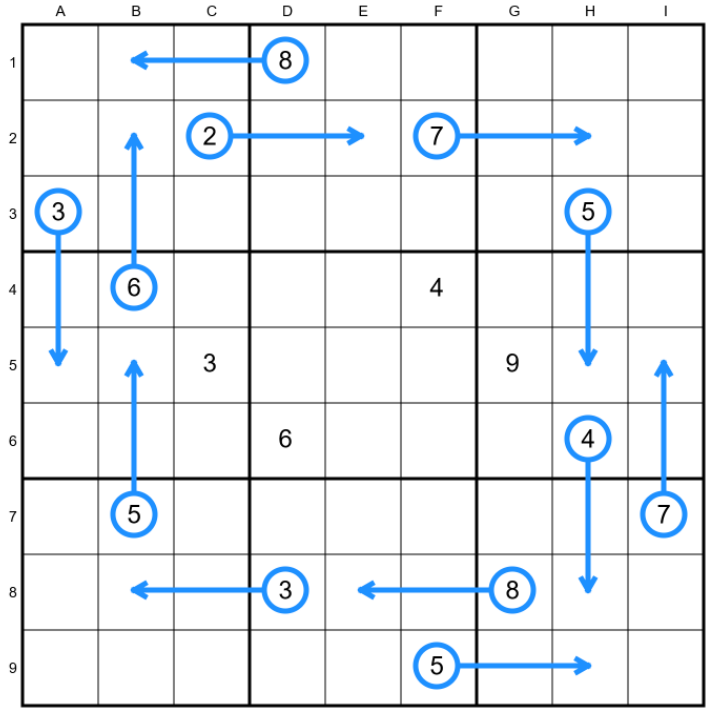
9.8.5 Model

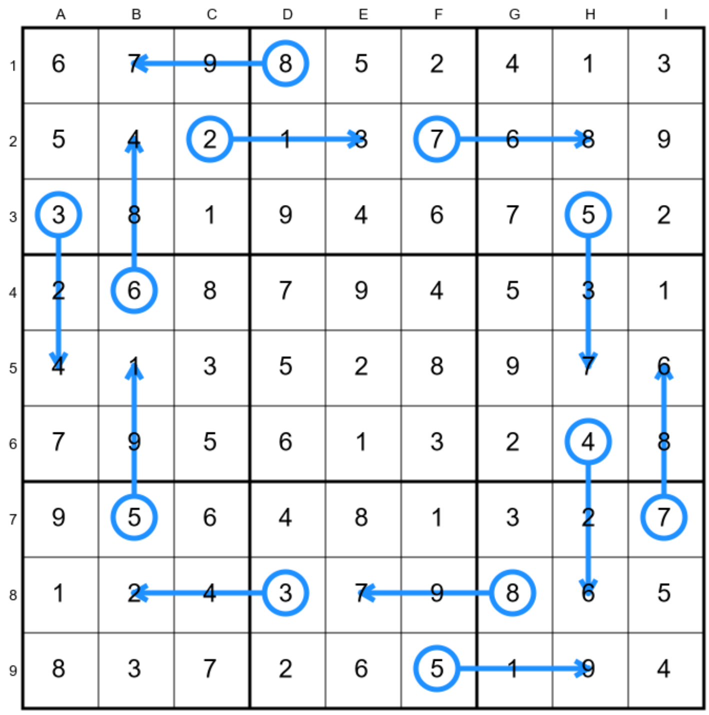
Model description not finished

9.8.6 See Also

9.8.7 averageArrow example timotab90: "Average Arrows Sudoku II" by Bill Murphy

Video: `https://youtu.be/s94ayTkpAdY?si=VTnkH3kWTx8FlnK5` (9:52)





Rules: standard (Section 3.13) averageArrow (Section 9.8)

9.9 Constraint between

9.9.1 Category: Line

9.9.2 Description

The extreme values on the line are in the first and last cells. All cells in between must be greater than the minimum and smaller than the maximum values. All values on the line must be pairwise different. The sequence <7, 2, 4, 5, 1> would form a valid between line.

9.9.3 Arguments

between List(Cell)

9.9.4 JSON Format

```
"lines":[{"between":[2,3,4,5]},...],
```

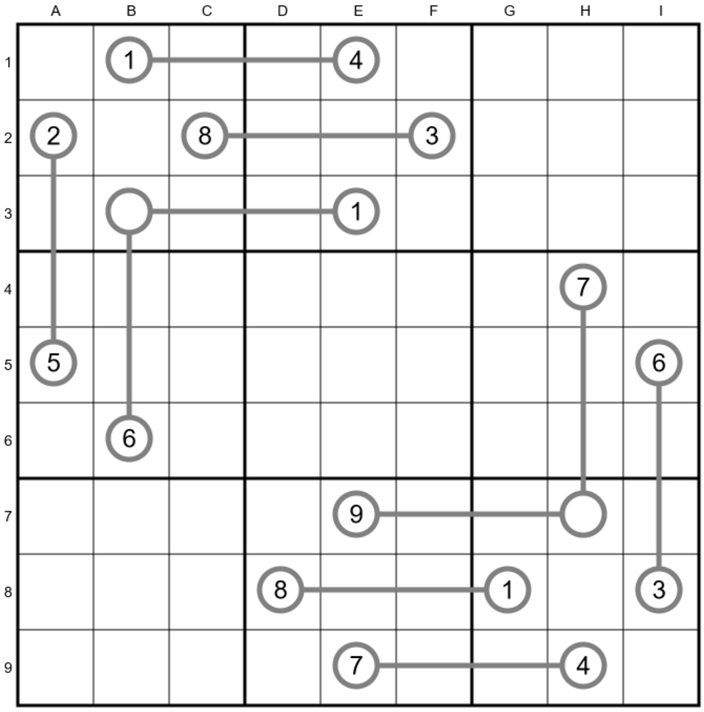
9.9.5 Model

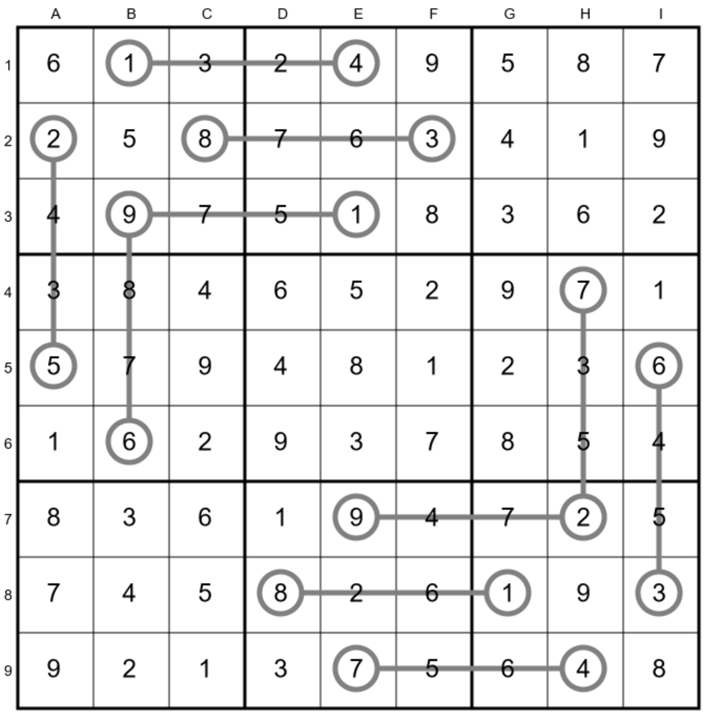
Model description not finished

9.9.6 See Also

9.9.7 between example timotab32: "Immurement" by clover

Video: `https://www.youtube.com/watch?v=SDfPR560HwU&t=48s`
Video2: `https://youtu.be/0VPk5OCBfHw?si=ecbFm1Pw4z0ILqu0`





Rules: standard (Section 3.13) between (Section 9.9)

9.10 Constraint betweenNValue

9.10.1 Category: Line

9.10.2 Description

The two endpoints contain the same value, which is the number of distinct values occurring on the line, including the endpoints. The line may consist of only the two endpoints, there obviously is no constraint that cells on the line must be distinct.

9.10.3 Arguments

betweenNValue List(Cell)

9.10.4 JSON Format

```
"lines":[{"betweenNValue":[4,12]},...],
```

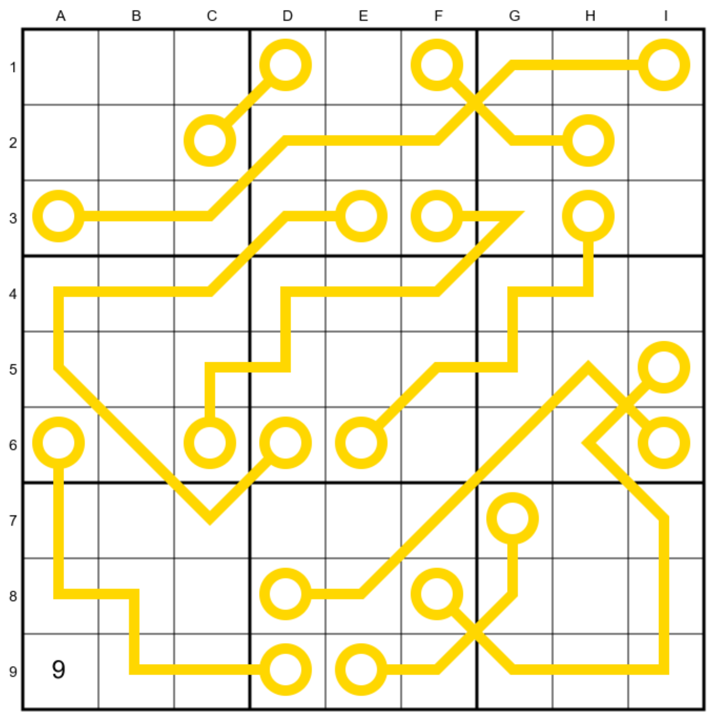
9.10.5 Model

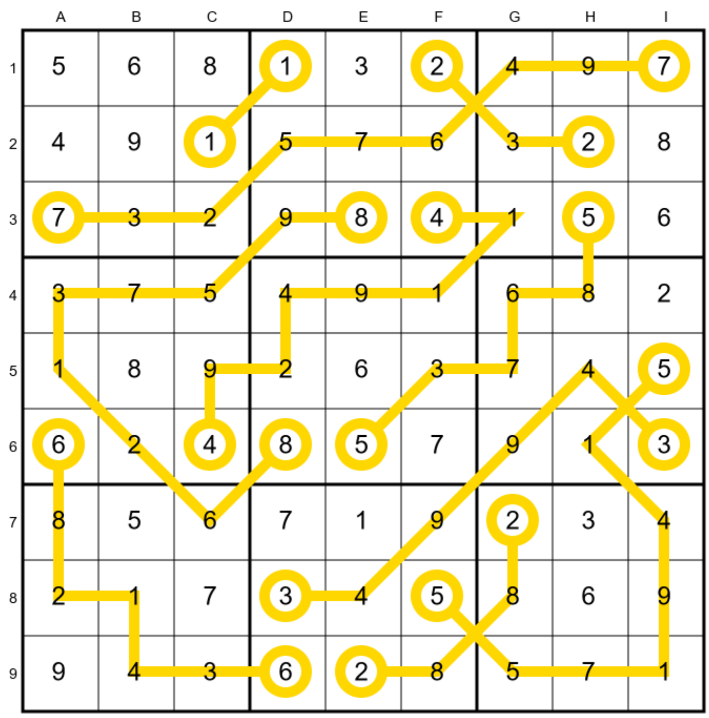
Model description not finished

9.10.6 See Also

9.10.7 betweenNValue example ctc57: "Circuit Board" by Jaze

Video: `https://www.youtube.com/watch?v=Ikrmeuq9G1M`





Rules: standard (Section 3.13) betweenNValue (Section 9.10)

9.11 Constraint betweenSum

9.11.1 Category: Line

9.11.2 Description

For each pair of circles connected by a line, the sum of the two digits in the circles must be equal to the sum of the digits on the line connecting them.

9.11.3 Arguments

betweenSum List(Cell)

9.11.4 JSON Format

```
"lines":[{"betweenSum":[1,2,3,4]},...],
```

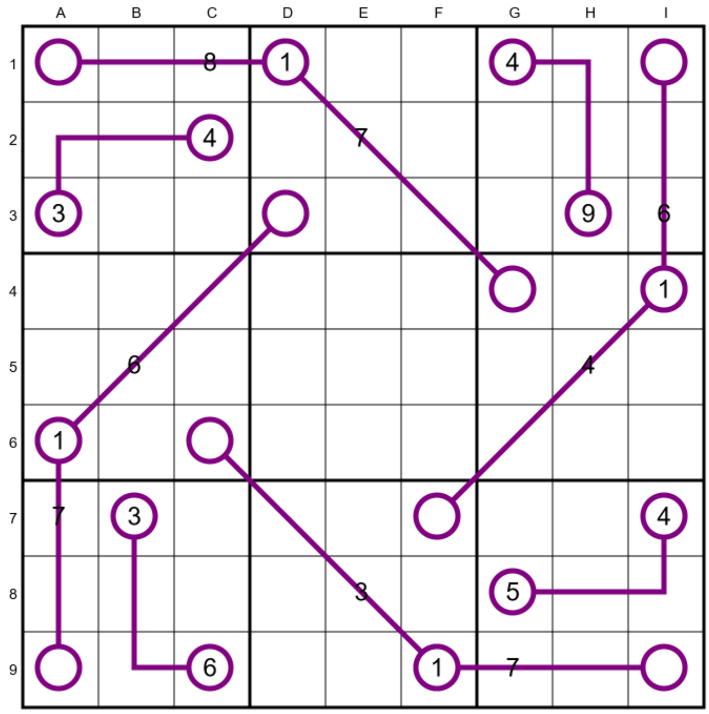
9.11.5 Model

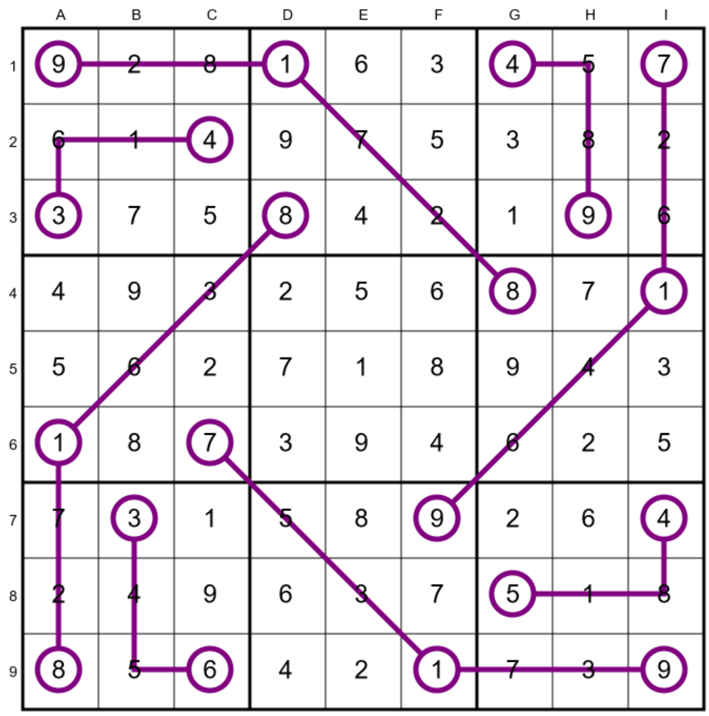
Model description not finished

9.11.6 See Also

9.11.7 betweenSum example timotab7: "Kaboom!" by clover

Video: `https://www.youtube.com/watch?v=aQrY1R-dlGQ`





Rules: standard (Section 3.13) betweenSum (Section 9.11)

9.12 Constraint consecutiveLine

9.12.1 Category: Line

9.12.2 Description

Adjacent cells on the line must be consecutive.

9.12.3 Arguments

consecutiveLine List(Cell)

9.12.4 JSON Format

```
"lines":[{"consecutiveLine":[1,11,21]},...]
```

9.12.5 Model

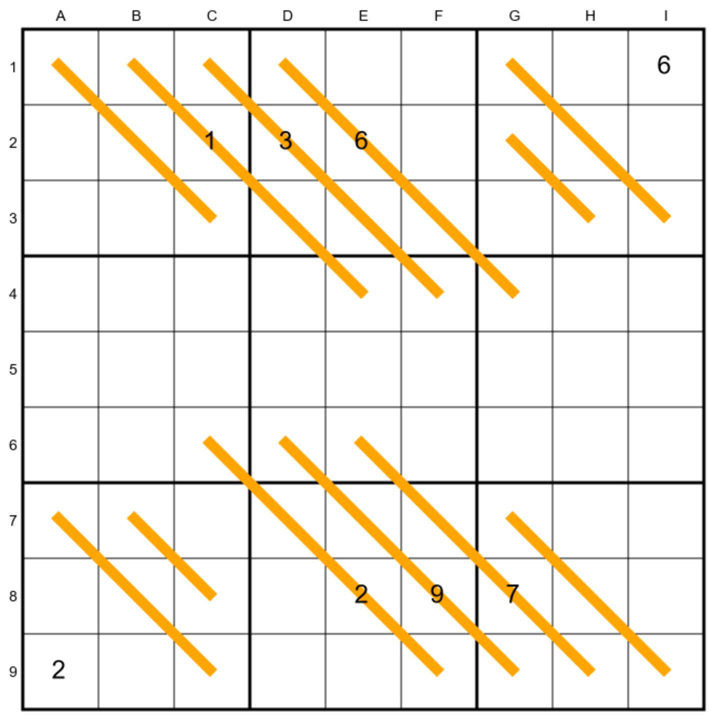
Model description not finished

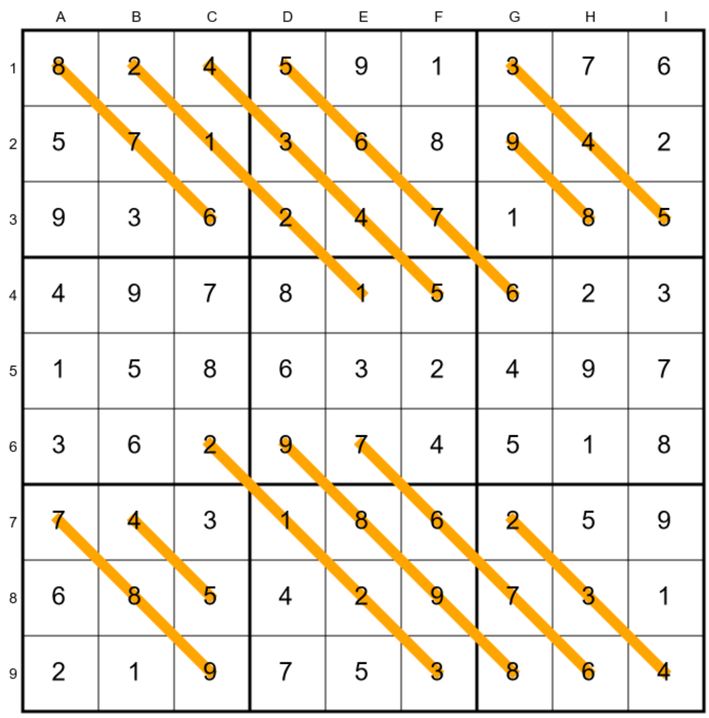
9.12.6 See Also

whiteDot (Section 6.10)

9.12.7 consecutiveLine example timotab60: "Diagonal Consecutive" by Bill Murphy

Video: `https://www.youtube.com/watch?v=buEuezbkRVk`





Rules: standard (Section 3.13) consecutiveLine (Section 9.12)

9.13 Constraint differenceLine

9.13.1 Category: Line

9.13.2 Description

All adjacent pairs of values on the line differ by the same amount.

9.13.3 Arguments

differenceLine List(Cell)

9.13.4 JSON Format

```
"lines":[{"differenceLine":[4,13,22]},...],
```

9.13.5 Model

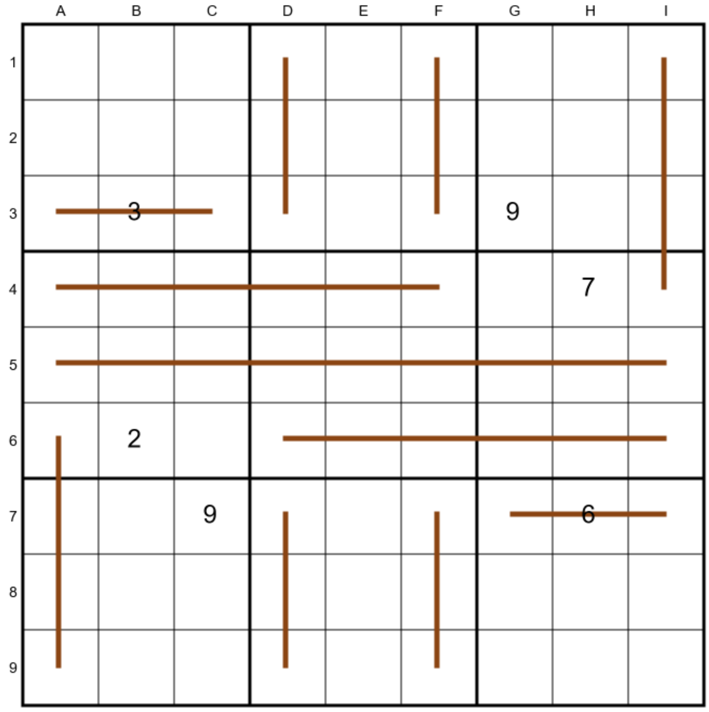
Model description not finished

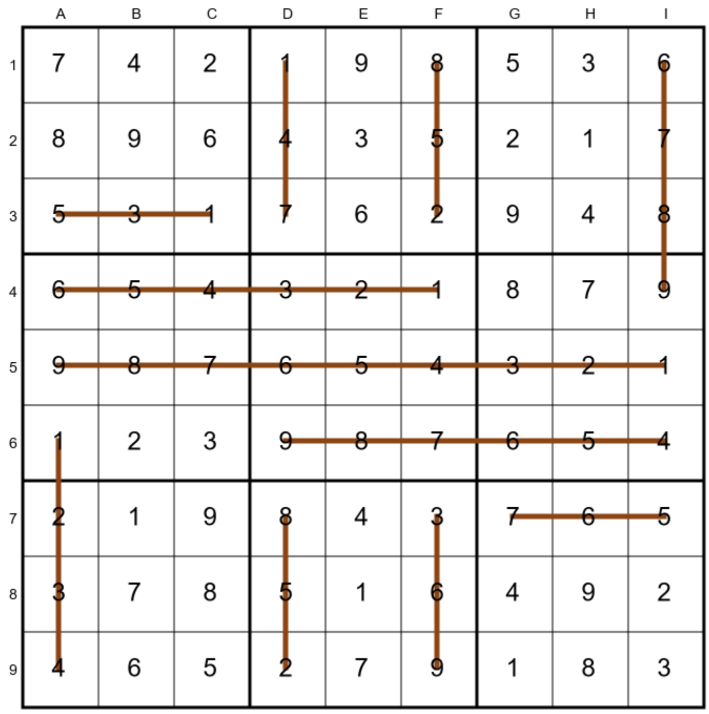
9.13.6 See Also

differencePair (Section 6.2)

9.13.7 differenceLine example gas57: "Easy As 17-42-67" by Philip Newman

Video2: `https://www.youtube.com/watch?v=7P7Jx1L0IRU&t=372s`





Rules: standard (Section 3.13) differenceLine (Section 9.13)

9.14 Constraint doubleArrow

9.14.1 Category: Line

9.14.2 Description

The sum of the digits on the inside of the line must be equal to the sum of the two endpoints.

9.14.3 Arguments

doubleArrow List(Cell)

9.14.4 JSON Format

```
"lines":[{"doubleArrow":[3,11,19,28,38,46,55,65,75]},...],
```

9.14.5 Model

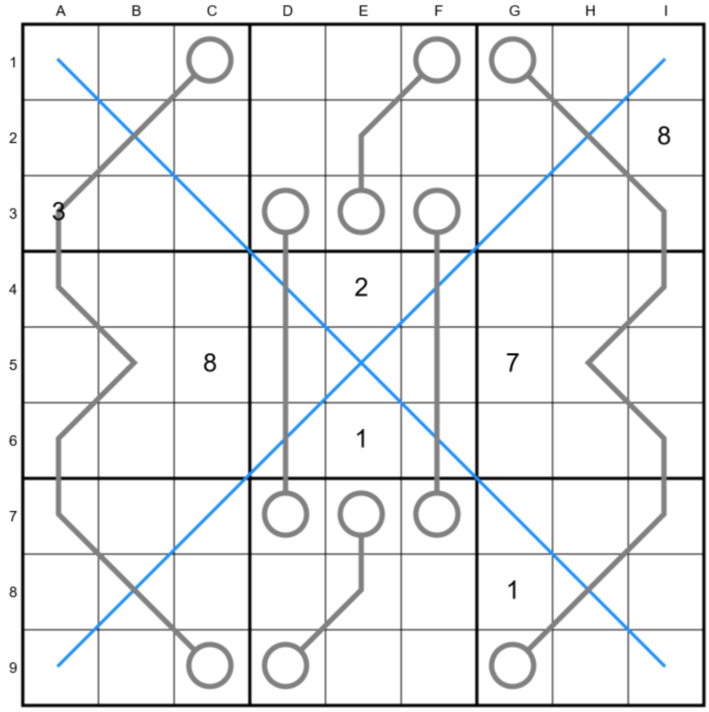
Model description not finished

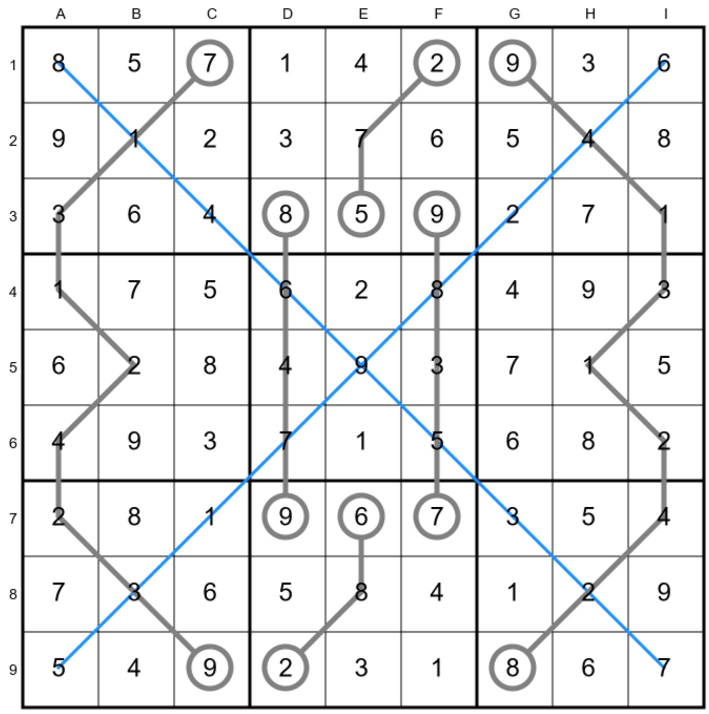
9.14.6 See Also

arrow (Section 9.2)

9.14.7 doubleArrow example ctc116: "Lynx" by Blobz

Video: `https://www.youtube.com/watch?v=z54K1xbazxg`





Rules: standard (Section 3.13) doubleArrow (Section 9.14) mainDiagonal (Section 4.9) minorDiagonal (Section 4.10)

9.15 Constraint dutchWhisper

9.15.1 Category: Line

9.15.2 Description

Neighbouring cells on the line must differ by at least four.

9.15.3 Arguments

dutchWhisper List(Cell)

9.15.4 JSON Format

```
"lines":[{"dutchWhisper":[11,12,21,20,11]},...],
```

9.15.5 Model

Let $[x_1, x_2, ..., x_k]$ be the set variables corresponding to the given list of cells. Consecutive variables must differ by at least four, i.e.

$$\forall_{i \in 1..k-1}: \quad |x_{i+1} - x_i| \geq 4 \tag{21}$$

We can achieve better propagation by using a table constraint

$$\forall_{i \in 1..k-1}: \quad \text{table}([x_i, x_{i+1}], \text{dutchWhisper}_2) \tag{22}$$

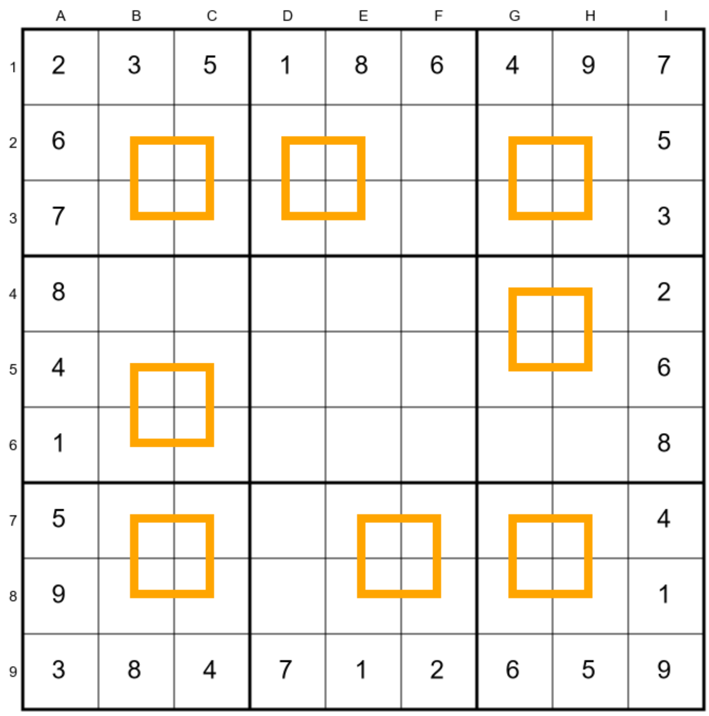
with tuples dutchWhisper_2 enumerating all pairs of values which satisfy the constraint (21).

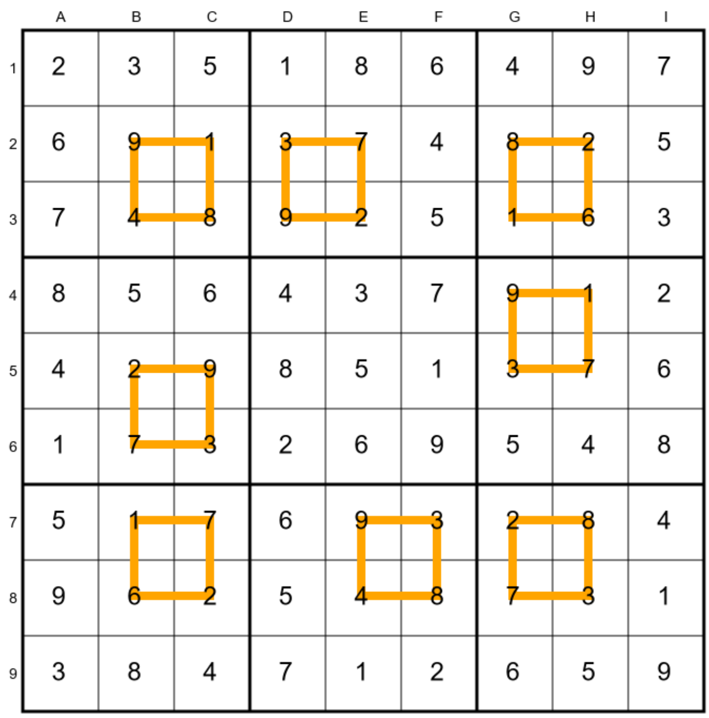
9.15.6 See Also

germanWhisper (Section 9.19)

9.15.7 dutchWhisper example timotab69: "Fangless" by Bill Murphy

Video: `https://www.youtube.com/watch?v=O464qmXvrlg`





Rules: standard (Section 3.13) dutchWhisper (Section 9.15)

9.16 Constraint echoLine

9.16.1 Category: Line

9.16.2 Description

The line is filled with a repeated sequence of digits. The pattern must repeat at least twice, and must repeat completely. Different lines may have different sequences.

9.16.3 Arguments

echoLine List(Cell)

9.16.4 JSON Format

```
"lines":[{"echoLine":[2,3,13,14]},...],
```

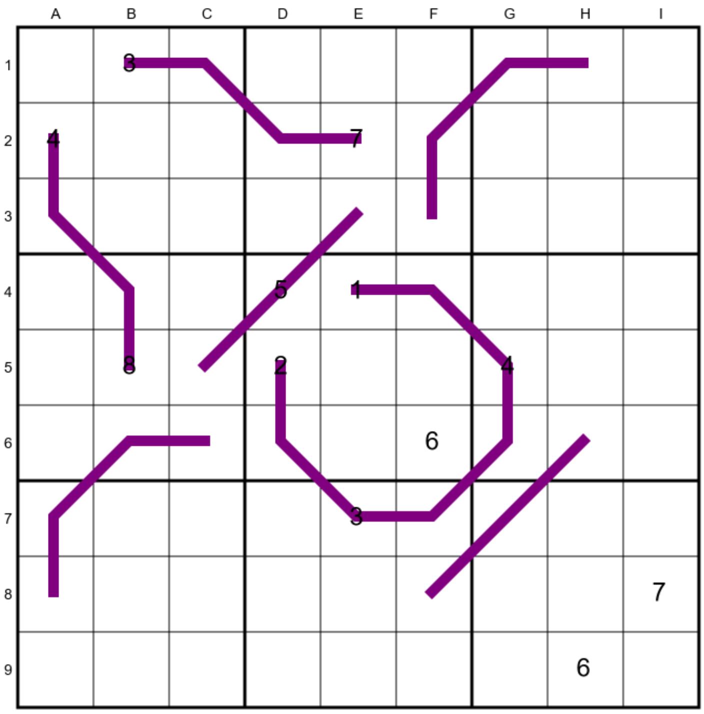
9.16.5 Model

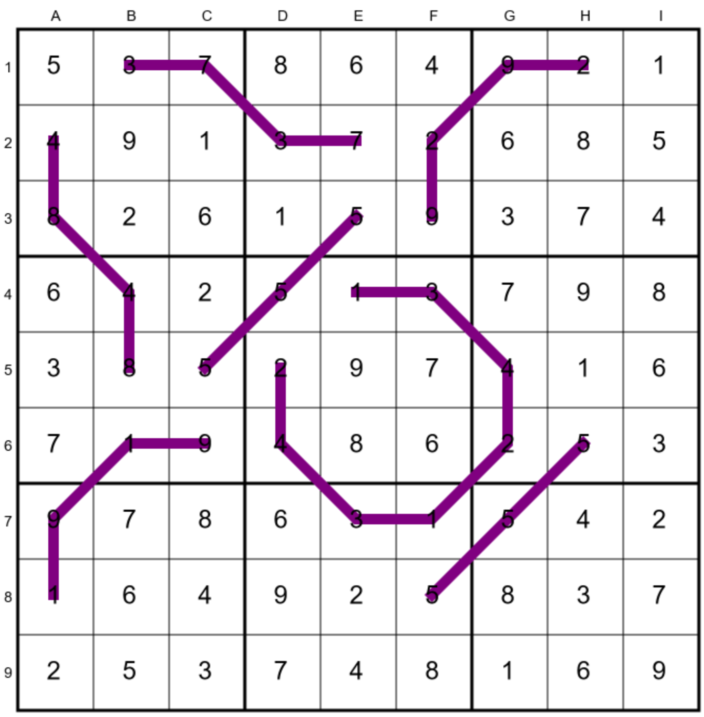
Model description not finished

9.16.6 See Also

9.16.7 echoLine example timotab10: "Echo Lines" by Bill Murphy

Video: `https://www.youtube.com/watch?v=wNzIsWGibyk`
Video2: `https://youtu.be/dToF189EzQk?si=C2VDtUq7EiQJNy6s`





Rules: standard (Section 3.13) echoLine (Section 9.16)

9.17 Constraint eightLine

9.17.1 Category: Line

9.17.2 Description

The line can be partitioned into non-overlapping segments which each add up to eight. Values may repeat on the line, if allowed by other Sudoku rules.

9.17.3 Arguments

eightLine List(Cell)

9.17.4 JSON Format

```
"lines":[{"eightLine":[4,12,11,20,29]},...],
```

9.17.5 Model

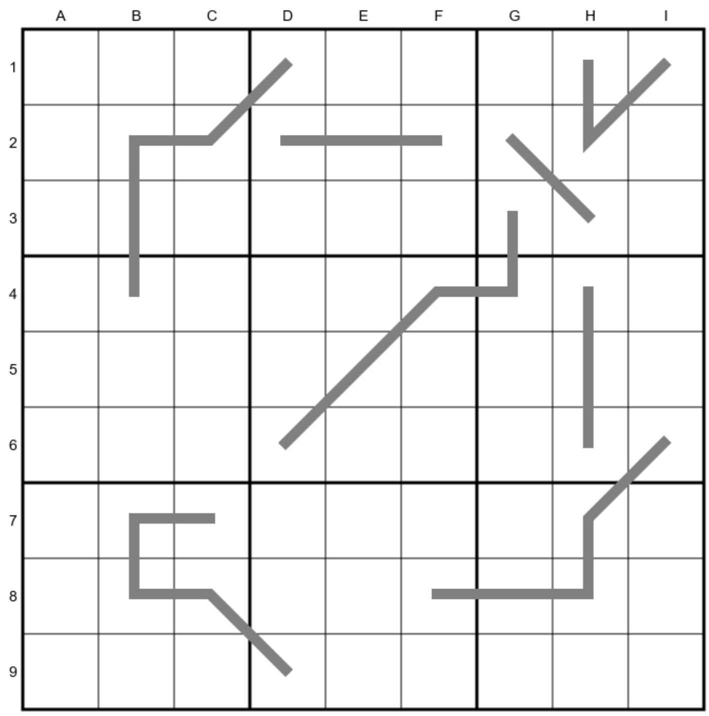
Model description not finished

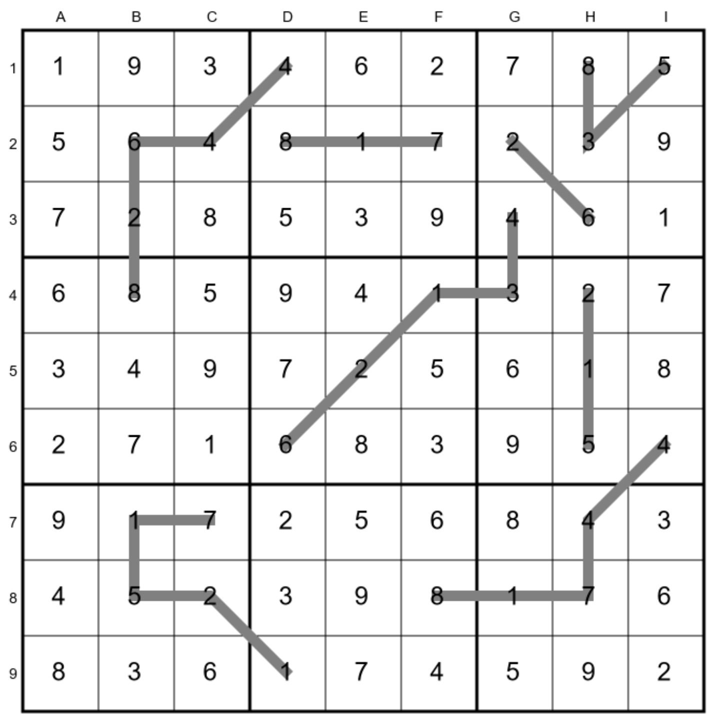
9.17.6 See Also

tenLine (Section 9.36)

9.17.7 eightLine example logiclemurgaming28: "Dutch Flat Mates: Eight Lines" by Flinty

Video: `https://www.youtube.com/watch?v=ElpCKSGhAmw`





Rules: standard (Section 3.13) dutchFlatMates (Section 4.6) eightLine (Section 9.17)

9.18 Constraint entropicLine

9.18.1 Category: Line

9.18.2 Description

Every subsequence of three consecutive cells on the line consists of one each of small (1-3), middle (4-6), and high (7-9) values.

9.18.3 Arguments

entropicLine List(Cell)

9.18.4 JSON Format

```
"lines":[{"entropicLine":[12,3,2,1,10,19]},...],
```

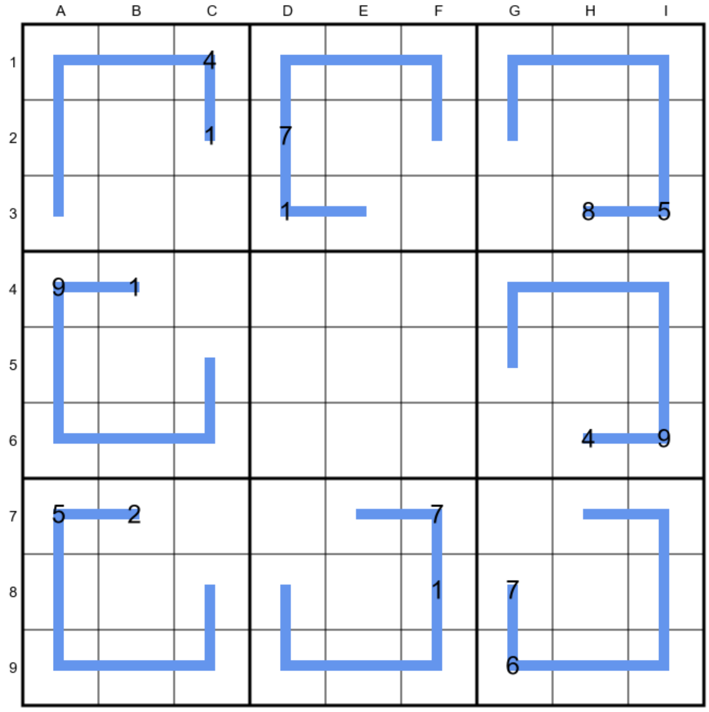
9.18.5 Model

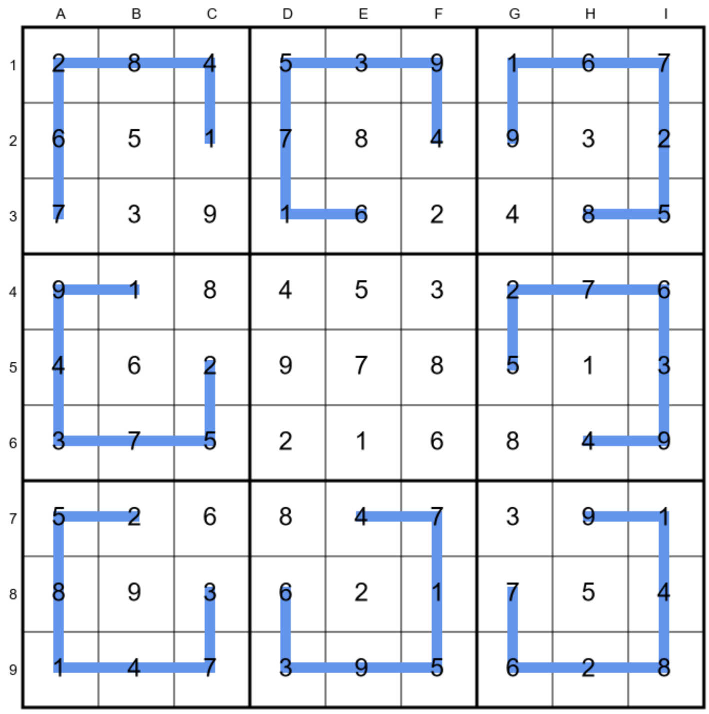
Model description not finished

9.18.6 See Also

9.18.7 entropicLine example gas40: "Numeric Keypad" by clover

Video: `https://youtu.be/2Mdp0G4_0Ns?si=MtM7sQiu962rcRdd`





Rules: standard (Section 3.13) entropicLine (Section 9.18)

9.19 Constraint germanWhisper

9.19.1 Category: Line

9.19.2 Description

Neighbouring cells on the line must differ by at least five. The value five cannot appear on the line. The sequence <1, 6, 1, 8, 2, 7> would be a legal German Whisper line.

9.19.3 Arguments

germanWhisper List(Cell)

9.19.4 JSON Format

"lines":[{"germanWhisper":[41,42,34,24,23,22,30,39,48,58,59,60,61,53,45,36,27,18,8,7,6,5,4,3,11,19,28

9.19.5 Model

Let $[x_1, x_2, ..., x_k]$ be the set variables corresponding to the given list of cells. Consecutive variables must differ by at least five, i.e.

$$\forall_{i \in 1..k-1}: \quad |x_{i+1} - x_i| \geq 5 \tag{23}$$

We can achieve better propagation by using a table constraint

$$\forall_{i \in 1..k-1}: \quad \text{table}([x_i, x_{i+1}], \text{germanWhisper}_2) \tag{24}$$

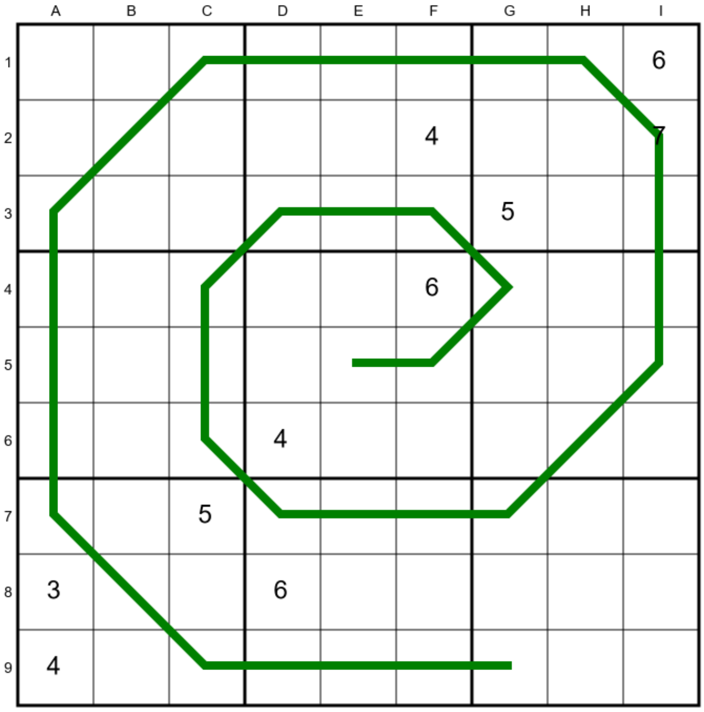
with tuples germanWhisper_2 enumerating all pairs of values which satisfy the constraint (23).

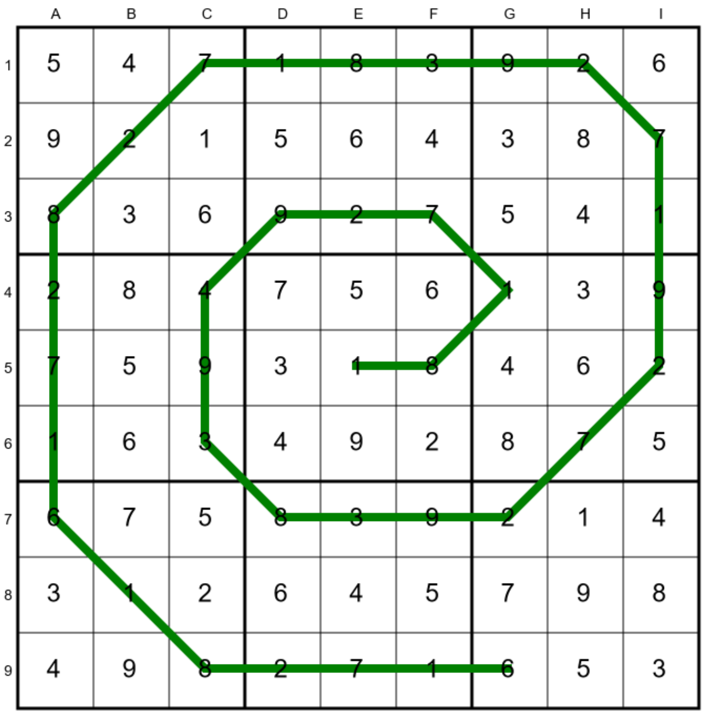
9.19.6 See Also

dutchWhisper (Section 9.15)

9.19.7 germanWhisper example timotab9: "This Murmur Coil" by Philip Newman

Video: `https://www.youtube.com/watch?v=bPohCwlGb44`
Video2: `https://youtu.be/YiwW8jETz70?si=ShwLgAB4MDjtbmG7`





Rules: standard (Section 3.13) germanWhisper (Section 9.19)

9.20 Constraint greatestLine

9.20.1 Category: Line

9.20.2 Description

The largest value on the line is greater than the sum of any two other cells on the line

9.20.3 Arguments

greatestLine List(Cell)

9.20.4 JSON Format

```
"lines":[{"greatestLine":[8,17,18]},...],
```

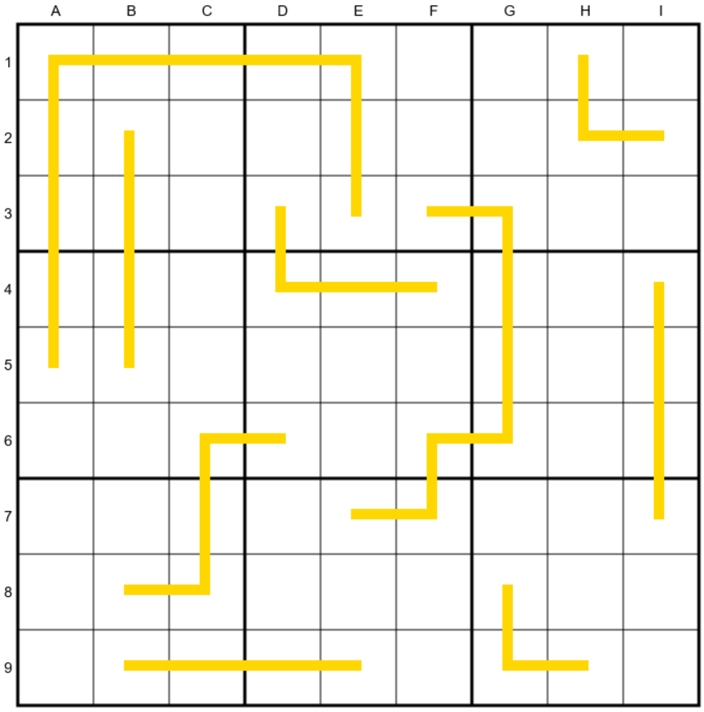
9.20.5 Model

Model description not finished

9.20.6 See Also

9.20.7 greatestLine example ctc78: "The Greatest Line" by Dorlir

Video: `https://www.youtube.com/watch?v=0Z8XXgfiVB4&t=2358s`



	A	B	C	D	E	F	G	H	I
1	9	4	3	2	1	6	7	5	8
2	2	5	6	7	4	8	9	1	3
3	1	8	7	5	3	9	2	6	4
4	4	1	8	3	9	2	5	7	6
5	3	2	5	6	8	7	1	4	9
6	7	6	9	4	5	1	3	8	2
7	5	7	4	8	2	3	6	9	1
8	6	3	1	9	7	4	8	2	5
9	8	9	2	1	6	5	4	3	7

Rules: standard (Section 3.13) greatestLine (Section 9.20)

9.21 Constraint highLowLine

9.21.1 Category: Line

9.21.2 Description

The values on the line are either all low (1-5) or all high (5-9) values.

9.21.3 Arguments

highLowLine List(Cell)

9.21.4 JSON Format

```
"lines":[{"highLowLine":[4,5,6,7]},...],
```

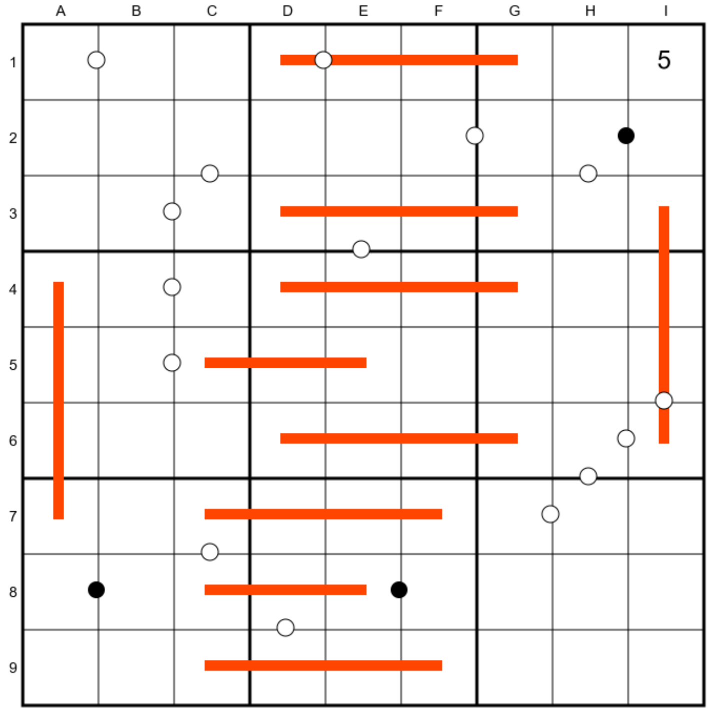
9.21.5 Model

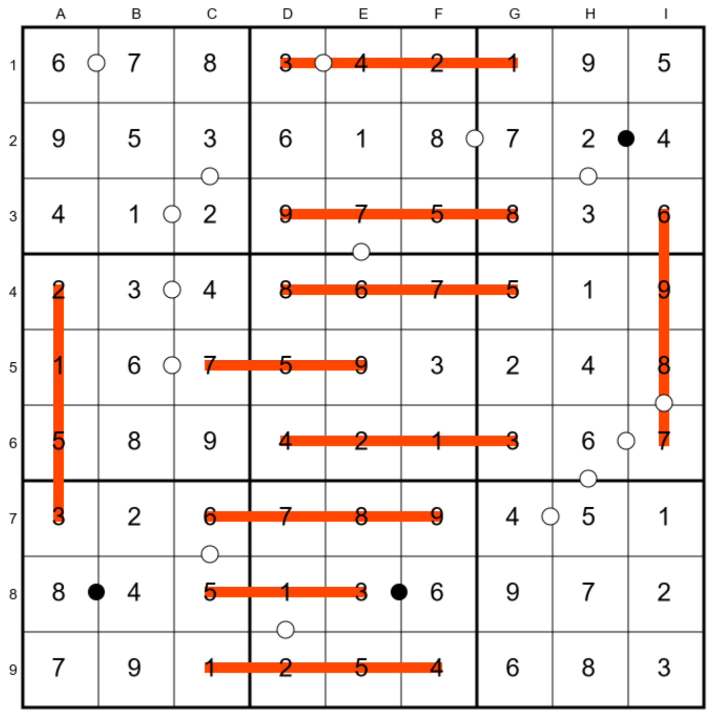
Model description not finished

9.21.6 See Also

9.21.7 highLowLine example BremSter49: "High - Low" by The Bard

Video: `https://www.youtube.com/watch?v=ltZWMsFRbJw`





Rules: standard (Section 3.13) blackDot (Section 6.1) whiteDot (Section 6.10) highLowLine (Section 9.21)

9.22 Constraint increasingRegionSumLine

9.22.1 Category: Line

9.22.2 Description

The line is split into segments by the block borders, the sums of the values in the segments are strictly increasing or decreasing.

9.22.3 Arguments

increasingRegionSumLine List(Cell)

9.22.4 JSON Format

```
"lines":[{"increasingRegionSumLine":[65,56,48,39,38,37,28,20,12,13,23,31,32,33,25,26,...],...],
```

9.22.5 Model

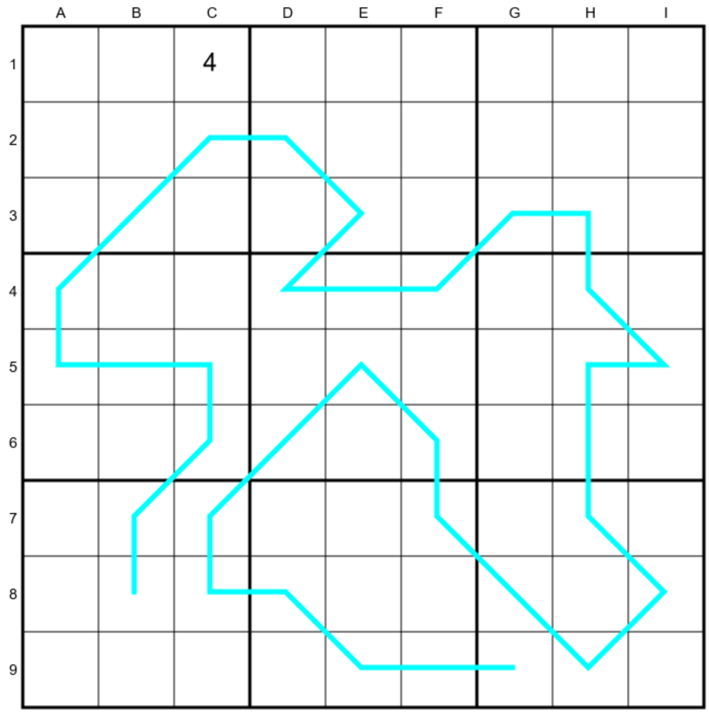
Model description not finished

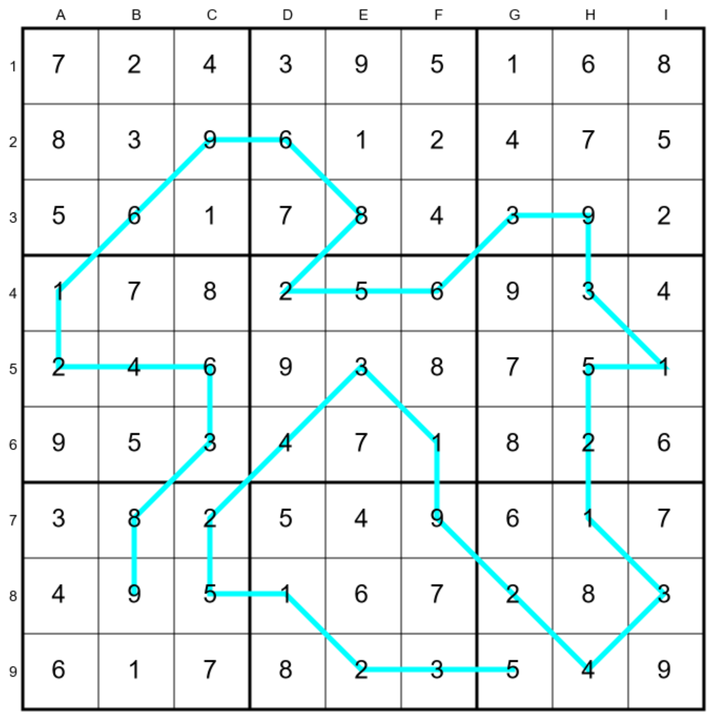
9.22.6 See Also

regionSumLine (Section 9.31)

9.22.7 increasingRegionSumLine example ctc7: "Increasing Sum Line" by Florian Wortmann

Video: `https://www.youtube.com/watch?v=m3IzqXmL4qw`





Rules: standard (Section 3.13) increasingRegionSumLine (Section 9.22)

9.23 Constraint indexLine

9.23.1 Category: Line

9.23.2 Description

The digit in the nth position on the line (counting from 1), indicates the position of value n on the the line.

9.23.3 Arguments

indexLine List(Cell)

9.23.4 JSON Format

```
"lines":[{"indexLine":[13,5,15,6,7]},...],
```

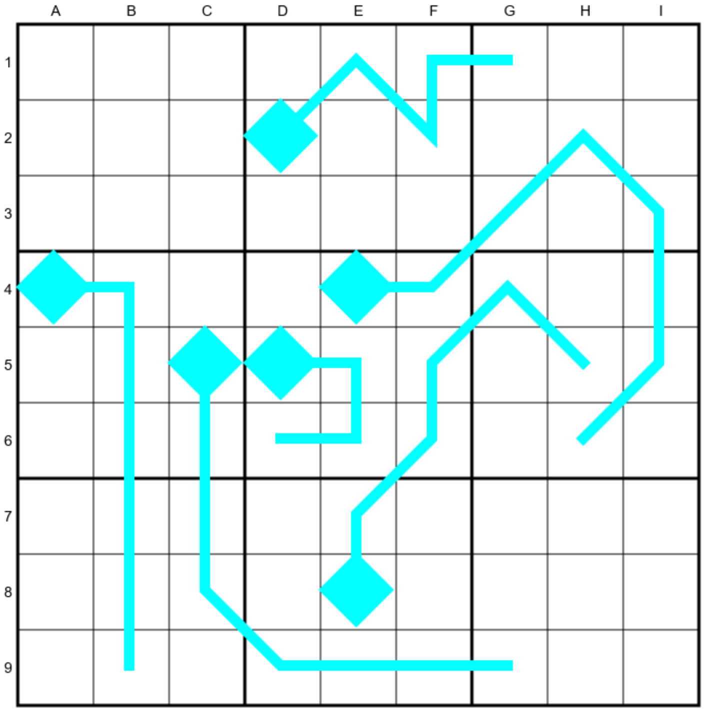
9.23.5 Model

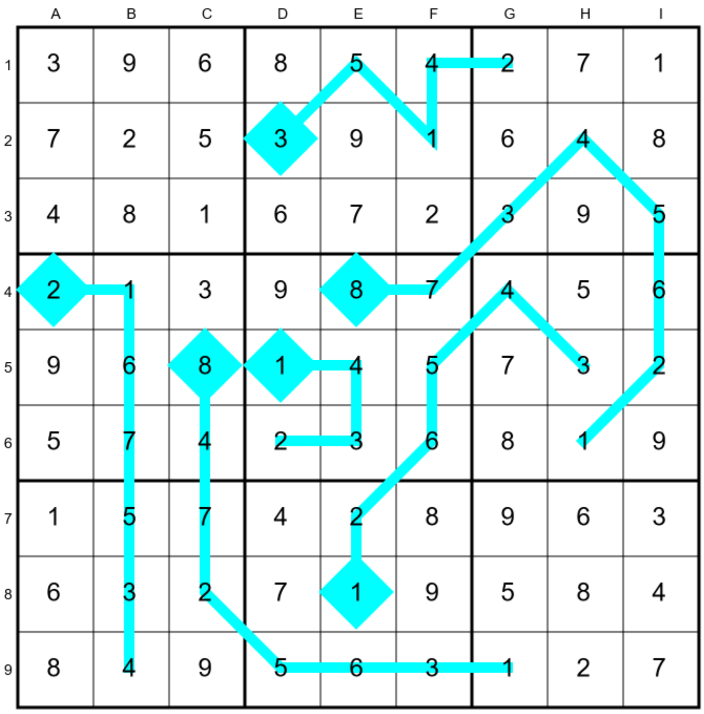
Model description not finished

9.23.6 See Also

9.23.7 indexLine example timotab62: "Index Lines" by Richard

Video: `https://www.youtube.com/watch?v=MqK5g3EhRBw`





Rules: standard (Section 3.13) indexLine (Section 9.23)

9.24 Constraint lockoutLine

9.24.1 Category: Line

9.24.2 Description

The values at the end of the line define an Interval, they must be at least 4 apart. All other digits on the line must be outside the interval, they are either smaller than the minimum of the end points, or larger than the maximum of the end points.

9.24.3 Arguments

lockoutLine List(Cell)

9.24.4 JSON Format

```
"lines":[{"lockoutLine":[3,13,5,15,7]},...],
```

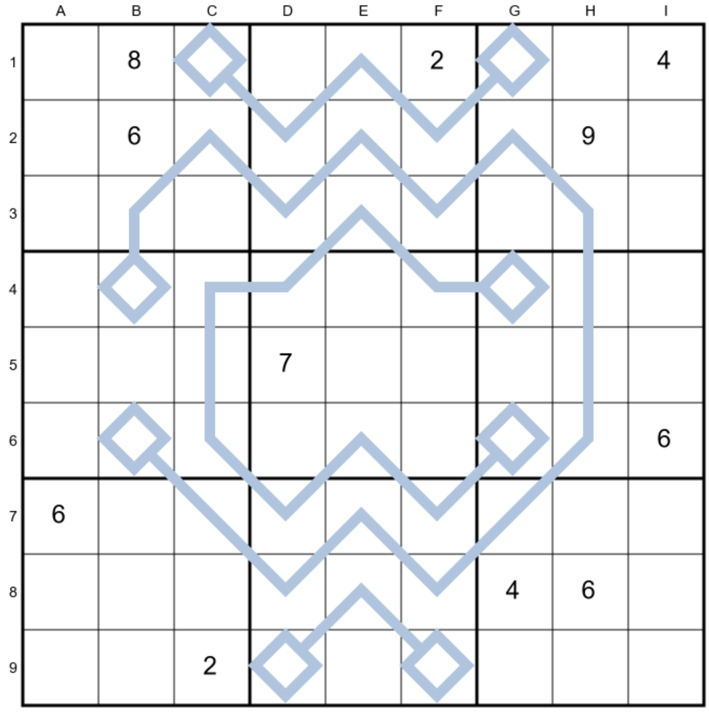
9.24.5 Model

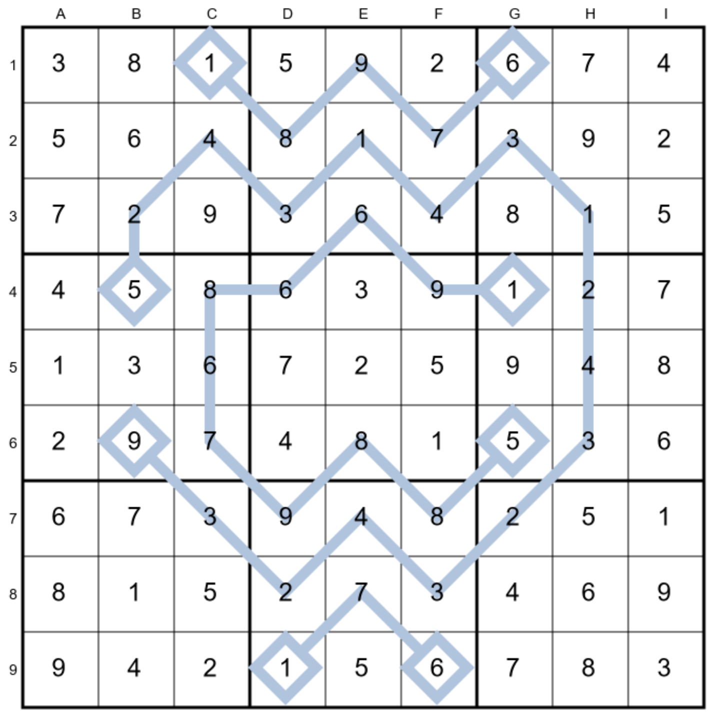
Model description not finished

9.24.6 See Also

9.24.7 lockoutLine example ctc81: "Lockout Lines turn 4" by Riffclown

Video: `https://www.youtube.com/watch?v=YC7dPnq8bcM`





Rules: standard (Section 3.13) lockoutLine (Section 9.24)

9.25 Constraint lookSayLine

9.25.1 Category: Line

9.25.2 Description

The line contains the digits given in the count list a certain time. The sum of the counts does not always add up to the length of the line.

9.25.3 Arguments

lookSayLine List(Cell)

counts List(Pair(Integer))

9.25.4 JSON Format

```
"lines":[{"lookSayLine":[1,2,3,4,5,6,7,8,...],"counts":[[8,7],[6,5],[4,3],[2,1]]},
```

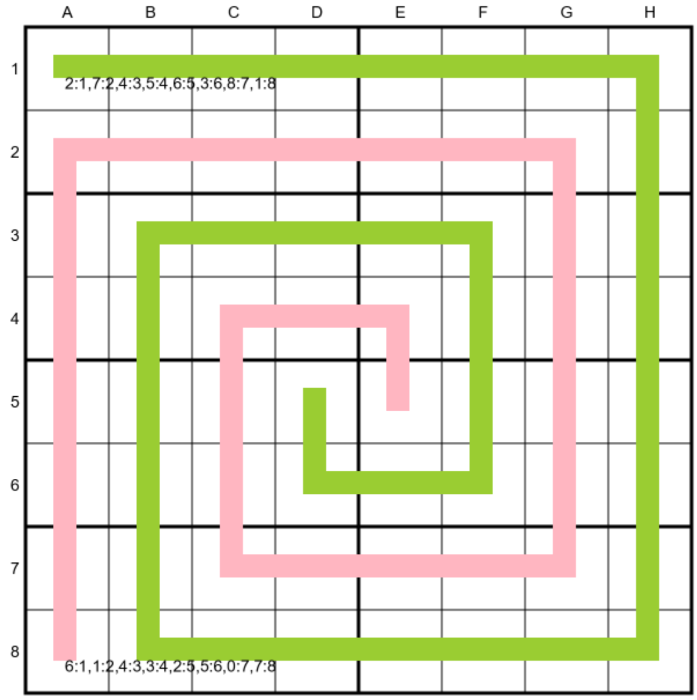
9.25.5 Model

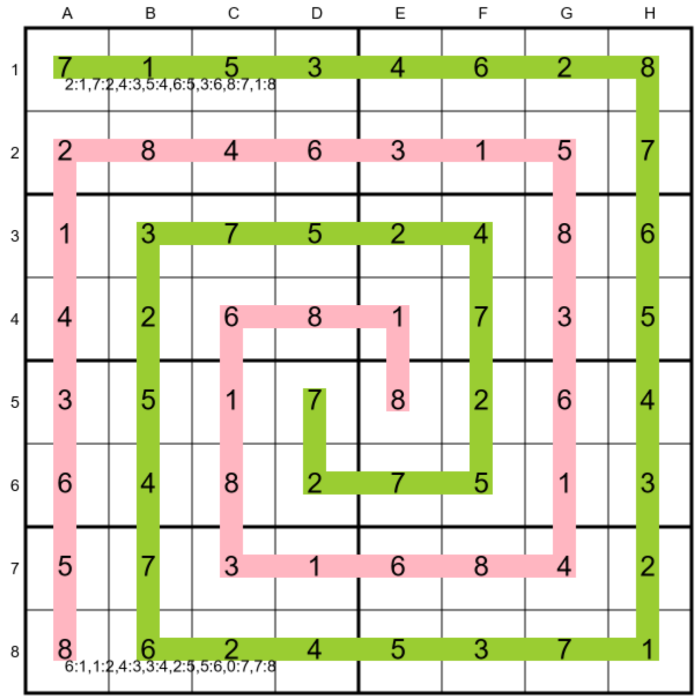
Model description not finished

9.25.6 See Also

9.25.7 lookSayLine example ctc35: "Hypnotic Suggestion" by Atticus837

Video: `https://www.youtube.com/watch?v=ddnLr8jWNeY`





Rules: standard (Section 3.13) 8x8 (Section 3.2) lookSayLine (Section 9.25)

9.26 Constraint modLine

9.26.1 Category: Line

9.26.2 Description

Every subsequence of three consecutive cells on the line consists of one each from the three sets (1,4,7), (2,5,8), and (3,6,9), i.e. all three cells have a different remainder modulo 3.

9.26.3 Arguments

modLine List(Cell)

9.26.4 JSON Format

```
"lines":[{"modLine":[12,20,29,39]},...],
```

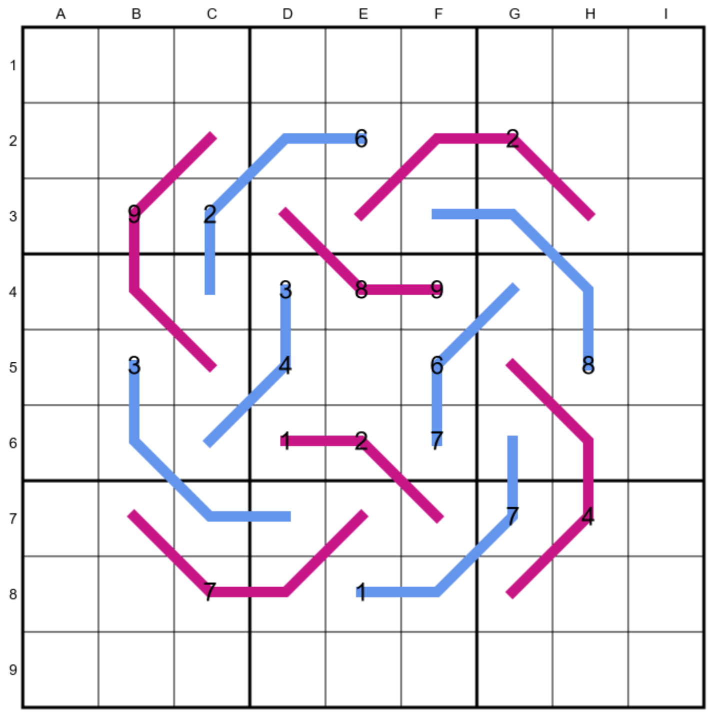
9.26.5 Model

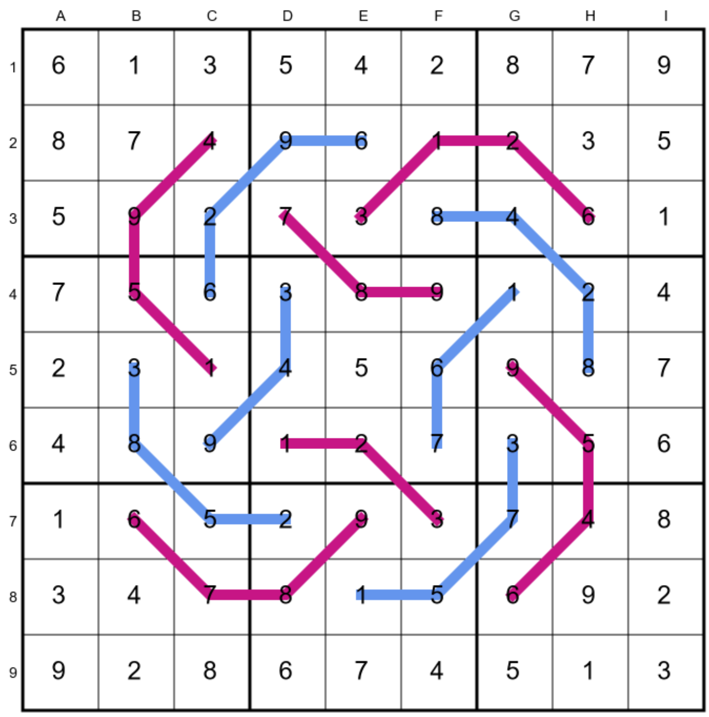
Model description not finished

9.26.6 See Also

9.26.7 modLine example gas5: "Birbs of Paradise" by Philip Newman

Video: `https://www.youtube.com/watch?v=hC7b4p37PBg`
Video2: `https://www.youtube.com/watch?v=JcgA42-05tI&t=52s`





Rules: standard (Section 3.13) entropicLine (Section 9.18) modLine (Section 9.26)

9.27 Constraint nValueLine

9.27.1 Category: Line

9.27.2 Description

The number of values used on the line must be exactly the value given as parameter.

9.27.3 Arguments

nValueLine List(Cell)

value Integer

9.27.4 JSON Format

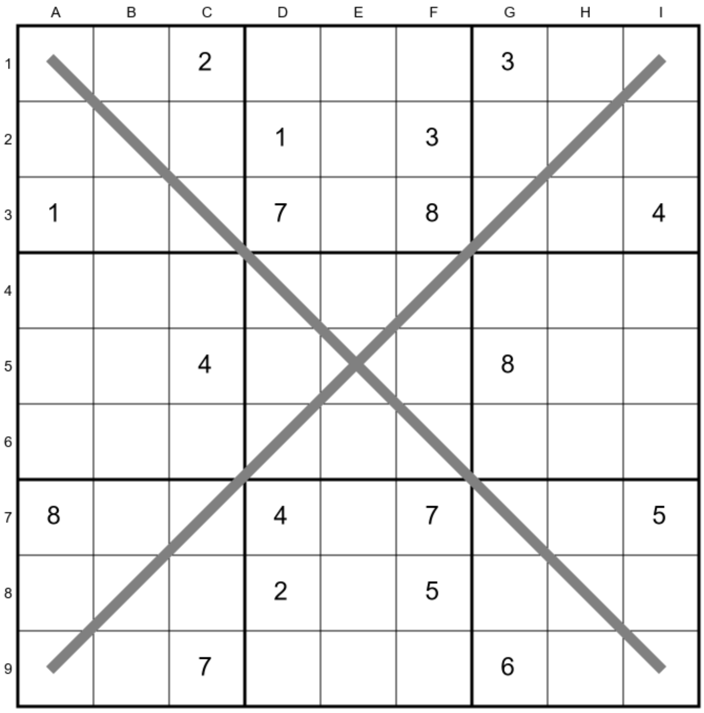
```
"lines":[{"nValueLine":[1,11,21,31,41,51,61,71,81],"value":3},...],
```

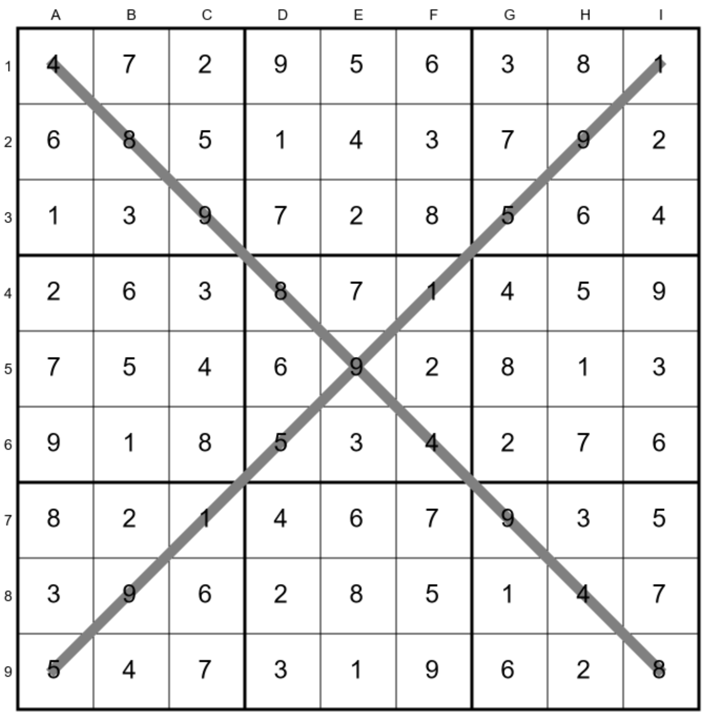
9.27.5 Model

Model description not finished

9.27.6 See Also

9.27.7 nValueLine example wpf gp2025r7c6: "Sudoku GP 2025 Round 7" by





Rules: standard (Section 3.13) nValueLine (Section 9.27)

9.28 Constraint nabLine

9.28.1 Category: Line

9.28.2 Description

The values on the line must be different from each other. Any two digits on a Nab (Nabner) line cannot be consecutive, regardless of their position on the line.

9.28.3 Arguments

nabLine List(Cell)

9.28.4 JSON Format

```
"lines":[{"nabLine":[1,11,21,31]},...],
```

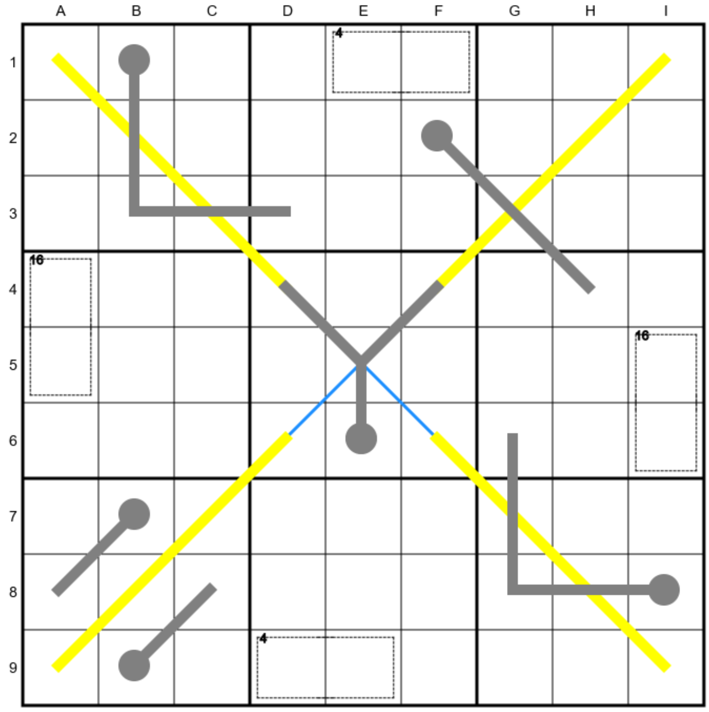
9.28.5 Model

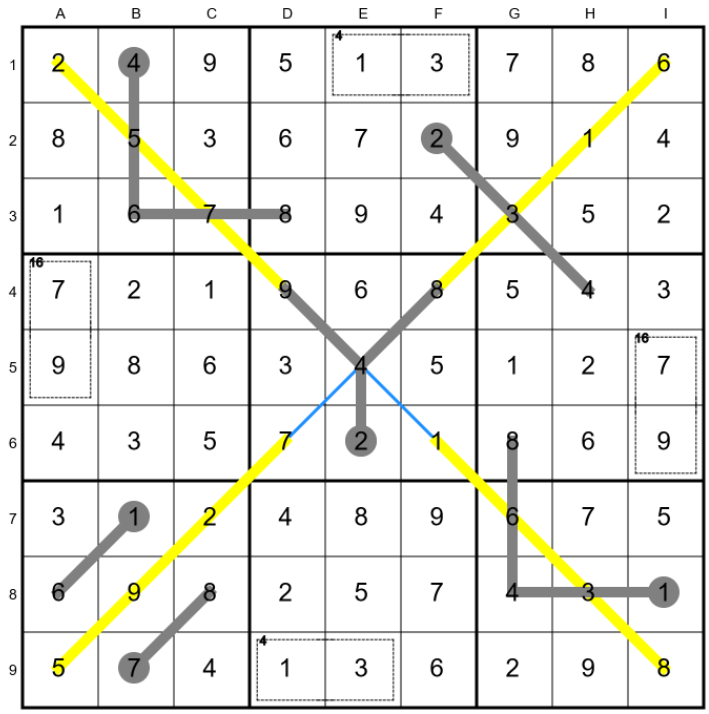
Model description not finished

9.28.6 See Also

9.28.7 nabLine example BremSter47: "Century" by James Sinclair

Video: `https://www.youtube.com/watch?v=w0n7EAxz00k`





Rules: standard (Section 3.13) mainDiagonal (Section 4.9) minorDiagonal (Section 4.10) cage (Section 11.4) thermo (Section 9.38) nabLine (Section 9.28)

9.29 Constraint oddEven

9.29.1 Category: Line

9.29.2 Description

The values on the line alternate between odd and even numbers.

9.29.3 Arguments

oddEven List(Cell)

9.29.4 JSON Format

```
"lines":[{"oddEven":[1,2,3,4]},...],
```

9.29.5 Model

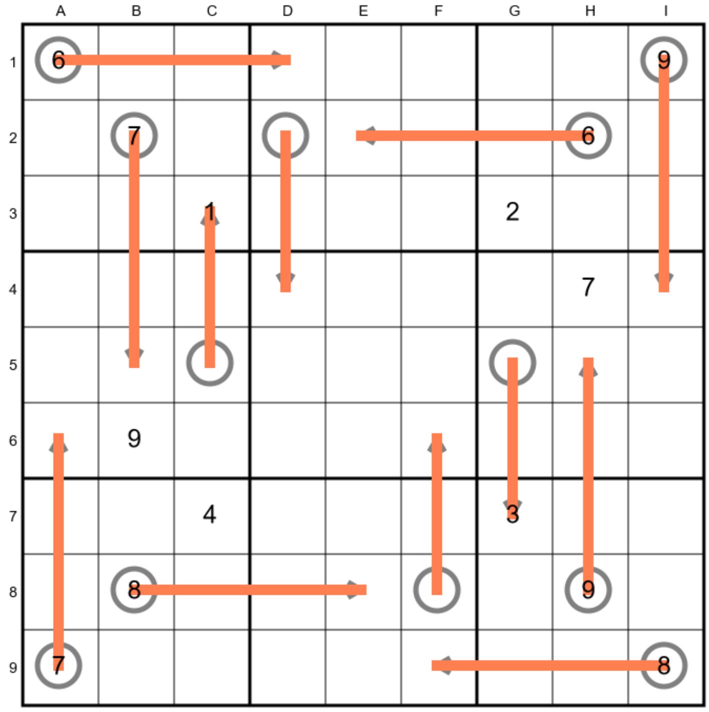
Model description not finished

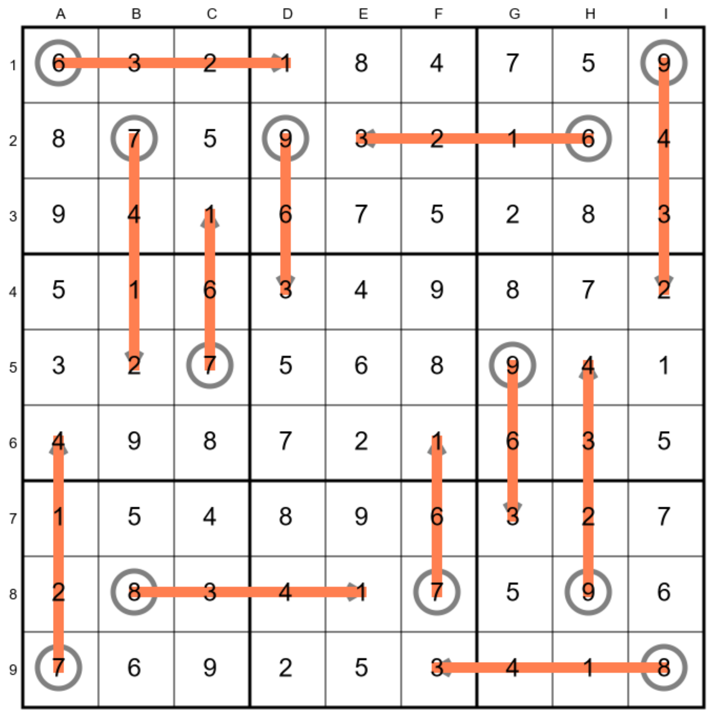
9.29.6 See Also

odd (Section 5.9) , even (Section 5.5)

9.29.7 oddEven example timotab53: "Not an Arrow Sudoku" by clover

Video: `https://www.youtube.com/watch?v=ugmoSsqzbb4`





Rules: standard (Section 3.13) arrow (Section 9.2) oddEven (Section 9.29)

9.30 Constraint palindrome

9.30.1 Category: Line

9.30.2 Description

The sequence of values must be the same in both directions. The sequence <3, 4, 6, 4, 3> would be a valid palindrome of length five.

9.30.3 Arguments

palindrome List(Cell)

9.30.4 JSON Format

```
"lines":[{"palindrome":[2,3,13,14]},...],
```

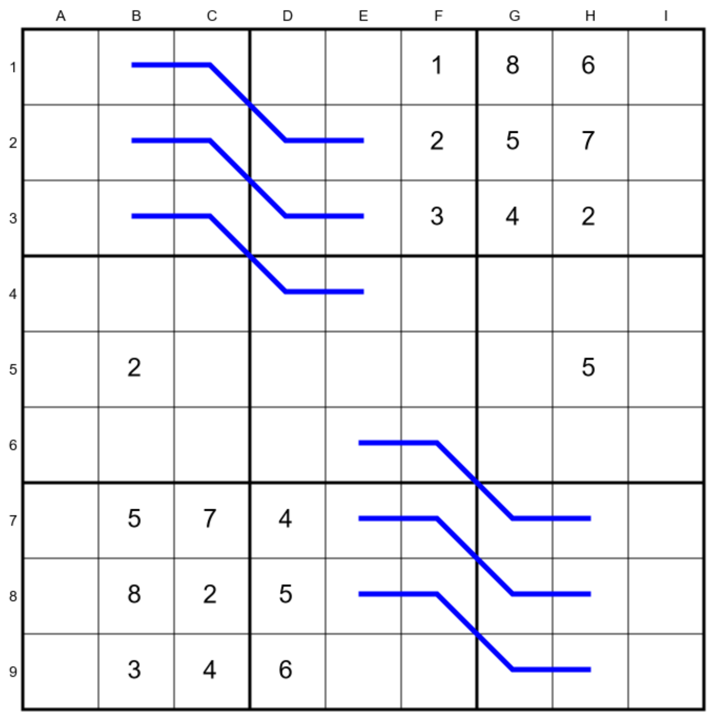
9.30.5 Model

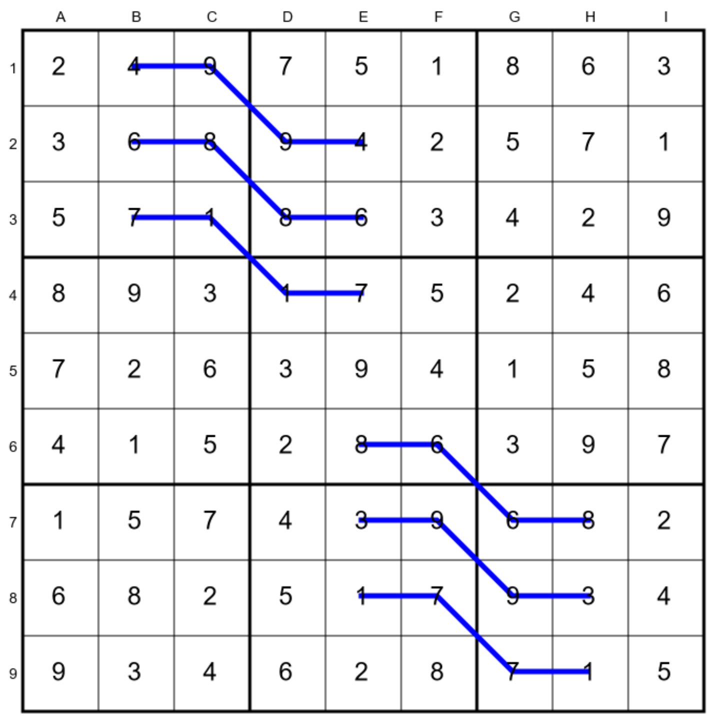
Model description not finished

9.30.6 See Also

9.30.7 palindrome example gas73: "Spike Island" by Bill Murphy

Video: `https://youtu.be/DOHwYUHsCKI?si=f-1LYySGvj6DQe-E`





Rules: standard (Section 3.13) palindrome (Section 9.30)

9.31 Constraint regionSumLine

9.31.1 Category: Line

9.31.2 Description

The line is split into segments by the block borders. The numbers on each segment add up to the same total. Different lines can have different sums.

9.31.3 Arguments

regionSumLine List(Cell)

9.31.4 JSON Format

```
"lines":[{"regionSumLine":[14,15,16,26,27]},...],
```

9.31.5 Model

Let $[x_1, x_2, ..., x_k]$ be the set variables corresponding to the given list of cells. The line is split into regions $[R_1, ..., R_j]$ by the borders of the blocks.

The values of the cells in each region sum up to the same value s, which needs to be determined. The first point on the line, variable x_1, belongs to the first region R_1, the last point x_k belongs to the last region R_j. Two consecutive variables x_i and x_{i+1} belong to the same region R_j if they belong to the same block B ($\{i, j\} \subseteq B$), otherwise they belong to regions R_j and R_{j+1}. Note that all variables in the same region belong to the same block, they are therefore pairwise different.

$$\exists_{s \in N} \forall_{R \in [R_1,...,R_j]} : \quad s = \sum_{i \in R} x_i \tag{25}$$

If a region only contains a single cell, then its value must be equal to s.

$$\forall_{j \in 1..k} \forall_{R \in [R_1,...,R_j] \text{ s.t. } |R|=1 \wedge x_j \in R} : \quad x_j = s \tag{26}$$

We can also express the sum in (25) using table constraints:

$$\forall_{R \in [R_1,...,R_j] \text{ s.t. } \{x_j ...,x_{j+|R|}\} \subseteq R} : \quad \text{table}([x_j, ..., x_{j+|R|}, s], \text{sums}_{|R|}) \tag{27}$$

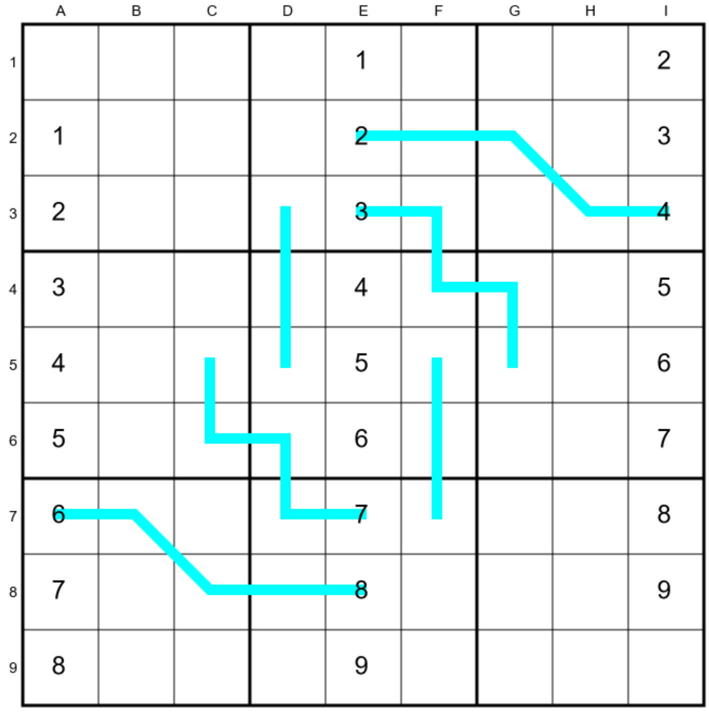
using the same sums_k table as for the cage constraint, Equation (42).

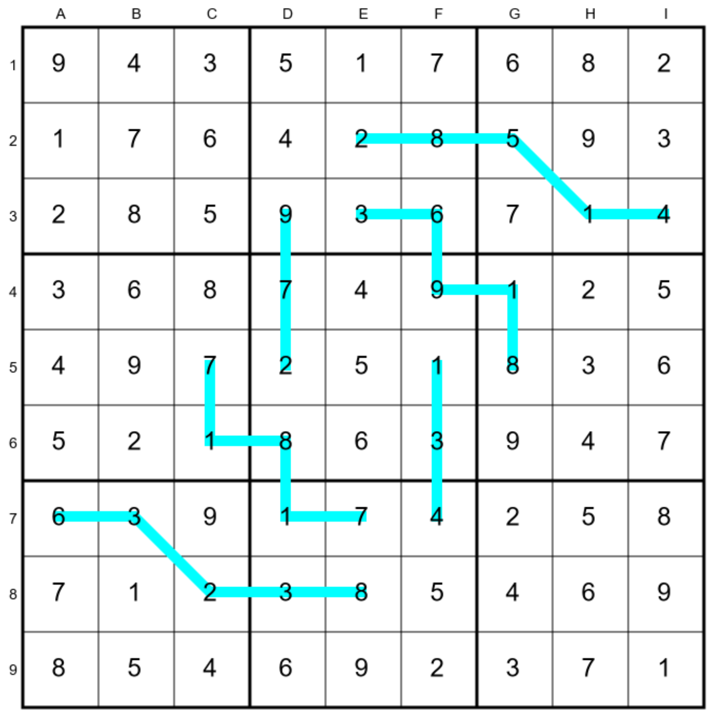
9.31.6 See Also

increasingRegionSumLine (Section 9.22)

9.31.7 regionSumLine example gas68: "The Waiting" by Philip Newman

Video: `https://www.youtube.com/watch?v=Q1ZvMtd21Pk`





Rules: standard (Section 3.13) regionSumLine (Section 9.31)

9.32 Constraint renban

9.32.1 Category: Line

9.32.2 Description

The values on the line of length n must consist of n consecutive integers, but they can appear in any order. The values <3, 5, 4, 2, 6> for example form a valid renban line of length five.

9.32.3 Arguments

renban List(Cell)

9.32.4 JSON Format

```
"lines":[{"renban":[4,13,22]},...],
```

9.32.5 Model

Let $[x_1, x_2, ..., x_k]$ be the set variables corresponding to the given list of cells. The values must form a consecutive set of integers, not necessarily in order. This can be expressed by a global constraint

$$\text{alldifferent\_consecutive\_values}([x_1, x_2, ..., x_k]) \tag{28}$$

If that constraint is not available, a naive decomposition can be achieved by

$$\text{alldifferent}([x_1, x_2, ..., x_k]) \tag{29}$$

$$\min([x_1, x_2, ..., x_k], \min) \tag{30}$$

$$\max([x_1, x_2, ..., x_k], \max) \tag{31}$$

$$\max - \min = k - 1 \tag{32}$$

We can achieve better propagation by using a table constraint

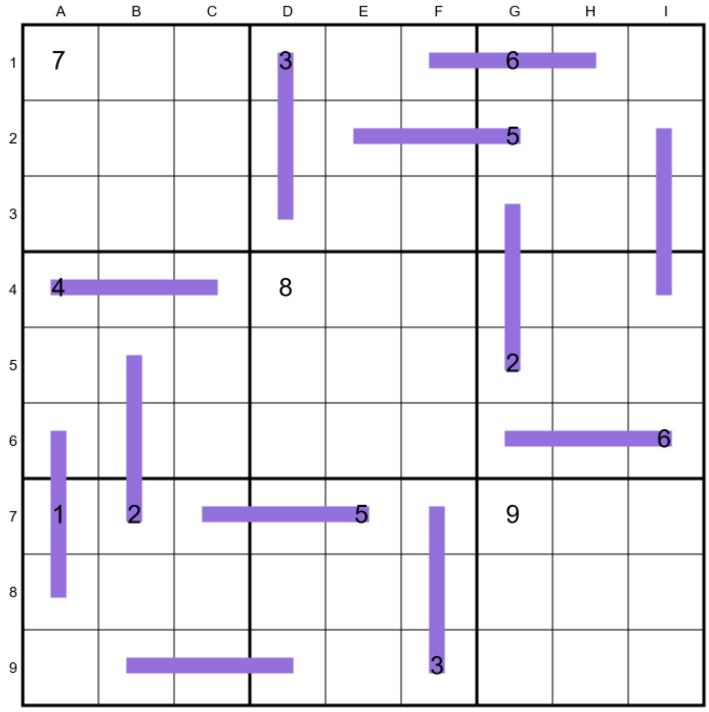
$$\text{table}([x_1, x_2, ..., x_k], \text{renban}_k) \tag{33}$$

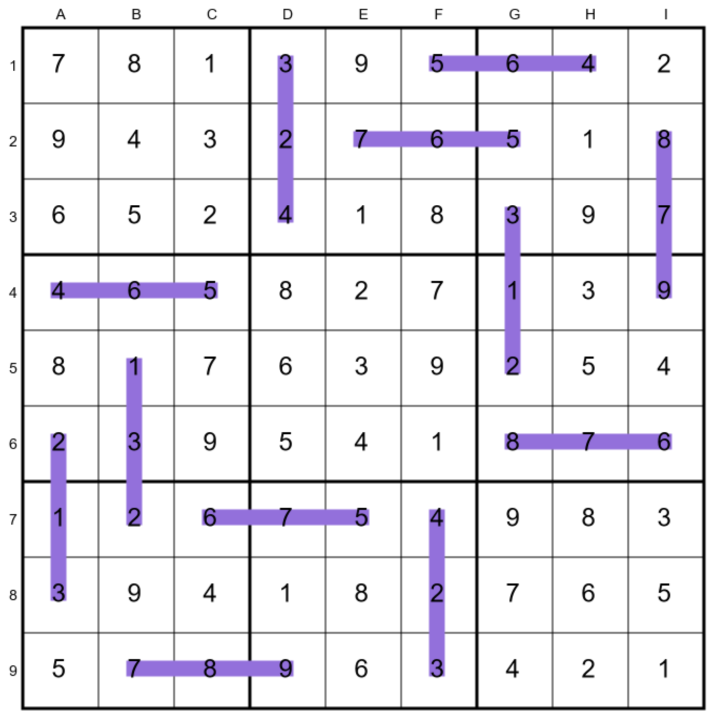
with tuples renban_k enumerating all tuples which satisfy the constraints (29)-(32).

9.32.6 See Also

9.32.7 renban example timotab89: "Eight Nine, Seven Did" by Philip Newman

Video: `https://youtu.be/kHFam4o8iN8?si=_WgcFwqbVKPOgcdH` (12:53)





Rules: standard (Section 3.13) renban (Section 9.32)

9.33 Constraint slowThermo

9.33.1 Category: Line

9.33.2 Description

The values on the line must be (not necessarily strictly) increasing, starting from the lowest value in the circle.

9.33.3 Arguments

slowThermo List(Cell)

9.33.4 JSON Format

```
"lines":[{"slowThermo":[35,44,53,61,69,68,67,57,47,38,29,21,13,14,15,25]},...],
```

9.33.5 Model

Let $[x_1, x_2, ..., x_k]$ be the set variables corresponding to the given list of cells. The values must be increasing, starting from the bulb, which represents variable x_1. This can be expressed with a global constraint

$$\text{increasing}([x_1, x_2, ..., x_k]) \tag{34}$$

It can also be expressed by a conjunction of binary constraints, i.e.

$$\forall_{i \in 1..k-1} : \quad x_{i+1} \geq x_i \tag{35}$$

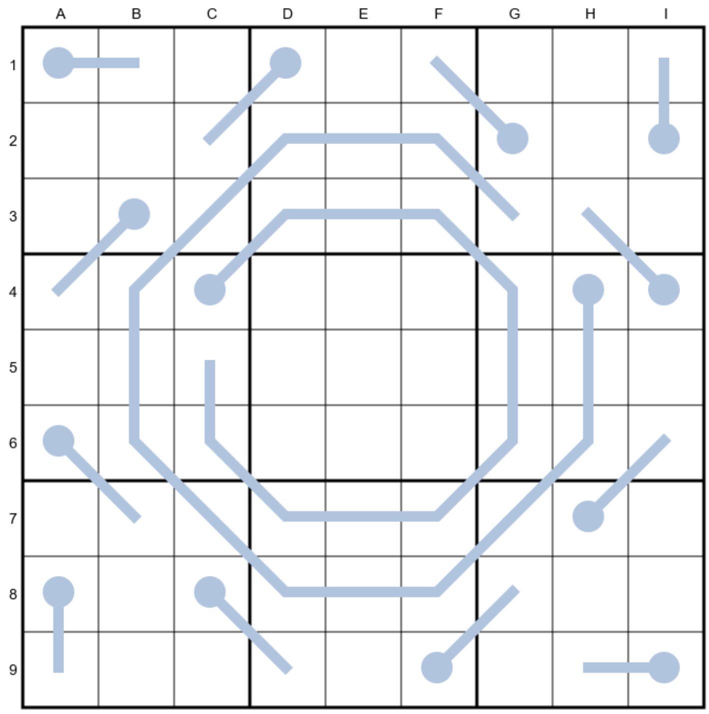
The difference to the thermo constraint (Section 9.38) is that here consecutive variables on the line can be equal, if other problem constraint allow that. We can improve propagation by considering which variables are in the same row block, or column, and can therefore not be equal. This leads to redundant constraints

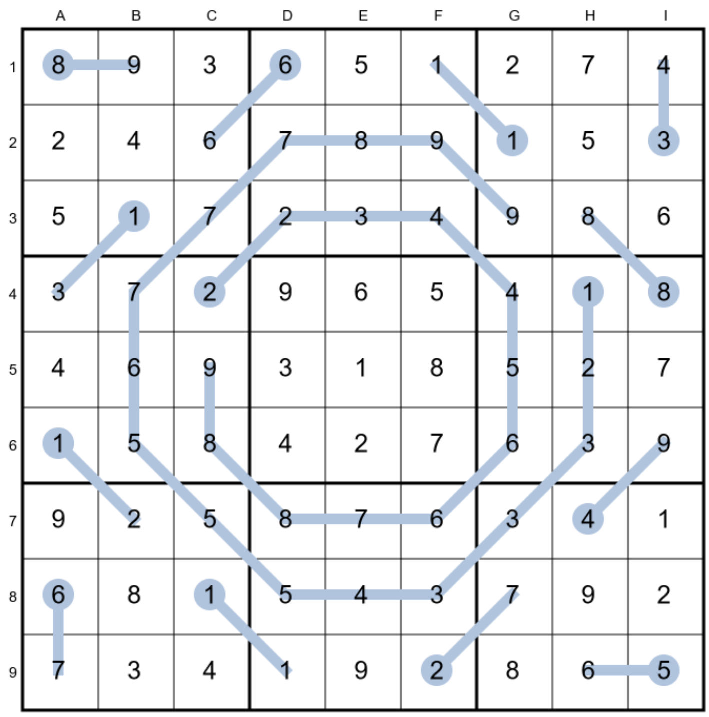
$$x_j > x_i \text{if} j > i \wedge \exists_{H \in Houses} s.t. c_i \in H, c_j \in H \tag{36}$$

9.33.6 See Also

9.33.7 slowThermo example timotab17: "Slow Solvers Playing" by clover

Video: `https://www.youtube.com/watch?v=r4ZzqkfDOCE`
Video2: `https://www.youtube.com/watch?v=z3H8kLBwvFo`





Rules: standard (Section 3.13) slowThermo (Section 9.33)

9.34 Constraint splitPeaLine

9.34.1 Category: Line

9.34.2 Description

The sum of the values between the two circles must add up to a two-digit number made up from the digits on the circled end points of the line. The 'tens' digit can be on either end. Digits can repeat on the line if allowed by other rules.

9.34.3 Arguments

splitPeaLine List(Cell)

9.34.4 JSON Format

```
"lines":[{"splitPeaLine":[3,4,5,6,7,16]},...],
```

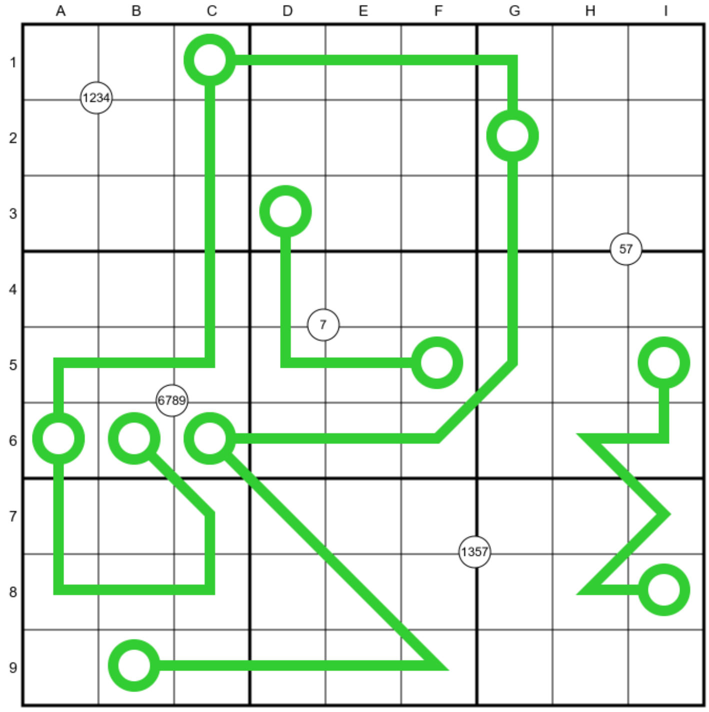
9.34.5 Model

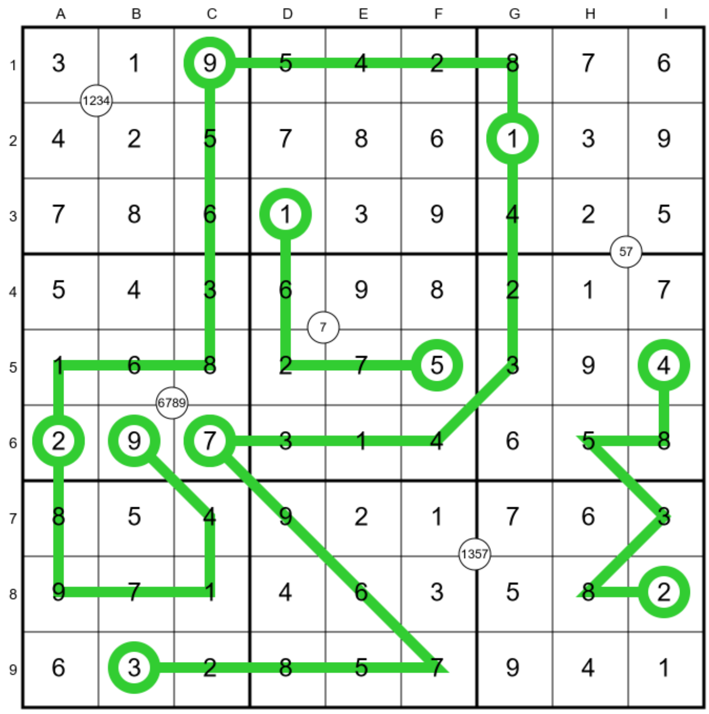
Model description not finished

9.34.6 See Also

9.34.7 splitPeaLine example BremSter52: "PEArates of the QUADribbean" by Calvinjoy

Video: `https://www.youtube.com/watch?v=HOE3E_K_6cs`





Rules: standard (Section 3.13) quad (Section 8.8) splitPeaLine (Section 9.34)

9.35 Constraint sumSquareMaxLine

9.35.1 Category: Line

9.35.2 Description

Digits on the line sum to the square of the largest digit on the line.

9.35.3 Arguments

sumSquareMaxLine List(Cell)

9.35.4 JSON Format

```
"lines":[{"sumSquareMaxLine":[4,3,2,11,10,19,28]},...],
```

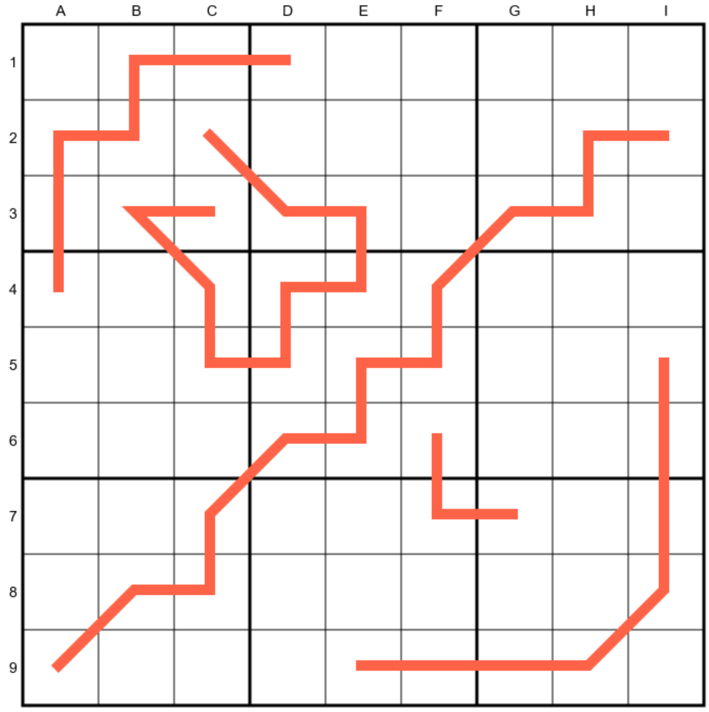
9.35.5 Model

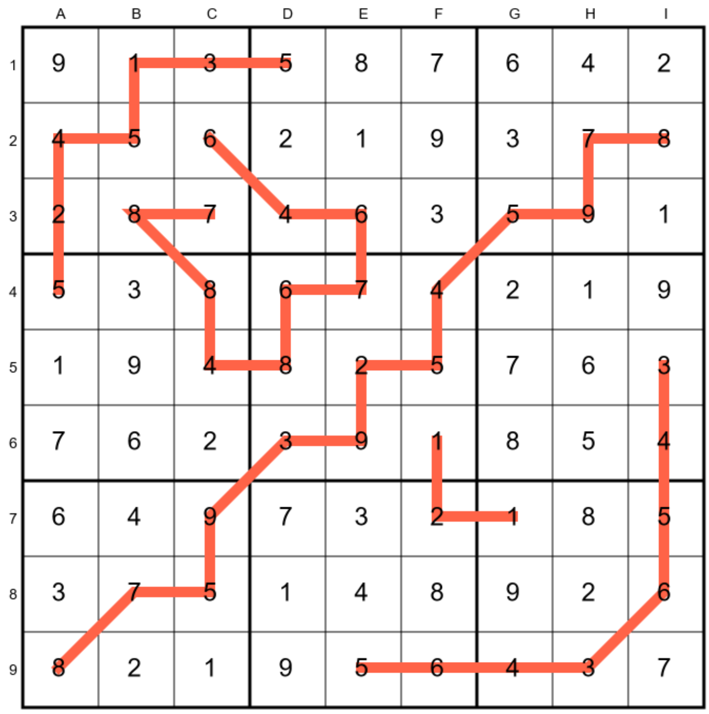
Model description not finished

9.35.6 See Also

9.35.7 sumSquareMaxLine example ctc85: "Most Squares" by henter

Video: `https://www.youtube.com/watch?v=_Rf4tFPFPqU`





Rules: standard (Section 3.13) sumSquareMaxLine (Section 9.35)

9.36 Constraint tenLine

9.36.1 Category: Line

9.36.2 Description

The line can be partitioned into non-overlapping segments which each add up to ten. Values may repeat on the line, if allowed by other Sudoku rules.

9.36.3 Arguments

tenLine List(Cell)

9.36.4 JSON Format

```
"lines":[{"tenLine":[3,2,1,10,19]},...],
```

9.36.5 Model

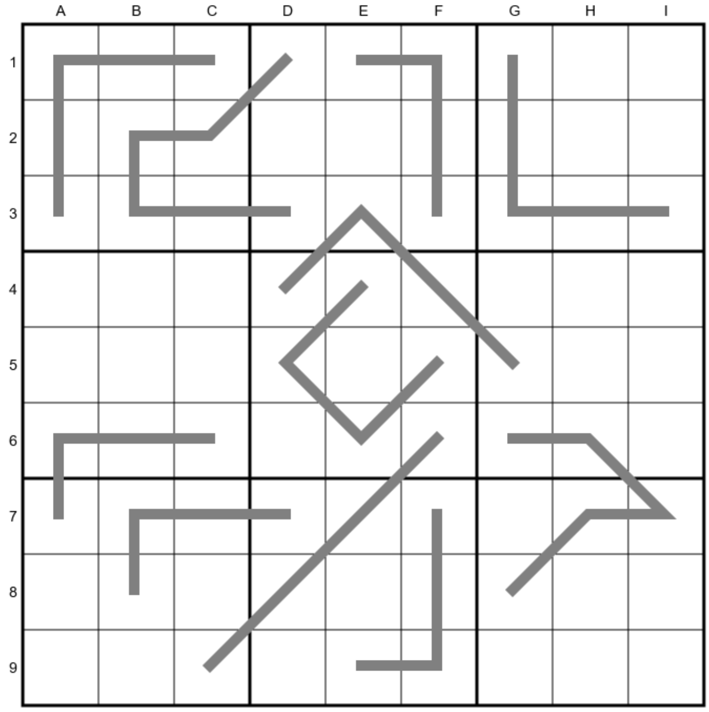
Model description not finished

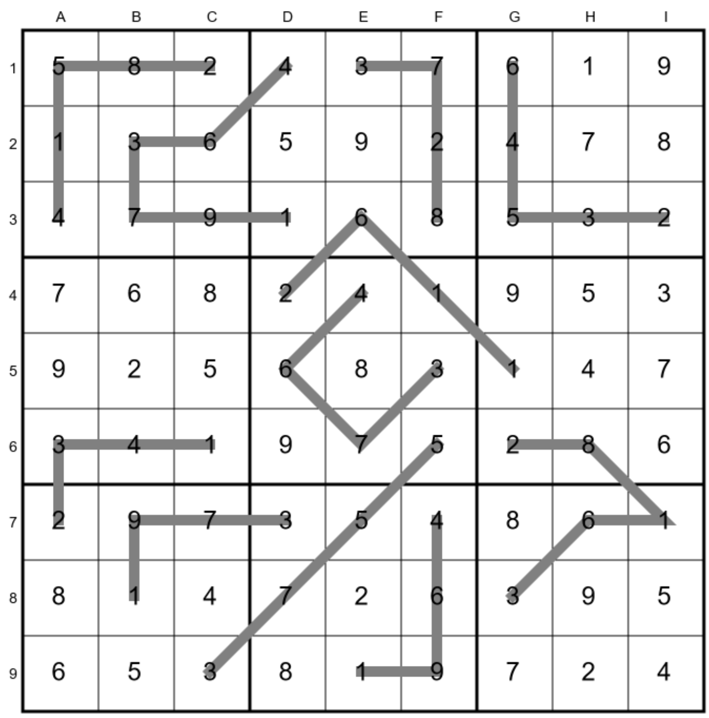
9.36.6 See Also

eightLine (Section 9.17)

9.36.7 tenLine example sudokupad1: "10 Lines" by zetamath

Video: `sudokupad.app/DMp47hdP2D`





Rules: standard (Section 3.13) tenLine (Section 9.36)

9.37 Constraint tenOrElevenLine

9.37.1 Category: Line

9.37.2 Description

Any two consecutive digits on the line add to 10 or 11.

9.37.3 Arguments

tenOrElevenLine List(Cell)

9.37.4 JSON Format

```
"lines":[{"tenOrElevenLine":[5,4,13,21,20,11]},...],
```

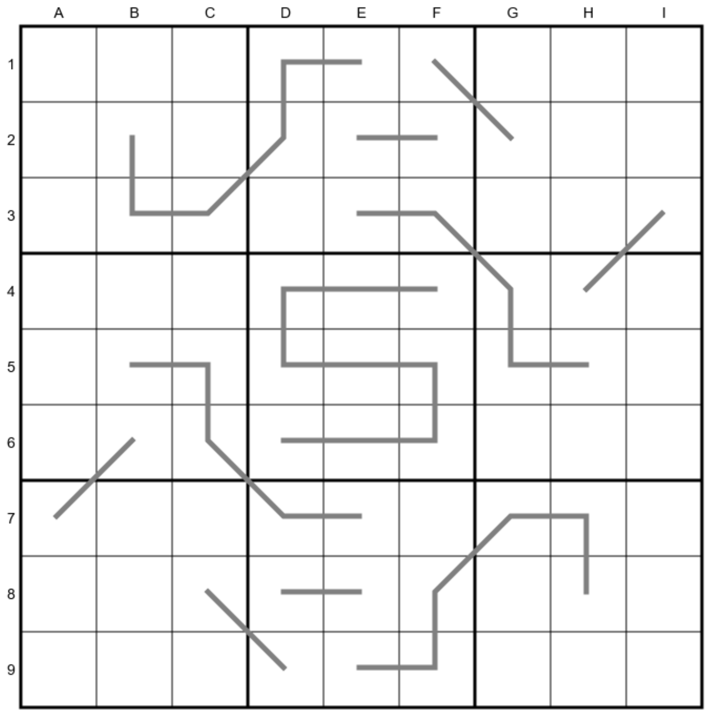
9.37.5 Model

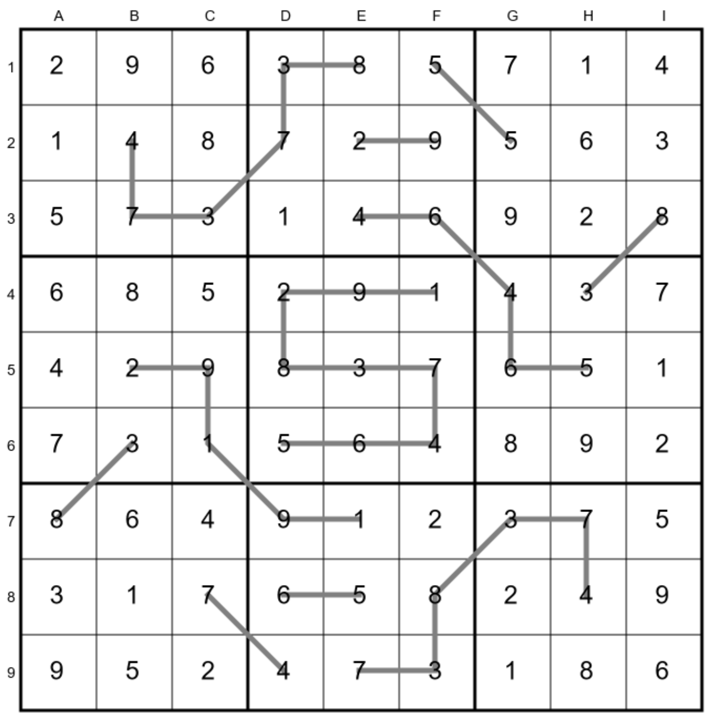
Model description not finished

9.37.6 See Also

9.37.7 tenOrElevenLine example BremSter88: "Dutch Flat Mates - Ten or Eleven" by Marty Sears

Video: `https://www.youtube.com/watch?v=bI4T6s-xxns`





Rules: standard (Section 3.13) dutchFlatMates (Section 4.6) tenOrElevenLine (Section 9.37)

9.38 Constraint thermo

9.38.1 Category: Line

9.38.2 Description

The values on the line must be strictly increasing, starting with the lowest value in the circle. The sequence <2, 4, 6, 7> would be a legal thermo line.

9.38.3 Arguments

thermo List(Cell)

9.38.4 JSON Format

```
"lines":[{"thermo":[2,3,4,5,6,7,8,9]},...],
```

9.38.5 Model

Let $[x_1, x_2, ..., x_k]$ be the set variables corresponding to the given list of cells. The values must be strictly increasing, starting from the bulb, which represents variable x_1. This can be expressed with a global constraint

$$\text{strictly\_increasing}([x_1, x_2, ..., x_k]) \tag{37}$$

It can also be expressed by a conjunction of binary constraints, i.e.

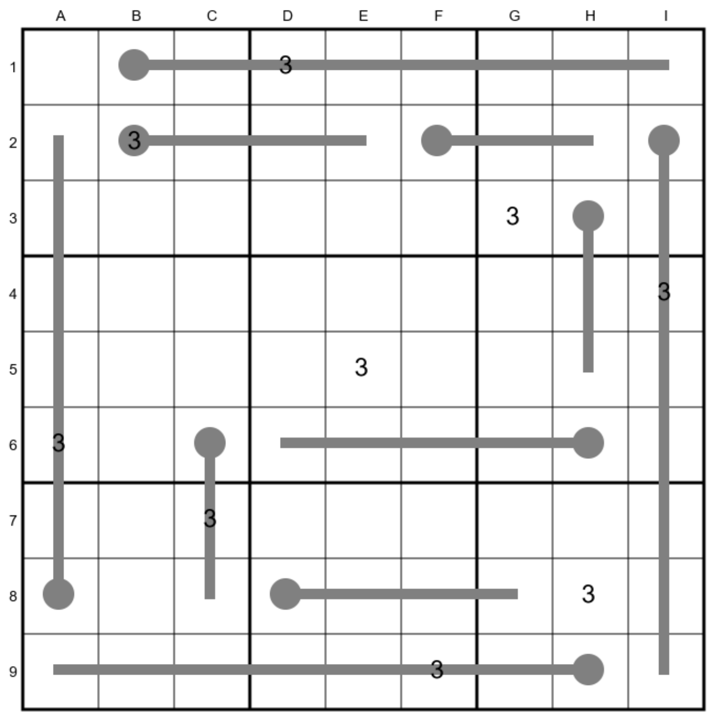
$$\forall_{i \in 1..k-1} : \quad x_{i+1} > x_i \tag{38}$$

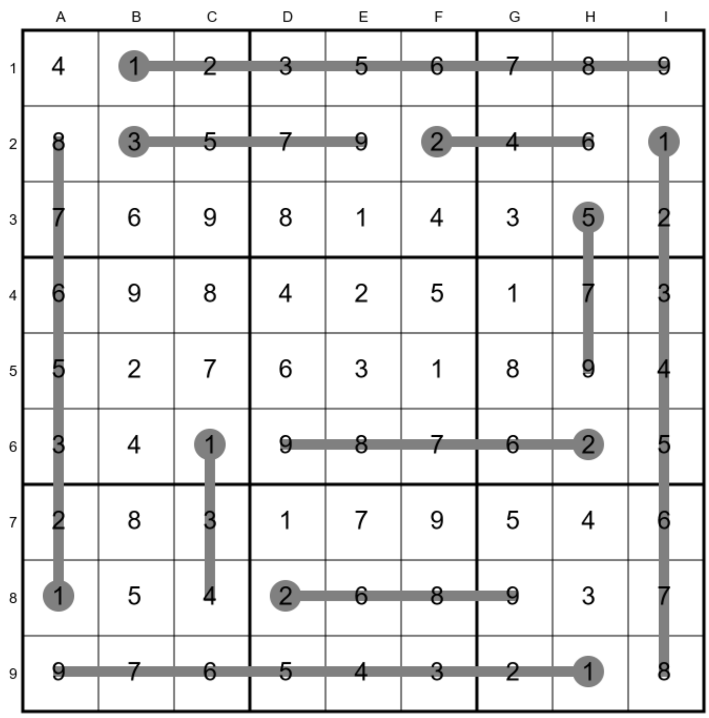
9.38.6 See Also

slowThermo (Section 9.33)

9.38.7 thermo example timotab61: "Free Threes" by Philip Newman

Video: `https://www.youtube.com/watch?v=6sAPiXW4gVg`





Rules: standard (Section 3.13) thermo (Section 9.38)

9.39 Constraint tinder

9.39.1 Category: Line

9.39.2 Description

The line contains exactly one repeated digit, all other entries occur only once.

9.39.3 Arguments

tinder List(Cell)

9.39.4 JSON Format

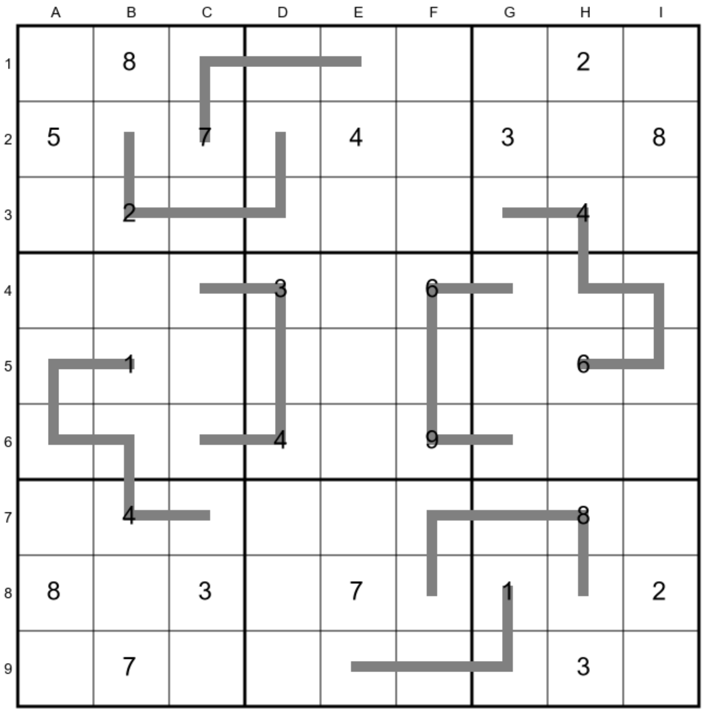
```
"lines":[{"tinder":[5,4,3,12]},...],
```

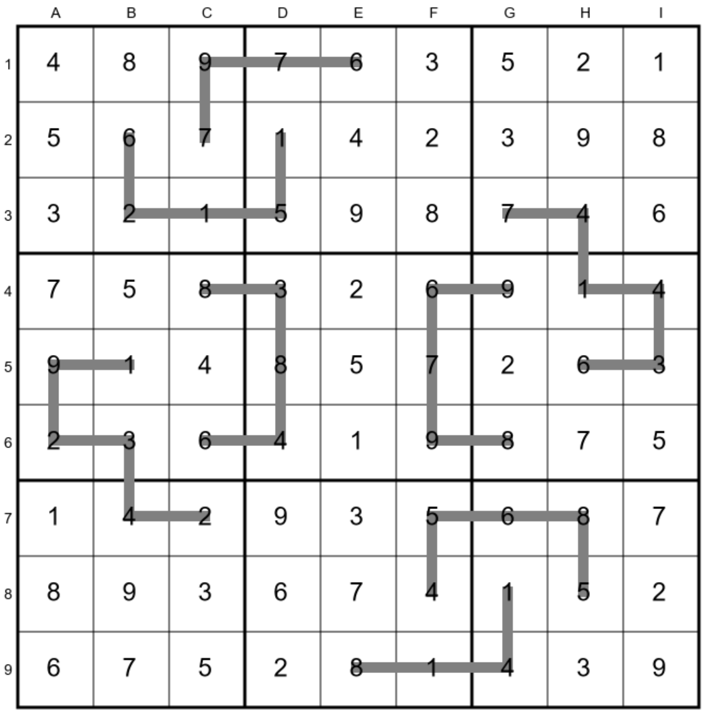
9.39.5 Model

Model description not finished

9.39.6 See Also

9.39.7 tinder example wpf gp2025r5c15: "Sudoku GP 2025 Round 5" by





Rules: standard (Section 3.13) tinder (Section 9.39)

9.40 Constraint trio24

9.40.1 Category: Line

9.40.2 Description

the three digits in the triple must can be written as an arithmetic expression with two operators from +,*,-,/ which evaluate to 24. An example is 3*5+9=24. The division must be exact, i.e. without remainder. Digits cannot repeat in the triple.

9.40.3 Arguments

trio24 List(Cell)

9.40.4 JSON Format

```
"lines":[{"trio24":[4,13,22]},...],
```

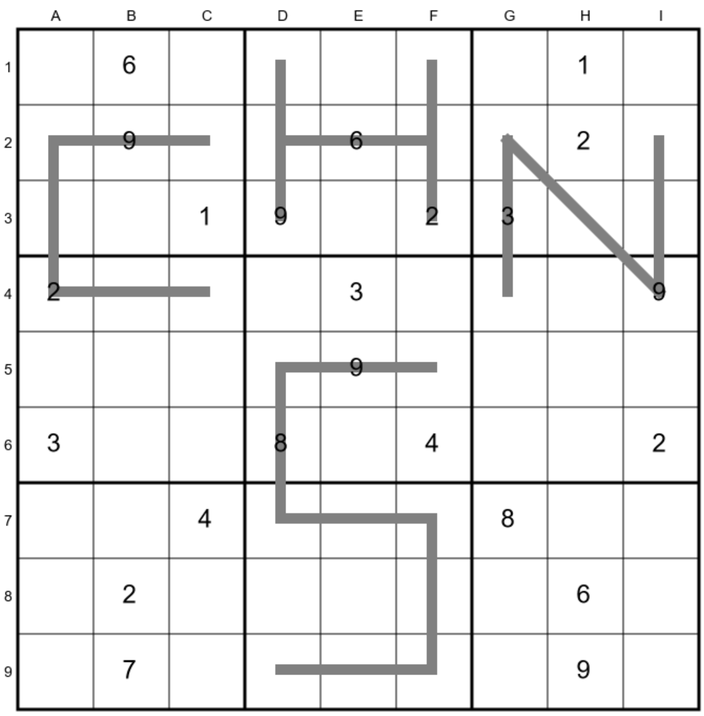
9.40.5 Model

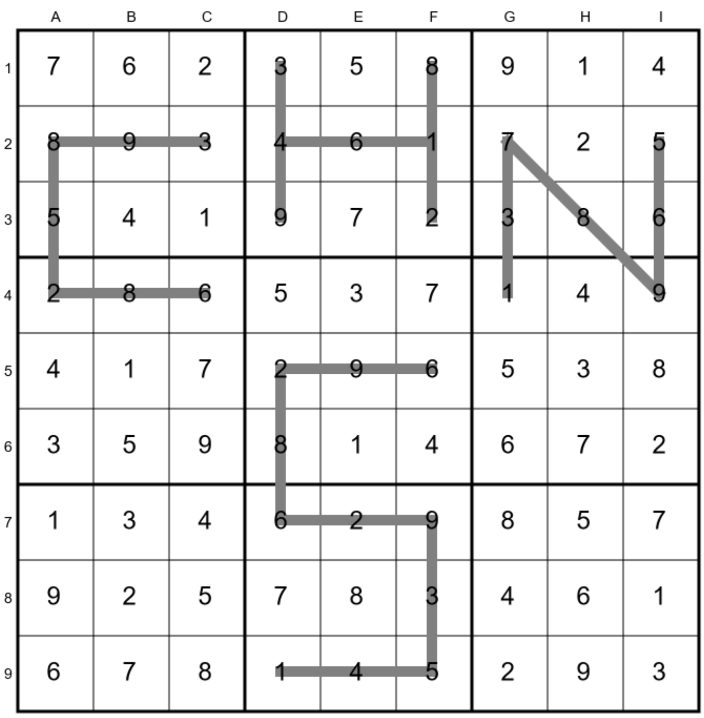
Model description not finished

9.40.6 See Also

9.40.7 trio24 example BremSter12: "wfp GP 2016 round 5" by

Video: `https://www.youtube.com/watch?v=r8FIah7nE_4`





Rules: standard (Section 3.13) trio24 (Section 9.40)

9.41 Constraint zipLine

9.41.1 Category: Line

9.41.2 Description

The line has an odd number of cells. Points the same distance from the centre of the line sum up to the same value, which is also the value of the centre point.

9.41.3 Arguments

zipLine List(Cell)

9.41.4 JSON Format

```
"lines":[{"zipLine":[7,16,25]},...],
```

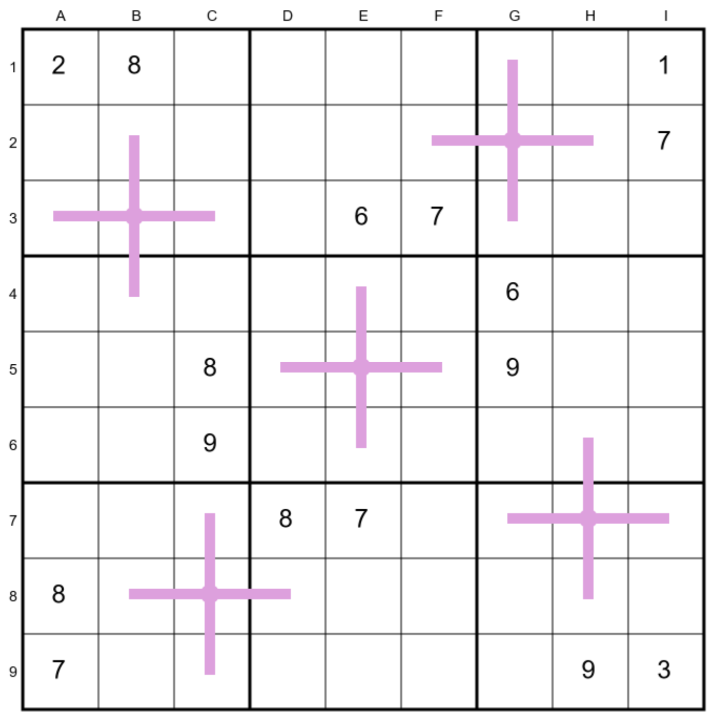
9.41.5 Model

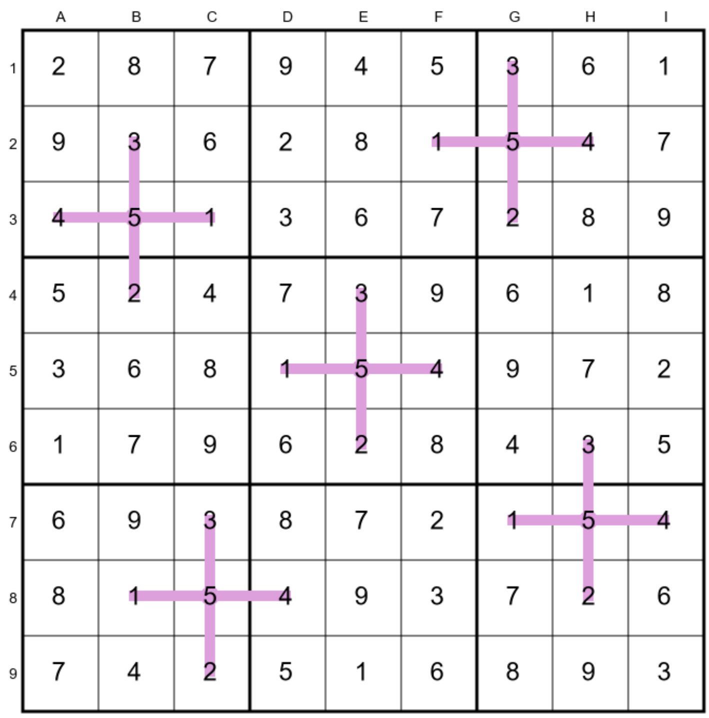
Model description not finished

9.41.6 See Also

9.41.7 zipLine example timotab37: "Going, Going, Gone" by Bill Murphy

Video: `https://www.youtube.com/watch?v=gcfm76vRUIA`





Rules: standard (Section 3.13) zipLine (Section 9.41)

10 MultiLine Constraints

10.1 Constraint multiArrows

10.1.1 Category: MultiLine

10.1.2 Description

The number in the bulb indicates how many unique digits are on all arrows originating from the bulb combined. The value in the bulb is not considered in the count.

10.1.3 Arguments

multiArrows Integer

lines List(List(Cell))

10.1.4 JSON Format

```
"multiLines":[{"multiArrows":2,"lines":[[3,4,5],[11,20,29]]},...],
```

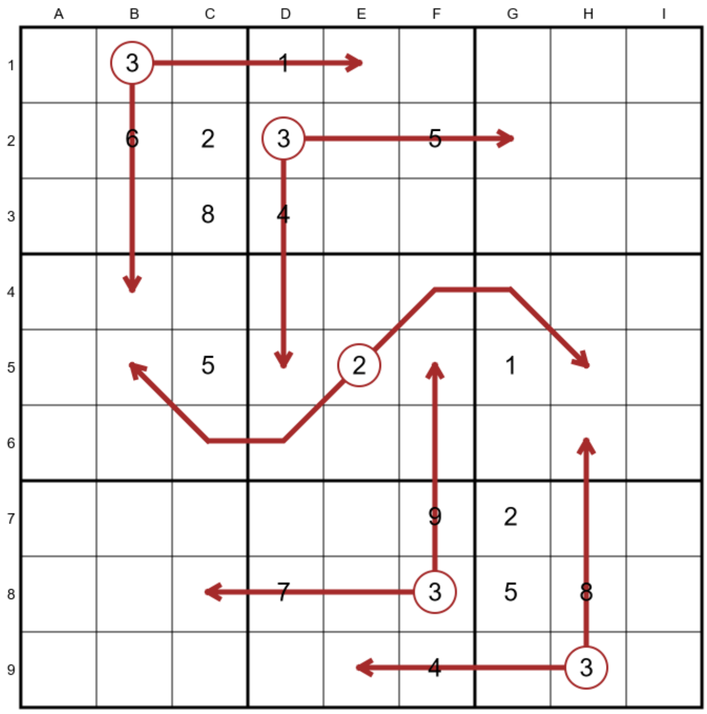
10.1.5 Model

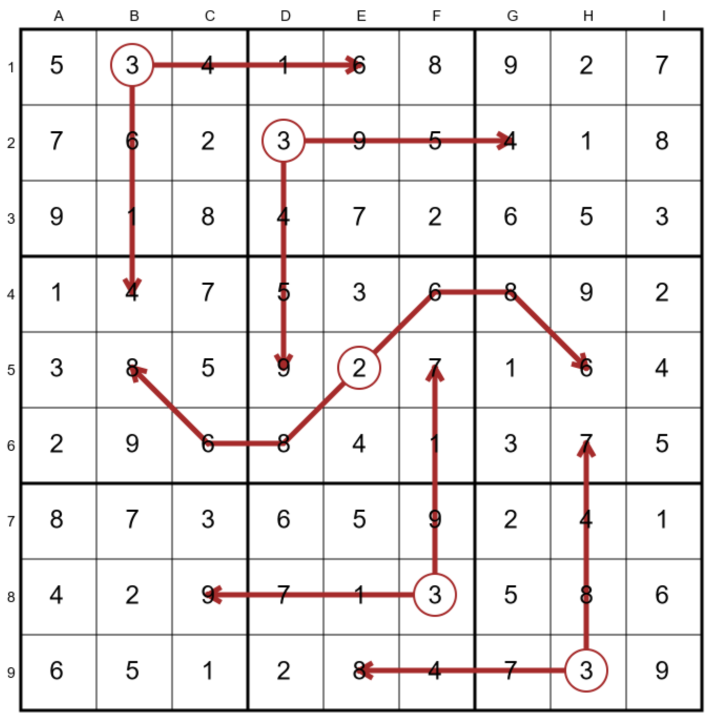
Model description not finished

10.1.6 See Also

10.1.7 multiArrows example gas6: "Count Different (Multiple Arrows)" by Bill Murphy

Video: `https://www.youtube.com/watch?v=6APLg5Wz9oI&t=13s`
Video2: `https://www.youtube.com/watch?v=XRgUp7fnM8k`





Rules: standard (Section 3.13) multiArrows (Section 10.1)

10.2 Constraint sumLineAlldifferent

10.2.1 Category: MultiLine

10.2.2 Description

The sums of the cell in all lines are alldifferent.

10.2.3 Arguments

sumLineAlldifferent List(List(Cell))

10.2.4 JSON Format

```
"sumLineAlldifferent":[[1,2,3],[4,5,6],[7,8,9],...],
```

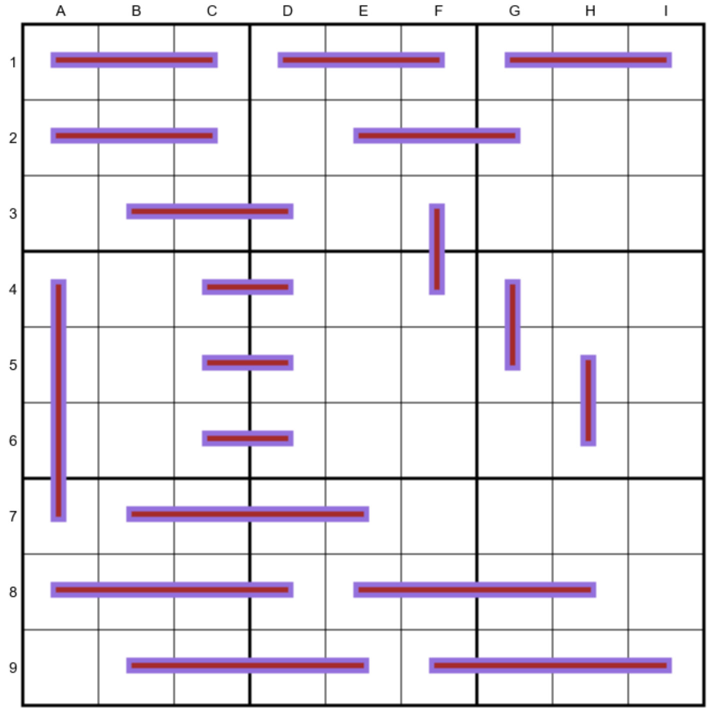
10.2.5 Model

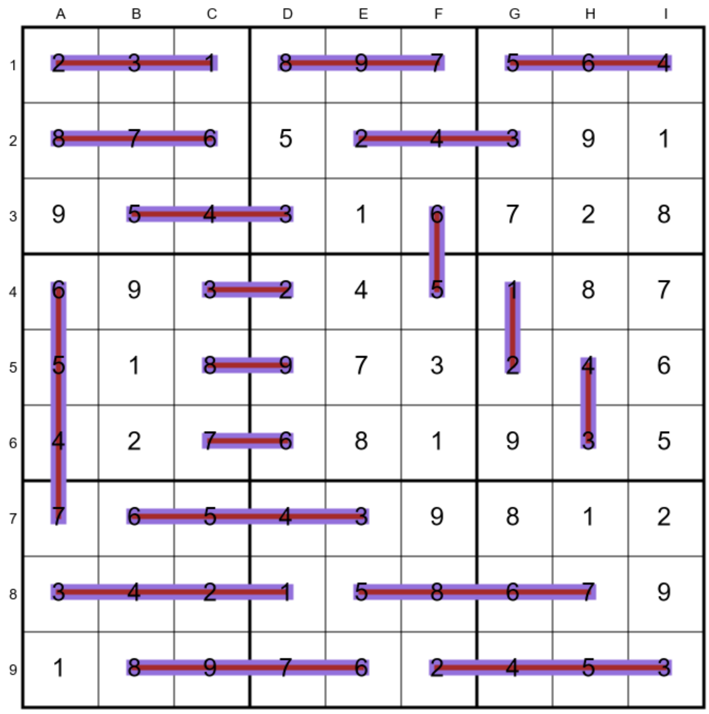
Model description not finished

10.2.6 See Also

10.2.7 sumLineAlldifferent example ctc91: "" by Kaktuslav

Video: `https://www.youtube.com/watch?v=XQzSE16SMfE`





Rules: standard (Section 3.13) renban (Section 9.32) sumLineAlldifferent (Section 10.2)

11 Area Constraints

11.1 Constraint allOddOrEven

11.1.1 Category: Area

11.1.2 Description

The cells in the area must either be all odd or all even.

11.1.3 Arguments

allOddOrEven List(Cell)

11.1.4 JSON Format

```
"areas":[{"allOddOrEven":[11,19,20]},...]
```

11.1.5 Model

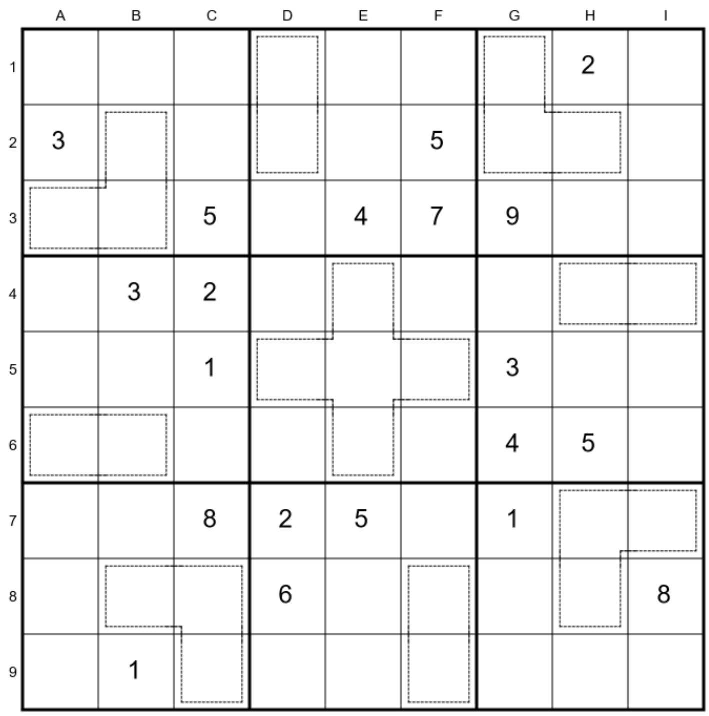
Model description not finished

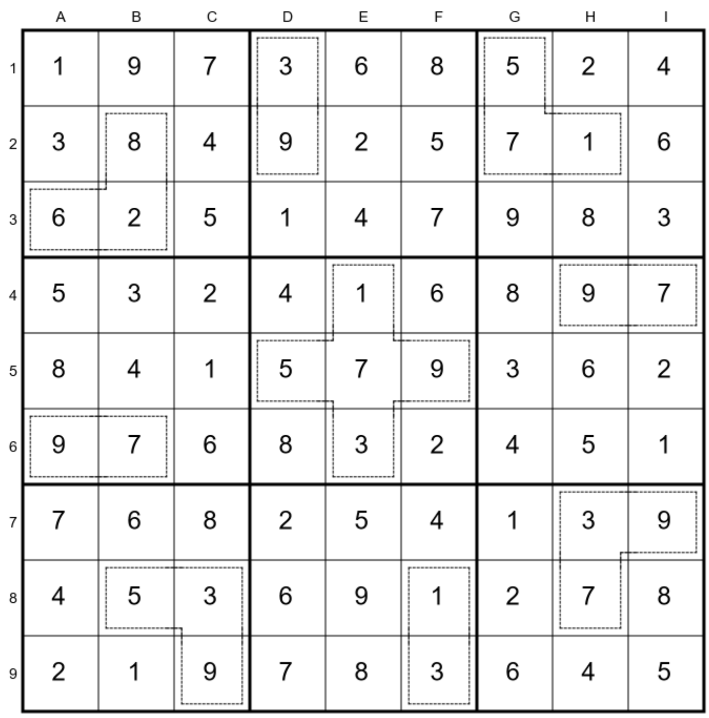
11.1.6 See Also

odd (Section 5.9) , even (Section 5.5)

11.1.7 allOddOrEven example timotab11: "Never Odd Or Even" by clover

Video: `https://www.youtube.com/watch?v=ZSPAvK-Jky8`





Rules: standard (Section 3.13) allOddOrEven (Section 11.1)

11.2 Constraint antiFortress

11.2.1 Category: Area

11.2.2 Description

Each value in an antiFortress must be smaller than the value on any orthogonal neighbours outside the antiFortress.

11.2.3 Arguments

antiFortress List(Cell)

11.2.4 JSON Format

```
"areas":[{"antiFortress":[15,16,17,24,25,26,33,34,35]},...],
```

11.2.5 Model

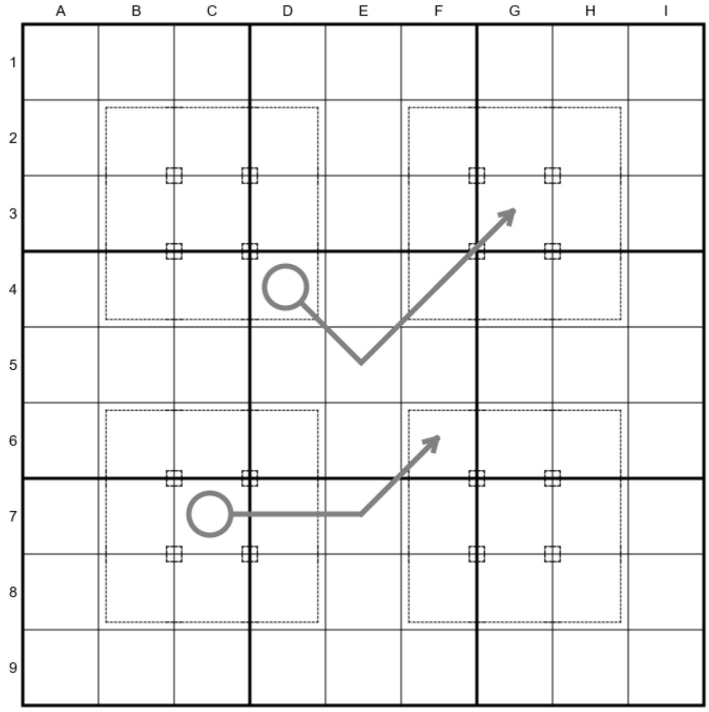
Model description not finished

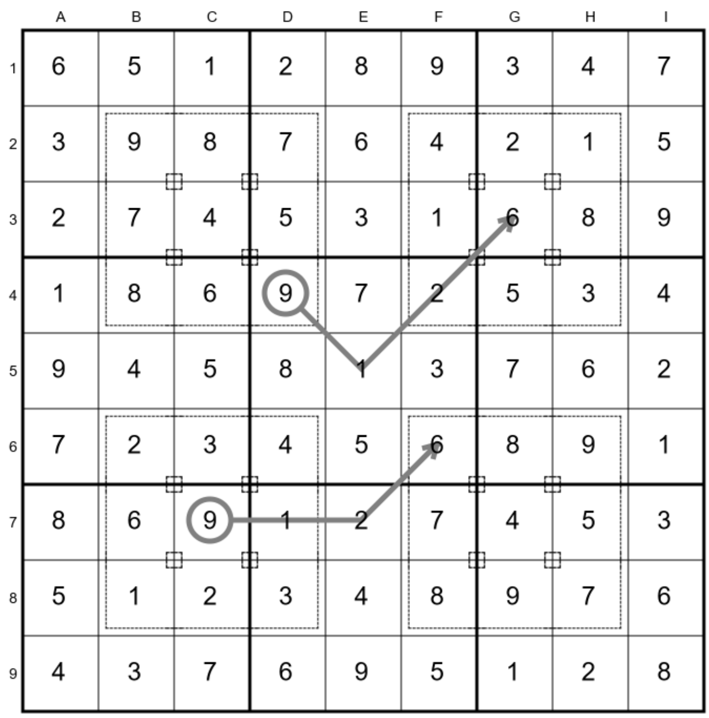
11.2.6 See Also

fortress (Section 11.10)

11.2.7 antiFortress example ctc80: "King's Siege" by Sniglett

Video: `https://www.youtube.com/watch?v=ZwaxrFBHjPk&t=673s`





Rules: standard (Section 3.13) fortress (Section 11.10) antiFortress (Section 11.2) arrow (Section 9.2) antiKing (Section 4.1)

11.3 Constraint box

11.3.1 Category: Area

11.3.2 Description

All values inside the box must be different from each other. The box is indicated by its outline.

11.3.3 Arguments

box List(Cell)

11.3.4 JSON Format

```
"areas":[{"box":[12,13,14,21,22,23,30,31,32]},...],
```

11.3.5 Model

Let $[x_1, x_2, ..., x_k]$ be the set variables corresponding to the given list of cells. The values of the variables must be pairwise different

$$\text{alldifferent}([x_1, x_2, ..., x_k]) \tag{39}$$

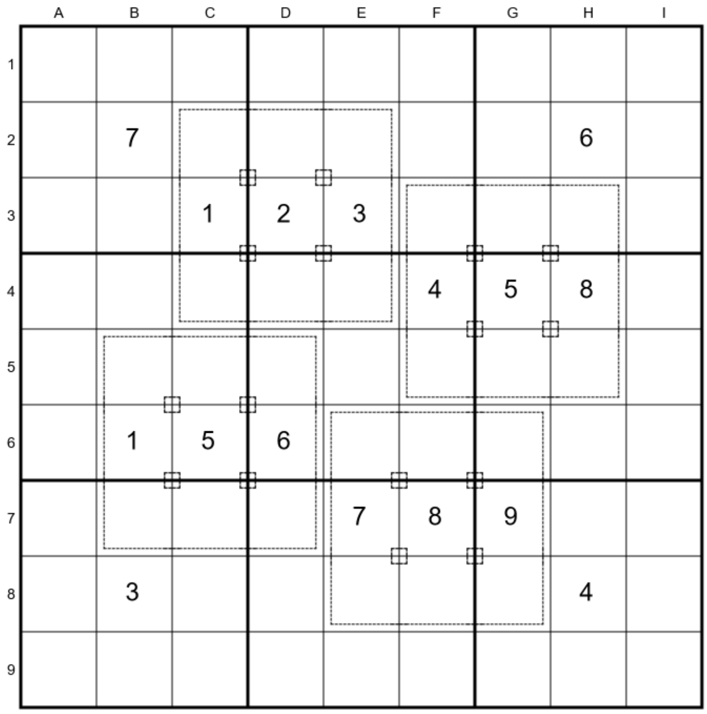
A box constraint specifies an additional alldifferent constraint on the grid cells. That constraint can interact with other alldifferent constraints, which may allow additional reasoning.

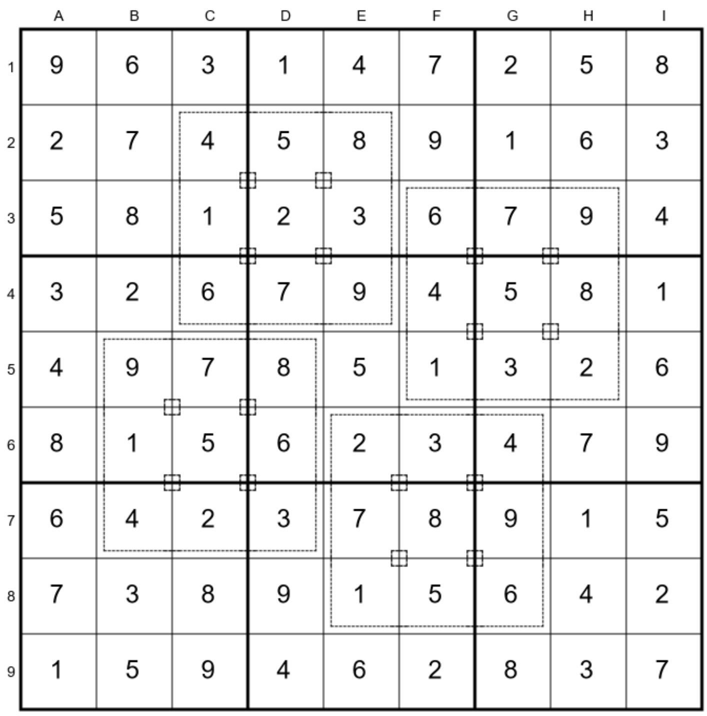
11.3.6 See Also

cage (Section 11.4)

11.3.7 box example gas7: "This Is Hardcore" by Bill Murphy

Video: `https://www.youtube.com/watch?v=XpIceI-W21Q`





Rules: standard (Section 3.13) box (Section 11.3)

11.4 Constraint cage

11.4.1 Category: Area

11.4.2 Description

The values inside the cage must add to the given total. All values in the cage must be different from each other.

11.4.3 Arguments

cage List(Cell)

value Integer

11.4.4 JSON Format

"areas":[{"cage":[2,3,4,5,6,7,8],"value":42},...],

11.4.5 Model

Let $[x_1, x_2, ..., x_k]$ be the set variables corresponding to the given list of cells, let v be the given value for the cage. The values of the variables must be pairwise different

$$\text{alldifferent}([x_1, x_2, ..., x_k]) \tag{40}$$

The sum of all cells must be equal to the given value:

$$v = \sum_{i=1}^{k} x_i \tag{41}$$

We achieve better propagation by creating a table constraint

$$\text{table}([x_1, x_2, ..., x_k, v], \text{sums}_k) \tag{42}$$

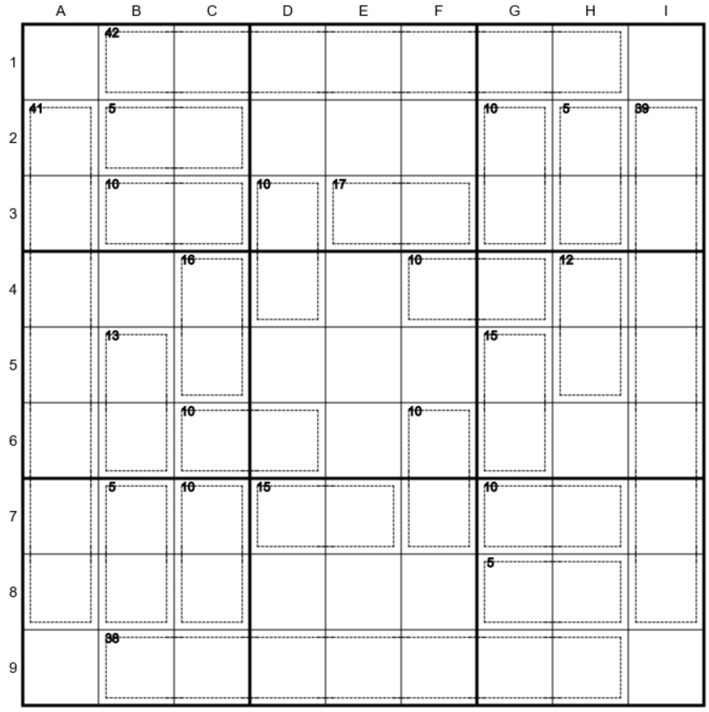
with tuples sums_k enumerating all combinations of pairwise different values for x_i that sum to v.

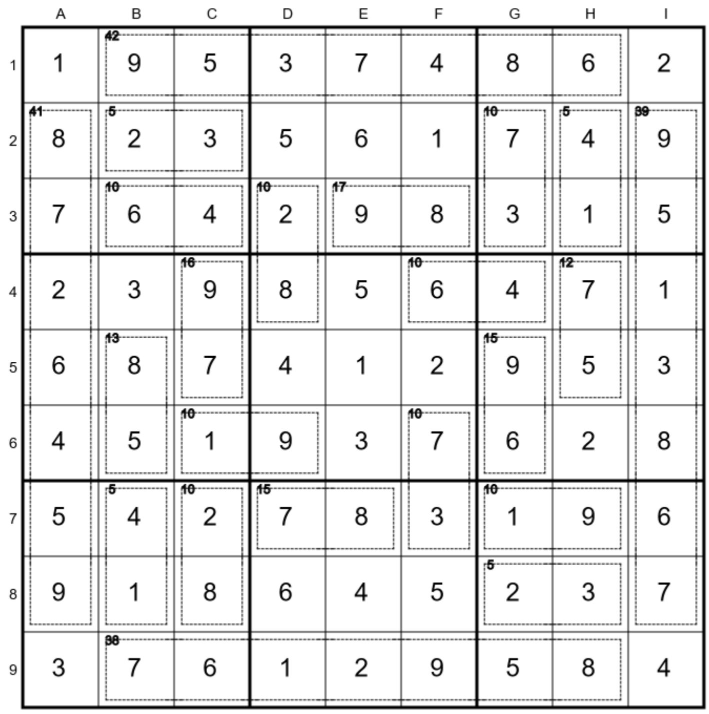
11.4.6 See Also

box (Section 11.3)

11.4.7 cage example timotab16: "Killer Vibes" by clover

Video: `https://www.youtube.com/watch?v=6lMYaWjE1go`





Rules: standard (Section 3.13) cage (Section 11.4)

11.5 Constraint chain19

11.5.1 Category: Area

11.5.2 Description

The digits in the 3x3 block must form a chain from 1 to 9, where two consecutive values are orthogonally connected.

11.5.3 Arguments

chain19 List(Cell)

11.5.4 JSON Format

```
"areas":[{"chain19":[1,2,3,10,11,12,19,20,21]},...],
```

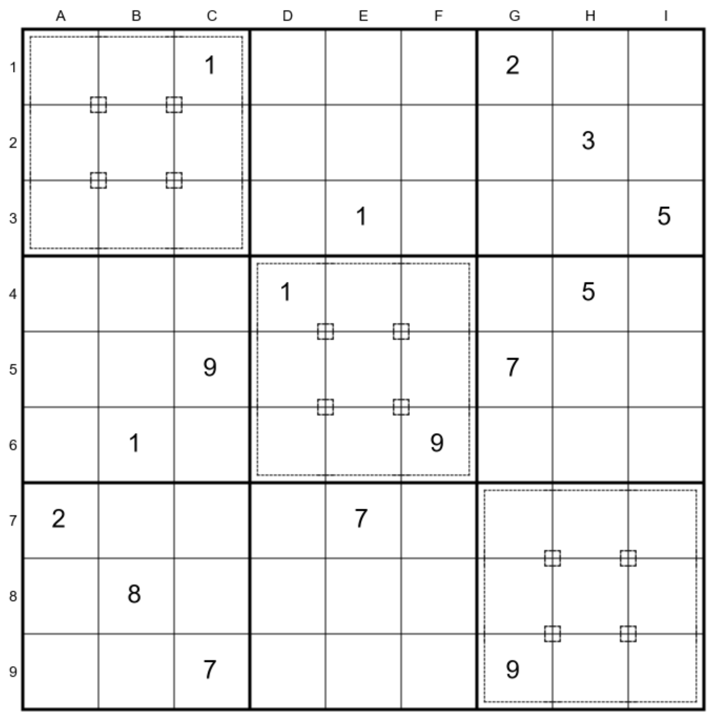
11.5.5 Model

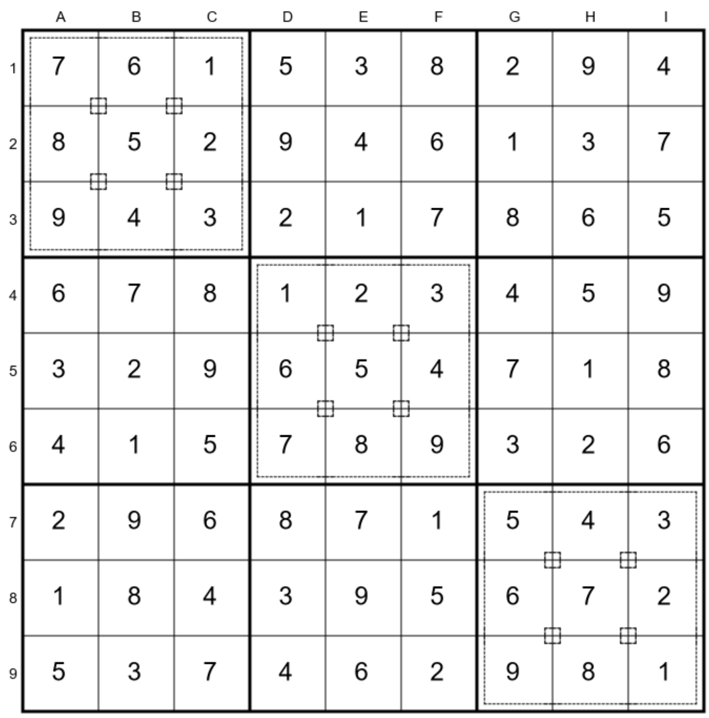
Model description not finished

11.5.6 See Also

11.5.7 chain19 example timotab38: "Nothing to Lose" by clover

Video: `https://www.youtube.com/watch?v=-nOaTUbb1fg`





Rules: standard (Section 3.13) chain19 (Section 11.5)

11.6 Constraint circleCount

11.6.1 Category: Area

11.6.2 Description

The cells marked with a circle form a collection. A number inside a circled cell indicates how many circles cells contain this number in total. A value which does not occur in any circled cell is not present in the collection.

11.6.3 Arguments

circleCount List(Cell)

11.6.4 JSON Format

```
{"circleCount":[1,11,14,21,22,25,28,34,36,40,42,48,54,57,60,61,68,71,74,81]},
```

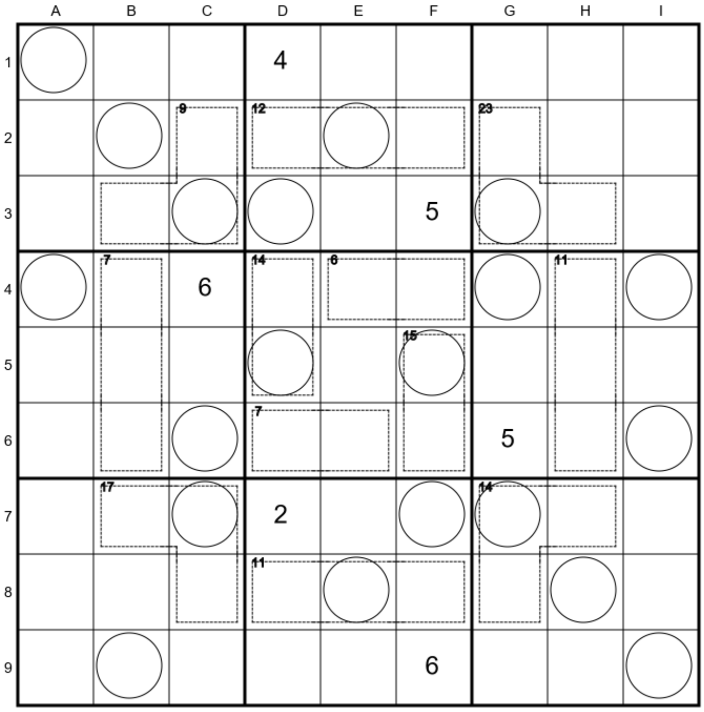
11.6.5 Model

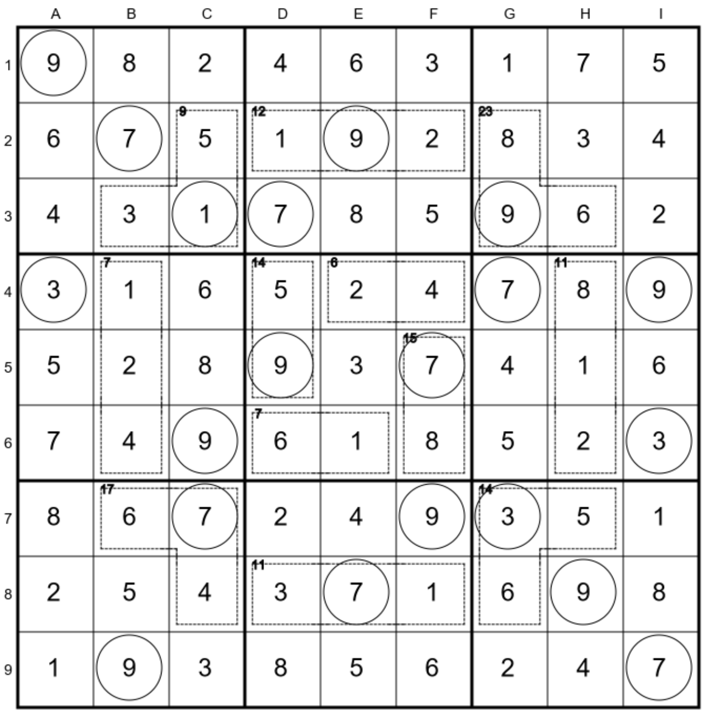
Model description not finished

11.6.6 See Also

11.6.7 circleCount example logiclemurgaming18: "The Rings of Power" by Blobz

Video: `https://www.youtube.com/watch?v=HJ13jOmWKac`





Rules: standard (Section 3.13) circleCount (Section 11.6) cage (Section 11.4)

11.7 Constraint clock

11.7.1 Category: Area

11.7.2 Description

This four cell area must read as a valid 24 hour clock value in the format HHMM.

11.7.3 Arguments

clock List(Cell)

11.7.4 JSON Format

```
"areas":[{"clock":[10,11,12,13]},...],
```

11.7.5 Model

Model description not finished

11.7.6 See Also

11.7.7 clock example BremSter11: "wpf 2026 r6" by

Video: `https://www.youtube.com/watch?v=GfJqulHALJQ`

	A	B	C	D	E	F	G	H	I
1							2	6	
2						7			3
3		6							7
4			5						
5				9		3			
6							8		
7	5							3	
8	9			6					
9		1	7						

	A	B	C	D	E	F	G	H	I
1	7	3	9	4	8	1	2	6	5
2	1	5	4	2	6	7	9	8	3
3	8	6	2	3	5	9	4	1	7
4	4	9	5	7	1	8	3	2	6
5	6	7	8	9	2	3	5	4	1
6	3	2	1	5	4	6	8	7	9
7	5	8	6	1	9	4	7	3	2
8	9	4	3	6	7	2	1	5	8
9	2	1	7	8	3	5	6	9	4

Rules: standard (Section 3.13) clock (Section 11.7)

11.8 Constraint clone

11.8.1 Category: Area

11.8.2 Description

Each cell in a copy must have the same value as the corresponding cell in the original.

11.8.3 Arguments

clone List(Cell)

copies List(Cell)

11.8.4 JSON Format

```
"areas":[{"clone":[2,3,11,12,20,21],"copies":[50]},...],
```

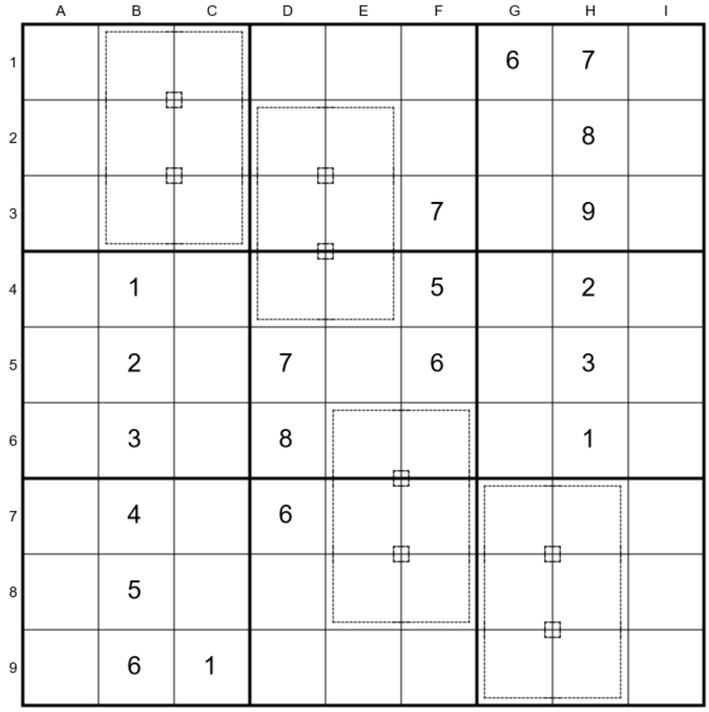
11.8.5 Model

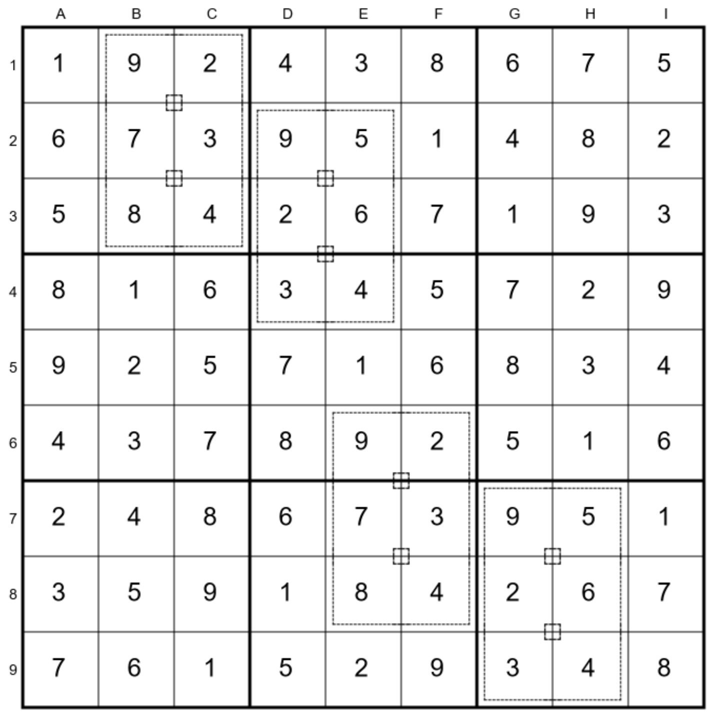
Model description not finished

11.8.6 See Also

11.8.7 clone example timotab52: "Legs in a Snare" by Bill Murphy

Video: `https://www.youtube.com/watch?v=Q72z2d18L2A`





Rules: standard (Section 3.13) clone (Section 11.8)

11.9 Constraint digitSum

11.9.1 Category: Area

11.9.2 Description

A series of numbers made from cells in the positional notation must add to the given total.

11.9.3 Arguments

digitSum List(List(Cell))

value Integer

11.9.4 JSON Format

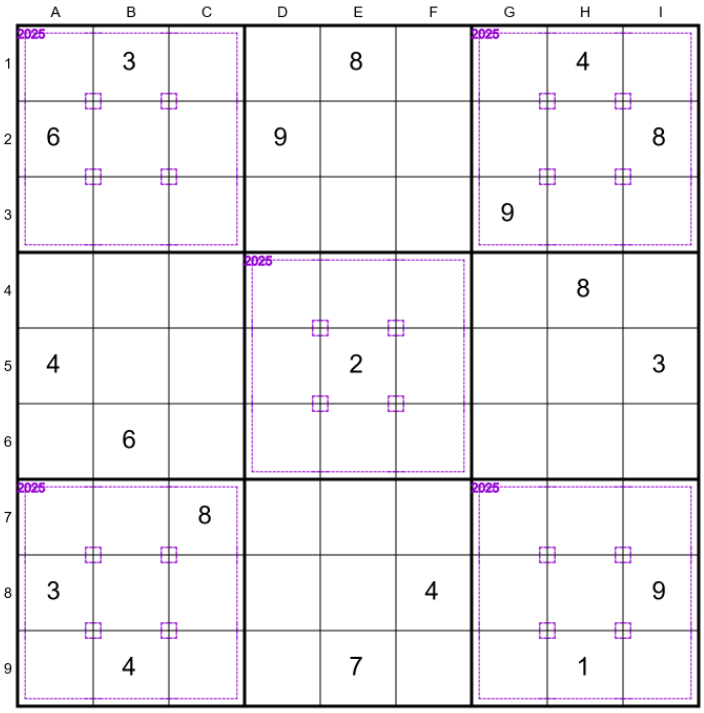
```
"areas":[{"digitSum":[[1,2,3],[10,11,12],[19,20,21]],"value":2025},...],
```

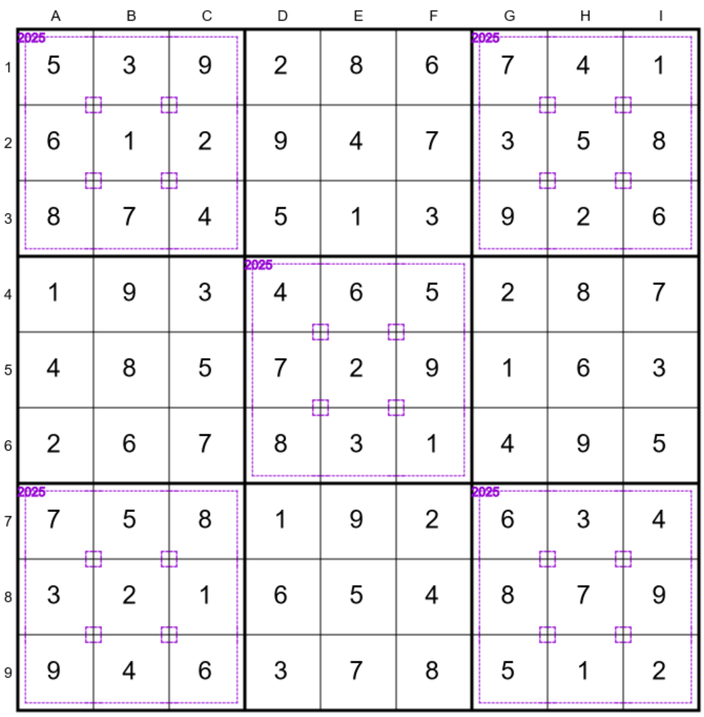
11.9.5 Model

Model description not finished

11.9.6 See Also

11.9.7 digitSum example wpf gp2025r1c6: "Sudoku GP 2025 Round 1" by





Rules: standard (Section 3.13) digitSum (Section 11.9)

11.10 Constraint fortress

11.10.1 Category: Area

11.10.2 Description

Each value in a fortress must be bigger than the value on any orthogonal neighbours outside the fortress.

11.10.3 Arguments

fortress List(Cell)

11.10.4 JSON Format

```
"areas":[{"fortress":[23,40,41,42,57,58,59,60,61,74,75,76,77,78,79,80]},],
```

11.10.5 Model

Model description not finished

11.10.6 See Also

antiFortress (Section 11.2)

11.10.7 fortress example timotab47: "Tower of Babel" by Bill Murphy

Video: `https://www.youtube.com/watch?v=zWEUCYM-EgQ`

	A	B	C	D	E	F	G	H	I
1	2	6						4	3
2	7	1			8			2	6
3									
4				3	2	8			
5									
6			3	1	5	4	2		
7									
8		4	6	7	3	1	5	8	
9									

	A	B	C	D	E	F	G	H	I
1	2	6	9	5	1	7	8	4	3
2	7	1	5	4	8	3	9	2	6
3	8	3	4	2	9	6	1	5	7
4	5	7	1	3	2	8	4	6	9
5	4	8	2	6	7	9	3	1	5
6	6	9	3	1	5	4	2	7	8
7	1	2	8	9	6	5	7	3	4
8	9	4	6	7	3	1	5	8	2
9	3	5	7	8	4	2	6	9	1

Rules: standard (Section 3.13) fortress (Section 11.10)

11.11 Constraint inCage

11.11.1 Category: Area

11.11.2 Description

The values in each cage must be different from each other, and can only take values given in the hint.

11.11.3 Arguments

inCage List(Cell)

values List(Integer)

11.11.4 JSON Format

```
"areas":[{"inCage":[2,3],"values":[3,4,5]},...]
```

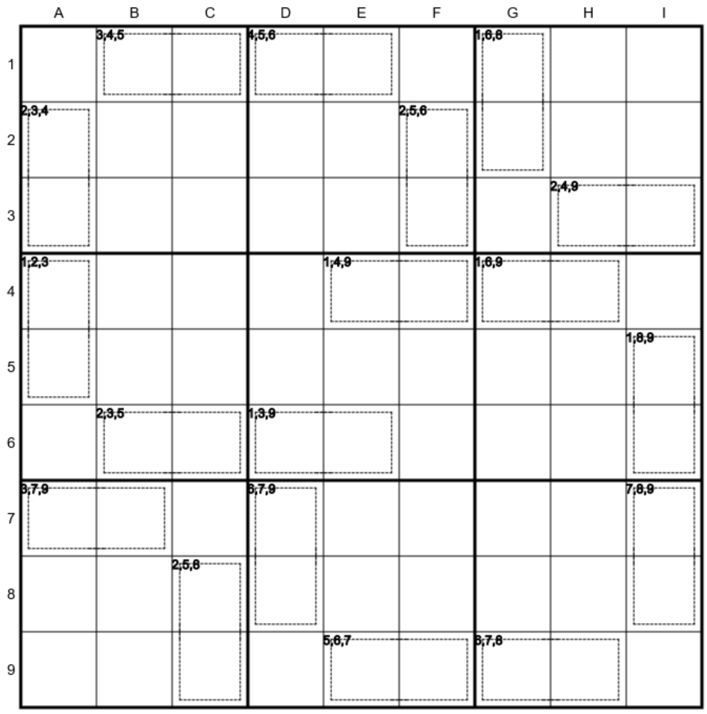
11.11.5 Model

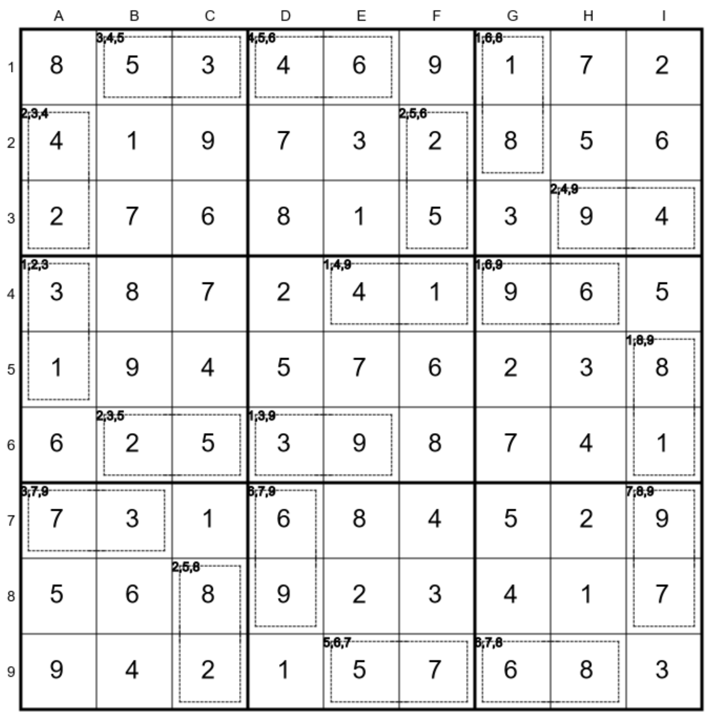
Model description not finished

11.11.6 See Also

11.11.7 inCage example timotab33: "Liar Zone II" by Bill Murphy

Video: `https://www.youtube.com/watch?v=6y2IQpwewEc&t=168s`





Rules: standard (Section 3.13) inCage (Section 11.11)

11.12 Constraint magicSquare

11.12.1 Category: Area

11.12.2 Description

The 3x3 block forms a magic square, where all digits are different from each other, and each row, column and both diagonals add up to the same number. There are more deductions possible, e.g. the sum must be 15, the number 5 must be in the centre, etc.

11.12.3 Arguments

magicSquare List(Cell)

11.12.4 JSON Format

```
"areas":[{"magicSquare":[5,6,7,14,15,16,23,24,25]},...],
```

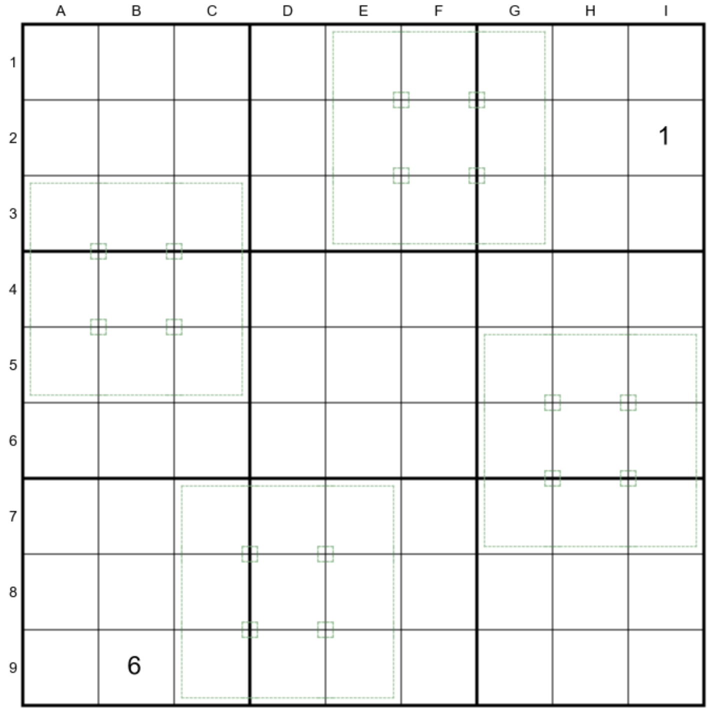
11.12.5 Model

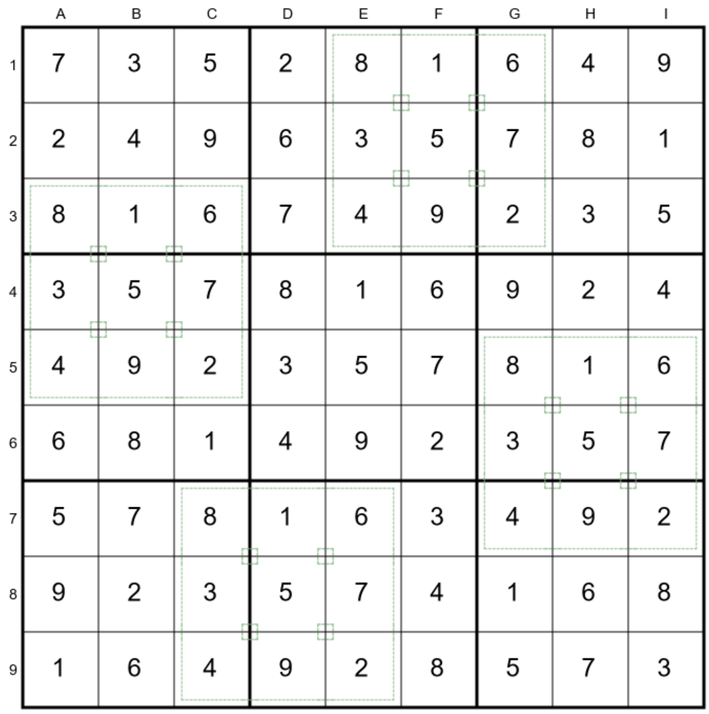
Model description not finished

11.12.6 See Also

11.12.7 magicSquare example ctc54: "Four Times the Magic" by Simon Ferré

Video: `https://www.youtube.com/watch?v=AhtOf4iVicY`





Rules: standard (Section 3.13) magicSquare (Section 11.12)

11.13 Constraint nValueArea

11.13.1 Category: Area

11.13.2 Description

The cells in the area contain exactly value different values.

11.13.3 Arguments

nValueArea List(Cell)

value Integer

11.13.4 JSON Format

```
"areas":[{"nValueArea":[11,12,13,20,21,22,29,30,31],"value":4},...],
```

11.13.5 Model

Model description not finished

11.13.6 See Also

nValueLine (Section 9.27)

11.13.7 nValueArea example gas26: "Passing Open Windows" by clover

Video: `https://youtu.be/ozkf7_koysI?si=I4TZMKVGh06ctiTI`

	A	B	C	D	E	F	G	H	I
1				6	7	3			
2		1	2				8		
3		3	4					9	
4	5								2
5	3								9
6	8								1
7		9					2	4	
8			8				3	5	
9				2	6	7			

	A	B	C	D	E	F	G	H	I
1	9	8	5	6	7	3	1	2	4
2	6	1	2	4	5	9	8	7	3
3	7	3	4	1	2	8	6	9	5
4	5	4	1	3	9	6	7	8	2
5	3	2	7	5	8	1	4	6	9
6	8	6	9	7	4	2	5	3	1
7	1	9	6	8	3	5	2	4	7
8	2	7	8	9	1	4	3	5	6
9	4	5	3	2	6	7	9	1	8

Rules: standard (Section 3.13) nValueArea (Section 11.13)

11.14 Constraint renbanArea

11.14.1 Category: Area

11.14.2 Description

The values in the n cells of the area must consist of n consecutive integers, but they can appear in any order.

11.14.3 Arguments

renbanArea List(Cell)

11.14.4 JSON Format

```
"areas":[{"renbanArea":[2,10,11,12,20]},...],
```

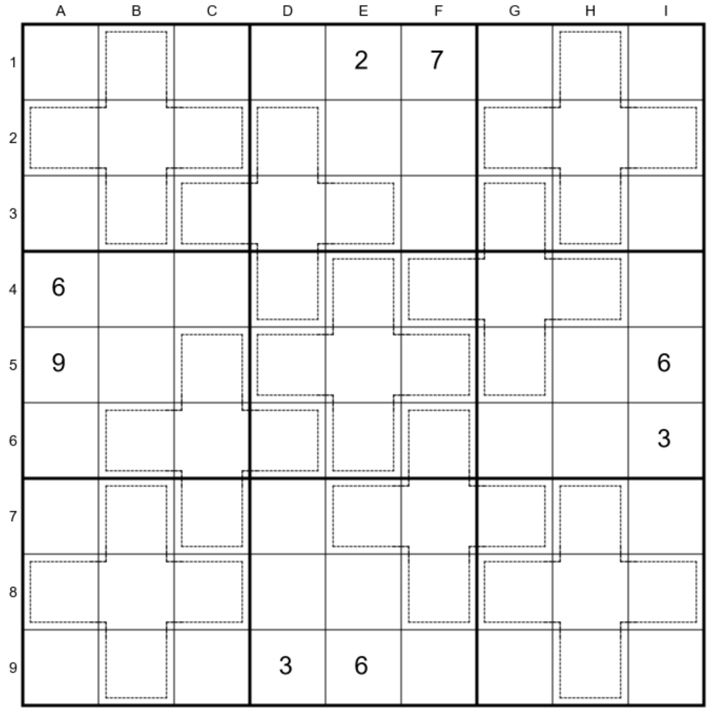
11.14.5 Model

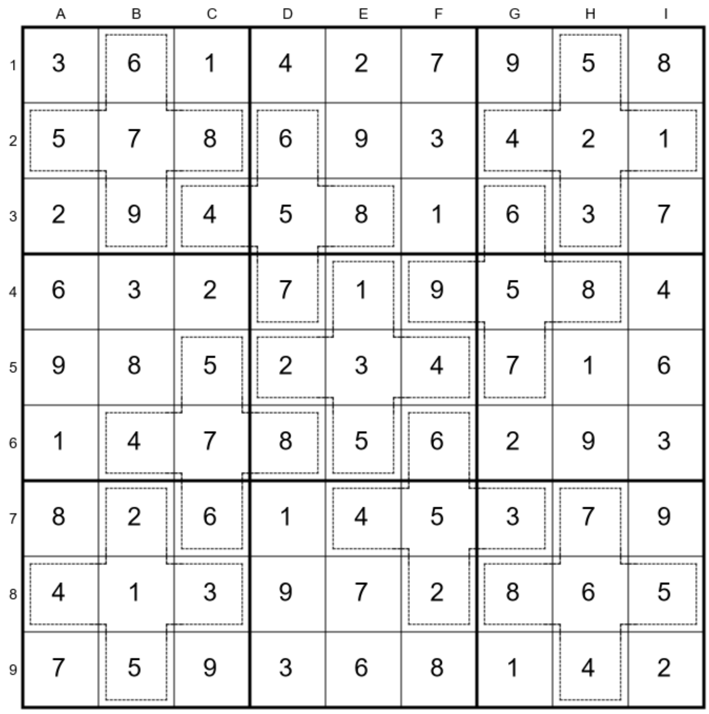
Model description not finished

11.14.6 See Also

11.14.7 renbanArea example ctc55: "Renban Sudoku" by from German GP

Video: `https://www.youtube.com/watch?v=9a_5dcBJPw4`





Rules: standard (Section 3.13) renbanArea (Section 11.14)

11.15 Constraint roundedCage

11.15.1 Category: Area

11.15.2 Description

The values in the cage are read as a two-digit number. The number in the cage is that number rounded to a multiple of 10 (values 0-4 are rounded down, values 5-9 are rounded up

11.15.3 Arguments

roundedCage List(Cell)

value Integer

11.15.4 JSON Format

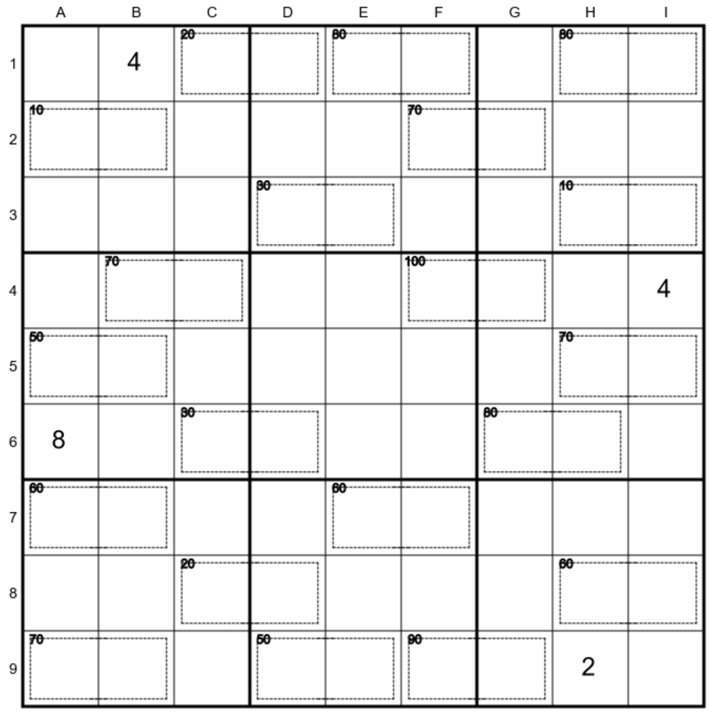
```
"areas":[{"roundedCage":[3,4],"value":20},...],
```

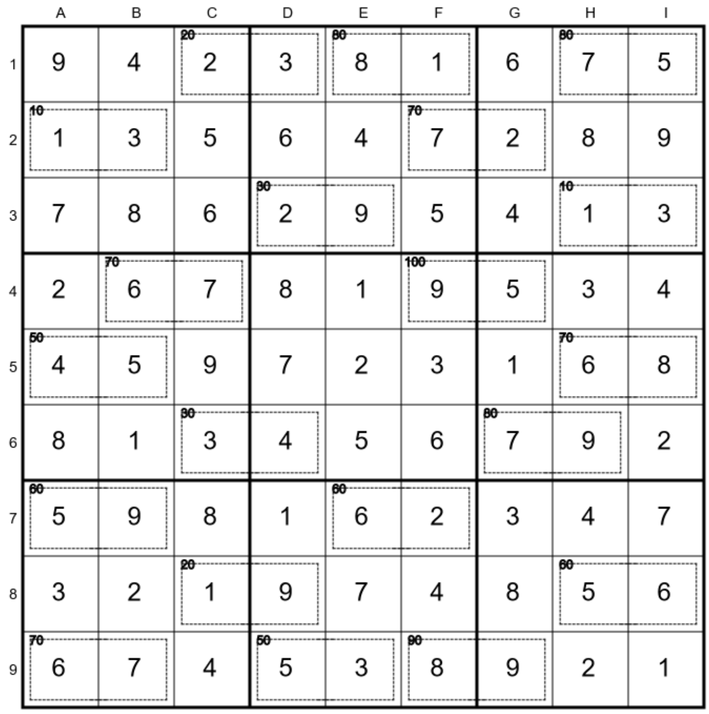
11.15.5 Model

Model description not finished

11.15.6 See Also

11.15.7 roundedCage example wpf gp2025r8c8: "Sudoku GP 2025 Round 8" by





Rules: standard (Section 3.13) roundedCage (Section 11.15)

12 Outside Constraints

12.1 Constraint bouncingXSum

12.1.1 Category: Outside

12.1.2 Description

The clues outside a row/column indicate the sum of the first x cells in the given direction, where x is the value of the first digit placed in that direction. If we hit an edge of the grid, we continue at a 90 degree angle.

12.1.3 Arguments

pos Position

nr Integer

dir Direction

value Integer

12.1.4 JSON Format

```
"bouncingXSum":[{"pos":"top","nr":3,"dir":"sw","value":9},...],
```

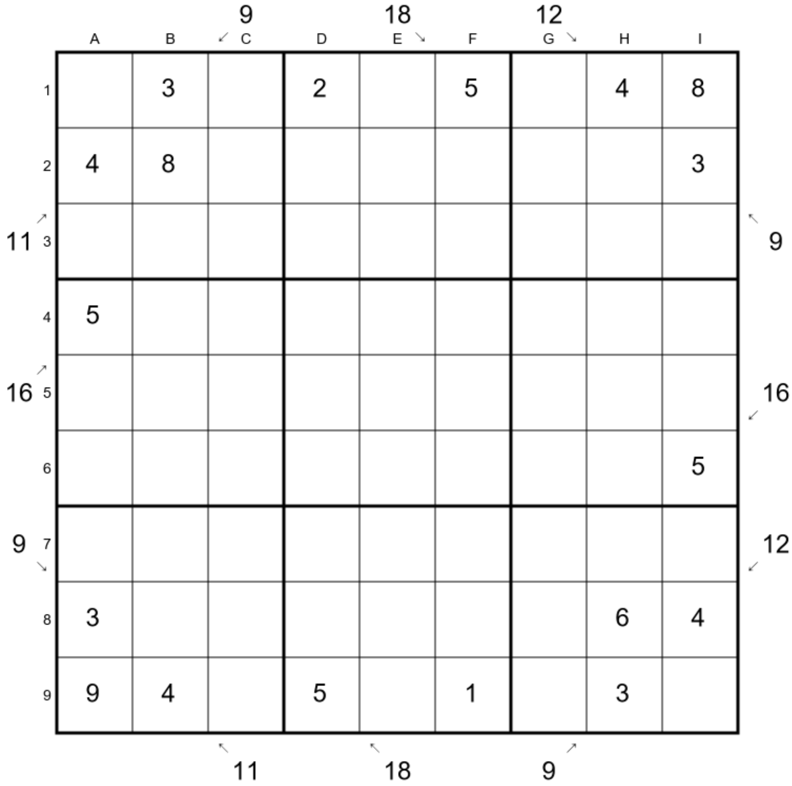
12.1.5 Model

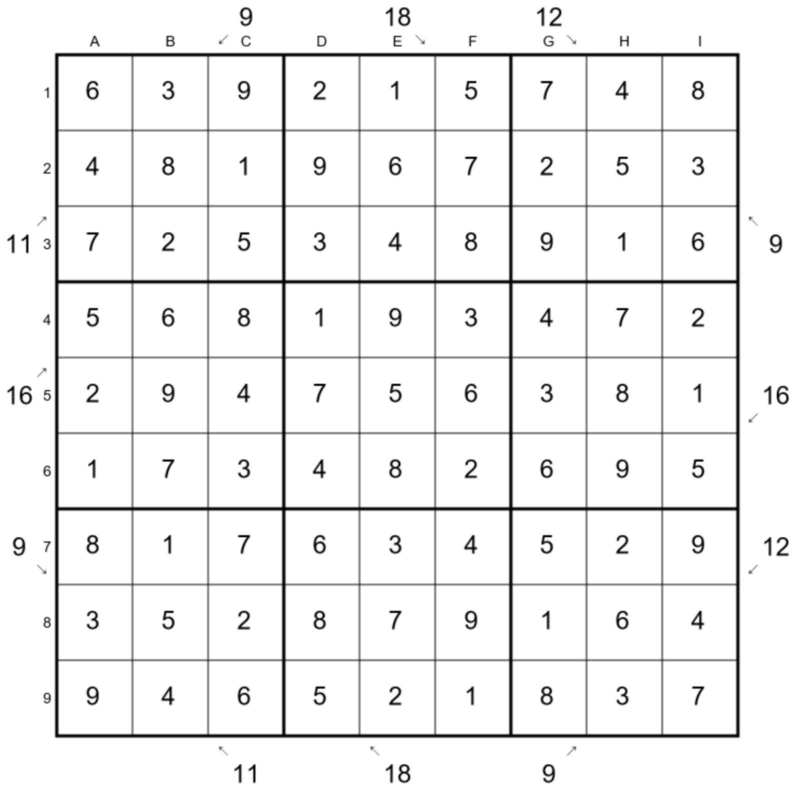
Model description not finished

12.1.6 See Also

12.1.7 bouncingXSum example gas60: "Bouncing Baby Puzz" by clover

Video: `https://www.youtube.com/watch?v=JvcLQlVkrak`
Video2: `https://www.youtube.com/watch?v=k5fLhl7cgEs&t=35s`





Rules: standard (Section 3.13) bouncingXSum (Section 12.1)

12.2 Constraint countingConsecutive

12.2.1 Category: Outside

12.2.2 Description

The clue outside the row/column gives the number of adjacent, consecutive value pairs in the row/column.

12.2.3 Arguments

pos Position

nr Integer

value Integer

12.2.4 JSON Format

```
"countingConsecutive":[{"pos":"left","nr":1,"value":0},...],
```

12.2.5 Model

Model description not finished

12.2.6 See Also

12.2.7 countingConsecutive example gas56: "Countsecutive" by clover

Video: `https://www.youtube.com/watch?v=73FZbu6rQAw`
Video2: `https://www.youtube.com/watch?v=CE5ChD_jgII`

		7				4				7
		A	B	C	D	E	F	G	H	I
0	1	9			5		3			2
	2									
0	3				6		8			
	4									
0	5			2		5		3		
	6									
0	7				4		2			
	8									
0	9	8			7		6			1

		7				4				7
		A	B	C	D	E	F	G	H	I
0	1	9	4	8	5	7	3	1	6	2
	2	1	6	5	9	2	4	8	7	3
0	3	2	7	3	6	1	8	5	9	4
	4	3	1	7	2	6	9	4	8	5
0	5	4	9	2	8	5	7	3	1	6
	6	5	8	6	3	4	1	9	2	7
0	7	6	3	1	4	9	2	7	5	8
	8	7	2	4	1	8	5	6	3	9
0	9	8	5	9	7	3	6	2	4	1

Rules: standard (Section 3.13) countingConsecutive (Section 12.2)

12.3 Constraint distances

12.3.1 Category: Outside

12.3.2 Description

Clues outside the grid indicate the distance (in steps) between the digits in the corresponding row or column. Digits must be placed in order of appearance.

12.3.3 Arguments

pos Position

nr Integer

a Integer

b Integer

distance Integer

12.3.4 JSON Format

```
"distances":[{"pos":"top","nr":1,"a":3,"b":5,"distance":7},...],
```

12.3.5 Model

Model description not finished

12.3.6 See Also

12.3.7 distances example wpf gp2025r3c18: "Sudoku GP 2025 Round 3" by

	3-5:7 A	2-5:3 B	3-8:5 C	4-1:3 D	6-9:7 E	6-7:3 F	4-5:7 G	6-7:5 H	9-2:5 I
5-7:6 1					1				
9-5:5 2									
8-2:2 3									
2-5:7 4									
7-4:5 5					8				
6-2:5 6				9		3			
1-3:7 7			4				9		
2-4:3 8									
7-1:5 9									

	3-5:7 A	2-5:3 B	3-8:5 C	4-1:3 D	6-9:7 E	6-7:3 F	4-5:7 G	6-7:5 H	9-2:5 I
5-7:6 1	3	6	5	4	1	8	2	9	7
9-5:5 2	9	2	7	3	6	5	4	1	8
8-2:2 3	4	8	1	2	7	9	3	6	5
2-5:7 4	2	7	3	1	4	6	8	5	9
7-4:5 5	1	5	9	7	8	2	6	3	4
6-2:5 6	8	4	6	9	5	3	7	2	1
1-3:7 7	6	1	4	5	2	7	9	8	3
2-4:3 8	5	9	2	8	3	4	1	7	6
7-1:5 9	7	3	8	6	9	1	5	4	2

Rules: standard (Section 3.13) distances (Section 12.3)

12.4 Constraint elementOr

12.4.1 Category: Outside

12.4.2 Description

It is either that the xth entry is y, or the yth entry is x, reading the row/column in the correct sequence (left: left to right, right: right to left. Indices are counted from 1.

12.4.3 Arguments

pos Position

nr Integer

a Integer

b Integer

12.4.4 JSON Format

```
"elementOr":[{"pos":"top","nr":1,"a":1,"b":6},...],
```

12.4.5 Model

Model description not finished

12.4.6 See Also

12.4.7 elementOr example wpf gp2025r8c10: "Sudoku GP 2025 Round 8" by

		16	25	36	49	64	81	25	49	81	
		A	B	C	D	E	F	G	H	I	
16	1										
25	2										49
36	3										25
49	4										81
64	5										
81	6										36
16	7										36
36	8										25
64	9										
			49	64	81		25	81	16		

		16	25	36	49	64	81	25	49	81	
		A	B	C	D	E	F	G	H	I	
16	1	6	3	1	7	5	8	4	2	9	
25	2	4	5	2	1	9	6	7	8	3	49
36	3	9	7	8	2	4	3	1	5	6	25
49	4	5	2	4	9	6	7	3	1	8	81
64	5	1	6	7	8	3	4	2	9	5	
81	6	8	9	3	5	1	2	6	7	4	36
16	7	2	4	9	3	8	1	5	6	7	36
36	8	7	8	6	4	2	5	9	3	1	25
64	9	3	1	5	6	7	9	8	4	2	
			49	64	81		25	81	16		

Rules: standard (Section 3.13) elementOr (Section 12.4)

12.5 Constraint evenSandwich

12.5.1 Category: Outside

12.5.2 Description

The digit field tells which digits in the row/column are sandwiched by even numbers in the indicated row/column. An empty list indicates that there is no such digit. If no hint is given, then there is no constraint for that row/column.

12.5.3 Arguments

pos Position

nr Integer

digits List<Integer>

12.5.4 JSON Format

```
"evenSandwich":[{"pos":"top","nr":2,"digits":[1,3,5]},...],
```

12.5.5 Model

Model description not finished

12.5.6 See Also

12.5.7 evenSandwich example gas32: "Don't Get Odd, Get Even Sandwich" by clover

Video: `https://youtu.be/awlTDquutoA?si=2_1froPXsdQQ2-mf`

			1,3,5	1	4,8	3,7,9	2,6	1	3,5,7	
		A	B	C	D	E	F	G	H	I
	1	8								2
	2		2		7		6		4	
	3			3		8		1		
2	4		4						6	
	5			8				4		
2	6		6						8	
	7			1		2		9		
	8		8		6		1		2	
	9	2								8

			1,3,5	1	4,8	3,7,9	2,6	1	3,5,7	
		A	B	C	D	E	F	G	H	I
	1	8	7	6	3	1	4	5	9	2
	2	1	2	9	7	5	6	8	4	3
	3	4	5	3	9	8	2	1	7	6
2	4	3	4	5	1	7	8	2	6	9
	5	7	1	8	2	6	9	4	3	5
2	6	9	6	2	4	3	5	7	8	1
	7	6	3	1	8	2	7	9	5	4
	8	5	8	4	6	9	1	3	2	7
	9	2	9	7	5	4	3	6	1	8

Rules: standard (Section 3.13) evenSandwich (Section 12.5)

12.6 Constraint expressionXSum

12.6.1 Category: Outside

12.6.2 Description

The clues outside a row/column indicate the sum of the first x cell in the given direction, where x is the value of the first digit placed in that direction. The clue from an arithmetic expression using variables A-Z, and operators +,-,*,/.

12.6.3 Arguments

pos Position

nr Integer

expr Expression

12.6.4 JSON Format

```
"expressionXSum":[{"pos":"top","nr":1,"expr":"X-K"},...],
```

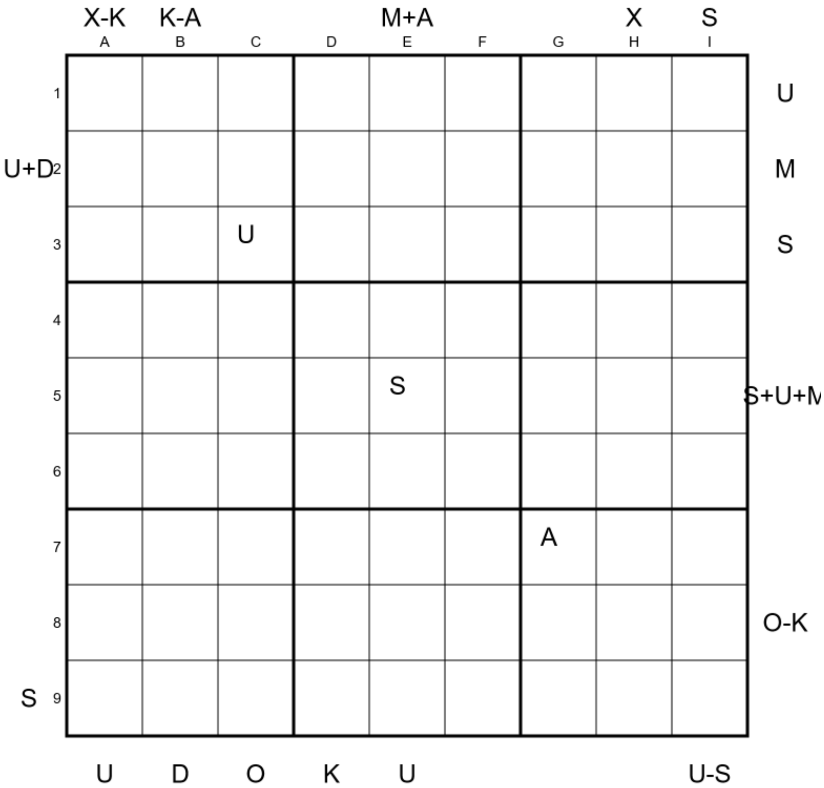
12.6.5 Model

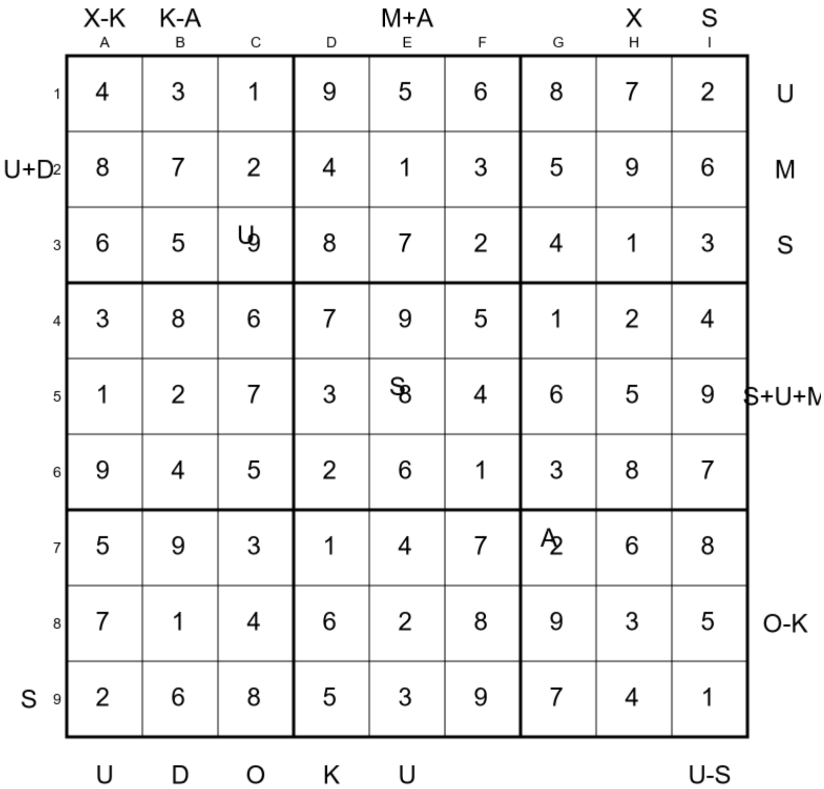
Model description not finished

12.6.6 See Also

xSum (Section 12.26)

12.6.7 expressionXSum example wpf gp2025r4c10: "Sudoku GP 2025 Round 4" by





Rules: standard (Section 3.13) expressionXSum (Section 12.6)

12.7 Constraint germanSum

12.7.1 Category: Outside

12.7.2 Description

A clue on the outside of a row/column gives the sum of the initial cells in the row/column starting from the position of the clue, and including digits as long as they satisfy a germanWhisper rule, i.e. each value is at least five apart from the values in the neighbouring cells. All cells that satisfy the germanWhisper must be included in the sum. The first cell next to the clue is always included.

12.7.3 Arguments

pos Position

nr Integer

value Integer

12.7.4 JSON Format

```
"germanSum":[{"pos":"left","nr":4,"value":20},...],
```

12.7.5 Model

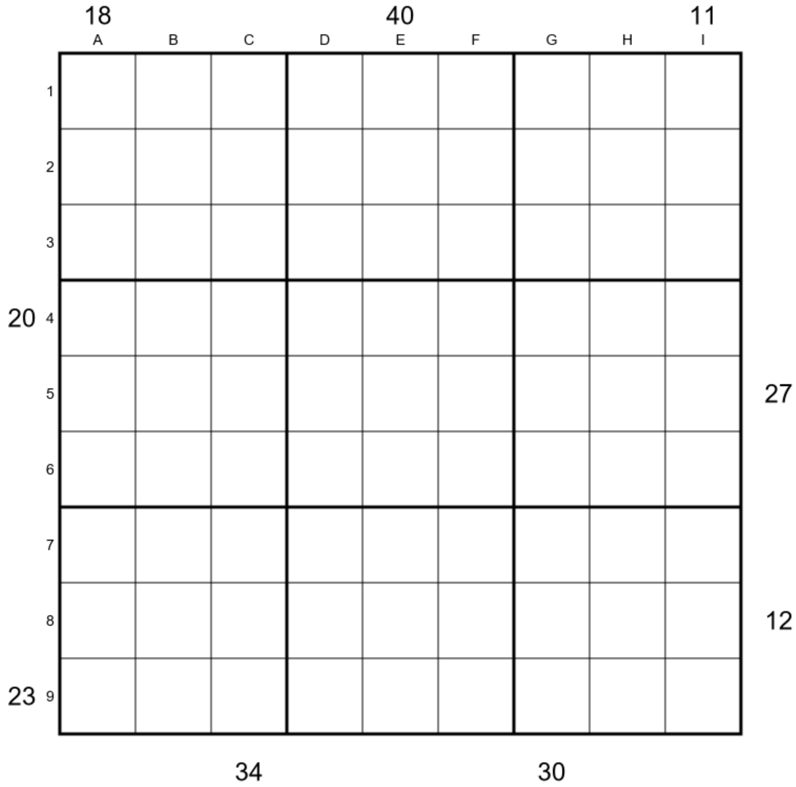
Model description not finished

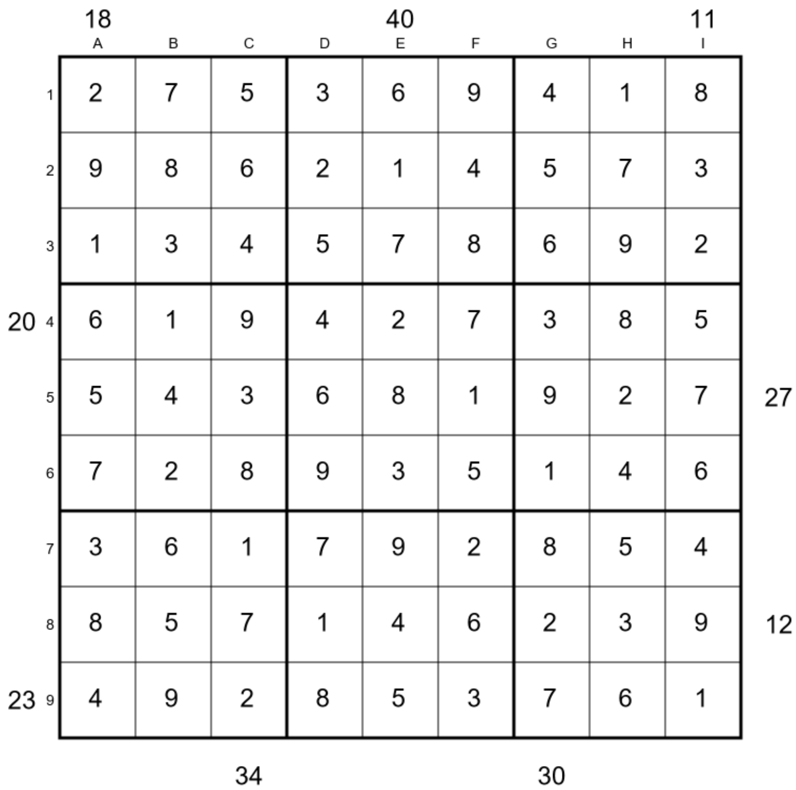
12.7.6 See Also

germanWhisper (Section 9.19)

12.7.7 germanSum example logiclemurgaming15: "German Sums Introduction" by SennyK

Video: `https://www.youtube.com/watch?v=Pey3n2jo4QQ`





Rules: standard (Section 3.13) germanSum (Section 12.7)

12.8 Constraint incDec

12.8.1 Category: Outside

12.8.2 Description

The direction of the arrow in the hint indicates the direction that the three first cells in the row/column next to the hint increase. For example, on the rows/columns, an arrow pointing east states that the cell on the left is larger than the cells next to it, which is itself larger than the third cell on the side of the hint.

12.8.3 Arguments

pos Position

nr Integer

dir Direction

12.8.4 JSON Format

```
"incDec":[{"pos":"left","nr":1,"dir":"e"},...],
```

12.8.5 Model

Model description not finished

12.8.6 See Also

12.8.7 incDec example gas51: "The Pointy End" by clover

Video: `https://www.youtube.com/watch?v=83yzKVl4uKQ`

	A	B	C	D	E	F	G	H	I
1			3	5			8		
2									
3	7		4				2		6
4					7				9
5				2		3			
6	9				8				
7	4		1				5		7
8									
9			8			6	4		

	A	B	C	D	E	F	G	H	I
1	1	2	3	5	6	9	8	7	4
2	6	8	9	7	4	2	3	1	5
3	7	5	4	8	3	1	2	9	6
4	8	3	2	1	7	4	6	5	9
5	5	1	6	2	9	3	7	4	8
6	9	4	7	6	8	5	1	2	3
7	4	9	1	3	2	8	5	6	7
8	3	6	5	4	1	7	9	8	2
9	2	7	8	9	5	6	4	3	1

Rules: standard (Section 3.13) incDec (Section 12.8)

12.9 Constraint largestDigit

12.9.1 Category: Outside

12.9.2 Description

Clues indicate the largest digit of the first three cells in the grid next to the clue.

12.9.3 Arguments

pos Position

nr Integer

value Integer

12.9.4 JSON Format

```
"largestDigit":[{"pos":"left","nr":3,"value":5},...],
```

12.9.5 Model

Model description not finished

12.9.6 See Also

12.9.7 largestDigit example gas62: "Highball Sudoku" by Bill Murphy

Video: `https://www.youtube.com/watch?v=_9tsQAbNab8`

		A	B	C	D	E	F	G	H	I	
		3		7	4				6		
	1		8								5
	2	2		6				4		7	
5	3		4								
5	4			4			6				
	5										
	6				1			5			9
	7								7		
	8	6		3				1		5	
8	9								3		4
			5				4	8		6	

		A	B	C	D	E	F	G	H	I	
		3		7	4				6		
	1	1	8	7	4	6	9	3	5	2	5
	2	2	9	6	3	5	8	4	1	7	
5	3	3	4	5	2	1	7	9	6	8	
5	4	5	3	4	9	2	6	7	8	1	
	5	9	7	1	8	4	5	6	2	3	
	6	8	6	2	1	7	3	5	4	9	9
	7	4	1	9	5	3	2	8	7	6	
	8	6	2	3	7	8	4	1	9	5	
8	9	7	5	8	6	9	1	2	3	4	4
			5				4	8		6	

Rules: standard (Section 3.13) largestDigit (Section 12.9)

12.10 Constraint littleKiller

12.10.1 Category: Outside

12.10.2 Description

The clue outside the grid gives the sum of the cell values inside the grid in the direction of the arrow.

12.10.3 Arguments

pos Position

nr Integer

dir Direction

value Integer

12.10.4 JSON Format

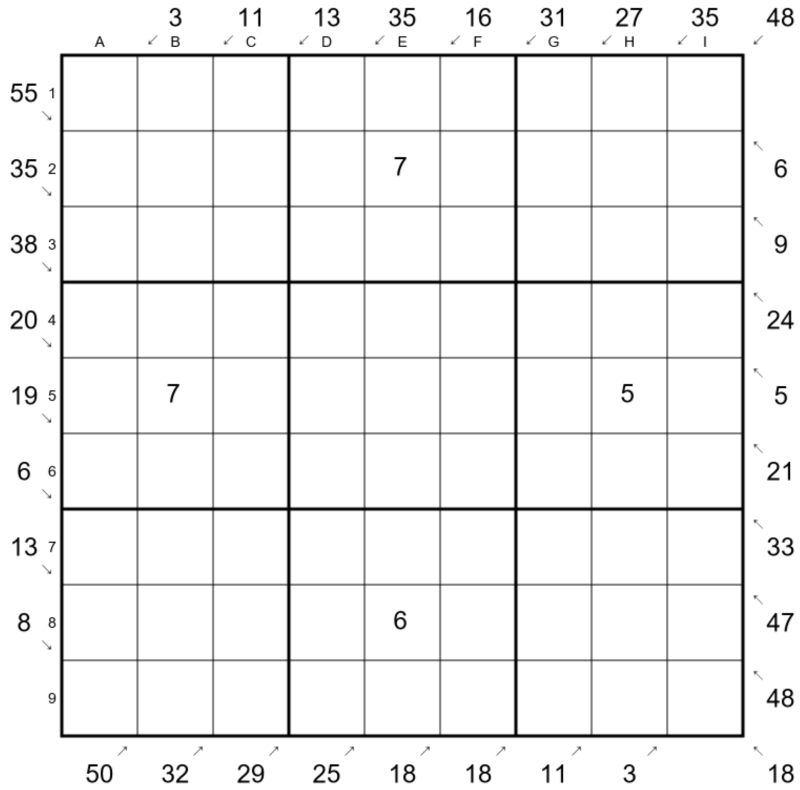
```
"littleKiller":[{"pos":"top","dir":"sw","nr":2,"value":3},...],
```

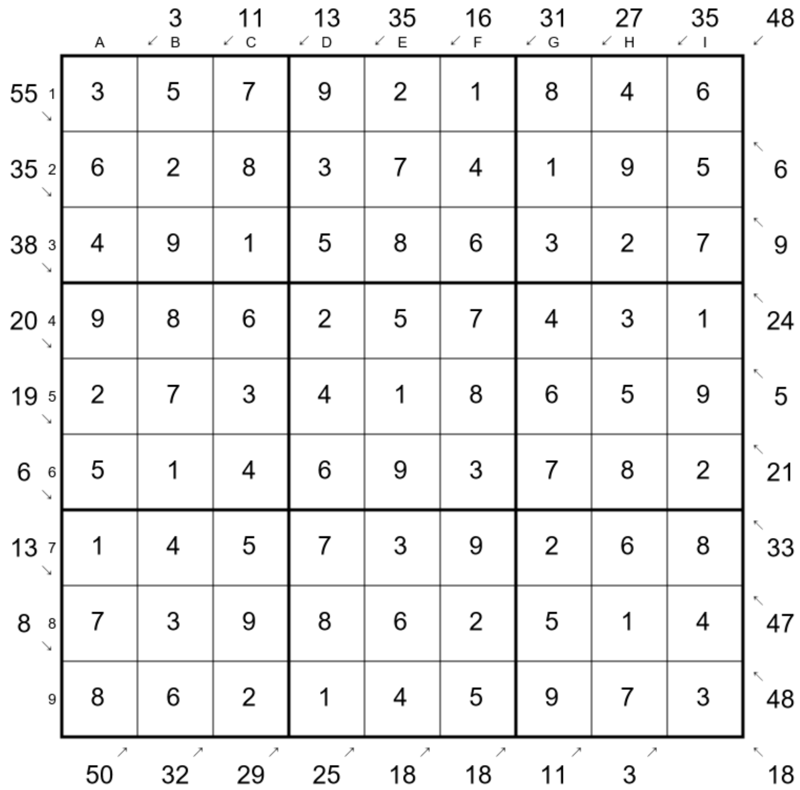
12.10.5 Model

Model description not finished

12.10.6 See Also

12.10.7 littleKiller example wpf gp2025r3c13: "Sudoku GP 2025 Round 3" by





Rules: standard (Section 3.13) littleKiller (Section 12.10)

12.11 Constraint middleDigit

12.11.1 Category: Outside

12.11.2 Description

Clues indicate the middle digit out of the first three cells in the grid next to the clue. One of the first three cells must be greater, and one smaller than the clue given.

12.11.3 Arguments

pos Position

nr Integer

value Integer

12.11.4 JSON Format

```
"middleDigit":[{"pos":"top","nr":1,"value":8},...],
```

12.11.5 Model

Model description not finished

12.11.6 See Also

12.11.7 middleDigit example gas30: "Middle Digit Syndrome" by clover

Video: `https://youtu.be/pM1022GmYfQ?si=PlDEEp-NAUH2r6E9`

		A	B	C	D	E	F	G	H	I	
		8	4	2	2	6	4	4	2	8	
	1			1				3			
	2				2		4				
5	3	7								8	6
	4		1						4		6
	5										
4	6		7						8		
5	7	3								1	7
	8				5		9				
	9	2		7				8		4	
				5	5		8	8			

		A	B	C	D	E	F	G	H	I	
		8	4	2	2	6	4	4	2	8	
	1	9	4	1	8	6	7	3	2	5	
	2	8	3	6	2	5	4	7	1	9	
5	3	7	5	2	1	9	3	4	6	8	6
	4	5	1	9	3	8	2	6	4	7	6
	5	6	2	8	7	4	5	1	9	3	
4	6	4	7	3	9	1	6	5	8	2	
5	7	3	6	5	4	2	8	9	7	1	7
	8	1	8	4	5	7	9	2	3	6	
	9	2	9	7	6	3	1	8	5	4	
				5	5		8	8			

Rules: standard (Section 3.13) middleDigit (Section 12.11)

12.12 Constraint nextToNineSum

12.12.1 Category: Outside

12.12.2 Description

Clues outside the grid indicate the sum of the number(s) immediately adjacent to the 9 in that row or column. If there is only one number next to 9, the outside clue gives the value of that number.

12.12.3 Arguments

pos Position

nr Integer

value Integer

12.12.4 JSON Format

```
"nextToNineSum":[{"pos":"top","nr":1,"value":8},...],
```

12.12.5 Model

Model description not finished

12.12.6 See Also

12.12.7 nextToNineSum example gas33: "Sums Are My Favourite Things" by clover

Video: `https://youtu.be/fFtBne34TeI?si=8Cu_CsqNLjbWCqfQ`

		8	7	14		6		15	4	7
		A	B	C	D	E	F	G	H	I
6	1							4		
3	2		9							
	3							5		1
15	4			9						
4	5									
15	6							9		
	7	4		5						
6	8								9	
7	9			1						

		8	7	14		6		15	4	7
		A	B	C	D	E	F	G	H	I
6	1	5	3	7	8	1	2	4	6	9
3	2	1	9	2	4	6	5	3	8	7
	3	6	4	8	9	7	3	5	2	1
15	4	2	8	9	7	4	6	1	5	3
4	5	7	5	6	3	9	1	8	4	2
15	6	3	1	4	5	2	8	9	7	6
	7	4	2	5	6	3	9	7	1	8
6	8	8	6	3	1	5	7	2	9	4
7	9	9	7	1	2	8	4	6	3	5

Rules: standard (Section 3.13) nextToNineSum (Section 12.12)

12.13 Constraint numberedRoom

12.13.1 Category: Outside

12.13.2 Description

The value of a clue in a row/column must occur in the nth cell of the row/column, where n is the value in the first cell next to the clue, and the counting is done starting at the first cell of the clue.

12.13.3 Arguments

pos Position

nr Integer

value Integer

12.13.4 JSON Format

```
"numberedRoom":[{"pos":"left","nr":2,"value":4},...],
```

12.13.5 Model

Model description not finished

12.13.6 See Also

12.13.7 numberedRoom example gas88: "Moonstruck" by Bill Murphy

Video: `https://youtu.be/8B1LWprpWgO?si=N6PCCdc79LranuSj` (7:40)

					6		2		5		
		A	B	C	D	E	F	G	H	I	
	1										
4	2		3		1		7		8		5
	3										
8	4		1		7		6		9		
	5										
	6		7		4		9		1		3
	7										
2	8		5		9		1		2		6
	9										
			8		8		4				

					6		2		5		
		A	B	C	D	E	F	G	H	I	
	1	4	8	9	3	2	5	1	7	6	
4	2	5	3	6	1	4	7	2	8	9	5
	3	1	2	7	6	9	8	5	3	4	
8	4	3	1	8	7	5	6	4	9	2	
	5	9	4	5	8	1	2	3	6	7	
	6	6	7	2	4	3	9	8	1	5	3
	7	7	6	3	2	8	4	9	5	1	
2	8	8	5	4	9	7	1	6	2	3	6
	9	2	9	1	5	6	3	7	4	8	
			8		8		4				

Rules: standard (Section 3.13) numberedRoom (Section 12.13)

12.14 Constraint oneTwoOrTwoThree

12.14.1 Category: Outside

12.14.2 Description

The clue outside a row/column gives the sum of either the first and second, or second and third elements in the row/column next to the clue.

12.14.3 Arguments

pos Position

nr Integer

value Integer

12.14.4 JSON Format

```
"oneTwoOrTwoThree":[{"pos":"left","nr":1,"value":4},...],
```

12.14.5 Model

Model description not finished

12.14.6 See Also

12.14.7 oneTwoOrTwoThree example timotab49: "Highball Sudoku II" by Bill Murphy

Video: `https://www.youtube.com/watch?v=U1SHasTAoaI`

		A	B	C	D	E	F	G	H	I	
		10			8			15	8		
4	1			2							
	2										
12	3	6						7			7
	4				4						14
	5										
6	6						1				
13	7			3						4	8
	8										
	9							8			16
			12	5			13			10	

		A	B	C	D	E	F	G	H	I	
		10			8			15	8		
4	1	3	1	2	6	7	5	9	4	8	
	2	7	9	5	2	4	8	6	3	1	
12	3	6	4	8	1	3	9	7	5	2	7
	4	5	3	7	4	9	2	1	8	6	14
	5	1	8	9	7	6	3	4	2	5	
6	6	4	2	6	5	8	1	3	7	9	
13	7	8	5	3	9	1	7	2	6	4	8
	8	9	7	4	8	2	6	5	1	3	
	9	2	6	1	3	5	4	8	9	7	16
			12	5			13			10	

Rules: standard (Section 3.13) oneTwoOrTwoThree (Section 12.14)

12.15 Constraint oneTwoThree

12.15.1 Category: Outside

12.15.2 Description

The clue outside the row/column gives the sum of the first three elements in the row/column next to the clue.

12.15.3 Arguments

none

12.15.4 Model

Model description not finished

12.15.5 See Also

12.15.6 oneTwoThree example Sudoku Variants58: "Frame" by Steven Anderson

		A	B	C	D	E	F	G	H	I	
		17	19	9	13	15	17	19	14	12	
18	1										6
14	2										17
13	3										22
22	4										7
6	5										20
17	6										18
9	7										16
23	8										11
13	9										18
		16	9	20	8	20	17	9	16	20	

		A	B	C	D	E	F	G	H	I	
		17	19	9	13	15	17	19	14	12	
18	1	9	4	5	6	8	7	2	3	1	6
14	2	6	7	1	3	2	9	8	4	5	17
13	3	2	8	3	4	5	1	9	7	6	22
22	4	7	9	6	8	3	5	4	1	2	7
6	5	1	3	2	9	6	4	7	5	8	20
17	6	4	5	8	7	1	2	6	9	3	18
9	7	3	2	4	5	9	6	1	8	7	16
23	8	8	6	9	1	7	3	5	2	4	11
13	9	5	1	7	2	4	8	3	6	9	18
		16	9	20	8	20	17	9	16	20	

Rules: standard (Section 3.13) oneTwoThree (Section 12.15)

12.16 Constraint outsideGT

12.16.1 Category: Outside

12.16.2 Description

The value in a marked cell must be greater than the hint outside on the left of the row, if present, and greater than the hint at the top of the column, if present.

12.16.3 Arguments

pos Position

nr Integer

value Integer

cells List(Cell)

12.16.4 JSON Format

```
"outsideGT":[{"pos":"left","nr":1,"value":7,"cells":[2,8]},...],
```

12.16.5 Model

Model description not finished

12.16.6 See Also

12.16.7 outsideGT example timotab48: "Out in the Cold" by clover

Video: `https://www.youtube.com/watch?v=zcCUlyn2FeM`

		6	4	4		5		5	5	7
		A	B	C	D	E	F	G	H	I
7	1	4								1
4	2		9						2	
5	3			1				3		
	4				1		3			
3	5					9				
	6				8		2			
4	7			4				1		
5	8		1						5	
6	9	9								3

		6	4	4		5		5	5	7
		A	B	C	D	E	F	G	H	I
7	1	4	8	3	6	2	7	5	9	1
4	2	7	9	5	3	4	1	6	2	8
5	3	2	6	1	9	5	8	3	7	4
	4	5	2	9	1	7	3	8	4	6
3	5	6	3	8	4	9	5	2	1	7
	6	1	4	7	8	6	2	9	3	5
4	7	3	5	4	7	8	9	1	6	2
5	8	8	1	6	2	3	4	7	5	9
6	9	9	7	2	5	1	6	4	8	3

Rules: standard (Section 3.13) outsideGT (Section 12.16)

12.17 Constraint outsideMustContain

12.17.1 Category: Outside

12.17.2 Description

The first three cells in the row/column next to the clue must contain the values given in the clue, in any order

12.17.3 Arguments

pos Position

nr Integer

dir Direction

digits List(Integer)

12.17.4 JSON Format

```
"outsideMustContain":[{"pos":"left","nr":2,"dir":"se","digits":[2,5]},...],
```

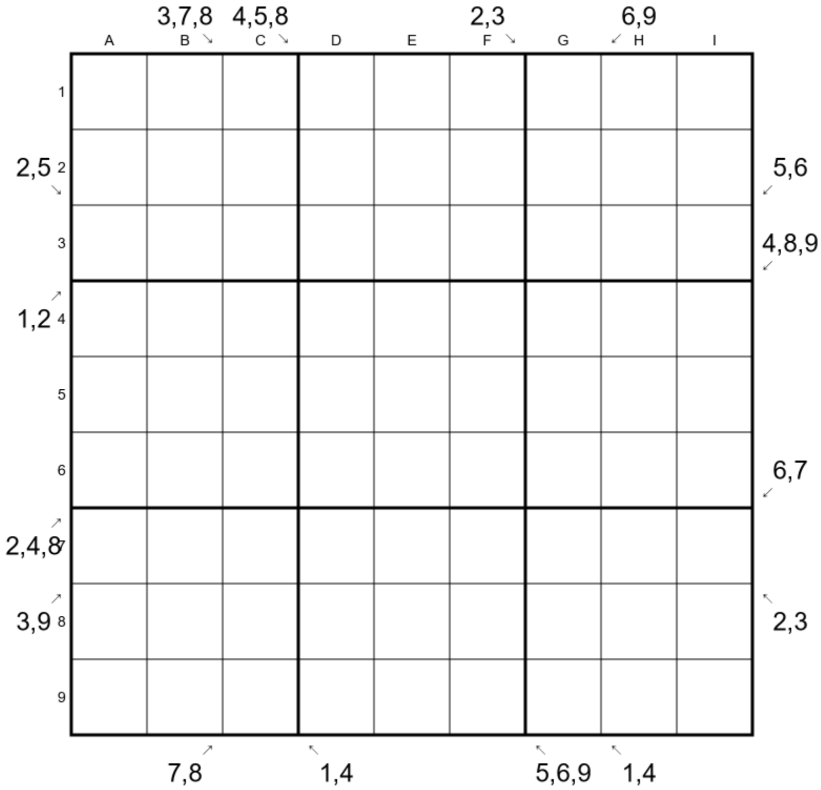
12.17.5 Model

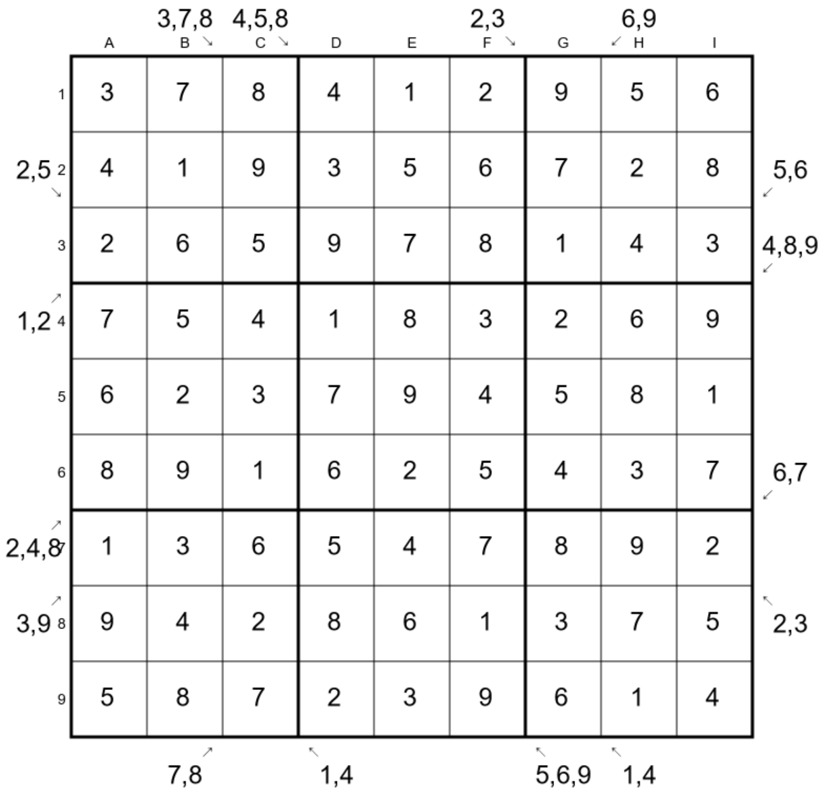
Model description not finished

12.17.6 See Also

12.17.7 outsideMustContain example gas69: "Diagonal Outside III" by Bill Murphy

Video: `https://youtu.be/5DBK_AaaIhI?si=NrjqZzNf62euOEFb`





Rules: standard (Section 3.13) outsideMustContain (Section 12.17)

12.18 Constraint pSum

12.18.1 Category: Outside

12.18.2 Description

A clue N outside a row/column means that any two cells in the row/column whose position values (starting from 1) add to N, also contain digits that sum to N. This applies to all pairs of cells whose position numbers add to N.

12.18.3 Arguments

pos Position

nr Integer

value Integer

12.18.4 JSON Format

```
"pSum":[{"pos":"left","nr":2,"value":14},...],
```

12.18.5 Model

Model description not finished

12.18.6 See Also

12.18.7 pSum example BremSter55: "P-Sum" by psninn

Video: `https://www.youtube.com/watch?v=rp2ElzgYvls`

			15		13		7		9	
		A	B	C	D	E	F	G	H	I
	1									
14	2									
11	3									
	4									
10	5									
	6									
	7	8								2
4	8									
	9									

			15		13		7		9	
		A	B	C	D	E	F	G	H	I
	1	9	2	6	3	8	5	1	4	7
14	2	7	4	3	1	9	6	2	8	5
11	3	1	5	8	2	7	4	9	3	6
	4	4	1	7	8	6	3	5	2	9
10	5	6	3	2	9	5	1	8	7	4
	6	5	8	9	7	4	2	3	6	1
	7	8	9	5	6	3	7	4	1	2
4	8	3	6	1	4	2	9	7	5	8
	9	2	7	4	5	1	8	6	9	3

Rules: standard (Section 3.13) pSum (Section 12.18)

12.19 Constraint paritySum

12.19.1 Category: Outside

12.19.2 Description

The clue outside the grid gives the sum of the initial cells in the grid, up to and including the first parity change. A one-digit outside clue could reference just the first cell, since that would be the first odd/even number in that direction.

12.19.3 Arguments

pos Position

nr Integer

value Integer

12.19.4 JSON Format

```
"paritySum":[{"pos":"top","nr":1,"value":5},...],
```

12.19.5 Model

Model description not finished

12.19.6 See Also

12.19.7 paritySum example gas24: "A Long Time Coming" by clover

Video: `https://youtu.be/w_UXe1qKW70?si=evci0uyayt3F8bTQ`

		A	B	C	D	E	F	G	H	I	
		5	13		3						
3	1	1						6	7		
13	2									9	
	3				6	8				1	
3	4			6							
	5			7		9		8			
	6							5			4
	7	2				7	5				
	8	8									12
	9		9	5						2	3
							4		12	5	

		A	B	C	D	E	F	G	H	I	
		5	13		3						
3	1	1	2	8	3	4	9	6	7	5	
13	2	4	6	3	5	1	7	2	8	9	
	3	7	5	9	6	8	2	3	4	1	
3	4	3	4	6	2	5	8	1	9	7	
	5	5	1	7	4	9	3	8	2	6	
	6	9	8	2	7	6	1	5	3	4	4
	7	2	3	4	1	7	5	9	6	8	
	8	8	7	1	9	2	6	4	5	3	12
	9	6	9	5	8	3	4	7	1	2	3
							4		12	5	

Rules: standard (Section 3.13) paritySum (Section 12.19)

12.20 Constraint positionSum

12.20.1 Category: Outside

12.20.2 Description

The hints give two pieces of information. The first is the sum of the first two entries Called a and b) in the row/column, the second is the sum of the ath and bth entries in the row column, where ath is the digit in position a of the row/column.

12.20.3 Arguments

pos Position

nr Integer

a Integer

b Integer

12.20.4 JSON Format

```
"positionSum":[{"pos":"top","nr":3,"a":6,"b":13},...],
```

12.20.5 Model

Model description not finished

12.20.6 See Also

12.20.7 positionSum example wpf gp2025r7c11: "Sudoku GP 2025 Round 7" by

			6 13	16 0	15 10		6 8	6 4	14 13
	A	B	C	D	E	F	G	H	I
1									
2									
10 15 3									
7 0 4									
15 9 5									
6									
6 10 7									
12 6 8									
10 10 9									

			6 13	16 0	15 10		6 8	6 4	14 13
	A	B	C	D	E	F	G	H	I
1	2	3	8	5	9	7	6	1	4
2	7	4	5	6	1	8	2	3	9
10 15 3	9	6	1	2	4	3	8	5	7
7 0 4	1	9	7	3	2	4	5	8	6
15 9 5	4	5	2	7	8	6	3	9	1
6	6	8	3	9	5	1	4	7	2
6 10 7	8	2	6	1	3	9	7	4	5
12 6 8	5	1	4	8	7	2	9	6	3
10 10 9	3	7	9	4	6	5	1	2	8

Rules: standard (Section 3.13) positionSum (Section 12.20)

12.21 Constraint sandwich

12.21.1 Category: Outside

12.21.2 Description

The value in the clue is the sum of the digits placed between the occurrences of 1 and 9 in the row/column.

12.21.3 Arguments

pos Position

nr Integer

value Integer

12.21.4 JSON Format

```
"sandwich":[{"pos":"top","nr":7,"value":9},...],
```

12.21.5 Model

Model description not finished

12.21.6 See Also

12.21.7 sandwich example timotab28: "Pop Quiz" by Philip Newman

Video: `https://youtu.be/Ri29q9uVBzU?si=GfKc4y6lVK09zV6O`

		A	B	C	D	E	F	G	H	I	
								9			
	1				7				4	9	
	2						8				
3	3										
6	4										
	5										
	6										11
	7										5
	8				2						
	9	1	6				3				
				10							

		A	B	C	D	E	F	G	H	I	
								9			
	1	6	8	5	7	2	1	3	4	9	
	2	4	9	7	3	5	8	1	2	6	
3	3	2	1	3	9	6	4	5	8	7	
6	4	3	2	1	6	9	7	4	5	8	
	5	8	5	6	4	3	2	9	7	1	
	6	9	7	4	1	8	5	2	6	3	11
	7	7	4	9	5	1	6	8	3	2	5
	8	5	3	8	2	7	9	6	1	4	
	9	1	6	2	8	4	3	7	9	5	
				10							

Rules: standard (Section 3.13) xSum (Section 12.26) sandwich (Section 12.21)

12.22 Constraint sandwich18

12.22.1 Category: Outside

12.22.2 Description

The value in the clue is the sum of the digits placed between the occurrences of 1 and 8 in the row/column.

12.22.3 Arguments

pos Position

nr Integer

value Integer

12.22.4 JSON Format

```
"sandwich18":[{"pos":"left","nr":1,"value":4},...],
```

12.22.5 Model

Model description not finished

12.22.6 See Also

12.22.7 sandwich18 example timotab97: "Nineless" by clover

Video: `https://www.youtube.com/watch?v=Tz9k-sEocPs` (11:18)

	5 A	2 B	2 C	5 D	4 E	3 F	3 G	2 H
4 1								
4 2		1						
6 3						1		
3 4								
2 5								
5 6			8					
7 7							8	
3 8								

	5 A	2 B	2 C	5 D	4 E	3 F	3 G	2 H
4 1	2	6	5	3	7	8	4	1
4 2	7	1	4	8	6	3	5	2
6 3	4	2	7	5	3	1	6	8
3 4	6	8	3	1	2	5	7	4
2 5	3	7	6	4	8	2	1	5
5 6	1	5	8	2	4	6	3	7
7 7	5	4	2	6	1	7	8	3
3 8	8	3	1	7	5	4	2	6

Rules: standard (Section 3.13) 8x8 (Section 3.2) sandwich18 (Section 12.22)

12.23 Constraint singleParitySkyscraper

12.23.1 Category: Outside

12.23.2 Description

How many cells are visible from outside the grid in the given direction, if we see the value in the cell as the height of a skyscraper building in the cell. Taller buildings hide smaller building behind them. The hint gives either the number of even height buildings visible, or the number of odd height buildings.

12.23.3 Arguments

pos Position

nr Integer

value Integer

12.23.4 JSON Format

```
"singleParitySkyscraper":[{"pos":"top","nr":2,"value":5},...],
```

12.23.5 Model

Model description not finished

12.23.6 See Also

12.23.7 singleParitySkyscraper example wpf gp2025r4c8: "Sudoku GP 2025 Round 4" by

		A	B	C	D	E	F	G	H	I	
			5			1			3		
	1										
3	2										4
	3					6					
	4		6						9		
3	5			8		3		1			2
	6					5					
	7			7		2		4			
4	8				4		9				2
	9										
			2			1			5		

		A	B	C	D	E	F	G	H	I	
			5			1			3		
	1	8	1	9	3	4	5	6	2	7	
3	2	2	3	6	8	9	7	5	4	1	4
	3	7	5	4	2	6	1	9	8	3	
	4	4	6	2	7	1	8	3	9	5	
3	5	5	7	8	9	3	2	1	6	4	2
	6	3	9	1	6	5	4	2	7	8	
	7	6	8	7	1	2	3	4	5	9	
4	8	1	2	5	4	7	9	8	3	6	2
	9	9	4	3	5	8	6	7	1	2	
			2			1			5		

Rules: standard (Section 3.13) singleParitySkyscraper (Section 12.23)

12.24 Constraint skyscraper

12.24.1 Category: Outside

12.24.2 Description

How many cells are visible from outside the grid in the given direction, if we see the value in the cell as the height of a skyscraper building in the cell. Taller buildings hide smaller building behind them. The hint gives the number of buildings visible.

12.24.3 Arguments

pos Position

nr Integer

value Integer

12.24.4 JSON Format

```
"skyscraper":[{"pos":"left","nr":2,"value":6},...],
```

12.24.5 Model

Model description not finished

12.24.6 See Also

singleParitySkyscraper (Section 12.23)

12.24.7 skyscraper example gas91: "Pick Up" by Bill Murphy

Video: `https://youtu.be/c3duTNth84E?si=aJWz_VptRq-R_gUi` (11:46)

		A	B	C	D	E	F	G	H	I	
	1										
6	2		5		7		9		3		4
	3			3				9			
2	4		9		6		3		1		5
	5										
7	6		3		5		7		9		2
	7			9				7			
3	8		6		9		4		2		6
	9										

		A	B	C	D	E	F	G	H	I	
	1	9	8	2	3	4	5	1	6	7	
6	2	1	5	6	7	8	9	4	3	2	4
	3	4	7	3	2	6	1	9	8	5	
2	4	8	9	7	6	2	3	5	1	4	5
	5	6	1	5	4	9	8	2	7	3	
7	6	2	3	4	5	1	7	8	9	6	2
	7	5	2	9	1	3	6	7	4	8	
3	8	7	6	8	9	5	4	3	2	1	6
	9	3	4	1	8	7	2	6	5	9	

Rules: standard (Section 3.13) skyscraper (Section 12.24)

12.25 Constraint starProduct

12.25.1 Category: Outside

12.25.2 Description

The hint in the row/column is the value of product of all cells in the row/column which are marked by a star.

12.25.3 Arguments

pos Position

nr Integer

value Integer

12.25.4 JSON Format

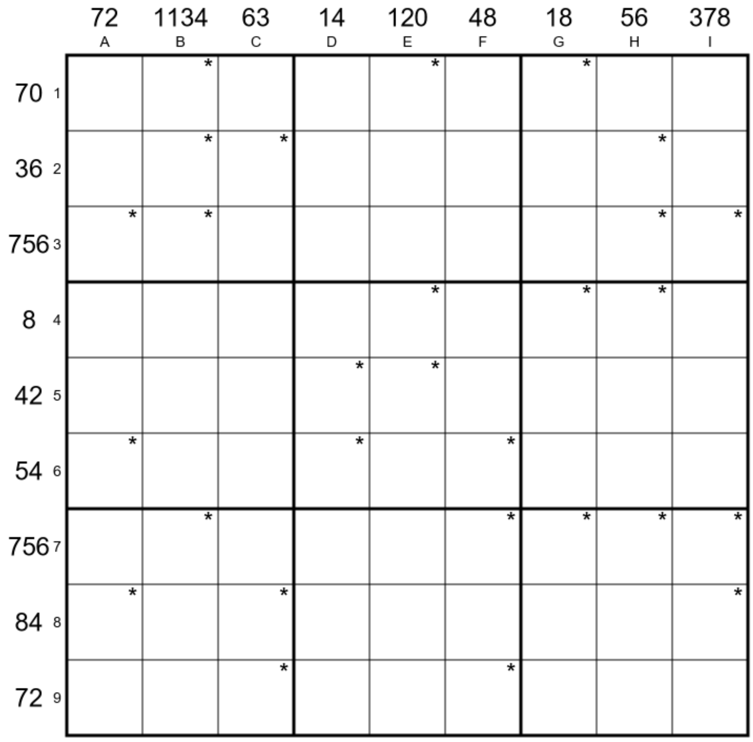
```
"starProduct":[{"pos":"top","nr":1,"value":72},...]
```

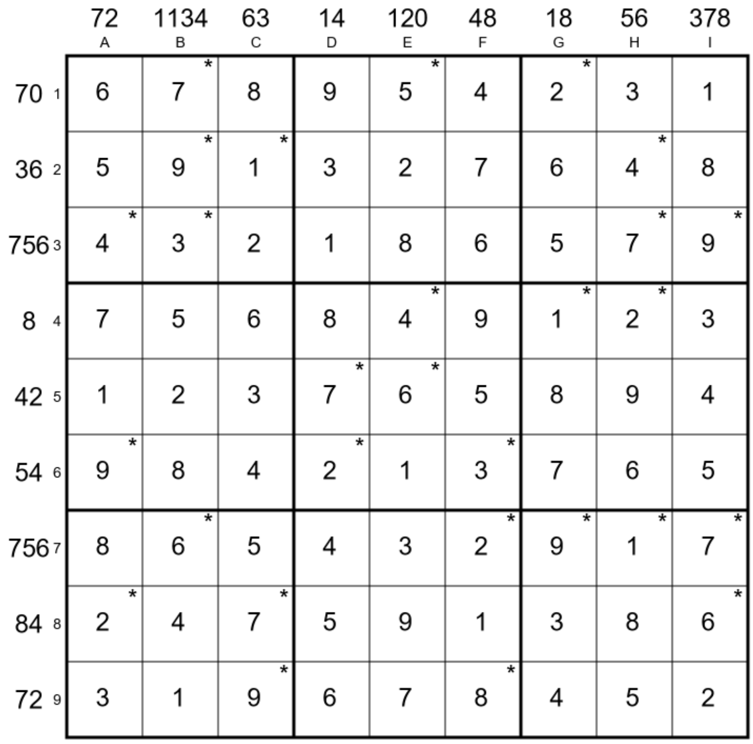
12.25.5 Model

Model description not finished

12.25.6 See Also

12.25.7 starProduct example wpf gp2025r3c12: "Sudoku GP 2025 Round 3" by





Rules: standard (Section 3.13) starProduct (Section 12.25)

12.26 Constraint xSum

12.26.1 Category: Outside

12.26.2 Description

The clues outside a row/column indicate the sum of the first x cells in the given direction, where x is the value of the first digit placed in that direction.

12.26.3 Arguments

pos Position

nr Integer

value Integer

12.26.4 JSON Format

```
"xSum":[{"pos":"top","nr":7,"value":9},...],
```

12.26.5 Model

Model description not finished

12.26.6 See Also

12.26.7 xSum example timotab28: "Pop Quiz" by Philip Newman

Video: `https://youtu.be/Ri29q9uVBzU?si=GfKc4y6lVK09zV6O`

								9			
		A	B	C	D	E	F	G	H	I	
	1				7				4	9	
	2						8				
3	3										
6	4										
	5										
	6										11
	7										5
	8				2						
	9	1	6				3				
				10							

								9			
		A	B	C	D	E	F	G	H	I	
	1	6	8	5	7	2	1	3	4	9	
	2	4	9	7	3	5	8	1	2	6	
3	3	2	1	3	9	6	4	5	8	7	
6	4	3	2	1	6	9	7	4	5	8	
	5	8	5	6	4	3	2	9	7	1	
	6	9	7	4	1	8	5	2	6	3	11
	7	7	4	9	5	1	6	8	3	2	5
	8	5	3	8	2	7	9	6	1	4	
	9	1	6	2	8	4	3	7	9	5	
				10							

Rules: standard (Section 3.13) xSum (Section 12.26) sandwich (Section 12.21)

13 Meta Constraints

13.1 Constraint liarOddEven

13.1.1 Category: Meta

13.1.2 Description

n of the odd/even hints are wrong. It is required to find out which hints cannot be correct.

13.1.3 Arguments

lies Integer

odd List(Cell)

even List(Cell)

13.1.4 JSON Format

```
"liarOddEven":{"lies":1,"odd":[12,26,32,40,42,50,56,70],"even":[4,6,28,36,46,54,76,78]},
```

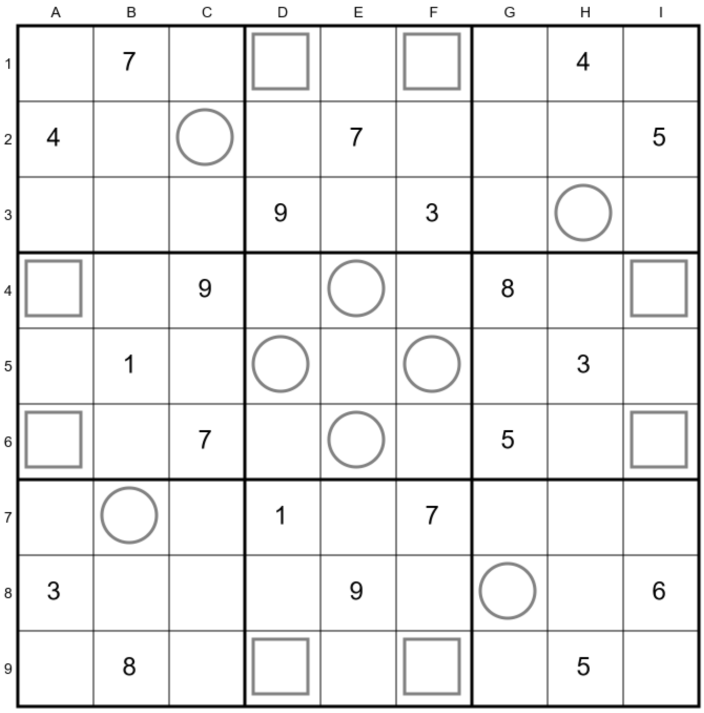
13.1.5 Model

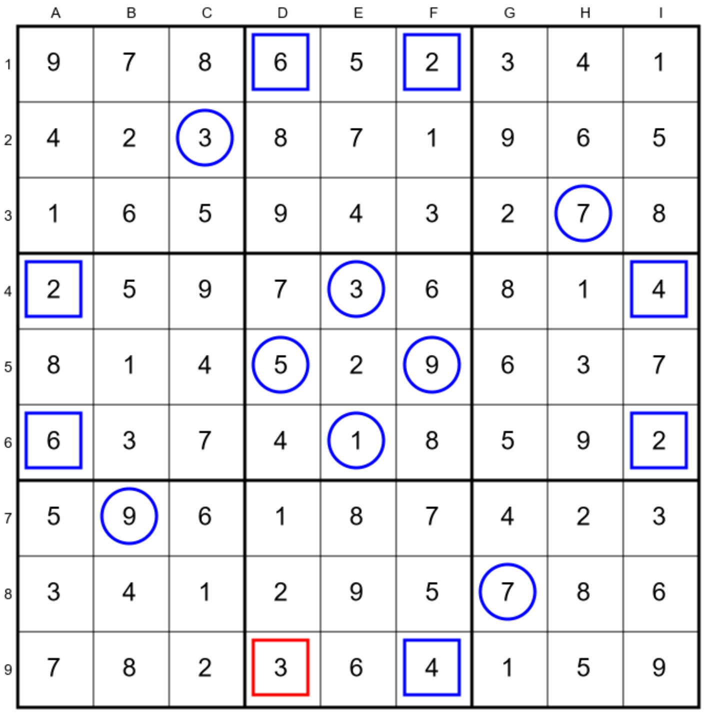
Model description not finished

13.1.6 See Also

odd (Section 5.9) , even (Section 5.5)

13.1.7 liarOddEven example wpf gp2025r4c6: "Sudoku GP 2025 Round 4" by





Rules: standard (Section 3.13) liarOddEven (Section 13.1)

14 Other Constraints

14.1 Constraint cryptic

14.1.1 Category: Other

14.1.2 Description

A set of letters are given, which can take values from 1 to 9. Different letters denote different numbers, and multiple occurrences of the same letter in the grid denote the same value.

14.1.3 Arguments

letter Letter

cells List(Cell)

14.1.4 JSON Format

```
"cryptic":[{"letter":"S","cells":[36,73]},...],
```

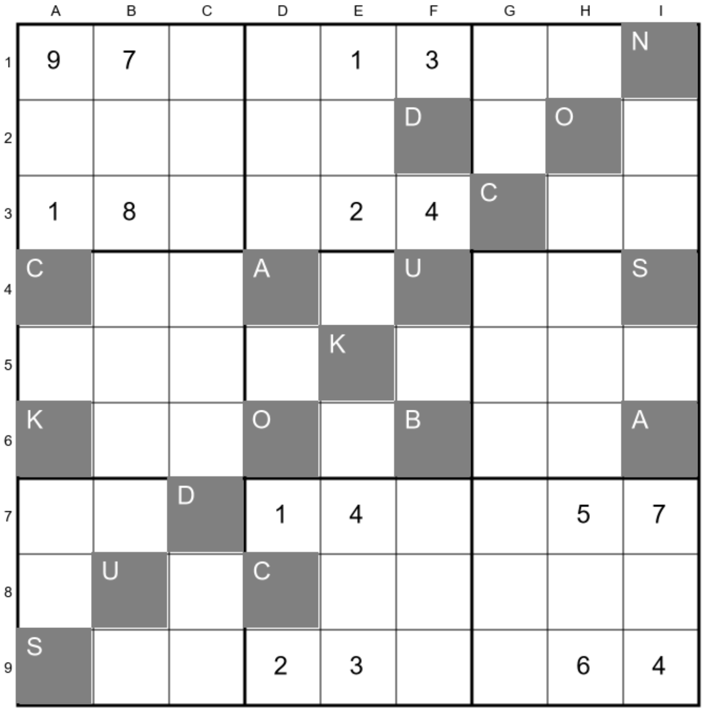
14.1.5 Model

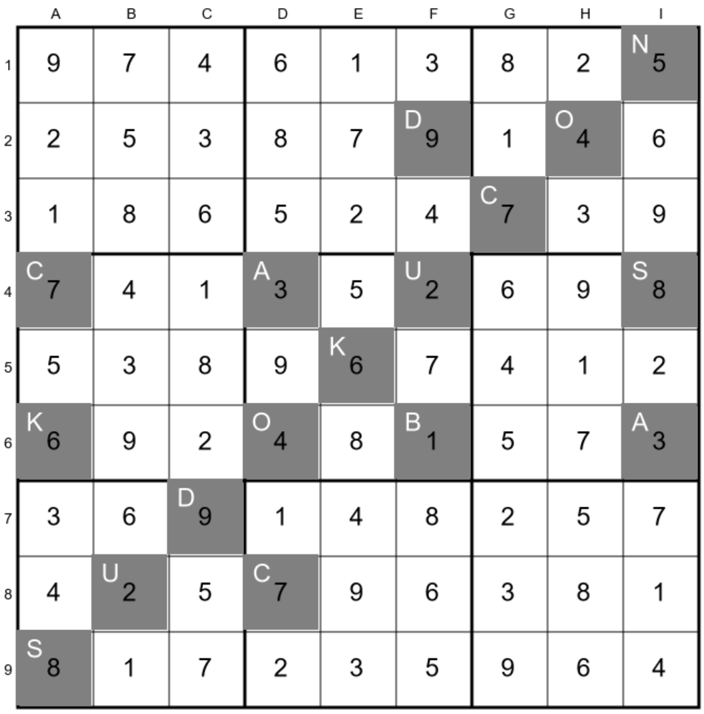
Model description not finished

14.1.6 See Also

14.1.7 cryptic example BremSter26: "Cryptic Sudoku: SUDOKUCON" by clover

Video: `https://www.youtube.com/watch?v=4iFbkFZMKZk`





Rules: standard (Section 3.13) cryptic (Section 14.1)

14.2 Constraint indexing

14.2.1 Category: Other

14.2.2 Description

The values in the given columns (1-9) indicate where the corresponding values are found in the row.

14.2.3 Arguments

indexing List(Integer)

14.2.4 JSON Format

```
"indexing":[1,5,9],
```

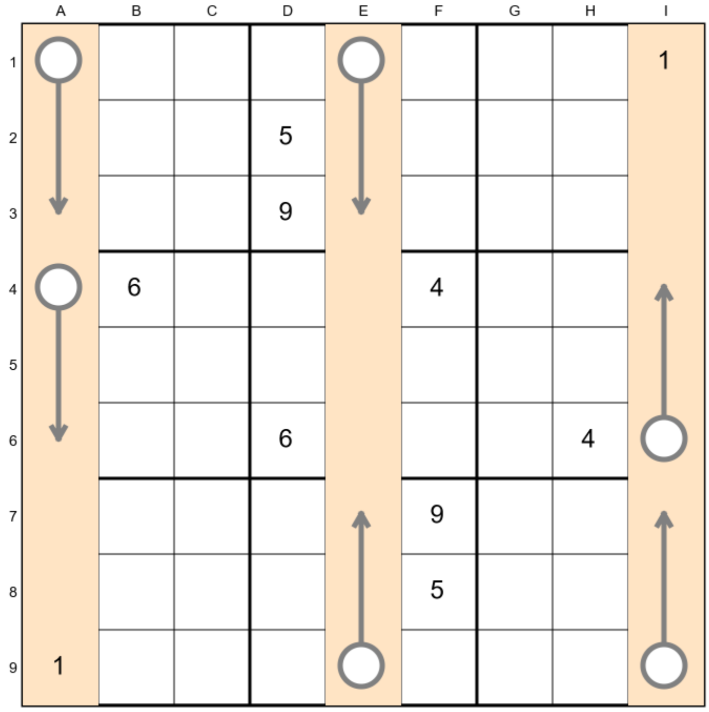
14.2.5 Model

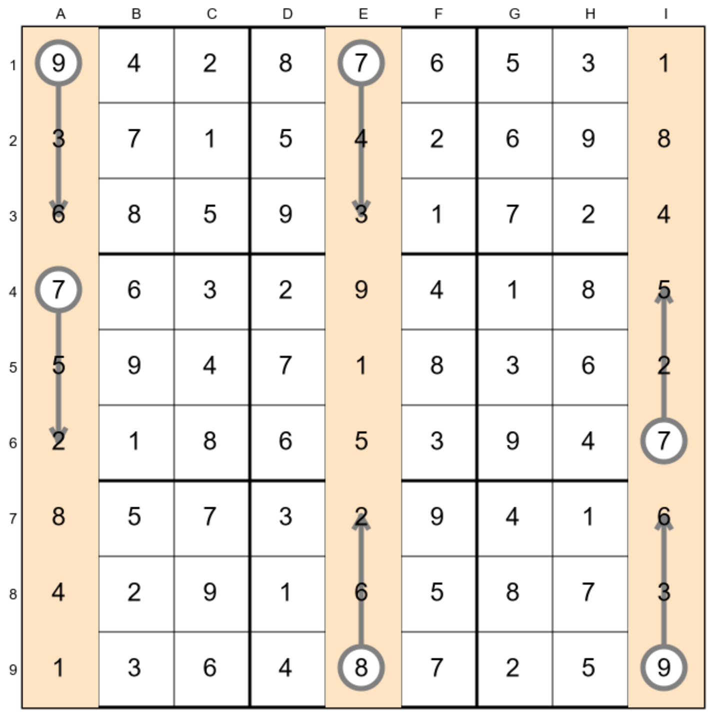
Model description not finished

14.2.6 See Also

14.2.7 indexing example timotab4: "Column Like I See 'Em" by Philip Newman

Video: `https://youtu.be/5H1O6dhAfpQ?si=RfHJEw_JnZOBb30N`





Rules: standard (Section 3.13) arrow (Section 9.2) indexing (Section 14.2)

14.3 Constraint sameSequence

14.3.1 Category: Other

14.3.2 Description

The values in matching entries in the two sequences must be the same.

14.3.3 Arguments

a List(Cell)

b List(Cell)

14.3.4 JSON Format

```
"sameSequence":[{"a":[3,12,21,30,39,48,57,66,75],"b":[65,74,73,64,55,56,57,66,75]},...],
```

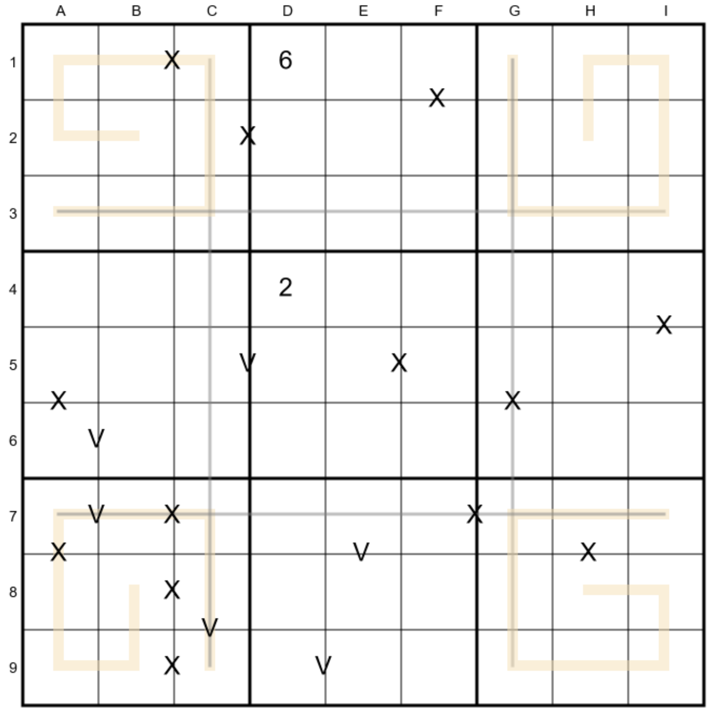
14.3.5 Model

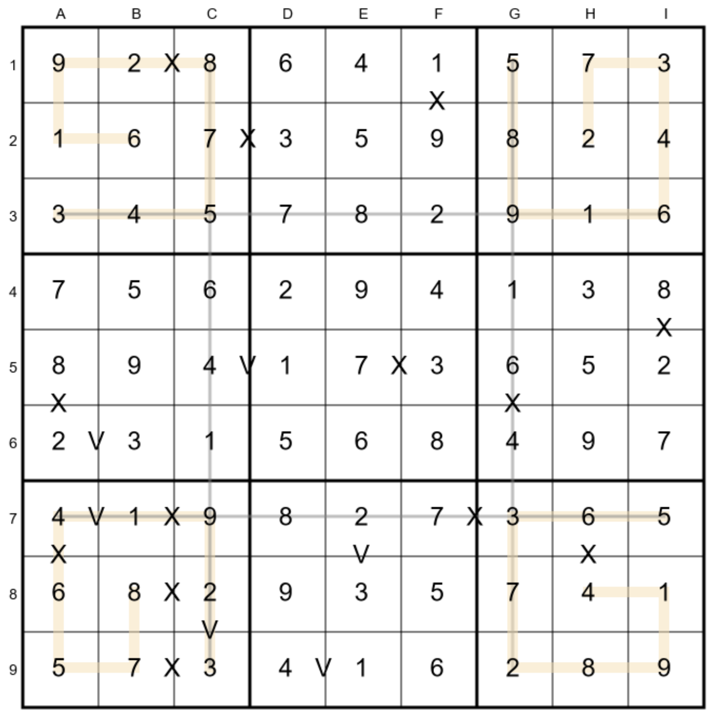
Model description not finished

14.3.6 See Also

14.3.7 sameSequence example ctc37: "Roll Out The Red Carpets" by Arlo Lipof

Video: `https://www.youtube.com/watch?v=xg0jC9uGrZ0`





Rules: standard (Section 3.13) sumV (Section 6.8) sumX (Section 6.9) sameSequence (Section 14.3)

A Missed Constraints

Table 14: Missed Constraints

Name	Category	Nr Problems
block	Base	0
column	Base	0
parquet	Base	1
row	Base	0
sudoku3D	Base	1
tripledoku1x1	Base	0
blockConsecutive	Global	1
entropyQuad	Global	0
globalEntropyQuad	Global	1
parityMirror	Global	1
rotationalSymmetrySum10	Global	1
strictBlockConsecutive	Global	1
strictEntropyQuad	Global	0
evenNeighbors	Node	0
orthoValue	Node	0
preset	Node	0
sameParityDigits	Node	2
gt	Pair	0
killerLine	Line	1
nValueArrow	Line	0
productSum	Line	1
skyscraperLine	Line	1
equalSplitLine	MultiLine	1
sloe	Outside	1
smallestDigit	Outside	0
irregularBuilder	Meta	1
blockIndexing	Other	0
boxBlackDot	Other	1
boxWhiteDot	Other	1
loop	Other	1

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